



Article No. 714

Chemistry⁰⁶²⁰

Paper-4

Topical Workbook with Mark Scheme

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Unit 1

THE PARTICULAR OF MATTER

IGCSE Chemistry P-4
Topical Workbook
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Unit 1: The Particulate Nature of Matter

1. O/N 21/P42/Q1

This question is about states of matter.

- (a) Complete the table, using ticks (✓) and crosses (x), to describe the properties of gases, liquids and solids.

state of matter	particles are touching	particles have random movement	particles are regularly arranged
gas	X	✓	X
liquid	✓	✓	X
solid	✓	✓	✓

[3]

- (b) Substances can change state.

- (i) Boiling and evaporation are two ways in which a liquid changes into a gas.

Describe **two** differences between boiling and evaporation.

1 Evaporation happens higher than freezing point while boiling happens at b.p

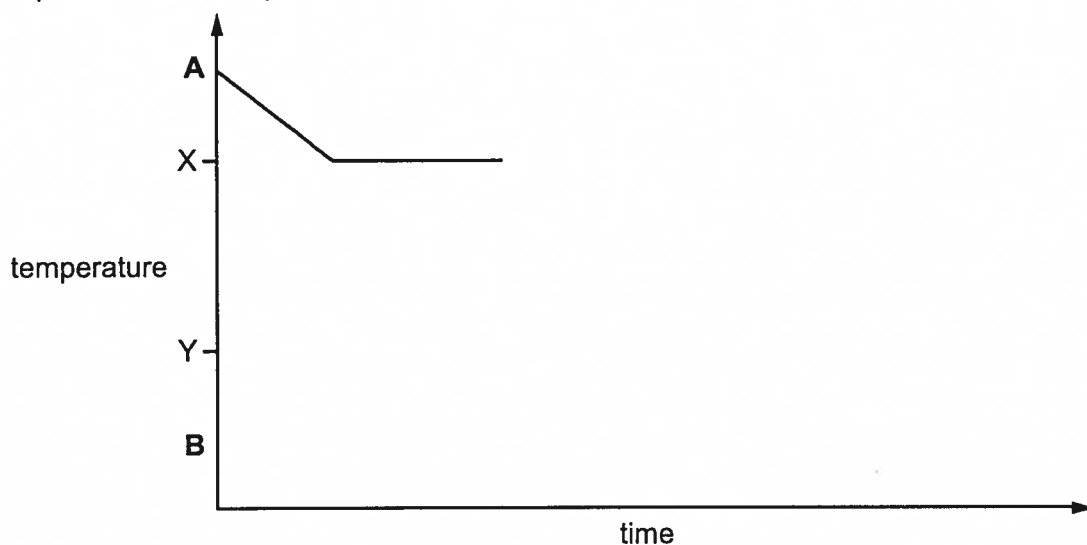
2 Evaporation happens at surface and boiling happens all over liquid..... [2]

- (ii) Name the change of state when:

- a gas becomes a liquid Condensation.....
- a solid becomes a gas. Sublimation..... [2]

- (c) A substance boils at temperature X and melts at temperature Y.

Complete the graph to show the change in temperature over time as the substance cools from temperature A to temperature B.



[2]

(d) A solution is a mixture of a solute and a solvent.

(i) Name the process when a solid substance mixes with a solvent to form a solution.

..... Dissolving[1]

(ii) Name the type of reaction when two solutions react to form an insoluble substance.

..... Precipitation[1]

2. MAR 21/P42/Q2

The elements shown are gases at room temperature and pressure.

hydrogen
nitrogen
oxygen
chlorine

(a) State which **one** of these gases is green.

..... Chlorine[1]

(b) The gases shown exist as diatomic molecules.

State the name of **another** element which has diatomic molecules and is a gas at room temperature and pressure.

..... Oxygen[1]

(c) When separate samples of each of these gases are placed in a container they will diffuse.

(i) Describe why these gases diffuse.

..... The air goes from high concentration area to lower concentration area.....[1]

(ii) State which of these four gases has the highest rate of diffusion.

Explain your answer.

gas hydrogen and nitrogen

explanation They have lower molar mass.....[2]

(d) Nitrogen, oxygen and other substances are found in clean, dry air.

(i) State the percentage of nitrogen in clean, dry air.

..... 78% approximetly[1]

(ii) Other than nitrogen and oxygen, identify another element found in clean, dry air.

..... hydrogen[1]

(iii) Identify a compound found in clean, dry air.

..... CO2[1]

(iv) Nitrogen and oxygen can be separated from liquid air.

State the name of this process.

..... Fractional distillation[2]

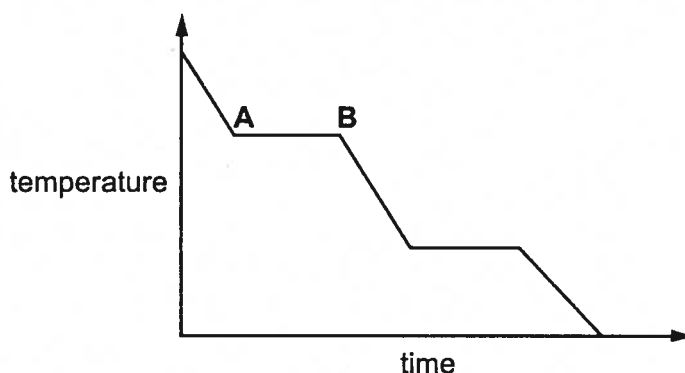
3. M/J 20/P43/Q5

(a) Complete the table about solids, liquids and gases.

	particle separation	particle arrangement	type of motion
solid	touching	regular	vibrate only
liquid	touching	irregular	random
gas	apart	random	random

[3]

(b) The graph shows the change in temperature as a sample of a gas is cooled.



Name the change of state taking place between A and B.

.....Condensation..... [1]

(c) A bottle of liquid perfume is left open at the front of a room.
After some time, the perfume is smelt at the back of the room.
Name the **two** physical processes taking place.

1 Diffusion

2 Brownian(bande ka nam tha jis ne microscope ke neche micro molecules dekhne ke motion)

[2]

4. M/J 19/P42/Q6/b(ii)

(a) When chlorine reacts with aqueous potassium bromide a displacement reaction occurs.

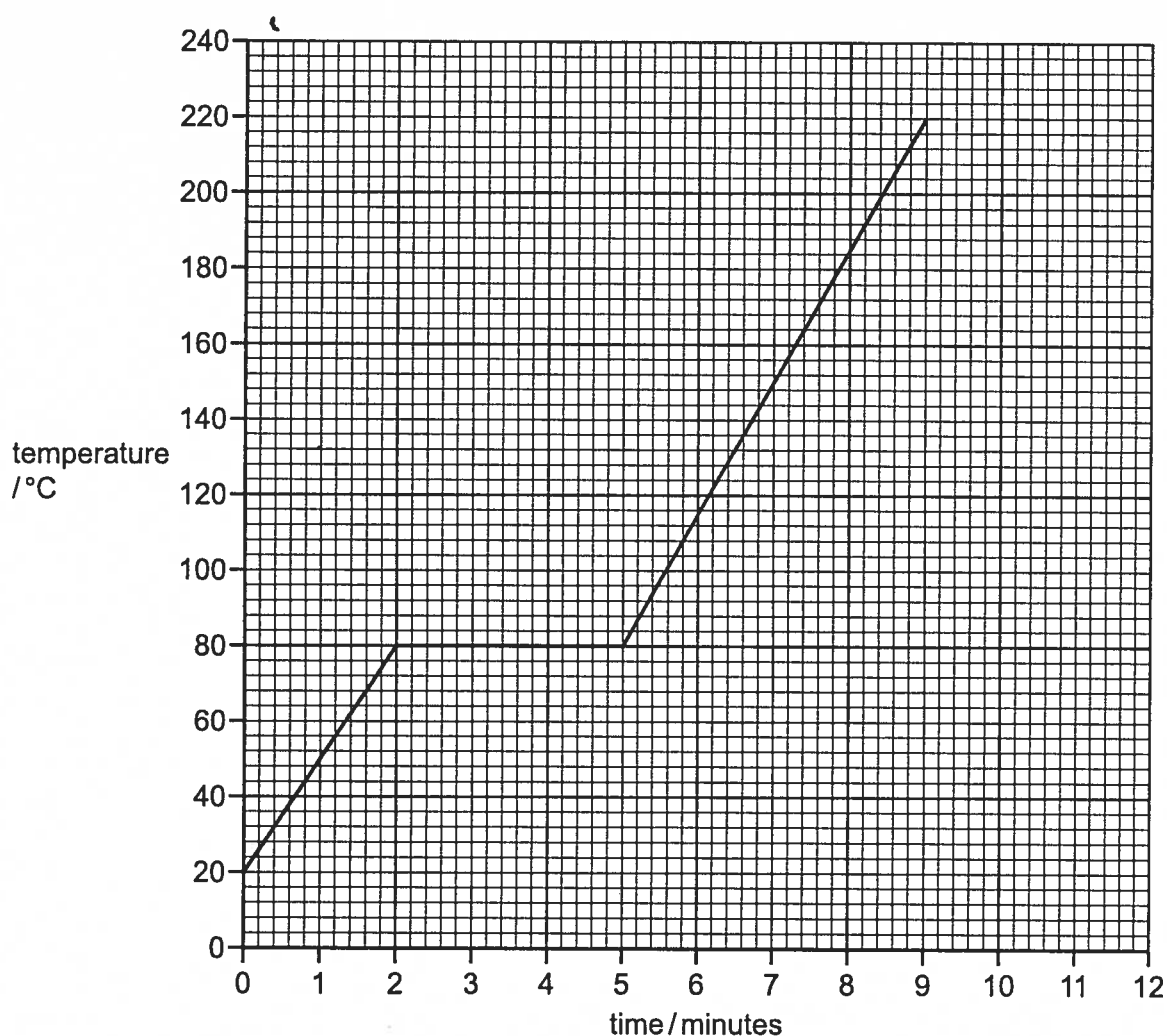
(i) Write a chemical equation for this reaction.

..... Displacement reaction [2]

5. M/J 19/P41/Q2

Z is a covalent substance. In an experiment, a sample of pure solid Z was continually heated for 11 minutes.

The graph shows how the temperature of the sample of pure Z changed during the first 9 minutes.



(a) What is the melting point of pure Z?

.....80..... °C [1]

(b) The sample of pure Z began to boil at 9 minutes. It was boiled for 2 minutes.

Use this information to sketch on the grid how the temperature of the sample of pure Z changed between 9 minutes and 11 minutes. [1]

(c) The sample of pure Z was continually heated between 2 minutes and 5 minutes.

Explain, in terms of attractive forces, why there was no increase in the temperature of the sample of pure Z between 2 minutes and 5 minutes.

This is because the z solid was melting and changing into liquid

.....
..... [2]

(d) Describe how the motion of particles of pure Z changed from 0 minutes to 2 minutes.

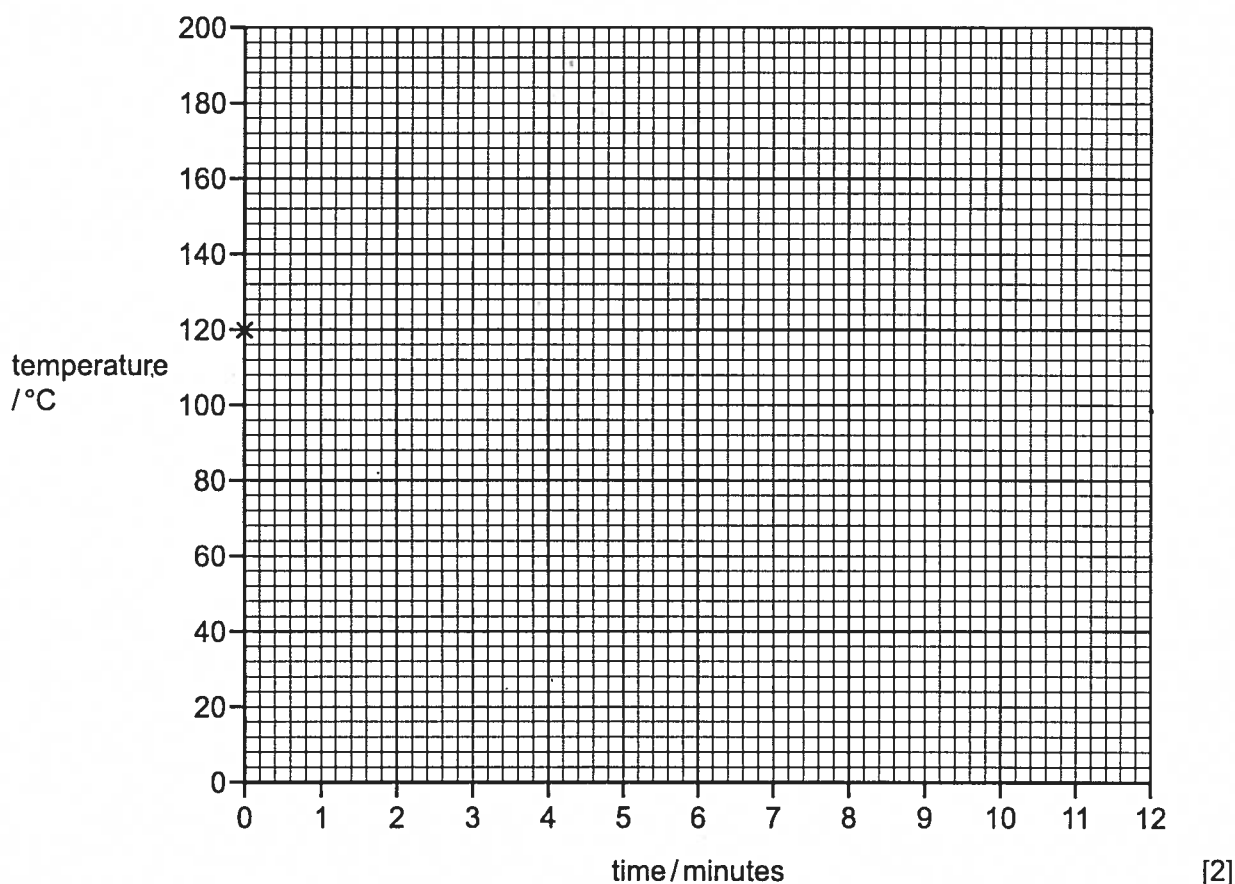
Random
..... [2]

- (e) The experiment was repeated using a solid sample of **impure Z**.
Suggest the differences, if any, in the melting point and boiling point of the sample of **impure Z** compared to the sample of **pure Z**.

melting point 75

boiling point 230 [2]

- (f) A sample of **pure Z** was allowed to cool from 120 °C to 20 °C. The total time taken was 8 minutes. Starting from point **x**, sketch on the grid how the temperature of the sample of **pure Z** changed between 0 minutes and 8 minutes.



6. MAR 19/P42/Q1/a

Period 3 of the Periodic Table is shown.

sodium	magnesium	aluminium	silicon	phosphorus	sulfur	chlorine	argon
--------	-----------	-----------	---------	------------	--------	----------	-------

Answer the following questions using only these elements.
Each element may be used once, more than once or not at all.

State which element:

- (a) is a gas at room temperature and pressure

...chlorine..... [1]

7. MAR 19/P42/Q3/a,e(iv)

Fluorine is a Group VII element. Fluorine forms compounds with metals and non-metals.

- (a) Predict the physical state of fluorine at room temperature and pressure.

idk

[1]

- (b) Tetrafluoromethane and lead(II) fluoride are fluorides of Group IV elements. Some properties of tetrafluoromethane and lead(II) fluoride are shown in the table.

property	tetrafluoromethane	lead(II) fluoride
formula	CF_4	
melting point/ $^{\circ}\text{C}$	-184	855
boiling point/ $^{\circ}\text{C}$	-127	1290
conduction of electricity when solid	non-conductor	non-conductor
conduction of electricity when molten	non-conductor	good conductor

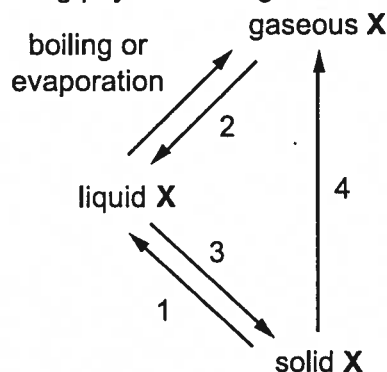
- (i) Explain, in terms of attractive forces between particles, why lead(II) fluoride has a much higher melting point than tetrafluoromethane.
In your answer refer to the types of attractive forces between particles and their relative strengths.

.....There is ionic bond in lead(2) fluorine. It has a much higher melting point.....
than that of tetrafluoromethane which has covalent bond.

[3]

8. O/N 18/P42/Q1

Element X can undergo the following physical changes.



- (a) (i) Give the scientific name for each of the numbered physical changes.

- 1 Melting
 2 Condensation
 3 Freezing
 4 Sublimation [4]

- (ii) Explain why the changes shown are physical changes.

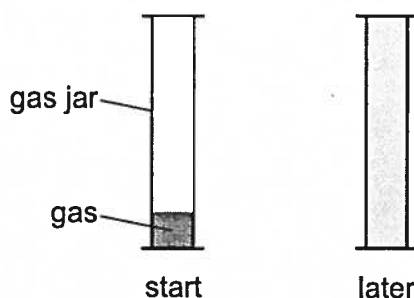
..... They can be done physically. [1]

- (iii) One difference between boiling and evaporation is the rate at which the processes occur.
State **one** other difference between boiling and evaporation.

- Evaporation happens at surface while boiling point happens all over liquid [1]
- (b) Describe the separation, arrangement and motion of particles of element X in the solid state.
- separation IDK [1]
- arrangement Particular arrangement [1]
- motion RANDOM [3]
- (c) Element X is a Group I metal. It burns in air to form an oxide X_2O .
Write a chemical equation for this reaction.
- Na [2]

9. O/N 17/P42/Q1

- (a) Dust particles in the air move around in a random way.
- (i) What term describes the random movement of the dust particles?
- Diffusion [1]
- (ii) Identify the particles in the air which cause the random movement of the dust particles.
- random [2]
- (iii) Explain why the dust particles move in this way.
- Diffusion and they have random motion [2]
- (b) When chlorine gas, Cl_2 , is put into a gas jar, it spreads out to fill the gas jar.
When bromine gas, Br_2 , is put into a gas jar, it also spreads out to fill the gas jar.
The process takes longer for bromine gas than for chlorine gas.



- (i) What term describes the way that the gas particles spread out?
- Diffusion [1]
- (ii) Use data from the Periodic Table to explain why bromine gas takes longer to fill a gas jar than chlorine gas.
- Because bromine is more molar mass than chlorine [2]

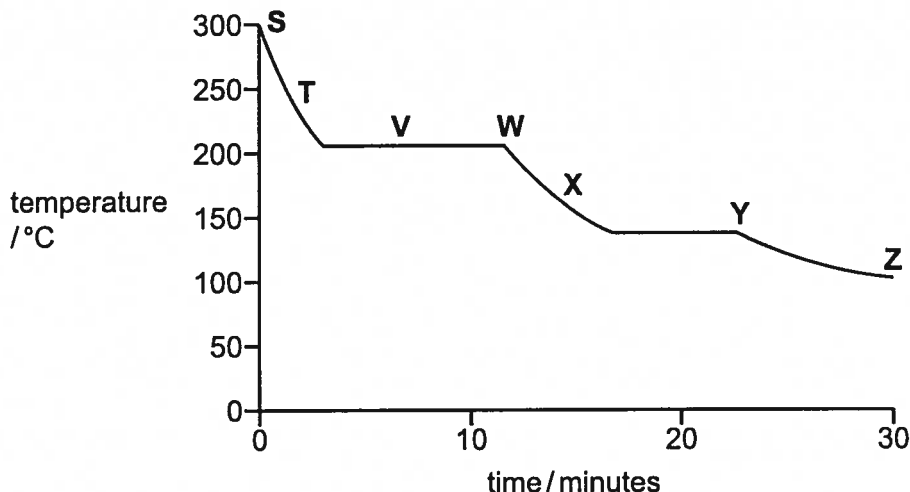
- (iii) Explain why increasing the temperature increases the rate at which the gas particles spread out.

This is due to increasing of kinetic energy.

[1]

10. O/N 17/P41/Q2

The graph shows how the temperature of a substance changes as it is cooled over a period of 30 minutes. The substance is a gas at the start.



Each letter on the graph may be used once, more than once or not at all.

- (a) Which letter, S, T, V, W, X, Y or Z, shows when

- (i) the particles in the substance have the most kinetic energy,

S [1]

- (ii) the particles in the substance are furthest apart,

S [1]

- (iii) the substance exists as both a gas and a liquid?

W [1]

- (b) Use the graph to estimate the freezing point of the substance.

100 °C [1]

- (c) Name the change of state directly from a solid to a gas.

Sublimation [1]

- (d) When smoke is viewed through a microscope, the smoke particles in the air appear to jump around.

- (i) What term describes this movement of the smoke particles?

Diffusion and random motion [1]

- (ii) Explain why the smoke particles move in this way.

Elbert said they move in random motion

[2]

11. O/N 16/P42/Q1

Particles behave differently when in different physical states.

(a) Solids have a fixed volume and a definite shape.

Gases have no fixed volume and take the shape of the container.

Describe the volume and shape of liquids.

..... [1]

(b) Complete the table to show the separation, arrangement and movement of particles in each physical state.

state	separation of particles	arrangement of particles	movement of particles
solid	touching	regular	
liquid	touching one another	randomly arranged	move over one another
gas	away		

[6]

(c) Name the following changes of state.

(i) Ice turning into water.

..... [1]

(ii) Solid carbon dioxide turning directly into gaseous carbon dioxide at room temperature.

..... [1]

12. O/N 16/P42/Q5/a,b

Chlorine, bromine and iodine are halogens.

(a) Chlorine can be made in the laboratory by heating manganese(IV) oxide with concentrated hydrochloric acid.



Calculate the volume of $8.00 \text{ mol/dm}^3 \text{ HCl}(\text{aq})$ needed to react with 3.48 g of MnO_2 .

- moles of MnO_2 used

- moles of HCl needed

..... mol

- volume of HCl needed

..... mol

..... cm^3 [4]

(b) A student bubbled chlorine gas into a test-tube containing aqueous potassium bromide.

(i) Describe the colour change seen in the test-tube.

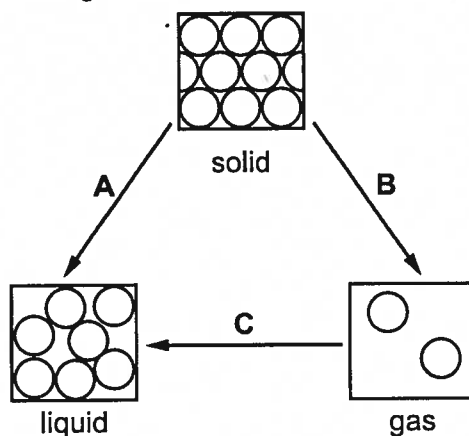
from to [2]

(ii) Complete the **ionic** equation for this reaction.
Include state symbols.



13. O/N 16/P41/Q2

Matter can exist as solid, liquid or gas. The arrows show some changes of state.



(a) Name the changes of state represented on the diagram.

(i) A [1]

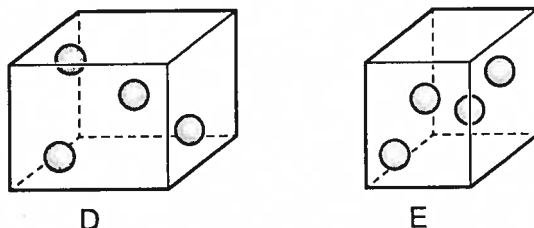
(ii) B [1]

(iii) C [1]

(b) Explain why energy has to be supplied to turn a liquid into a gas.

..... [1]

(c) The diagrams represent the same number of particles of a gas in two containers, D and E, which have different volumes. The two containers are at the same temperature.



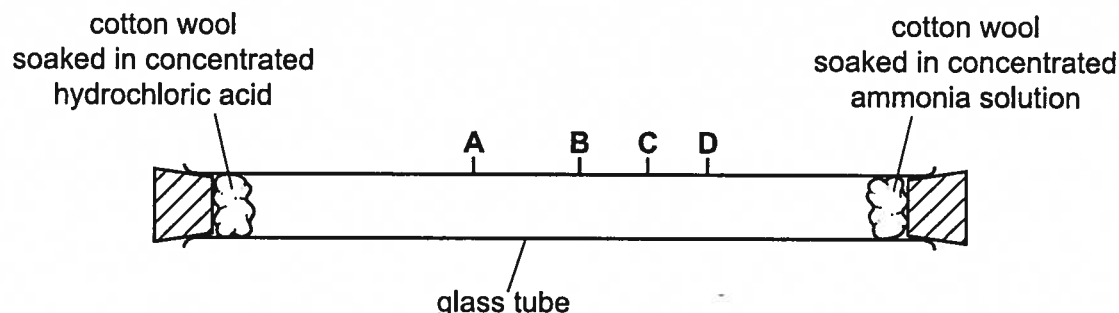
In which container will the pressure be higher? Explain your answer.

..... [1]

14. M/J 16/P43/Q6

Concentrated ammonia solution gives off ammonia gas. Concentrated hydrochloric acid gives off hydrogen chloride gas. Ammonia, NH_3 , and hydrogen chloride, HCl , are both colourless gases. Ammonia reacts with hydrogen chloride to make the white solid ammonium chloride.

Apparatus is set up as shown.



After ten minutes a white solid forms in the tube where the gases meet.

(a) (i) Write the chemical equation for the reaction of ammonia with hydrogen chloride.

..... [1]

(ii) Name the process by which the ammonia and hydrogen chloride gases move in the tube.

..... [1]

(iii) At which point, A, B, C or D, does the white solid form? Explain why the white solid forms at that point.

the solid forms at

explanation

..... [3]

(iv) The experiment was repeated at a higher temperature.

Predict how the results of the experiment would be different. Explain your answer.

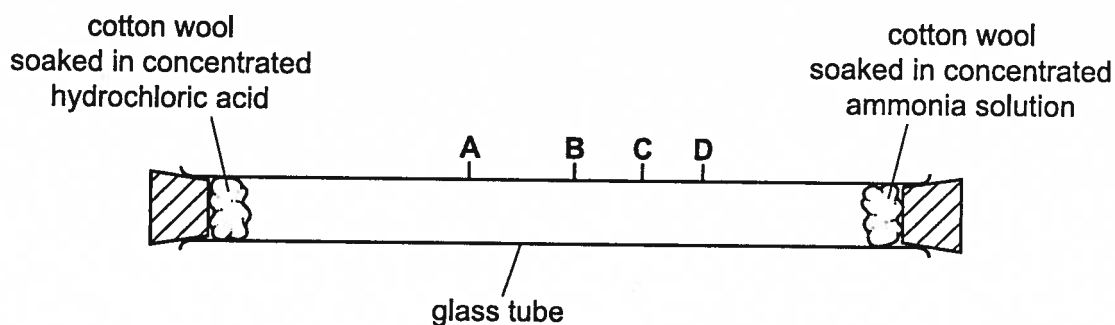
.....

.....

..... [3]

Concentrated ammonia solution gives off ammonia gas. Concentrated hydrochloric acid gives off hydrogen chloride gas. Ammonia, NH_3 , and hydrogen chloride, HCl , are both colourless gases. Ammonia reacts with hydrogen chloride to make the white solid ammonium chloride.

Apparatus is set up as shown.



After ten minutes a white solid forms in the tube where the gases meet.

(b) Some of the white solid is removed from the tube and dissolved in water.

Describe how the white solid could be tested to show it contains,

(i) ammonium ions,

test

result

..... [3]

(ii) chloride ions.

test

result

..... [3]

Answers Section

1. O/N 21/P42/Q1

(a) 1 mark for each correct row

State	touching	random movement	regularly arranged
Gas		✓	
Liquid	✓	✓	
Solid	✓		✓

3

(b) (i) boiling happens at a specific temperature (1)
boiling has bubbles (1)

2

(ii) condensation (1)
sublimation (1)

2

(c) one horizontal line level with Y (1)
two separate decreases before and after horizontal line (1)

2

(d) (i) dissolving

1

(ii) precipitation

1

2. MAR 21/P42/Q2

(a) chlorine

1

(b) fluorine

1

(c) (i) random motion of **molecules / particles**

1

(ii) hydrogen

lowest (relative) molecular mass

2

(d) (i) 78

1

(ii) argon / Ar

1

(iii) carbon dioxide

1

(iv) fractional (1)

distillation (1)

2

3. M/J 20/P43/Q5

(a)

	particle separation	particle arrangement	type of motion
solid	touching		
liquid		random	
gas			random

3

(b) condensing

1

(c) evaporation
diffusion

2

4. M/J 19/P42/Q6/b(ii)

(a) (i) $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$

OR

 $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$

M1 all formulae (1)

- M2 equation balanced correctly (1)** 2
- 5. M/J 19/P41/Q2**
- (a) 80(°C) (1) 1
 - (b) horizontal line from end of graph at minute 9 to minute 11 (1) 1
 - (c) energy is used to break bonds / overcome attraction (1)
between molecules (1) 2
 - (d) vibrations (1)
increase (1) 2
 - (e) melting point decreases (1)
boiling point increases (1) 2
 - (f) decrease from 120 °C to 80 °C and horizontal line at 80 °C (1)
decrease from horizontal line to finish at 20 °C at 8 mins (1) 2
- 6. MAR 19/P42/Q1/a**
- (a) chlorine / argon 1
- 7. MAR 19/P42/Q3/a,e(iv)**
- (a) gas / gaseous 1
 - (b) (i) **M1**(It or lead(II) fluoride) forces of attraction between ions / ionic bonds(1)
M2(tetrafluoromethane) forces of attraction between molecules(1)
M3 ionic bonds stronger than attractive forces between molecules /
ionic bonds need more energy to break than attractive
forces between molecules(1) 3
- 8. O/N 18/P42/Q1**
- (a) (i) **M1** Melting
M2 Condensing
M3 Freezing
M4 Sublimation 4
 - (ii) No new substances are made
or
The change can be reversed (by a physical process) 1
 - (iii) Boiling happens at a specific temperature
or
Evaporation happens over a range of temperatures 1
 - (b) **M1** Separation: Touching
M2 Arrangement: Regular
M3 Movement: Vibrate 3
 - (c) $4X + O_2 \rightarrow 2X_2O$
M1 Species
M2 Balance 2
- 9. O/N 17/P42/Q1**
- (a) (i) Brownian (motion) 1
 - (ii) molecules 1
nitrogen / N_2 / N OR oxygen / O_2 / O 1
 - (iii) nitrogen OR oxygen (particles) collide with / bombard / hit the dust (particles)
(the bombarding particles) move randomly 1
 - (b) (i) diffusion 1
 - (ii) Br_2 has an M_r of 160 AND Cl_2 has an M_r of 71 / bromine has an A_r of 80 AND

- chlorine has an A_r of 35.5 1
 (heavier) bromine (molecules / particles) diffuses more slowly 1
 (iii) particles have more energy / move faster 1
10. O/N 17/P41/Q2
- (a) (i) S 1
 (ii) S 1
 (iii) V 1
 (b) any value in the range 130–145 °C 1
 (c) sublimation 1
 (d) (i) Brownian motion 1
 (ii) nitrogen / oxygen / carbon dioxide / air molecules hit / bombard the smoke particles 1
 (the bombarding particles) move randomly 1
11. O/N 16/P42/Q1
- (a) fixed volume **AND** take the shape of the container 1
- (b)
- | | | | |
|--------|--------------|---------|---------|
| solid | touching | regular | vibrate |
| liquid | | | |
| gas | not touching | random | random |
- 6
- (c) (i) melting 1
 (ii) sublimation 1
12. O/N 16/P42/Q5/a,b
- (a) 20 cm³
 M1 M_r of MnO₂: 87
 M2 moles of MnO₂ used: $3.48 / 87 = 0.04$
 M3 moles of HCl needed: $0.04 \times 4 = 0.16$
 M4 volume of HCl needed: $(0.16 / 8.0) \times 1000$ **AND** 20 cm³ 4
- (b) (i) from colourless
 to yellow / orange / brown
 (ii) $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{Cl}^-(\text{aq})$ 2
 M1 (aq) as state symbols for the two products given
 M2 correct products
 M3 balancing 3
13. O/N 16/P41/Q2
- (a) (i) melt(ing) 1
 (ii) sublimation/sublime 1
 (iii) condensing/condensation 1
 (b) overcome/break the attractive forces 1
 (c) **E AND** particles hit the walls (of the container) more often 1
14. M/J 16/P43/Q6
- (a) (i) $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$; 1
 (ii) diffusion; 1

- (iii) solid forms at: A;
explanation: ammonia molecules/particles have a smaller mass; (and so) move/
diffuse faster; 3
- (iv) M1 solid forms in less time/faster/quicker;
M2 particles/molecules have more energy;
M3 (and so) move faster/diffuse faster; 3
- (b) (i) test: add sodium hydroxide (solution and warm); [1]
result: test gas/ammonia with (red) litmus/Universal Indicator/pH paper; indicator
turns blue/ammonia produced; [2]
- (ii) test: add silver nitrate (solution); [1]
result: add (dilute) nitric acid; white precipitate; [2]

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Unit 2

EXPERIMENTAL TECHNIQUES

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

2.1 Purity

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Unit 2: Experimental Techniques

2.1: Purity

1. M/J 21/P41/Q1

Give the name of the process that is used:

- (a) to produce ammonia from nitrogen

habber process [1]

- (b) to separate nitrogen from liquid air

fractional distillation [1]

- (c) to produce bromine from molten lead(II) bromide

..... [1]

- (d) to separate an undissolved solid from an aqueous solution

sieve [1]

- (e) to produce amino acids from proteins

hydrolosis [1]

- (f) to separate a mixture of amino acids.

..... [1]

2. M/J 21/P43/Q1

Give the name of the process that is used:

- (a) to produce large molecules from monomers

..... [1]

- (b) to separate oxygen from liquid air

fractional distillation [1]

- (c) to make ethanol from glucose

..... [1]

- (d) to separate water from aqueous sodium chloride

filtration [1]

- (e) to produce aluminium from aluminium oxide in molten cryolite

..... [1]

- (f) to separate the products of hydrolysis of long chain carbohydrates

..... [1]

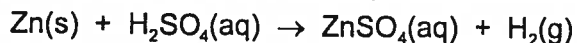
- (g) to separate an aqueous solution from an undissolved solid.

sieve [1]

3. O/N 20/P43/Q4/c(i,ii)

Zinc is manufactured from zinc blende. Zinc blende is an ore which consists mainly of zinc sulfide, ZnS.

- (a) Zinc reacts with dilute sulfuric acid to produce aqueous zinc sulfate.



Hydrated zinc sulfate crystals are made from aqueous zinc sulfate.

Step 1 Solid zinc is added to dilute sulfuric acid until zinc is in excess.

Step 2 Excess zinc is separated from aqueous zinc sulfate by filtration.

Step 3 Aqueous zinc sulfate is heated until the solution is saturated.

Step 4 The saturated solution is allowed to cool and crystallise.

Step 5 The crystals are removed and dried.

- (i) Name the residue in **step 2**.

zinc

[1]

- (ii) In **step 3**, a saturated solution is produced.

Describe what a saturated solution is.

Saturated solution consist of more solute

[2]

4. M/J 20/P42/Q1/a(i,ii,iii,v)

- (a) Give the name of the process that:

- (i) occurs when a gas turns into a liquid

condensation

[1]

- (ii) occurs when a solid turns into a gas without first forming a liquid

sublimation

[1]

- (iii) is used to separate a mixture of liquids with different boiling points

fractional distillation

[1]

- (iv) is used to separate a mixture of amino acids.

hydrolysis

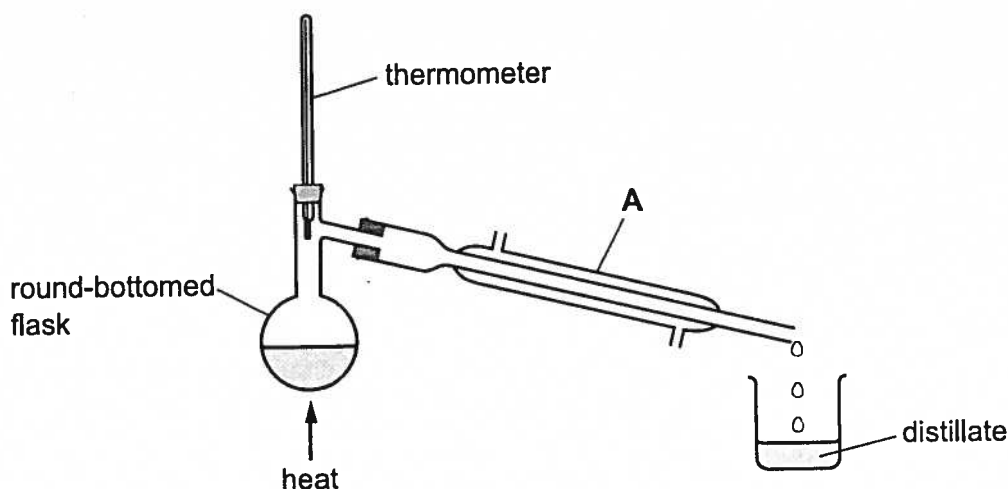
[1]

5. M/J 20/P41/Q5/c

- (a) Alcohols can be partially oxidised to form aldehydes.

Aldehydes are a homologous series of organic compounds.

Partial oxidation is achieved by reacting an alcohol with the oxidising agent in distillation apparatus as shown.



- (i) Name apparatus A.
 condenser [1]
- (ii) On the diagram, use **one** arrow to show where water enters apparatus A. [1]

6. M/J 20/P43/Q4

- (a) Filtration and chlorination are two stages in water treatment.

State the purpose of each stage.

filtration filters out any residue present in [1]

chlorination kills bacteria and fungi etc [2]

- (b) A student uses anhydrous copper(II) sulfate to test for the presence of water.

- (i) What colour change is seen if water is present?

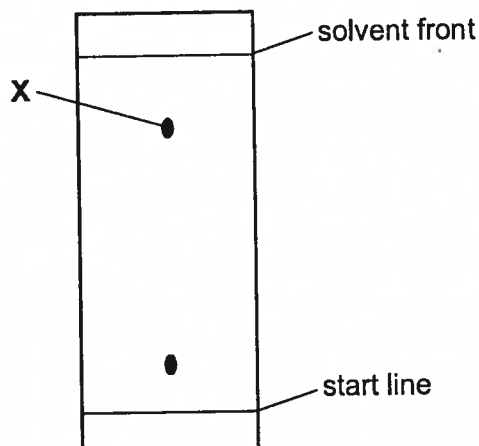
from to [2]

- (ii) The purity of a sample of water can be assessed by measuring its boiling point.

How is the boiling point of water affected by impurities?

..... [1]

- (c) Chromatography can be used to test the purity of substances.
 The diagram shows the chromatogram of a coloured substance.



(i) How does this chromatogram show that this substance is **not** pure?

..... [1]

(ii) Draw a circle round the correct R_f value for the spot labelled X.

0.2 0.4 0.8 1.2 [1]

(iii) State how a colourless substance can be made visible on a chromatogram.

..... [1]

7. M/J 19/P42/Q5/b,c,d,e,f,g

Copper(II) sulfate crystals, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, are hydrated.

Copper(II) sulfate crystals are made by reacting copper(II) carbonate with dilute sulfuric acid.

The equation for the overall process is shown.



step 1 Powdered solid copper(II) carbonate is added to 50.0 cm^3 of 0.05 mol/dm^3 sulfuric acid until the copper(II) carbonate is in excess.

step 2 The excess of copper(II) carbonate is separated from the aqueous copper(II) sulfate.

step 3 The aqueous copper(II) sulfate is heated until the solution is saturated.

step 4 The solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

(a) **Steps 1–5** were done correctly but the mass of crystals obtained was less than the maximum mass.

Explain why.

..... [1]

(b) State **two** observations that would indicate that the copper(II) carbonate is in excess in **step 1**.

1

2 [2]

(c) When the reaction in **step 1** is done using lumps of copper(II) carbonate instead of powder, the rate of reaction decreases. All other conditions are kept the same.

Give a reason for this. Explain your answer in terms of particles.

.....

.....

..... [2]

(d) Name a different substance, other than copper(II) carbonate, that could be added to dilute sulfuric acid to produce copper(II) sulfate in **step 1**.

..... [1]

(e) Name the process used to separate the aqueous copper(II) sulfate from the excess of copper(II) carbonate in **step 2**

..... [1]

(f) The solution of aqueous copper(II) sulfate was heated until it was saturated in **step 3**.

(i) Suggest what is meant by the term *saturated solution*.

.....
..... [2]

(ii) What evidence would show that the solution was saturated in **step 3**?

..... [1]

(iii) Why should the aqueous copper(II) sulfate **not** be heated to dryness in **step 3**?

..... [1]

8. M/J 19/P41/Q4/d,e(i)

Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Both ethanoic acid and hydrochloric acid dissociate in aqueous solution.

(a) A student prepares crystals of magnesium chloride by adding an excess of magnesium carbonate to 50.00 cm³ of 2.00 mol/dm³ hydrochloric acid.

The student filters the mixture and rinses the residue.

(i) Why does the student add an **excess** of magnesium carbonate?

..... [1]

(ii) Why does the student rinse the residue?

..... [1]

(iii) Describe how the student would obtain pure crystals of magnesium chloride from the filtrate.

.....
.....
..... [3]

(b) Silver chloride, AgCl, is insoluble. It can be made by a precipitation reaction between aqueous barium chloride and a suitable aqueous silver salt.

(i) What is meant by the term *precipitate*?

.....
..... [2]

9. M/J 19/P43/Q3/d(iii)

Lead(II) azide is insoluble in water. Solid lead(II) azide can be made in a precipitation reaction between aqueous lead(II) nitrate and aqueous sodium azide.

Lead(II) azide has the formula Pb(N₃)₂.

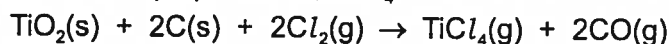
(a) (i) Describe how you could obtain a sample of lead(II) azide that is **not** contaminated with any soluble salts from the reaction mixture.

.....
..... [2]

10. MAR 19/P42/Q5/a(i),d(ii)

Titanium is extracted from an ore called rutile. Rutile is an impure form of titanium(IV)oxide, TiO_2 .

- (a) Rutile is mixed with coke and heated in a furnace through which chlorine gas is passed. The product is gaseous titanium(IV) chloride, TiCl_4 .



The gaseous titanium(IV)chloride produced is condensed into the liquid state. The titanium(IV) chloride is then separated from liquid impurities.

- (i) Suggest the name of the process by which liquid titanium(IV)chloride could be separated from the liquid impurities.

..... [1]

- (b) After titanium(IV) chloride is heated with magnesium, the unreacted magnesium is removed by adding an excess of dilute hydrochloric acid to the mixture.

The dilute hydrochloric acid also dissolves the magnesium chloride.

The dilute hydrochloric acid does **not** react with the titanium or dissolve it.

- (i) Name the process that is used to separate the titanium from the mixture after all the magnesium has been removed.

..... [1]

11. M/J 18/P42/Q1

Give the name of the process that is used:

- (a) to obtain water from aqueous sodium chloride

..... [1]

- (b) to produce lead from molten lead(II) bromide

..... [1]

- (c) to separate an insoluble solid from a liquid

..... [1]

- (d) to separate the components of petroleum

..... [1]

- (e) to separate a mixture of coloured dyes.

..... [1]

12. M/J 18/P41/Q1

Substances can be classified as elements, compounds or mixtures.

- (a) What is meant by the term *compound*?

Compound mean that it consist of two element chemically combine.

.....

..... [2]

- (b) Mixtures can be separated by physical processes.

A sequence of physical processes can be used to separate common salt (sodium chloride) from a mixture containing sand and common salt only.

Give the order and the correct scientific term for the physical processes used to separate the common salt from the mixture.

.....

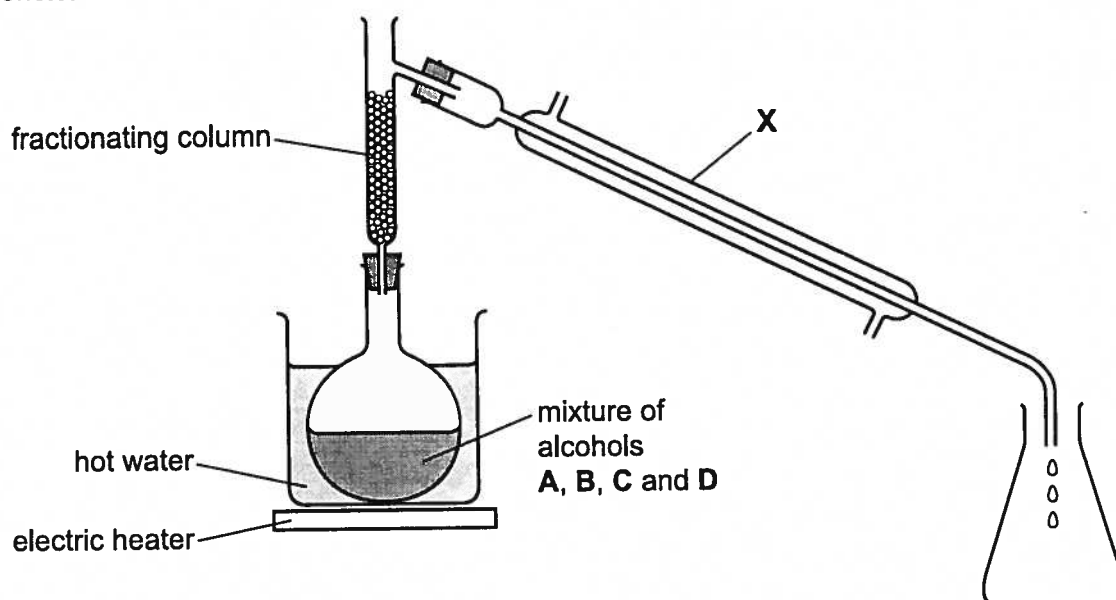
I F/D!

- 1 **Distillation**
- 2 **Distillation**
- 3 [4]

The boiling points of four different alcohols, **A**, **B**, **C** and **D**, are shown.

alcohol	A	B	C	D
boiling point/°C	56	78	122	160

- (c) A student suggested that the apparatus shown could be used to separate the mixture of alcohols.



- (i) Apparatus **X** needs to have cold water flowing through it.
- Draw an arrow on the diagram to show where the cold water enters apparatus **X**.
 - Name apparatus **X**.
- [2]
- (ii) Part of the fractionating column is missing. This means that the experiment will not work.
- Draw on the diagram the part of the fractionating column which is missing.
 - Explain why the experiment will **not** work with this part of the fractionating column missing.
- [2]
- (iii) Suggest why a Bunsen burner is **not** used to heat the flask.
- [1]
- (iv) A hot water bath cannot be used to separate alcohols **C** and **D**.
Explain why.

.....
..... [2]

13. MAR 18/P42/Q1/c

This question is about gases.

(a) Air is a mixture. Nitrogen and oxygen are the two most common gases in air.

(i) What is meant by the term *mixture*?

..... [1]

(ii) State the percentage of oxygen, to the nearest whole number, in clean dry air.

..... [1]

(iii) Describe the steps in the industrial process which enables nitrogen and oxygen to be separated from clean dry air.

Use scientific terms in your answer.

.....
.....
..... [3]

(iv) Which physical property of nitrogen and oxygen allows them to be separated?

..... [1]

14. MAR 18/P42/Q5/e

Alcohols are a 'family' of organic molecules which have the same general formula.

(a) Chromatography can be used to identify simple sugars in a mixture.

A student analysed a mixture of simple sugars by chromatography. All the simple sugars in the mixture were colourless.

(i) What is the name given to the type of substance used to identify the positions of the simple sugars on the chromatogram?

..... [1]

(ii) The student calculated the R_f value of a spot on the chromatogram.

Complete the expression for the R_f value of the spot.

$R_f =$ [1]

(iii) How could a student identify a simple sugar from its R_f value?

..... [1]

(iv) Sometimes not all the substances in a mixture can be identified from the chromatogram produced.

Explain why this may happen.

..... [1]

15. M/J 17/P42/Q1

(a) State the name of the process that is used to

(i) separate oxygen from liquid air,

..... [1]

(ii) separate the individual dyes in ink,

..... [1]

(iii) produce ethanol from simple sugars,

..... [1]

(iv) obtain water from aqueous sodium chloride,

..... [1]

(v) separate the precipitate formed when aqueous silver nitrate is added to aqueous sodium chloride.

..... [1]

(b) State what is meant by the terms

(i) *element*,

..... [1]

(ii) *compound*,

..... [1]

(iii) *ion*.

..... [1]

16. O/N 16/P42/Q7/c

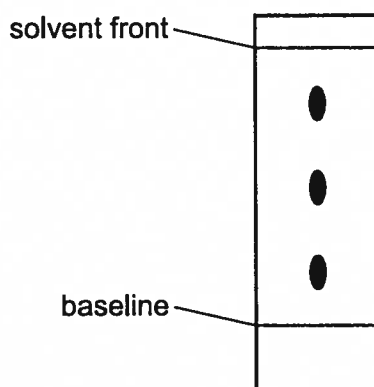
Proteins are a major constituent of food.

Proteins are polymers.

(a) A colourless mixture of amino acids was separated by chromatography.

Amino acid X has an R_f value of 0.8.

The chromatogram of the mixture after treatment with a locating agent is shown.



- (i) How is an R_f value calculated?

$$R_f = \qquad \qquad \qquad [1]$$

- (ii) On the diagram put a ring around the spot caused by amino acid X.

[1]

- (iii) Describe how you would perform a chromatography experiment to produce the chromatogram shown in (c). Assume you have been given the mixture of amino acids and a suitable locating agent. You are provided with common laboratory apparatus.

.....
.....
..... [3]

Answers Section

- | | |
|--|---|
| 1. M/J 21/P41/Q1 | 1 |
| (a) Haber (process) | 1 |
| (b) fractional distillation | 1 |
| (c) electrolysis | 1 |
| (d) filtration | 1 |
| (e) hydrolysis | 1 |
| (f) chromatography | 1 |
| 2. M/J 21/P43/Q1 | 1 |
| (a) polymerisation | 1 |
| (b) fractional distillation | 1 |
| (c) fermentation | 1 |
| (d) distillation | 1 |
| (e) electrolysis | 1 |
| (f) chromatography | 1 |
| (g) filtration | 1 |
| 3. O/N 20/P43/Q4/c(i,ii) | 1 |
| (a) (i) zinc | |
| (ii) (a solution containing the) maximum amount of solute
dissolved / no more solute can dissolve (1)
at a given temperature (1) | 2 |
| 4. M/J 20/P42/Q1/a(i,ii,iii,v) | 1 |
| (a) (i) condensation | 1 |
| (ii) sublimation | 1 |
| (iii) fractional distillation | 1 |
| (iv) chromatography | 1 |
| 5. M/J 20/P41/Q5/c | 1 |
| (a) (i) (Liebig) condenser | 1 |
| (ii) arrow at the lower inlet | |
| 6. M/J 20/P43/Q4 | 1 |
| (a) (filtration:) remove solids from water / remove insoluble substances
(chlorination:) sterilises / kill microbes / prevent illness | 2 |
| (b) (i) white to blue | 2 |
| (ii) higher boiling point / greater than 100°C | 1 |
| (c) (i) more than one spot | 1 |
| (ii) 0.8 (circled) | 1 |
| (iii) use a locating agent | 1 |
| 7. M/J 19/P42/Q5/b,c,d,e,f,g | 1 |
| (a) some copper(II) sulfate remains in solution / some copper(II) sulfate does not form crystals
OR some of the crystals decomposed OR some crystals lost in transfer | 1 |
| (b) M1 no more bubbling / fizzing / effervescence (1)
M2 solid or powder stops dissolving (1) | 2 |

- (c) **M1** (lumps have) smaller surface area **OR powder** has larger surface area (1)
M2 (lumps have) fewer collisions per unit time / less collision frequency **OR powder**
 has more collisions per unit time / more collision frequency 2
- (d) copper(II) oxide **or** copper(II) hydroxide 1
- (e) filtration 1
- (f) (i) **M1** containing the maximum amount of **dissolved** solute / no more solute can
dissolve (1)
M2 at any given temperature (1) 2
- (ii) when crystals form on a glass rod withdrawn from solution / on a sample
 of solution placed on microscope slide etc. 1
- (iii) (heating to dryness) would remove water **of crystallisation** 1
8. **M/J 19/P41/Q4/d,e(i)**
- (a) (i) to remove the acid / make sure all the acid is used up / no acid is left over 1
- (ii) to make sure all the filtrate / MgCl_2 / salt goes through / no MgCl_2 left behind 1
- (iii) evaporation mark (1)
 the starting of crystallisation mark (1)
 drying the crystals mark (1) 3
- (b) (i) a solid (1)
 which forms when two solutions are mixed / reacted / added (1) 2
9. **M/J 19/P43/Q3/d(iii)**
- (a) (i) **M1** filter
M2 wash with water 2
10. **MAR 19/P42/Q5/a(i),d(ii)**
- (a) (i) fractional distillation 1
- (b) (i) filtration 1
11. **M/J 18/P42/Q1**
- (a) distillation [1]
- (b) electrolysis [1]
- (c) filtration [1]
- (d) fractional distillation / fractionation [1]
- (e) chromatography [1]
12. **M/J 18/P41/Q1**
- (a) a substance made from two (or more) **elements** 1
 chemically combined 1
- (b) dissolving 1
 filtration 1
 evaporation / crystallisation 1
 three correct stages in the correct order 1
- (c) (i) condenser 1
 arrow pointing into lower aperture **only** 1
- (ii) stopper shown in diagram 1
 gases or vapours escape 1
- (iii) (mixture is) (in)flammable 1
- (iv) water bath cannot exceed 100 (°C) 1
C AND D have a boiling point above 100 (°C) 1
13. **MAR 18/P42/Q1/c**
- (a) (i) two (or more) substances not chemically combined 1

- (ii) 21(%) 1
- (iii) fractional distillation of liquid air
M1 air is made into a liquid
M2 (allow air to) boil or evaporate
M3 condense the vapours / collect the vapours in order (of evaporation)
fractional distillation gets M2 and M3 3
(iv) boiling points 1
14. MAR 18/P42/Q5/e 1
- (a) (i) locating ((re)agent) 1
- (ii) distance travelled by substance
distance travelled by solvent 1
1
- (iii) compare to known data 1
- (iv) similar R_f values 1
15. M/J 17/P42/Q1 1
- (a) (i) fractional distillation 1
- (ii) chromatography 1
- (iii) fermentation / ferment 1
- (iv) (simple) distillation / distil 1
- (v) filtration / decantation / centrifugation 1
- (b) (i) (substance that) cannot be split up /
broken down into (two or more) simpler substances by chemical means
OR
(substance) made of **atoms** with the same atomic number / number of protons /
proton number 1
- (ii) (two or more) elements joined or combined or bonded (together) 1
- (iii) (particle) containing different numbers of protons and electrons
OR
atom or group of atoms that has gained or lost an electron / electrons 1
16. O/N 16/P42/Q7/c
- (a) (i) distance moved by substance
distance moved by solvent (front) 1
1
- (ii) circle around top spot
- (iii) mixture of amino acids is placed as a spot onto a (pencil) baseline
placed into a (suitable) solvent / water
a locating agent is added to the (finished) chromatogram (to reveal spots) 3



Unit 3

ATOMS, ELEMENTS AND COMPOUNDS

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 3.1 Atomic structure and the periodic table
- 3.2 Chemical bonding

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Unit 3: Atoms, Elements And Compounds

3.1: Atomic structure and the periodic table

1. O/N 22/P43/Q1

Atoms and ions are made from small particles called electrons, neutrons and protons.

(a) Complete the table.

particle	relative charge	relative mass
electron	-1	$\frac{1}{1840}$
neutron		
proton		

[2]

(b) Information about atoms and ions, **A**, **B** and **C**, is shown in the table.
Complete the table.

atom or ion	number of electrons	number of neutrons	number of protons	symbol
A	18		20	$^{42}_{20}\text{Ca}^{2+}$
B		18		$^{35}_{17}\text{Cl}$
C	18	16	16	

[6]

2. M/J 22/P41/Q2

(a) Atoms are made of protons, neutrons and electrons. Atoms of the same element are known as isotopes.

(i) Complete the table.

particle	relative charge	relative mass
electron		$\frac{1}{1840}$
neutron		
proton	+1	

[2]

(ii) $^{24}_{12}\text{Mg}$ and $^{25}_{12}\text{Mg}$ are isotopes of magnesium.

Complete the table to show the numbers of electrons, neutrons and protons in these isotopes of magnesium.

isotope	number of electrons	number of neutrons	number of protons
$^{24}_{12}\text{Mg}$			
$^{25}_{12}\text{Mg}$			

[2]

(iii) Explain why magnesium ions have a charge of 2+.

..... [1]

(b) Mg^{2+} ions have the electronic structure 2,8.

Give the formula of the following particles which have the same electronic structure as Mg^{2+} ions.

- a cation (positive ion)

.....

- an anion (negative ion)

.....

- an atom

..... [3]

3. M/J 22/P43/Q2

(a) $^{32}_{16}\text{S}$ and $^{33}_{16}\text{S}$ are isotopes of sulfur.

Use your knowledge of protons, neutrons and electrons to answer the following questions.

(i) Describe how these isotopes of sulfur are the same and how they are different.

same

.....

different

..... [3]

(ii) Explain why each of these isotopes have an overall charge of zero.

..... [1]

(iii) Explain why both isotopes have the same chemical properties.

..... [1]

(b) Sulfide ions, S^{2-} , have the electronic structure 2,8,8.

(i) Explain why sulfide ions have a charge of 2-.

..... [1]

(ii) Give the formula of:

- an anion which has the same electronic structure as S^{2-}

.....

- a cation which has the same electronic structure as S^{2-} .

..... [2]

4. O/N 21/P42/Q3

Atoms contain protons, neutrons and electrons.

- (a) Complete the table to show the relative mass and the relative charge of a proton, a neutron and an electron.

	relative mass	relative charge
proton		
neutron		
electron	$\frac{1}{1840}$	

[3]

- (b) The table shows the number of protons, neutrons and electrons in some atoms and ions. Complete the table.

atom or ion	number of protons	number of neutrons	number of electrons
$^{32}_{16}\text{S}$			
$^{39}_{19}\text{K}^+$			
	35	44	36

[5]

5. O/N 21/P41/Q2/a

This question is about copper and its compounds.

- (a) Copper has two different naturally occurring atoms, ^{63}Cu and ^{65}Cu .

- (i) State the term used for atoms of the same element with different nucleon numbers.

..... [1]

- (ii) The atomic number of copper is 29.

Complete the table to show the number of protons, neutrons and electrons in the particles of copper shown.

	^{63}Cu	$^{65}\text{Cu}^{2+}$
protons		
neutrons		
electrons		

[3]

- (iii) Relative atomic mass is the average mass of naturally occurring atoms of an element. The percentage of the naturally occurring atoms in a sample of copper is shown.

^{63}Cu	^{65}Cu
70%	30%

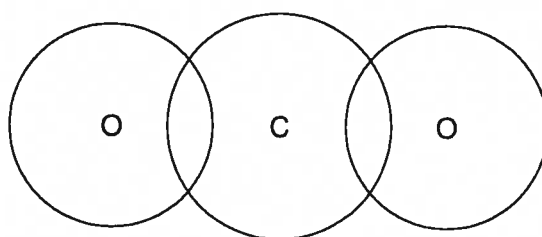
Deduce the relative atomic mass of copper in this sample.
Give your answer to **one** decimal place.

relative atomic mass = [2]

6. O/N 21/P43/Q3/f

(a) Lead(II) oxide and carbon dioxide are oxides of Group IV elements.

- (i) Complete the diagram to show the electron arrangement in one molecule of CO_2 .
Show only the outer electrons.



[2]

- (ii) The melting points of lead(II) oxide and carbon dioxide are shown.

	melting point/ $^{\circ}\text{C}$
lead(II) oxide	886
carbon dioxide	-56

Use your knowledge of structure and bonding to explain why lead(II) oxide has a much higher melting point than carbon dioxide.

Your answer should refer to:

- the types of particles involved
- the relative strength of the forces of attraction between the particles.

.....

 [3]

7. O/N 21/P43/Q4/a,b

Carbon is an important element.

- (a) Carbon exists as the isotopes $^{12}_6\text{C}$ and $^{13}_6\text{C}$.

Complete the table.

isotope	number of protons in one atom	number of electrons in one atom	number of neutrons in one atom
$^{12}_6\text{C}$			
$^{13}_6\text{C}$			

[2]

(b) Name **two** forms of the element carbon which have giant covalent structures.

..... and [1]

8. M/J 21/P42/Q2/a,b

Silver has an atomic number of 47.

(a) Naturally occurring atoms of silver are $^{107}_{47}\text{Ag}$ and $^{109}_{47}\text{Ag}$.

(i) State the name given to atoms of the same element with different nucleon numbers.

..... [1]

(ii) Complete the table to show the number of protons, neutrons and electrons in each atom and ion of silver shown.

	$^{107}_{47}\text{Ag}$	$^{109}_{47}\text{Ag}^+$
protons		
neutrons		
electrons		

[3]

(iii) Complete this definition of relative atomic mass.

Relative atomic mass is the mass of naturally occurring atoms of an element on a scale where the atom has a mass of exactly units. [3]

(iv) A sample of silver has a relative atomic mass of 108.0.

Deduce the percentage of ^{107}Ag present in this sample of silver.

..... [1]

(b) Silver nitrate is a salt of silver made by reacting silveroxide with an acid.

Write the formula of the acid which reacts with silver oxide to form silver nitrate.

..... [1]

9. M/J 21/P42/Q5/c

(a) Lithium nitride melts at 813°C . Nitrogen trifluoride melts at -206°C .

Explain in terms of attractive forces why lithium nitride has a much higher melting point than nitrogen trifluoride.

In your answer refer to the types of attractive forces between particles and their relative strengths.

.....

 [3]

10. M/J 21/P41/Q2

Complete the table to:

- deduce the number of protons, electrons and neutrons in the magnesium atom and copper ion shown
- identify the atom or ion represented by the final row.

	number of protons	number of electrons	number of neutrons
$^{25}_{12}\text{Mg}$	12		
$^{65}_{29}\text{Cu}^{2+}$			36
	17	18	20

[5]

11. MAR 21/P42/Q1The table shows the numbers of protons, neutrons and electrons in particles **A** to **I**.

particle	protons	neutrons	electrons
A	1	0	0
B	6	6	6
C	6	8	6
D	10	10	10
E	16	16	18
F	17	18	17
G	18	22	18
H	19	20	19
I	20	20	18

Answer the following questions about particles **A** to **I**. Each letter may be used once, more than once or not at all.**(a)** State which of the particles **A** to **I**:

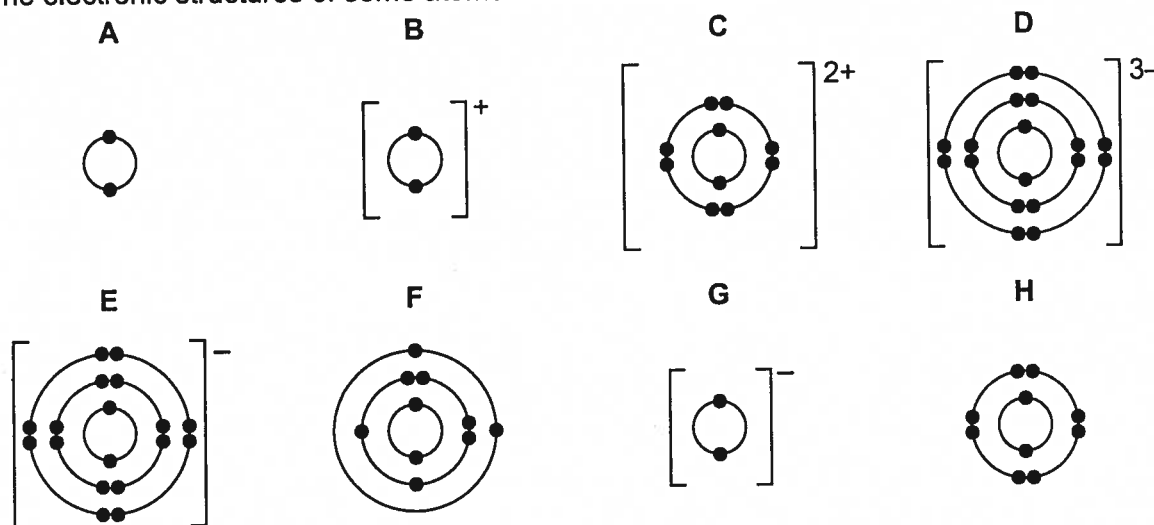
- (i) is an anion [1]
- (ii) are cations and [2]
- (iii) are noble gas atoms and [2]
- (iv) is a halogen atom [1]
- (v) is a Group I atom [1]
- (vi) have the same nucleon number and [1]
- (vii) causes acidity in aqueous solutions [1]

- (viii) is used to define the relative atomic mass of elements. [1]
 (b) Explain why **B** and **C** are isotopes of the same element.

.....
 [2]

12. O/N 20/P42/Q1

The electronic structures of some atoms and ions are shown.



- (a) Write the letters, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, of the electronic structures which show:

(i) atoms of two different noble gases and [2]

(ii) an ion of a Group I element [1]

(iii) an ion of a Group V element [1]

(iv) a pair of ions that could form a compound with the formula XY_2 and [1]

- (b) State which electronic structure, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, is incorrect.

Explain why.

incorrect electronic structure

explanation [2]

- (c) State how many protons are found in the nucleus of ion **C**. [1]

- (d) Use the Periodic Table to deduce:

(i) the chemical symbol for ion **G** [1]

(ii) the element which forms an ion with a 3+ charge and the same electronic structure as **H**.

..... [1]

13. O/N 20/P41/Q1/a(vi)

(a) This question is about elements.

aluminium
carbon
iron
hydrogen
oxygen
silicon
sodium
sulfur

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

(i) Name the element that has atoms with the electronic structure 2,6.

..... [1]

14. O/N 20/P43/Q1/d

The names of nine substances are shown.

aluminium oxide
ammonia
carbon monoxide
anhydrous cobalt(II) chloride
hydrated copper(II) sulfate
iron(III) oxide
nitrogen dioxide
silver
steel

Answer the following questions using these substances. Each substance may be used once, more than once or not at all.

Name the substance that is:

(a) an element [1]

15. O/N 20/P43/Q2/a,b,c

The table gives information about five particles, A, B, C, D and E.

particle	number of electrons	number of neutrons	number of protons
A	10	13	11
B	18	20	18
C	18	18	18
D	10	12	8
E	10	10	10

(a) State the atomic number of **A**.

..... [1]

(b) State the nucleon number of **B**.

..... [1]

(c) Write the electronic structure of **C**.

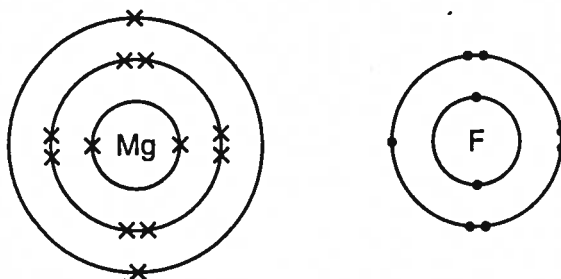
..... [1]

16. M/J 20/P42/Q2/a(ii,iii)

Fluorine forms both ionic and covalent compounds.

(a) Magnesium reacts with fluorine to form the ionic compound magnesium fluoride.

The electronic structures of an atom of magnesium and an atom of fluorine are shown.



(i) What is the formula of magnesium fluoride?

..... [1]

(ii) Magnesium fluoride does **not** conduct electricity when it is solid.

What can be done to solid magnesium fluoride to make it conduct electricity?

In your answer explain why magnesium fluoride conducts electricity when this change is made.

.....

..... [2]

17. M/J 20/P41/Q1/a,b(i)

This question is about elements **X**, **Y** and **Z**.

(a) An atom of element **X** is represented as ${}^{34}_{16}\text{X}$.

(i) Name the different types of particles found in the nucleus of this atom of **X**.

.....

..... [2]

(ii) What is the term for the total number of particles in the nucleus of an atom?

..... [1]

(iii) What is the total number of particles in the nucleus of an atom of ${}^{34}_{16}\text{X}$?

..... [1]

(iv) What is the electronic structure of the ion X^{2-} ?

..... [1]

(v) Suggest the formula of the compound formed between aluminium and X.

..... [1]

(b) (i) What term is used to describe atoms of the same element with different numbers of particles in the nucleus?

..... [1]

18. M/J 20/P43/Q3/a,b

Chlorine is in Group VII of the Periodic Table.

(a) Two isotopes of chlorine are chlorine-35 and chlorine-37.

(i) State why these two isotopes of chlorine have the same chemical properties.

.....
..... [2]

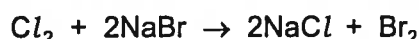
(ii) Complete the table to show the number of electrons, neutrons and protons in each atom and ion.

	number of electrons	number of neutrons	number of protons
$^{35}_{17}\text{Cl}$			
$^{37}_{17}\text{Cl}^-$			

[3]

(b) (i) Chlorine reacts with aqueous sodium bromide.

The equation for the reaction is shown.



State the type of reaction shown.

..... [1]

(ii) Why is there **no** reaction between iodine and aqueous sodium bromide?

..... [1]

19. O/N 19/P42/Q1/a

The Periodic Table is very useful to chemists.

Refer only to elements with atomic numbers 1 to 36 in the Periodic Table provided when answering **Question 1**.

(a) Use information from the Periodic Table provided to identify **one** element which:

(i) has atoms with exactly 9 protons [1]

(ii) has atoms with 0 neutrons [1]

- (iii) has atoms with exactly 23 electrons [1]
- (iv) has atoms with an electronic structure of 2,8,6 [1]
- (v) forms ions with a charge of 3⁻ containing 18 electrons [1]
- (vi) forms ions with a charge of 2⁺ containing 10 electrons [1]
- (vii) has a relative atomic mass that shows it has at least two isotopes. [1]

20. O/N 19/P41/Q2/a

- (a) Sulfur exists as a number of different isotopes.

What is meant by the term *isotopes*?

.....

..... [2]

21. O/N 19/P43/Q1

- (a) Atoms are made of smaller particles called electrons, neutrons and protons.

Complete the table.

particle	relative charge	relative mass
electron		$\frac{1}{1840}$
neutron		
proton	+1	

[2]

- (b) The table gives information about atoms and ions
- A**
- ,
- B**
- and
- C**
- .

Complete the table.

	number of electrons	number of neutrons	number of protons	symbol
A		14	13	${}^{27}_{13}\text{Al}$
B			12	${}^{25}_{12}\text{Mg}^{2+}$
C	10	10	9	

[6]

22. M/J 19/P42/Q2/a

- (a)
- ${}^{22}_{11}\text{Na}$
- ,
- ${}^{23}_{11}\text{Na}$
- and
- ${}^{24}_{11}\text{Na}$
- are isotopes of sodium.

- (i) Describe how these sodium isotopes are the same and how they are different in terms of the total number of protons, neutrons and electrons in each.

same

.....

different [3]

(ii) Why do all **three** isotopes have an overall charge of zero? [1]

(iii) Why do all **three** isotopes have the same chemical properties? [2]

(iv) Why do sodium ions have a charge of +1? [1]

23. M/J 19/P41/Q1/a,b,c

This question is about the structures of atoms and ions.

(a) Define the term *proton number*.

..... [2]

(b) (i) Complete the table to show the number of protons, neutrons and electrons present in atoms of $^{24}_{12}\text{Mg}$ and $^{26}_{12}\text{Mg}$.

	number of protons	number of neutrons	number of electrons
$^{24}_{12}\text{Mg}$			
$^{26}_{12}\text{Mg}$			

(ii) What term is used to describe atoms of the same element, such as $^{24}_{12}\text{Mg}$ and $^{26}_{12}\text{Mg}$? [1]

(iii) Explain why the chemical properties of $^{24}_{12}\text{Mg}$ and $^{26}_{12}\text{Mg}$ are the same. [2]

(c) Complete the table to identify the atoms and ions which have the following numbers of protons, neutrons and electrons.

	number of protons	number of neutrons	number of electrons
$^{23}_{11}\text{Na}^+$	11	12	10
	4	5	4
	17	20	18

[4]

24. M/J 19/P43/Q1

Atoms contain particles called electrons, neutrons and protons.

(a) Complete the table.

particle	where the particle is found in an atom	relative mass	relative charge
	orbiting the nucleus	$\frac{1}{1840}$	
			+1
	in the nucleus		

[3]

(b) How many electrons, neutrons and protons are there in the ion shown?



number of electrons

number of neutrons

number of protons

[3]

25. M/J 19/P43/Q2/a,b(i)

Magnesium exists as three isotopes, ${}_{12}^{24}\text{Mg}$, ${}_{12}^{25}\text{Mg}$ and ${}_{12}^{26}\text{Mg}$.

(a) State, in terms of the total numbers of electrons, neutrons and protons, **one** difference and **two** similarities between these magnesium isotopes.

difference

similarity 1

similarity 2 [3]

(b) All isotopes of magnesium react with dilute hydrochloric acid to make hydrogen and a salt.

(i) Why do all isotopes of magnesium react in the same way?

.....
..... [2]

26. MAR 19/P42/Q1/c,g

Period 3 of the Periodic Table is shown.

sodium	magnesium	aluminium	silicon	phosphorus	sulfur	chlorine	argon
--------	-----------	-----------	---------	------------	--------	----------	-------

Answer the following questions using only these elements.

Each element may be used once, more than once or not at all.

State which element:

(a) is made of atoms which have a full outer shell of electrons

..... [1]

(b) consists of diatomic molecules.

..... [1]

27. MAR 19/P42/Q2

(a) The table gives information about some atoms or ions, **A**, **B** and **C**.
Complete the table.

	number of protons	number of electrons	electronic structure	charge
A	11	10	2,8	
B		18		0
C		10	2,8	-1

[4]

(b) (i) Carbon is an element.
Define the term *element*.

..... [1]

(ii) $^{12}_6\text{C}$, $^{13}_6\text{C}$ and $^{14}_6\text{C}$ are isotopes of carbon.
Complete the table.

	number of protons	number of neutrons
$^{12}_6\text{C}$		
$^{13}_6\text{C}$		
$^{14}_6\text{C}$		

[2]

28. MAR 19/P42/Q3/e(iii)

(a) Tetrafluoromethane and lead(II) fluoride are fluorides of Group IV elements. Some properties of tetrafluoromethane and lead(II) fluoride are shown in the table.

property	tetrafluoromethane	lead(II) fluoride
formula	CF_4	
melting point/ °C	-184	855
boiling point / °C	-127	1290
conduction of electricity when solid	non-conductor	non-conductor
conduction of electricity when molten	non-conductor	good conductor

(i) What type of structure does solid lead(II) fluoride have?

..... [1]

29. O/N 18/P42/Q2/a,b

Magnesium, calcium and strontium are Group II elements.

(a) Complete the table to show the arrangement of electrons in a calcium atom.

shell number	1	2	3	4
number of electrons				

[1]

(b) Describe how the arrangement of electrons in a strontium atom is:

(i) similar to the arrangement of electrons in a calcium atom

.....

(ii) different from the arrangement of electrons in a calcium atom.

.....

..... [2]

30. M/J 18/P42/Q2/a

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

(a) forms an oxide with a macromolecular structure

..... [1]

31. M/J 18/P42/Q3

Complete the following table.

particle	number of protons	number of electrons	number of neutrons	number of nucleons
$^{23}_{11}\text{Na}$	11	11		23
$^{37}_{17}\text{Cl}^-$			20	
$^{56}_{26}\text{.....}$	26	24	30	56

[6]

32. M/J 18/P41/Q1/a

Substances can be classified as elements, compounds or mixtures.

(a) What is meant by the term *compound*?

.....

..... [2]

33. M/J 18/P41/Q2/d

Flerovium, Fl, atomic number 114, was first made in research laboratories in 1998.

- (a) Two isotopes of flerovium are ^{286}Fl and ^{289}Fl . The nuclei of both of these isotopes are unstable and emit energy when they split up.

(i) State the term used to describe isotopes with unstable nuclei.

..... [1]

- (ii) Complete the table to show the number of protons, neutrons and electrons in the atoms of the isotopes shown.

isotope	number of protons	number of neutrons	number of electrons
^{286}Fl			
^{289}Fl			

[2]

34. M/J 18/P43/Q2/a

- (a) ^{29}Al is a radioactive isotope of aluminium. The only non-radioactive isotope of aluminium is ^{27}Al .

- (i) Describe, in terms of protons, neutrons and electrons, how the isotopes ^{29}Al and ^{27}Al are similar and how they are different.

how they are similar

how they are different [2]

- (ii) Complete the table to show the number of nucleons, neutrons and electrons in an $^{27}_{13}\text{Al}^{3+}$ ion.

	number in $^{27}_{13}\text{Al}^{3+}$
nucleons	
neutrons	
electrons	

[3]

35. O/N 17/P42/Q2

- (a) Complete the table to show the electronic structure of the atoms and ions.

	electronic structure
F	2,7
Si	
Ca^{2+}	
N^{3-}	

[3]

- (b) Predict the formula of the compound formed between Ca^{2+} and N^{3-} .

..... [1]

- (c) Draw a dot-and-cross diagram to show the electron arrangements in the **two** ions present in lithium chloride, LiCl .
Show outer shell electrons only. Include the charges on the ions.

[3]

- (d) Sulfur dichloride, SCl_2 , is a covalent compound. It has the structure Cl-S-Cl .
Draw a dot-and-cross diagram to show the electron arrangement in a molecule of sulfur dichloride.
Show outer shell electrons only.

[3]

- (e) In terms of attractive forces, explain why LiCl has a higher melting point than SCl_2 .

.....

- [3]
- (f) Suggest the identity of a **covalent compound** with a higher melting point than LiCl .

..... [1]

36. O/N 17/P41/Q1

The table gives information about five particles. The particles are all atoms or ions.

particle	number of protons	number of neutrons	number of electrons
A	6	8	6
B	12	12	12
C	13	14	10
D	8	8	10
E	11	12	11

Answer the following questions using the information in the table.
Each particle may be used once, more than once or not at all.

(a) Which particle, **A**, **B**, **C**, **D** or **E**,

(i) is an atom with atomic number 12,

..... [1]

(ii) is an atom with nucleon number 14,

..... [1]

(iii) is an ion with a positive charge,

..... [1]

(iv) has only **one** electron in its outer shell?

..... [1]

(b) **D** is an ion of an element.

Identify the element and write the formula of **D**.

..... [2]

37. O/N 17/P43/Q1

Substances can be classified as elements, compounds or mixtures.

State whether each of the following is an element, a compound or a mixture.

(a) brass [1]

(b) gold [1]

(c) butane [1]

(d) air [1]

38. O/N 17/P43/Q2

(a) (i) Define the term *molecule*.

.....

..... [2]

(ii) Define the term *element*.

..... [1]

(b) The table shows the composition of four atoms or ions, **A**, **B**, **C** and **D**.

	number of protons	number of neutrons	number of electrons
A	10	10	10
B	10	12	10
C	12	10	10
D	13	14	10

(i) What is the atomic number of **A**?

..... [1]

(ii) What is the nucleon number of **B**?

..... [1]

(iii) Which of **A**, **B**, **C** and **D** are isotopes of each other?

..... [1]

(iv) Which of **A**, **B**, **C** and **D** are atoms?

..... [1]

(v) Which of **A**, **B**, **C** and **D** are positive ions?

..... [1]

(c) Complete the table.

	number of protons	number of electrons
Na		
S ²⁻		
Cl ₂		

[3]

39. M/J 17/P42/Q2/a,b

Carbon and silicon are elements in Group IV of the Periodic Table. Both carbon and silicon exist as more than one isotope.

(a) Define the term *isotopes*.

.....
..... [2]

(b) Complete the following table which gives information about carbon atoms and silicon atoms.

	carbon	silicon
proton number		
electronic structure		
nucleon number	12	28
number of neutrons in one atom		

[3]

40. O/N 16/P42/Q2/a,b,c,d

This question is about atoms, ions and isotopes.

(a) Define the term *nucleon number*.

.....

..... [2]

(b) Give the electronic structure of the following atom and ion.

Na

P³⁻

(c) State **one** medical use of radioactive isotopes. [2]

..... [1]

(d) What is meant by the term *relative atomic mass*?

.....

..... [2]

Answers Section

1. O/N 22/P43/Q1

(a)

M1 relative charge	M2 relative mass
0	1
+1	1

1 mark for each correct column

(b) M1 22 (1)

M2 17 (1)

M3 17 (1)

M4 32 and 16 (1)

M5 S (1)

M6 $2^{-} / ^{-}2 / ^{-}$ (1)

2. M/J 22/P41/Q2

(a) (i)

charge	relative mass
-1	
0	1
	1

(1)

(1)

Mark by column

(ii)

12	12	12 (1)
12	13	12 (1)

Mark by row

(iii) (they have) 2 more protons than electrons

OR

(they have) 2 fewer electrons than protons

OR

(they have) 12 protons and 10 electrons

(b) Na^{+} or Al^{3+} (1) F^{-} or O^{2-} or N^{3-} (1)

Ne (1)

3. M/J 22/P43/Q2

- (a) (i) number of protons (are the same) / 16 protons (1)
 number of electrons (are the same) / 16 electrons (1)
 number of neutrons (are different) / 16, 17 neutrons (1)

- (ii) **number** of protons is the same as (the number of) electrons 1
- (iii) same **number** of (outer shell) electrons 1
- (b) (i) (they have) **two** more electrons than protons 1
- (ii) P^{3-} OR Cl^{-} (1)
 K^{+} OR Ca^{2+} (1) 2

4. O/N 21/P42/Q3

(a) 1 mark for each correct row

Name	Relative mass	Relative charge
Proton	1	+1
Neutron	1	0
Electron	1 / 1840	-1

3

(b)

Particle	Number of protons	Number of neutrons	Number of electrons
$^{32}_{16}S$	16	16	16
$^{39}_{19}K^{+}$	19	20	18
$^{79}_{35}Br^{-}$	35	44	36

M1 = row 1 (1) M2 = row 2 (1) M3 = Br (1)

M4 = $^{79}_{35}$ (on left of any symbol) (1) M5 = charge (on any symbol) (1)

5

5. O/N 21/P41/Q2/a

(a) (i) isotopes

1

(ii) 1 mark for each correct row

	^{63}Cu	$^{65}Cu^{2+}$
p	29	29
n	34	36
e	29	27

3

(iii) **M1** = $(70 \times 63) + (30 \times 65)$ or $[(4410) + (1950)]$ or 6360 (1)**M2** = $M1 / 100 = 63.6$ (1)

OR

M1 = $(0.7(0) \times 63) + (0.3(0) \times 65)$ or $[(44.1(0)) + (19.5(0))]$ (1)**M2** = 63.6 (1)

2

6. O/N 21/P43/Q3/f

(a) (i) **M1** two double bonds (1)**M2** two pairs of non-bonding electrons on each oxygen and no non-bonding electrons on carbon (1)

2

- (ii) **M1** bonds between ions **or** ionic bonds in lead(II) oxide (1)
M2 attraction **between molecules** in carbon dioxide (1)
M3 weaker attraction (between particles) in carbon dioxide **ORA** (1)

3

7. O/N 21/P43/Q4/a,b

- (a) 1 mark for each row

6	6	6
6	6	7

2

- (b) diamond and graphite

1

8. M/J 21/P42/Q2/a,b

- (a) (i) isotopes

1

- (ii) mark by row

	^{107}Ag	$^{109}\text{Ag}^+$
p	47	47
n	60	62
e	47	46

3

- (iii) average / mean

 ^{12}C

12

3

- (iv) 50(%)

1

- (b)
- HNO_3

1

9. M/J 21/P42/Q5/c

- (a) ionic bonds in
- Li_3N
- (1)

attraction between molecules in NF_3 (1)weaker attraction (between particles) in NF_3 **ORA** (1)

3

10. M/J 21/P41/Q2

Mg: 12 and 13 (1)

 Cu^{2+} : 29 and 27 (1)

37(above)

and 17(below) (1)

Cl (1)

1- (1)

5

11. MAR 21/P42/Q1

- (a) (i) E

1

- (ii) A

I

2

- (iii) D

G

2

- (iv) F

1

- (v) H

1

- | | |
|--|---|
| (vi) G and I | 1 |
| (vii) A | 1 |
| (viii) B | 1 |
| (b) same proton number
different neutron number | 2 |
- 12. O/N 20/P42/Q1**
- | | |
|---|---|
| (a) (i) A (1)
H (1) | 2 |
| (ii) B | 1 |
| (iii) D | 1 |
| (iv) C and G OR C and E | 1 |
| (b) F (1)
third / outer shell is being filled before second shell is full; second shell has 6 electrons:
it should have 8 electrons (1) | 2 |
| (c) 12 | 1 |
| (d) (i) H ⁻ | 1 |
| (ii) aluminium / Al | 1 |
- 13. O/N 20/P41/Q1/a(vi)**
- | | |
|----------------|---|
| (a) (i) oxygen | 1 |
|----------------|---|
- 14. O/N 20/P43/Q1/d**
- | | |
|------------|---|
| (a) silver | 1 |
|------------|---|
- 15. O/N 20/P43/Q2/a,b,c**
- | | |
|-----------|---|
| (a) 11 | 1 |
| (b) 38 | 1 |
| (c) 2,8,8 | 1 |
- 16. M/J 20/P42/Q2/a(ii,iii)**
- | | |
|--|---|
| (a) (i) MgF ₂ | 1 |
| (ii) heat until molten or dissolve in water (1)
moving ions / mobile ions (1) | 2 |
- 17. M/J 20/P41/Q1/a,b(i)**
- | | |
|------------------------------------|---|
| (a) (i) protons | 1 |
| neutrons | 1 |
| (ii) nucleon number | 1 |
| (iii) 34 | 1 |
| (iv) 2 : 8 : 8 | 1 |
| (v) Al ₂ X ₃ | 1 |
| (b) (i) isotopes | 1 |
- 18. M/J 20/P43/Q3/a,b**
- | | |
|---|---|
| (a) (i) same number of electrons
same electronic configuration | 2 |
|---|---|

(ii)

	number of electrons	number of neutrons	number of protons
$^{35}_{17}\text{Cl}$	17	18	17
$^{37}_{17}\text{Cl}^-$	18	20	17

3
1
1

- (b) (i) displacement / redox
(ii) iodine is less reactive than bromine

19. O/N 19/P42/Q1/a

- (a) (i) fluorine / F
(ii) hydrogen / H
(iii) vanadium / V
(iv) sulfur / S
(v) phosphorus / P
(vi) magnesium / Mg
(vii) chlorine / Cl

1
1
1
1
1
1
1

20. O/N 19/P41/Q2/a

- (a) atoms with same number of protons or atoms of the same element or atoms with same atomic number (1)
atoms with different number of neutrons or atoms with different mass number or atoms with different nucleon number (1)

2

21. O/N 19/P43/Q1

(a)

particle	charge	relative mass
electron	M1 -1	
neutron	M2 0	M3 1
proton		M4 1

(1) (1)

2

Mark by column

(b)

number of electrons	number of neutrons	number of protons	symbol
M1 13 (1)			
M2 10 (1)	M3 13 (1)		
			M4 19 9 (1) M5 F (1) M6 - (1)

6

22. M/J 19/P42/Q2/a

- (a) (i) **M1** protons (are the same) / 11 protons (1)
M2 electrons (are the same) / 11 electrons (1)
M3 neutrons (are different) / 11, 12, 13 neutrons (1)

3

- (ii) same number of protons and electrons (1) 1
- (iii) **M1** same number of electrons (1)
M2 (same number of) electrons in outer shell (1) 2
- (iv) (they all have) 1 more proton than electrons / 11 protons and 10 electrons 1

23. M/J 19/P41/Q1/a,b,c

- (a) number of protons (1)
 protons in the nucleus (of an atom) (1) 2
- (b) (i) 12p 12n 12e (1)
 12p 14n 12e (1) 2
- (ii) isotope(s) 1
- (iii) same number of electrons (1)
 (same number) of electrons in the outer shell (1) 2
- (c) ${}^9_4\text{Be}$
 any element symbol with a single negative charge (1)
 use of Cl (1)
 use of ${}^{37}_{17}$ (1) 4

24. M/J 19/P43/Q1

(a)

particle	where found in an atom	relative mass	relative charge
electron	orbiting nucleus	1/1840	-1
proton	(in the) nucleus	1	+1
neutron	in the nucleus	1	0 / nil

3

- (b) **M1** electrons 18
M2 neutrons 24
M3 protons 20 3

25. M/J 19/P43/Q2/a,b(i)

- (a) difference: **M1** (number of) neutrons
 similarities: **M2** (number of) protons
M3 (number of) electrons 3
- (b) (i) **M1** same number of electrons
M2 (same number of) electrons in outer shell 2

26. MAR 19/P42/Q1/c,g

- (a) argon 1
- (b) chlorine 1

27. MAR 19/P42/Q2

(a)	number of protons	electrons	electronic structure	charge on particle
	11	10	2,8	M4 1+ / +1(1)
	M1 18(1)	18	M3 2,8,8(1)	0
	M2 9(1)	10	2,8	1-

4

(b) (i) *element***EITHER** (substance) made of **atoms** with the same atomic number / number of protons / proton number**OR** a substance that cannot be split up / broken down into two or more simple(r) substances by chemical means

1

(ii) M1 6 protons in all three rows(1)

M2 6,7 and 8 neutrons(1)

2

28. MAR 19/P42/Q3/e(iii)

(a) (i) giant ionic lattice

1

29. O/N 18/P42/Q2/a,b

(a) 2 : 8 : 8 : 2

1

(b) (i) M1 Same number of (or 2) outer electrons

(ii) M2 (Sr has) outer electrons are in the 5th shell

2

30. M/J 18/P42/Q2/a

(a) silicon / Si

1

31. M/J 18/P42/Q3

particles	number of protons	number of electrons	number of neutrons	number of nucleons
			12 (1)	
	17 (1)	18 (1)		37 (1)
Fe (1) 2+ (1)				

6

32. M/J 18/P41/Q1/a

(a) a substance made from two (or more) **elements**
chemically combined1
1

33. M/J 18/P41/Q2/d

(a) (i) radioisotopes

1

(ii) ^{286}Fl 114p 172n 114e

1

 ^{289}Fl 114p 175n 114e

1

34. M/J 18/P43/Q2/a

(a) (i) similarities: number of protons and electrons
differences: number of neutrons1
1

- (ii) nucleons: 27 1
neutrons: 14 1
electrons: 10 1
35. O/N 17/P42/Q2
- (a) Si: 2: 8 : 4 1
 Ca^{2+} : 2 : 8: 8 1
 N^{3-} : 2 : 8 1
- (b) Ca_3N_2 1
- (c) Li shown as having one shell with 2 electrons **OR** no electrons **OR** no outer shell 1
Cl shown as having an outer shell of 7 electrons of one type,
plus one different electron which matches Li electrons 1
'+' charge on Li **AND** '-' charge on Cl 1
- (d) two shared pairs of electrons 1
both Cl with complete outer shells 1
S with complete outer shell 1
- (e) SCl_2 has intermolecular forces (of attraction) 1
 LiCl has (electrostatic) forces (of attraction) between ions 1
intermolecular forces are weaker / less energy is needed to break intermolecular forces 1
- (f) silicon(IV) oxide 1
36. O/N 17/P41/Q1
- (a) (i) B 1
(ii) A 1
(iii) C 1
(iv) E 1
- (b) O^{2-}
M1 O
M2 $^{2-}$ 2
37. O/N 17/P43/Q1
- (a) mixture 1
(b) element 1
(c) compound 1
(d) mixture 1
38. O/N 17/P43/Q2
- (a) (i) (two or more) **atoms** 1
combined / joined / sharing electrons (by a covalent bond) / bonded 1
(ii) substance that cannot be split up / broken down / decomposed (into anything simpler) **OR**
(substance) made of **atoms** with the same atomic number /
number of protons / proton number 1
- (b) (i) 10 1
(ii) 22 1
(iii) A **AND** B 1
(iv) A **AND** B 1
(v) C **AND** D 1

(c)

	number of protons	number of electrons
Na	11	11
S ²⁻	16	18
Cl ₂	34	34

3

39. M/J 17/P42/Q2/a,b

- (a) atoms of the same element/ atoms with the same proton number/
atoms with the same atomic number
different neutron **number** / different nucleon **number** / different mass number

1

1

(b)

	carbon	silicon	
proton number	6	14	M1
electronic structure	2,4	2,8,4	M2
nucleon number	12	28	
number of neutrons in one atom	6	14	M3

3

40. O/N 16/P42/Q2/a,b,c,d

- (a) (total) number of protons and neutrons
in a nucleus (of an atom)
- (b) Na 2 : 8 : 1
P³⁻ 2 : 8 : 8
- (c) radiotherapy **OR** treatment of cancer
- (d) average mass of (naturally occurring) atom(s) (of an element)
(compared to an atom of) ¹²C

2

2

1

2

3.2: Chemical bonding

1. O/N 22/P42/Q1

Diamond and graphite are different solid forms of carbon. The carbon atoms in diamond and graphite are arranged in different ways.

- (a) State the number of covalent bonds each carbon atom has in diamond.

..... [1]

- (b) State the term used to describe the structure of diamond.

..... [1]

- (c) Name an oxide that has a similar structure to diamond.

..... [1]

- (d) Describe the arrangement of atoms in graphite.

.....
..... [2]

- (e) Explain how graphite conducts electricity.

..... [1]

- (f) Buckminsterfullerene is a simple molecular form of carbon.

The relative molecular mass of Buckminsterfullerene is 720.

Determine the number of carbon atoms in one molecule of Buckminsterfullerene.

..... [1]

- (g) All forms of carbon burn to produce carbon dioxide.

Name the substance used to test for carbon dioxide.

..... [1]

2. O/N 22/P42/Q2

Sodium is a reactive metal.

- (a) Suggest why sodium is stored under oil.

..... [1]

- (b) Sodium burns in air to form sodium oxide, Na_2O .

- (i) State the term given to a reaction in which a substance burns.

..... [1]

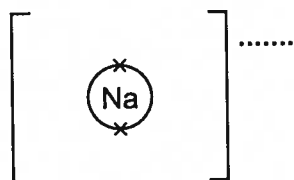
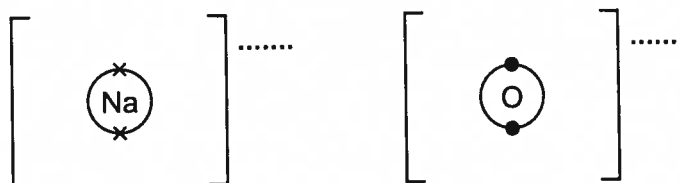
- (ii) State the colour of the flame seen when sodium burns.

..... [1]

- (iii) Write a chemical equation for the reaction which takes place when sodium burns in air to form sodium oxide.

.....[2]

- (iv) Complete the dot-and-cross diagram to show the electron arrangement and charges of the ions in sodium oxide.



[3]

- (c) Sodium reacts vigorously with water to form aqueous sodium hydroxide, NaOH, which is a strong base.

- (i) Explain in terms of proton transfer what is meant by a base.

.....[1]

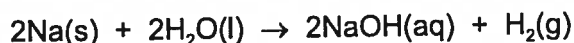
- (ii) State a pH number that indicates the presence of a strong alkali.

.....[1]

- (iii) State the colour of methyl orange in aqueous sodium hydroxide.

.....[1]

- (iv) The equation for the reaction is shown.



Calculate the concentration of NaOH(aq) formed, in g/dm³, when 0.345g of sodium is added to 50.0 cm³ of distilled water. Assume there is no change in volume.

Use the following steps.

- Calculate the number of moles of Na added.

=..... mol

- Determine the number of moles of NaOH formed.

=..... mol

- Calculate the concentration of NaOH in mol/dm³.

concentration of NaOH = mol/dm³

- Determine the M_r of NaOH and calculate the concentration of NaOH in g/dm³.

concentration of NaOH = g/dm³ [5]

(d) When NaOH(aq) is added to aqueous iron(III) chloride, FeCl₃(aq), a solid product is formed.

- (i) Name the type of reaction where a solid is formed from two solutions.

..... [1]

- (ii) State the colour of this solid product.

..... [1]

- (iii) Name this solid product.

..... [1]

- (iv) Write the ionic equation for the reaction. Include state symbols.

..... [3]

3. O/N 22/P41/Q2

Potassium is a Group I element.

- (a) Name and describe the bonding in potassium.

name

description

.....

..... [4]

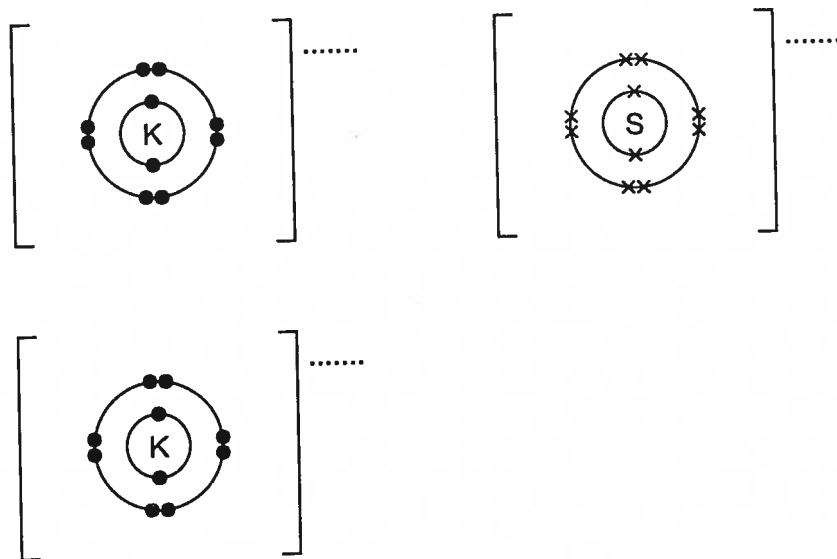
- (b) Potassium combines with sulfur to form an ionic compound, potassium sulfide, K₂S.

- (i) Give **two** physical properties of ionic compounds.

1

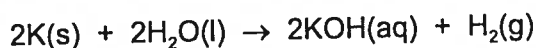
2 [2]

- (ii) Complete the dot-and-cross diagram to show the electron arrangement and charges of the ions in potassium sulfide.



[3]

- (c) When potassium is added to water, it reacts vigorously and a coloured flame is seen. The equation for the reaction is shown.



- (i) State the colour of the flame seen.

..... [1]

- (ii) The solution formed is potassium hydroxide, a strong alkali. State the formula of the ion responsible for alkalinity in a solution.

..... [1]

- (iii) State the colour of litmus in a strong alkali.

..... [1]

- (iv) Calculate the volume, in cm³, of hydrogen gas formed when 2.34 g of potassium is added to excess water at room temperature and pressure.

Use the following steps.

- Calculate the number of moles of potassium added.

= mol

- Determine the number of moles of hydrogen gas formed.

= mol

- Calculate the volume of hydrogen gas formed.

volume = cm³ [3]

(d) Aqueous potassium hydroxide reacts with a dilute acid to produce aqueous potassium chloride, KCl(aq), which is a salt.

(i) Name the dilute acid used.

..... [1]

(ii) State the type of reaction taking place.

..... [1]

(iii) Name the experimental technique used when salts are made by reacting a dilute acid with an aqueous alkali.

..... [1]

(e) When aqueous silver nitrate, AgNO₃(aq), is added to aqueous potassium chloride, a precipitate is formed.

(i) State the colour of the precipitate formed.

..... [1]

(ii) Name the precipitate formed.

..... [1]

(iii) Write the ionic equation for the reaction. Include state symbols.

..... [3]

4. M/J 22/P42/Q4

Fluorine and chlorine are halogens.

(a) Suggest the appearance of fluorine.

..... [1]

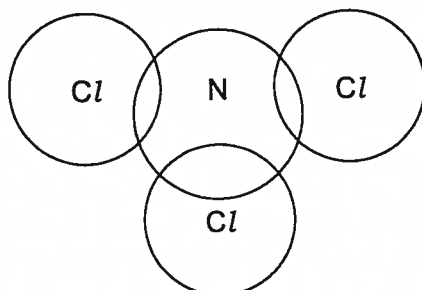
(b) Fluorine reacts with sulfur to form a compound which has 25.2% sulfur by mass and a relative molecular mass of 254.

Determine the molecular formula of this compound.

molecular formula = [3]

(c) Nitrogen trichloride, NCl_3 , is a covalent compound.

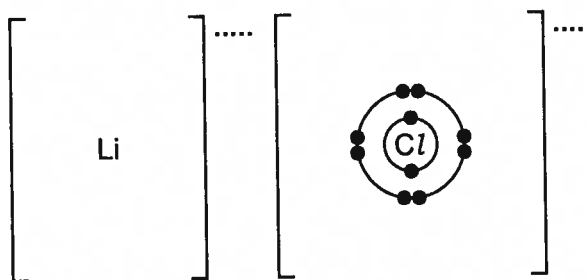
Complete the dot-and-cross diagram to show the electron arrangement in a molecule of NCl_3 .
Show outer electrons only.



[3]

(d) Lithium chloride, LiCl , is an ionic compound.

Complete the dot-and-cross diagram to show the electron arrangement and charges of the ions in lithium chloride.



[3]

(e) Explain, in terms of attractive forces between particles, why LiCl is a solid at room temperature but NCl_3 is a liquid with a relatively low boiling point.

.....

 [3]

5. M/J 21/P42/Q5/a

This question is about compounds of nitrogen.

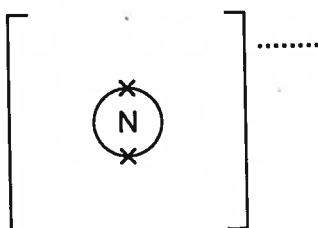
(a) Nitrogen reacts with lithium to form lithium nitride, Li_3N .

(i) Write the chemical equation for the reaction between lithium and nitrogen.

..... [2]

(ii) Lithium nitride is ionically bonded.

Complete the diagram to show the electronic structure of the nitride ion.
Show the charge on the nitride ion.



[2]

6. O/N 20/P42/Q2/f(i)

(a) Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

(i) Explain why, in terms of structure and bonding.

.....
.....
..... [3]

7. O/N 20/P41/Q1/a(ii)

(a) This question is about elements.

aluminium
carbon
iron
hydrogen
oxygen
silicon
sodium
sulfur

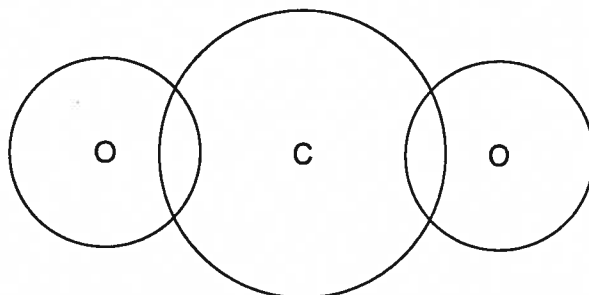
(i) Name the element that forms an oxide with a similar structure to diamond.

..... [1]

8. O/N 20/P41/Q4/c

(a) Carbon dioxide, CO_2 , is a covalent molecule.

Complete the diagram to show the electron arrangement in one molecule of CO_2 .
Show only the outer electrons.



[2]

9. O/N 20/P41/Q6/c(i,ii)

(a) Propane reacts with chlorine.

(i) Write the formula of the product which does not contain carbon.

..... [1]

(ii) Draw the structure of an organic product formed. Show all of the atoms and all of the bonds.

[1]

10. O/N 20/P43/Q2/d

The table gives information about five particles, **A, B, C, D** and **E**.

particle	number of electrons	number of neutrons	number of protons
A	10	13	11
B	18	20	18
C	18	18	18
D	10	12	8
E	10	10	10

(a) Give the letters of all the particles which are:

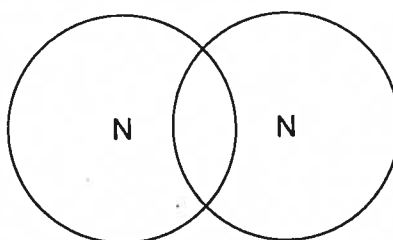
- (i) atoms [1]
- (ii) positive ions [1]
- (iii) negative ions [1]
- (iv) isotopes of each other. [1]

11. O/N 20/P43/Q3/a

This question is about nitrogen and some of its compounds.

(a) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of nitrogen, N_2 .

Show the outer shell electrons only.



[2]

12. M/J 20/P42/Q1/b(v,vi)

(a) The symbols of the elements in Period 2 of the Periodic Table are shown.

Li Be B C N O F Ne

For each of the following, give the symbol of an element from Period 2 which matches the description.

Each element may be used once, more than once or not at all.

Which element:

(i) is a monoatomic gas

..... [1]

(ii) is a soft metal stored in oil?

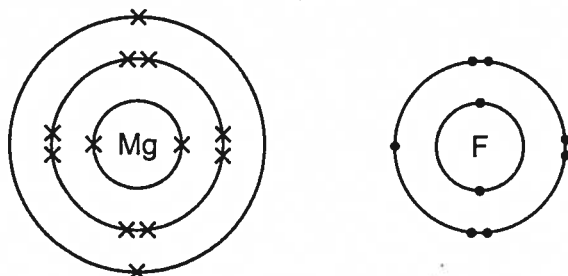
..... [1]

13. M/J 20/P42/Q2/a(i),b,c

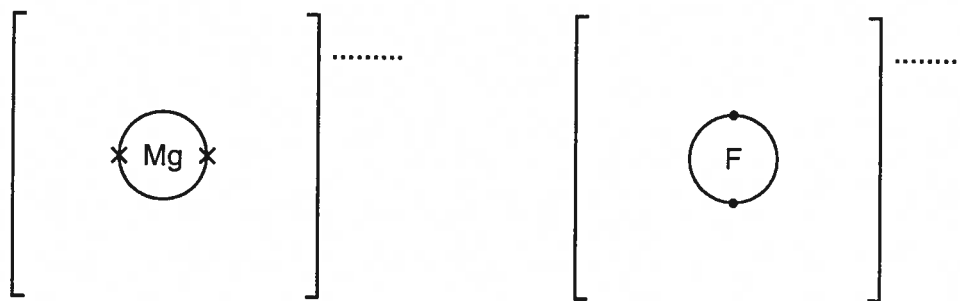
Fluorine forms both ionic and covalent compounds.

(a) Magnesium reacts with fluorine to form the ionic compound magnesium fluoride.

The electronic structures of an atom of magnesium and an atom of fluorine are shown.

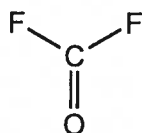


(i) Complete the dot-and-cross diagrams to show the electronic structures of one magnesium ion and one fluoride ion. Show the charges on the ions.

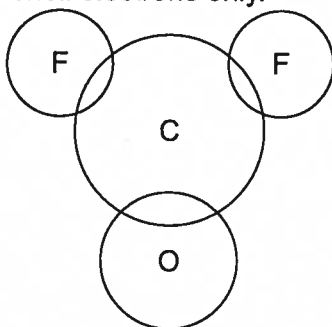


[3]

(b) Carbonyl fluoride, COF_2 , is a covalent compound. The structure of a molecule of COF_2 is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of carbonyl fluoride. Show outer shell electrons only.



[3]

(c) The melting points of magnesium fluoride and carbonyl fluoride are shown.

	melting point/ $^{\circ}\text{C}$
magnesium fluoride	1263
carbonyl fluoride	-111

(i) Explain, using your knowledge of structure and bonding, why magnesium fluoride has a high melting point.

.....

- [2]
- (ii) Explain, using your knowledge of structure and bonding, why carbonyl fluoride has a low melting point.
-
- [2]

14. M/J 20/P42/Q3/g(ii)

- (a) Barium sulfate is an insoluble salt.

Barium sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

- (i) Write an ionic equation for this precipitation reaction. Include state symbols.

..... [2]

15. M/J 20/P41/Q2

Magnesium is a metal.

- (a) Name and describe the bonding in magnesium.

name

description of bonding

.....

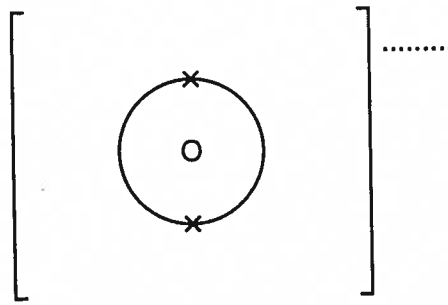
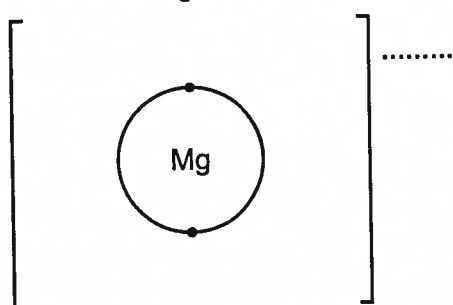
..... [4]

- (b) Magnesium oxide, MgO, is formed when magnesium burns in oxygen.

- (i) Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium oxide.

The inner shells have been drawn.

Give the charges on the ions.



[3]

- (ii) Write the chemical equation for the reaction that occurs when magnesium burns in oxygen.

..... [2]

- (c) Magnesium oxide also forms when magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, is heated strongly. This is an endothermic reaction.

- (i) Write the chemical equation for this reaction.

..... [2]

- (ii) What type of reaction is this?

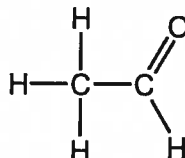
..... [1]

- (iii) Name **two** other compounds of magnesium that form magnesium oxide when heated.

.....
..... [2]

16. M/J 20/P41/Q5/e,f(ii)

- (a) The structural formula of ethanal is shown.

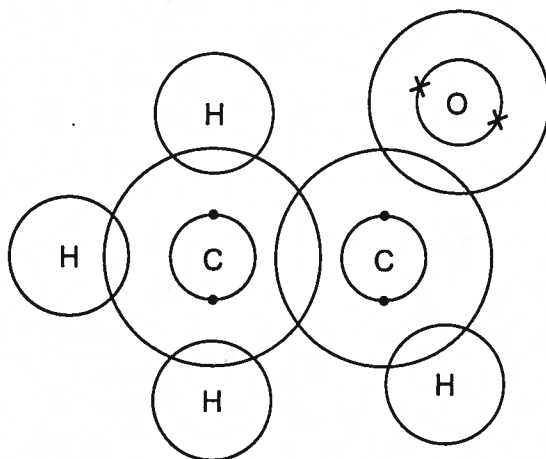


The C=O group in aldehydes is at the end of the carbon chain.
This is a reactive part of the molecule.

- (i) What is the name given to the reactive part of any organic molecule?

..... [1]

- (ii) Complete the dot-and-cross diagram to show the electron arrangement of a molecule of ethanal. Inner shells have been drawn.



[3]

- (b) Propanone belongs to a homologous series called ketones. Ketones have the same C=O group as aldehydes but the C=O group is not at the end of the carbon chain. Propanone has the same molecular formula as propanal, C₃H₆O.

- (i) Suggest the structure of propanone, C₃H₆O. Show all of the atoms and all of the bonds.

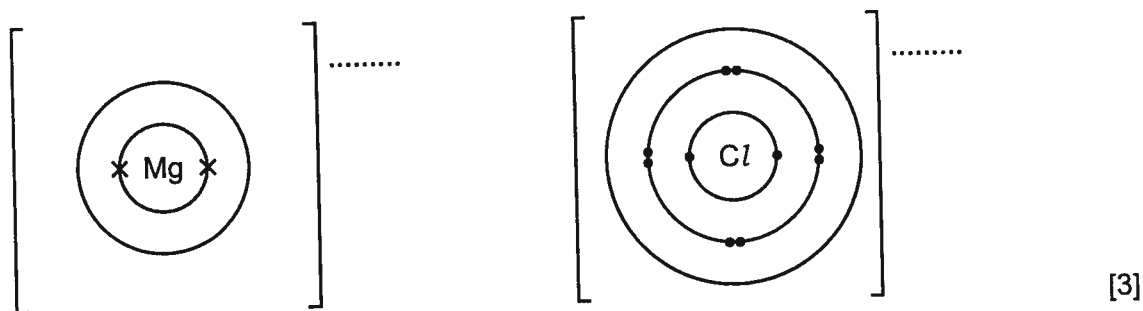
[2]

17. M/J 20/P43/Q3/c

- (a) Magnesium reacts with chlorine to form magnesium chloride.

Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium chloride. Give the charges on the ions.

The inner shells have been completed.

**18. MAR 20/P42/Q3/c,f**

(a) Group V chlorides are covalent molecules. The boiling points of some Group V chlorides are shown.

chloride	boiling point/°C
NCl_3	71
PCl_3	
AsCl_3	130
SbCl_3	283

(i) Suggest the approximate boiling point of PCl_3 .

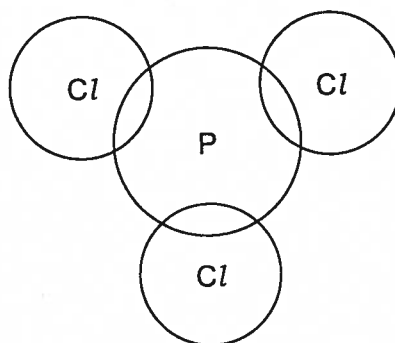
..... [1]

(ii) Explain the trend in boiling points in terms of attractive forces between particles.

..... [2]

(iii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of PCl_3 .

Show outer electrons only.



[3]

(b) Lithium fluoride has ionic bonding.

(i) What is an ionic bond?

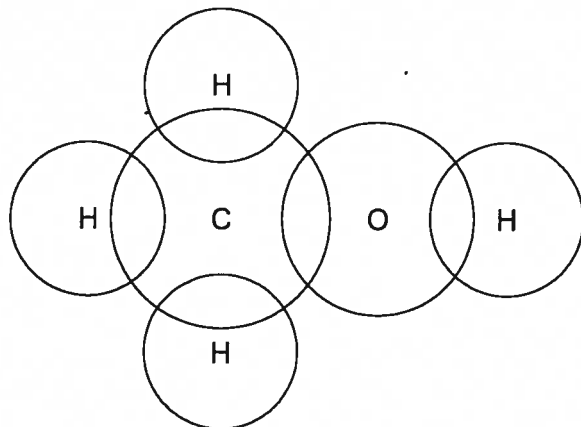
..... [2]

- (ii) Give **two** physical properties of ionic compounds.

.....
..... [2]

19. O/N 19/P42/Q5/b,c(iii)

- (a) Draw a dot-and-cross diagram to show the electron arrangement in a molecule of methanol. Show outer shell electrons only.



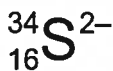
[2]

- (b) Methanol reacts with propanoic acid to form an ester with a molecular formula $C_4H_8O_2$.
(i) Draw the structure of an ester which is a structural isomer of the ester named in (c)(i). Show all of the atoms and all of the bonds.

[3]

20. O/N 19/P41/Q2/b

- (a) A sulfide ion has the symbol shown.



- (i) How many neutrons are contained in this sulfide ion?

..... [1]

- (ii) How is a sulfide ion, S^{2-} , formed from a sulfur atom?

..... [1]

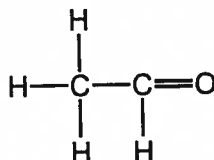
- (iii) Which element forms an ion with a 2+ charge that has the same number of electrons as a S^{2-} ion?

..... [1]

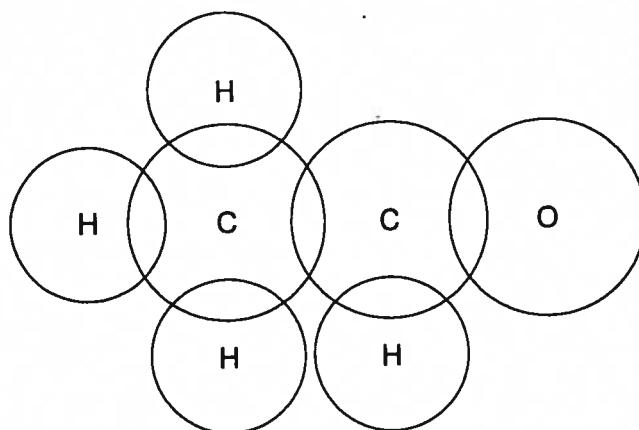
21. O/N 19/P41/Q7/c(ii)

(a) Ethanol can be oxidised by hydrogen peroxide to form ethanal, CH_3CHO . A catalyst for this reaction is Fe^{3+} .

(i) The structure of ethanal is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanal. Show outer shell electrons only.



[3]

22. O/N 19/P43/Q2

The table shows the melting points, boiling points and electrical conductivities of six substances D, E, F, G, H and I.

substance	melting point /°C	boiling point /°C	electrical conductivity when solid	electrical conductivity when liquid
D	1610	2230	non-conductor	non-conductor
E	801	1413	non-conductor	good conductor
F	-119	43	non-conductor	non-conductor

G	1535	2750	good conductor	good conductor
H	114	184	non-conductor	non-conductor
I	-210	-196	non-conductor	non-conductor

Choose substances from the table which match the following descriptions. Each substance may be used once, more than once or not at all.

(a) Which substance is a liquid at 25°C?[1]

(b) Which substance is a gas at 25°C?[1]

(c) Which **three** substances contain simple molecules?

.....[3]

(d) Which substance could be a metal? Give a reason for your answer.

substance

reason [2]

(e) Which substance has a macromolecular structure? Give **two** reasons for your answer.

substance

reason 1

reason 2 [3]

(f) Which substance is an ionic solid? Give **one** reason for your answer.

substance

reason [2]

23. O/N 19/P43/Q4/a,b

This question is about phosphorus and compounds of phosphorus.

(a) A phosphorus molecule contains four phosphorus atoms **only**.

What is the formula of a phosphorus molecule?

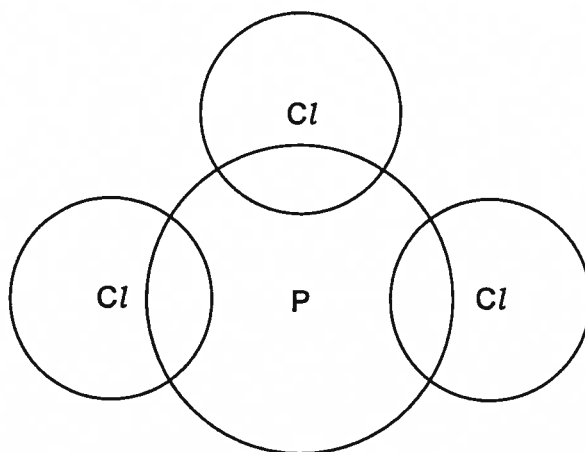
..... [1]

(b) Phosphorus reacts with chlorine gas to produce phosphorus(III) chloride, PCl_3 .

(i) Write a chemical equation for the reaction between phosphorus and chlorine to produce phosphorus(III) chloride, PCl_3 .

..... [2]

(ii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of phosphorus(III) chloride, PCl_3 . Show outer shell electrons only.



[2]

24. M/J 19/P42/Q1/f

The names of eight substances are given.

aluminium oxide

calcium oxide

ethanol

nitrogen

iron(III) oxide

methane

oxygen

silicon(IV) oxide

(a) a macromolecular solid.

.....[1]

25. M/J 19/P42/Q2/b(i)

(a) Carbon is an element which exists in different forms.

(i) Name **two** forms of the element carbon that have giant covalent structures.

..... and[1]

26. M/J 19/P42/Q3/a,d

This question is about phosphorus and compounds of phosphorus.

(a) Phosphorus has the formula P_4 . Some properties of P_4 are shown.

melting point/ $^{\circ}\text{C}$	45
boiling point/ $^{\circ}\text{C}$	280
electrical conductivity	non-conductor
solubility in water	insoluble

(i) Name the type of bonding that exists between the atoms in a P_4 molecule.

.....[1]

(ii) Explain, in terms of attractive forces between particles, why P_4 has a low melting point.

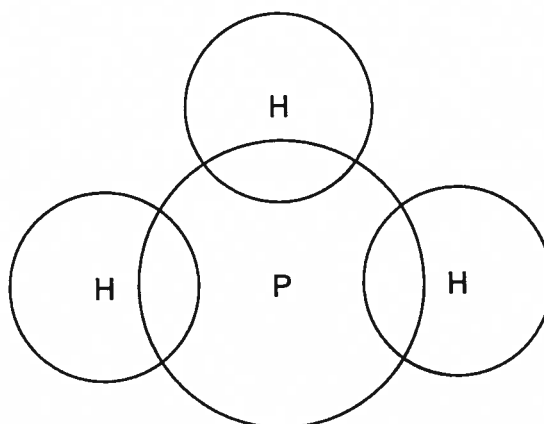
.....[1]

(iii) Explain why phosphorus is a non-conductor of electricity.

.....[1]

(b) Phosphine has the formula PH_3 .

Complete the dot-and-cross diagram to show the electron arrangement in a molecule of phosphine. Show outer shell electrons only.



[2]

27. M/J 19/P42/Q7/b

(a) The ionic equation for another displacement reaction is shown.



Write the **two** ionic half-equations for this reaction.

1

2 [2]

28. M/J 19/P41/Q1/d

(a) State the electronic structure of the following atom and ion.

Al

S²⁻ [2]

29. M/J 19/P43/Q2/c,d

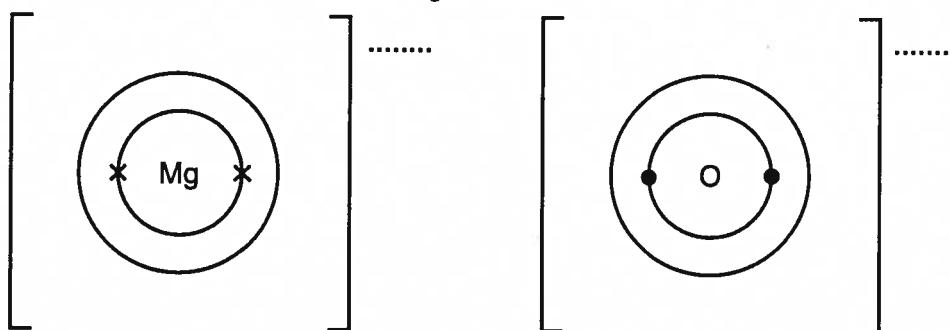
(a) Magnesium is a metal.

Describe the structure and bonding of metals. Include a labelled diagram in your answer.

.....
.....
..... [3]

(b) Magnesium reacts with oxygen to form the ionic compound magnesium oxide.

(i) Complete the dot-and-cross diagrams to show the electronic structures of the ions in magnesium oxide. Show the charges on the ions.



[3]

(ii) Magnesium oxide melts at 2853 °C.

Why does magnesium oxide have a high melting point?

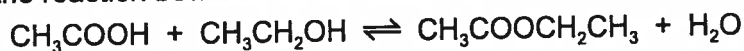
..... [1]

(iii) Explain why molten magnesium oxide can conduct electricity.

..... [1]

30. M/J 19/P43/Q5/a(ii)

Carboxylic acids react with alcohols to form esters. The reaction is reversible.
The equation for the reaction between ethanoic acid and ethanol is shown.



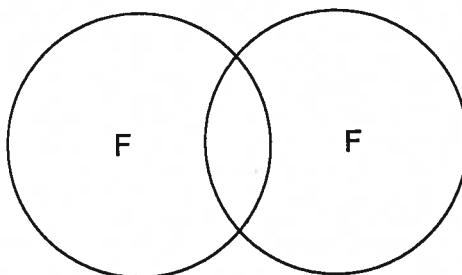
- (a) (i) Draw the structure of the ester formed. Show all of the atoms and all of the bonds.

[1]

31. MAR 19/P42/Q3/b,c,e(ii),f(iii)

- (a) Fluorine exists as diatomic molecules.

Complete the dot-and-cross diagram to show the electron arrangement in a molecule of fluorine. Show outer shell electrons only.



[2]

- (b) Write a chemical equation for the reaction between sodium and fluorine.

..... [2]

- (c) Tetrafluoromethane and lead(II) fluoride are fluorides of Group IV elements. Some properties of tetrafluoromethane and lead(II) fluoride are shown in the table.

property	tetrafluoromethane	lead(II) fluoride
formula	CF ₄	
melting point/°C	-184	855
boiling point/°C	-127	1290
conduction of electricity when solid	non-conductor	non-conductor
conduction of electricity when molten	non-conductor	good conductor

- (i) What type of bonding is present between the atoms in tetrafluoromethane?

..... [1]

- (d) Tetrafluoroethene is an unsaturated compound with the formula C₂F₄.

Tetrafluoroethene is the monomer used to make the polymer poly(tetrafluoroethene).

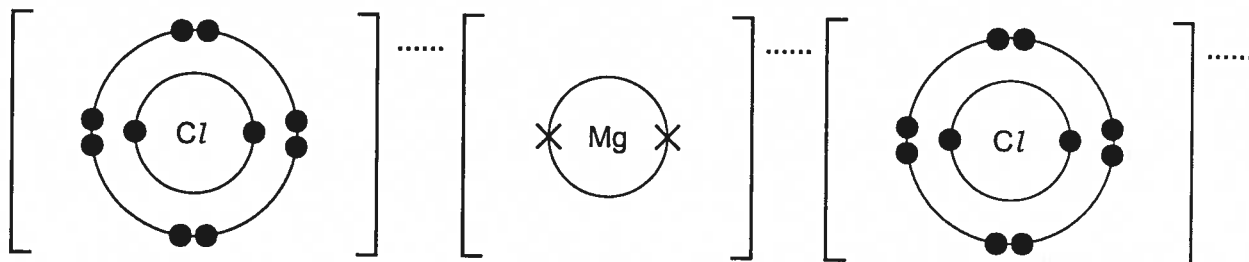
- (i) Draw the structure of a molecule of tetrafluoroethene. Show all of the atoms and all of the bonds.

[1]

32. O/N 18/P42/Q2/d

(a) Magnesium reacts with chlorine to form magnesium chloride, MgCl_2 . Magnesium chloride is an ionic compound.

(i) Complete the diagrams to show the electronic structures of the ions in magnesium chloride. Show the charges on the ions.



[3]

(ii) Give three physical properties that are typical of ionic compounds such as MgCl_2 .

1

2

3 [3]

33. O/N 18/P41/Q2

The table gives some information about four different particles, A, B, C and D.

particle	number of electrons	number of neutrons	number of protons	electronic structure	charge on particle
A	11	12	11	2,8,1	0
B		14	11	2,8,1	0
C	18	20		2,8,8	0
D	18	20	17		

(a) Complete the table. The first row has been done for you. [4]

(b) Give two particles from the table which are isotopes of each other. [1]

..... [1]

(c) Element Z is in the same group of the Periodic Table as A and is less reactive than A. State the identity of element Z. [1]

..... [1]

(d) C is unreactive. [1]

Use information from the table to explain why.

..... [1]

34. M/J 18/P42/Q4/a,b

Potassium reacts with bromine at room temperature to form potassium bromide.

(a) Write a chemical equation for this reaction. Include state symbols.

..... [3]

(b) Potassium bromide exists as an ionic lattice.

Potassium bromide does **not** conduct electricity when solid but does conduct electricity when molten.

(i) What is meant by the term *ionic lattice*?

..... [2]

(ii) Explain why potassium bromide does **not** conduct electricity when solid but does conduct electricity when molten.

..... [2]

35. M/J 18/P41/Q4/a,b

This question is about masses, volumes and moles.

(a) Which term is defined by the following statement?

The average mass of naturally occurring atoms of an element on a scale where the ^{12}C atom has a mass of exactly 12 units.

..... [1]

(b) Butane, C_4H_{10} , has a relative **molecular** mass of 58.

Potassium fluoride, KF , has a relative **formula** mass of 58.

Explain why the term relative molecular mass can be used for butane but **cannot** be used for potassium fluoride.

..... [2]

36. MAR 18/P42/Q1/b

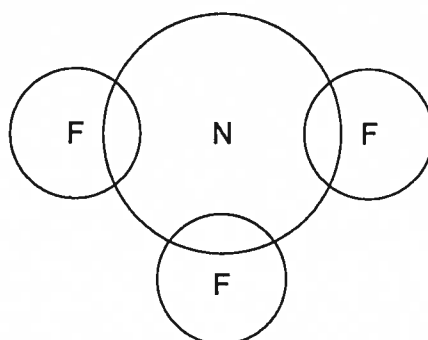
(a) NF_3 has covalent bonds.

(i) What is a covalent bond?

..... [2]

(ii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of NF_3 .

Show outer shell electrons only.



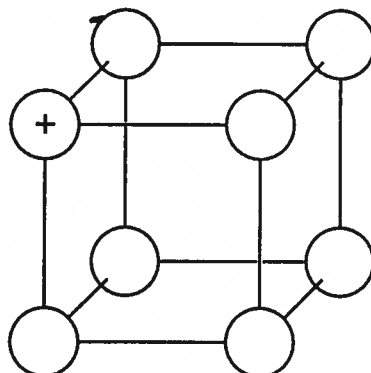
[3]

37. MAR 18/P42/Q2/a

Sodium chloride is a typical ionic compound.

(a) The diagram shows part of a lattice of sodium chloride.

- (i) Complete the diagram to show the ions present. Use '+' for Na^+ ions and '-' for Cl^- ions. One ion has been completed for you.



[2]

- (ii) How many electrons does a chloride ion have?

..... [1]

- (iii) Identify an element which has atoms with the same number of electrons as a sodium ion.

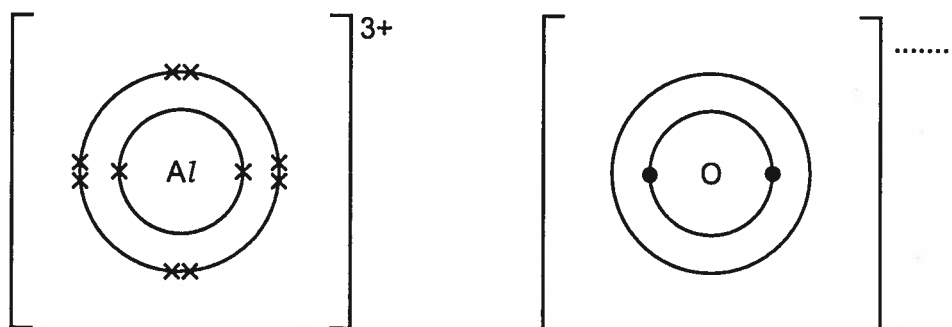
..... [1]

38. O/N 17/P41/Q6/b

Aluminium is extracted from aluminium oxide by electrolysis.

(a) Aluminium oxide is an ionic compound with a high melting point.

- (i) Complete the dot-and-cross diagram to show the electron arrangement in **one** of the oxide ions present in aluminium oxide. Include the charge on the oxide ion. One of the aluminium ions is shown.



[2]

- (ii) The melting point of aluminium oxide is above 2000 °C.

Explain why aluminium oxide has a high melting point.

.....
..... [2]

39. O/N 17/P43/Q1

Substances can be classified as elements, compounds or mixtures.

State whether each of the following is an element, a compound or a mixture.

(a) brass [1]

(b) gold [1]

(c) butane [1]

(d) air [1]

40. M/J 17/P42/Q1/b

- (a) State what is meant by the terms

(i) *element*,

..... [1]

(ii) *compound*,

..... [1]

(iii) *ion*.

..... [1]

41. M/J 17/P42/Q2/c,e

Carbon and silicon are elements in Group IV of the Periodic Table. Both carbon and silicon exist as more than one isotope.

- (a) Silicon has a giant structure which is similar to the structure of diamond.

(i) Name the type of bond which is present between silicon atoms in silicon.

..... [1]

- (ii) Suggest **two** physical properties of silicon.

Use your knowledge of structure and bonding to explain why silicon has these physical properties.

property 1

reason 1

property 2

reason 2 [4]

- (b) Carbon dioxide, CO_2 , is a gas at room temperature and pressure, whereas silicon(IV) oxide, SiO_2 , is a solid.

(i) Name the type of structure which the following compounds have.

carbon dioxide [1]

silicon(IV) oxide [1]

- (ii) Use your knowledge of structure and bonding to explain why carbon dioxide is a gas at room temperature and pressure, whereas silicon(IV) oxide is a solid.

.....

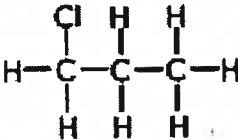
.....

..... [3]

Answers Section

- 1. O/N 22/P42/Q1**
- | | |
|----------------------------------|---|
| (a) 4 | 1 |
| (b) giant covalent | 1 |
| (c) silicon dioxide | 1 |
| (d) M1 layers | 2 |
| M2 hexagon(al) (rings of carbon) | 1 |
| (e) mobile electrons | 1 |
| (f) 60 | 1 |
| (g) limewater | 1 |
- 2. O/N 22/P42/Q2**
- | | |
|---|---|
| (a) (i) to prevent contact with air / oxygen and / or water | 1 |
| (b) (i) combustion | 1 |
| (ii) yellow | 1 |
| (iii) $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ | |
| M1 species (1) | |
| M2 balancing (1) | 2 |
| (iv) M1 eight crosses in second shell of both Na (1) | |
| M2 six dots and two crosses in second shell of O (1) | |
| M3 '+' charge on each Na ion on correct answer line and '2-' charge on O ion on correct answer line (1) | 3 |
| (c) (i) proton acceptor | 1 |
| (ii) pH 14 | 1 |
| (iii) yellow | 1 |
| (iv) M1 mol Na = $0.345 / 23 = 0.015(00)$ (1) | |
| M2 mol NaOH = M1 = $0.015(00)$ (1) | |
| M3 $M2 \times 1000 / 50 = 0.015(00) \times 20 = 0.3(00)$ (1) | |
| M4 M_r NaOH = 40(1) | |
| M5 $M4 \times M3 = 40 \times 0.3 = 12.0$ (g / dm ³) (1) | 5 |
| (d) (i) precipitation | 1 |
| (ii) red-brown | 1 |
| (iii) iron(III) hydroxide | 1 |
| (iv) $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$ | |
| M1 $\text{Fe}(\text{OH})_3$ (as only product) (1) | |
| M2 Fe^{3+} and 3OH^{-} (as only reactants) (1) | |
| M3 state symbols (1) | 3 |
- 3. O/N 22/P41/Q2**
- | | |
|--|---|
| (a) M1 metallic (1) | |
| M2 lattice of potassium ions (1) | |
| M3 sea of electrons (1) | |
| M4 attraction between potassium ions and electrons (1) | 4 |
| (b) (i) any two (one from each bullet point) | |
| • <i>physical constants</i> : high boiling point / melting point | |
| • <i>conductivity</i> : conduct electricity when aqueous / conduct electricity when molten | |
| • <i>solubility</i> : soluble in water | 2 |

- (ii) **M1** eight dots in third shell of both K (1)
M2 six crosses and two dots in third shell of S (1)
M3 '+' charge on each K on correct answer line and '2-' charge on S ion on correct answer line (1) 3
- (c) (i) lilac 1
(ii) OH⁻ 1
(iii) blue 1
(iv) **M1** mol of K = $2.34 / 39 = 0.06(00)$ (1)
M2 mol of H₂ = $0.06 / 2 = 0.03(00)$ (1)
M3 volume of H₂ = $0.03 \times 24\,000 = 720\text{ cm}^3$ (1) 3
- (d) (i) hydrochloric (acid) 1
(ii) neutralisation 1
(iii) titration 1
- (e) (i) white 1
(ii) silver chloride 1
(iii) $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
M1 AgCl (as *only* product) (1)
M2 Ag⁺ and Cl⁻ (as *only* reactants) (1)
M3 state symbols (1) 3
4. **M/J 22/P42/Q4**
(a) (pale) yellow and gas 1
(b) **Method 1**
M1 S 25.2 / 32 or 0.78/0.79 ...
and
F 74.8 / 19 or 3.93 / 3.94...
M2 ÷ both by 0.7875 = 1 : 5 or SF₅
M3 (254 ÷ 127 = 2 and) S₂F₁₀
Method 2
M1 $254 \times 25.2/100$ and $254 \times 74.8/100$ OR '64' and '190'
M2 64 / 32 and 190 / 19
M3 (2 and 10) to give S₂F₁₀ 3
- (c) N with 1 dot-cross bonding pair with each Cl
2 non-bonding dots for N
6 non-bonding crosses for Cl 3
- (d) two crosses in first shell of Li
7 dots and 1 cross in third shell of Cl
'+' charge on Li on correct answer line and '-' charge on Cl ion on correct answer line 3
- (e) ionic bonds in LiCl
attraction between molecules in NC₃
weaker attraction (between particles) in NC₃ ORA 3
5. **M/J 21/P42/Q5/a**
(a) (i) $6\text{Li} + \text{N}_2 \rightarrow 2\text{Li}_3\text{N}$
N₂ as reactant (1)
rest of equation (1) 2
(ii) new octet of 8 electrons consisting of 5 crosses and 3 dots in second shell (1)
charge of 3- (1) 2
6. **O/N 20/P42/Q2/f(i)**
(a) ions (1)
(ions) are fixed (in a lattice) (1)
ions are mobile (1) 3

7. **O/N 20/P41/Q1/a(ii)**
(a) silicon 1
8. **O/N 20/P41/Q4/c**
(a) 2 double bonds (1)
whole molecule correct (2 pairs of lone pairs on each O) (1) 2
9. **O/N 20/P41/Q6/c(i,ii)**
(a) (i) HCl 1
(ii)  1
10. **O/N 20/P43/Q2/d**
(a) (i) B, C and E 1
(ii) A 1
(iii) D 1
(iv) B and C 1
11. **O/N 20/P43/Q3/a**
(a) triple bond (1)
diagram completely correct (1) 2
12. **M/J 20/P42/Q1/b(v,vi)**
(a) (i) Ne 1
(ii) Li 1
13. **M/J 20/P42/Q2/a(i),b,c**
(a) (i) magnesium 2.8 (all crosses) (1)
fluorine 2.8 (seven dots and one cross in outer shell) (1)
Mg²⁺ and F⁻ (1) 3
(b) two single bonds (1)
one double bond (1)
six non-bonding electrons on both F atoms and four non-bonding electrons on O atom to complete the octet in each case (1) 3
(c) (i) forces of attraction between oppositely charged ions / ionic bonds (1)
strong / need a lot of energy to break / weaken (1) 2
(ii) forces of attraction between molecules (1)
weak / need a small of energy to break / weaken (1) 2
14. **M/J 20/P42/Q3/g(ii)**
(a) $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
formulae (1) state symbols (1) 2
15. **M/J 20/P41/Q2**
(a) metallic (bonding) 1
sea of electrons 1
positive ions 1
attraction between 1
(b) (i) Mg octet of eight dots 1
O octet of six crosses and two dots. 1
correct charges on both ions 1
(ii) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ 1

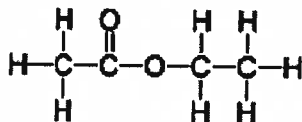
- M1 species
M2 balancing 2
- (c) (i) $2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$
M1 product species
M2 balancing 2
- (ii) (thermal) decomposition 1
- (iii) magnesium carbonate 1
magnesium hydroxide 1
16. M/J 20/P41/Q5/e,f(ii)
- (a) (i) functional group 1
- (ii) M1 $4 \times \text{C}-\text{H}$ dot cross bonds and 1 $\text{C}-\text{C}$ dot cross bond
M2 $1 \times \text{C}=\text{O}$ dot cross bond
M3 non-bonding electrons on O 3
- (b) (i) M1 any structure with correct valencies and formula of $\text{C}_3\text{H}_6\text{O}$
M2 $\text{C}=\text{O}$ bond on second carbon (of a chain of 3) 2
17. M/J 20/P43/Q3/c
- (a) magnesium ion has an outer shell with eight crosses
chloride ion has an outer shell with seven dots **and** one cross
chloride has a charge of 1^- **and** magnesium has a charge 2^+ 3
18. MAR 20/P42/Q3/c,f
- (a) (i) any number in the range $72 - 129^\circ\text{C}$ 1
- (ii) attraction increase (1)
between molecules (1) 2
- (iii) $3 \text{P}-\text{Cl}$ dot cross bonds (1)
2 (only) non-bonding electrons to make an octet on P (1)
6 (only) non-bonding electrons to make an octet on each Cl (1) 3
- (b) (i) oppositely charged ions
(ions) are attracted 2
- (ii) any two from:
physical constants: high boiling point / melting point
conductivity: conduct (electricity) when aqueous or conduct (electricity) when molten
solubility: soluble in water 2
19. O/N 19/P42/Q5/b,c(iii)
- (a) all bonding pairs correct (1)
H atoms have 2 electrons and C and O atoms have 8 electrons (1) 2
- (b) (i) molecular formula = $\text{C}_4\text{H}_8\text{O}_2$ (1)
correct ester link showing all atoms and all bonds (1)
rest of molecule (1) 3
20. O/N 19/P41/Q2/b
- (a) (i) 18 1
- (ii) gain of two electrons 1
- (iii) Ca / calcium 1
21. O/N 19/P41/Q7/c(ii)
- (a) 4 electrons in double bond between C and O (1)
all single bonds correct (1)
C and O each have 8 electrons in outer shell, all H have 2 electrons in outer shell (1) 3

22. O/N 19/P43/Q2
- (a) F 1
 - (b) I 1
 - (c) F (1)
H (1)
I (1) 3
 - (d) G (1)
good conductor when solid (1) 2
 - (e) D (1)
high melting point (1)
non-conductor of electricity when solid or liquid (1) 3
 - (f) E (1)
only conducts when liquid / conducts when liquid but not when solid (1) 2
23. O/N 19/P43/Q4/a,b
- (a) P₄ 1
 - (b) (i) $P_4 + 6Cl_2 \rightarrow 4PCl_3$
formulae correct (1) equation balanced (1) 2
 - (ii) 3 bonding pairs and 1 lone pair on P (1)
six non-bonding electrons on 3 chlorine atoms (1) 2
24. M/J 19/P42/Q1/f
- (a) silicon(IV) oxide 1
25. M/J 19/P42/Q2/b(i)
- (a) diamond / graphite / graphene 1
ANY TWO
26. M/J 19/P42/Q3/a,d
- (a) (i) covalent 1
 - (ii) forces of attraction between molecules AND
are weak / need a small amount of energy to break 1
1
 - (iii) no moving or flowing or mobile charged particles or ions or electrons
 - (b) M1 3 pairs of bonding electrons (1) 2
M2 only 1 lone pair on P (1)
27. M/J 19/P42/Q7/b
- (a) M1 $Pb \rightarrow Pb^{2+} + 2e^-$ / (1) 2
M2 $Ag^+ + e^- \rightarrow Ag$ (1)
28. M/J 19/P41/Q1/d
- (a) 2 8 3 (1)
2 8 8 (1) 2
29. M/J 19/P43/Q2/c,d
- (a) M1 (lattice of) positive ions / cations
M2 (delocalised / sea of) electrons
M3 attraction / attract between positive and negative 3
 - (b) (i) M1 magnesium ion
second shell shown containing 8 electrons shown as X
M2 oxide ion
second shell shown containing 8 electrons, two as X and six as ●
M3 charges:
magnesium ion as 2+ and oxide as 2- 3
(ii) strong forces of attraction (between oppositely charged ions) 1

(iii) ions / Mg^{2+} and O^{2-} / anions and cations can move (throughout the structure) 1

30. M/J 19/P43/Q5/a(ii)

(a) (i) correct structure of ethyl ethanoate showing all bonds



1

31. MAR 19/P42/Q3/b,c,e(ii),f(iii)

(a) M1 1 shared pair of electrons(1)

M2 6 non-bonding electrons on each atom to complete an octet(1)

2

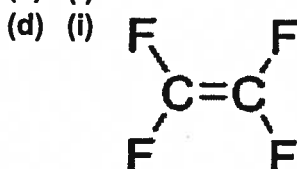
(b) $2\text{Na} + \text{F}_2 \rightarrow 2\text{NaF}$

M1 NaF anywhere(1) M2 equation fully correct(1)

2

(c) (i) covalent

1



1

32. O/N 18/P42/Q2/d

(a) (i) M1 Mg shown with new outer shell with 8 crosses;

M2 Both Cl atoms with a new outer shell with 7 dots and 1 cross;

M3 '2+' charge on Mg and '-' charge on each Cl;

3

(ii) M1 *Physical constants mark*

High melting point or high boiling point

M2 *Solubility mark*

Dissolve in water

M3 *Electrical conductivity mark*

Conduct (electricity) when molten

or

conduct (electricity) in aqueous solution

3

33. O/N 18/P41/Q2

(a) M1 11

M2 18

M3 2.8.8

M4 -1

4

(b) A and B

1

(c) Li / Lithium

1

(d) it has a complete or full or 8 electrons in the outer shell

1

34. M/J 18/P42/Q4/a,b

(a) $2\text{K(s)} + \text{Br}_2(\text{l}) \rightarrow 2\text{KBr(s)}$

1 mark for formulae all correct

1 mark for balancing

1 mark for state symbols

3

(b) (i) (ionic): made of, positive and negative ions / anions and cations /

oppositely charged ions / unlike charged ions / different charged ions

1

(lattice): regular / sequence / pattern / alternating / repeated / framework /

ordered / organised / network / uniform

1

- (ii) (in solid) ions don't move 1
(when molten) ions move / ions mobile 1
35. M/J 18/P41/Q4/a,b 1
(a) relative atomic mass 1
(b) C_4H_{10} is covalent 1
KF is ionic 1
36. MAR 18/P42/Q1/b
(a) (i) a shared pair of electrons (between two atoms)
M1 shared electrons 2
M2 pair of / two electrons
(ii) M1 three correct bonding pairs from one N atom to each of three F atoms
M2 (3 pairs of) non-bonding electrons on each of three F atoms to complete an octet 3
M3 (1 pair of) non-bonding electrons on N atom to complete an octet
37. MAR 18/P42/Q2/a
(a) (i) M1 correct orientation of '+' and '-' on front four ions 2
M2 rest of structure 1
(ii) 18 1
(iii) Ne or Neon 1
38. O/N 17/P41/Q6/b
(a) (i) oxide ion has an outer shell with six dots and two crosses 1
oxide ion has a charge of 2^- 1
(ii) (electrostatic) forces of attraction between ions 1
(are) strong OR require lots of energy to overcome 1
39. O/N 17/P43/Q1
(a) mixture 1
(b) element 1
(c) compound 1
(d) mixture 1
40. M/J 17/P42/Q1/b
(a) (i) (substance that) cannot be split up /
broken down into (two or more) simpler substances by chemical means
OR
(substance) made of **atoms** with the same atomic number /
number of protons / proton number 1
(ii) (two or more) **elements** joined or combined or bonded (together) 1
(iii) (particle) containing different numbers of protons and electrons
OR
atom or group of atoms that has gained or lost an electron / electrons 1
41. M/J 17/P42/Q2/c,e 1
(a) (i) covalent
(ii) award 1 mark for each correct property and one mark for each correct matching reason.
property: high melting point / high boiling point
reason: bonds between atoms are strong OR covalent bonds are strong /
bonds need large amount of energy to break
property: non-conductor / poor conductor (of electricity) / insulator
reason: no moving charged particles / no moving ions / no moving electrons /
all (outer shell) electrons used in bonding

property: hard

reason: bonds between atoms are strong **OR** covalent bonds are strong

property: brittle

reason: bonds between atoms are strong **OR** covalent bonds are strong /
bonds are directional

property: insoluble

reason: does not form hydrogen bonds with water / no ions that can be hydrated

4

(b) (i) carbon dioxide: (simple) molecular/simple covalent

1

silicon(IV) dioxide: macromolecular/giant molecular/giant covalent/giant atomic

1

(ii) carbon dioxide: **weak (force of) attraction** between molecules /
weak intermolecular forces/**weak** van der Waals'

forces /**weak** dispersion forces/**weak** London forces

1

silicon(IV) dioxide: covalent bonds are **strong**/force of attraction between atoms is
strong / no weak bonds (are present)/**all** bonds are **strong**

1

(weak) **forces of attraction** in carbon dioxide need small amounts of
energy or heat to break/**less energy** or heat needed to break **forces of attraction**
in carbon dioxide

OR

(strong) **bonds** in silicon(IV) dioxide need large amounts of **energy** or heat to break /
more energy or heat needed to break **bonds** in silicon(IV) dioxide

1

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Unit 4

STOICHIOMETRY

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 4.1 Stoichiometry
- 4.2 The mole concept

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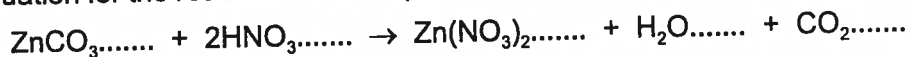
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Unit 4: Stoichiometry

4.1: Stoichiometry

1. O/N 21/P42/Q2/b

(a) The equation for the reaction between powdered zinc carbonate and dilute nitric acid is shown.



- (i) Complete the equation by adding state symbols. [2]
(ii) A student found that 2.5g of zinc carbonate required 20 cm³ of dilute nitric acid to react completely.

Calculate the concentration of dilute nitric acid using the following steps:

- calculate the mass of 1 mole of ZnCO₃

..... g

- calculate the number of moles of ZnCO₃ reacting

..... moles

- determine the number of moles of HNO₃ reacting

..... moles

- calculate the concentration of HNO₃.

..... mol/dm³ [4]

2. M/J 20/P41/Q1/b(ii,iii,iv),c,d

(a) (i) Identify the atom against which the relative masses of all other atoms are compared.

..... [1]

(ii) What is the name of the amount of any substance that contains 6.02×10^{23} particles?

..... [1]

(iii) The constant 6.02×10^{23} has a name.

What is the name of this constant?

..... [1]

(b) Part of the definition of relative atomic mass is 'the average mass of naturally occurring atoms of an element'.

Some relative atomic masses are not whole numbers.

Element Y has only two different types of atom, ^{69}Y and ^{71}Y .

The ratio of atoms present in element Y is shown.

$$^{69}\text{Y} : ^{71}\text{Y} = 3 : 2$$

- Calculate the relative atomic mass of element Y to one decimal place.

relative atomic mass =

- Identify element Y.

..... [3]

(c) Element Z is in Period 3 and Group V.

- (i) Identify element Z.

..... [1]

- (ii) Explain in terms of electron transfer why Z behaves chemically as a non-metal.

.....
..... [2]

3. MAR 20/P42/Q4/d

- (a) Deduce the charge on the iron ion in each of these compounds.

FeF_3

$\text{Fe}(\text{NO}_3)_3$ [2]

4. MAR 19/P42/Q3/f(v)

- (a) Tetrafluoroethene is an unsaturated compound with the formula C_2F_4 .

Tetrafluoroethene is the monomer used to make the polymer poly(tetrafluoroethene).

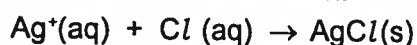
- (i) Deduce the empirical formula of:

tetrafluoroethene

poly(tetrafluoroethene). [2]

5. M/J 17/P42/Q5/b

- (a) A sample of vanadium chloride was weighed and dissolved in water. An excess of aqueous silver nitrate, acidified with dilute nitric acid, was added. A precipitate of silver chloride was formed. The ionic equation for this reaction is shown.



The mass of silver chloride formed was 2.87g.

- (i) State the colour of the precipitate of silver chloride.

..... [1]

- (ii) The relative formula mass of silver chloride, AgCl , is 143.5.
Calculate the number of moles in 2.87 g of AgCl .

moles of AgCl = mol [1]

- (iii) Use your answer to (b)(ii) and the ionic equation to deduce the number of moles of chloride ions, Cl^- , that produced 2.87 g of AgCl .

moles of Cl^- = mol [1]

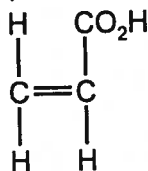
- (iv) The amount of vanadium chloride in the sample was 0.01 moles.
Use this and your answer to (b)(iii) to deduce the **whole number** ratio of moles of vanadium chloride : moles of chloride ions.
Deduce the formula of vanadium chloride.

moles of vanadium chloride : moles of chloride ions :

formula of vanadium chloride [2]

6. M/J 17/P42/Q6/c

- (a) The structure of prop-2-enoic (acrylic) acid is shown.



- (i) What would you see if prop-2-enoic acid were added to

aqueous bromine,

a solution of sodium carbonate. [2]

- (ii) Prop-2-enoic acid can be polymerised to form poly(acrylic acid).
Suggest the type of polymerisation that occurs and draw **one** repeat unit of the polymer.

type of polymerisation

repeat unit

7. O/N 16/P42/Q5/a,b,c

Chlorine, bromine and iodine are halogens.

- (a) Chlorine can be made in the laboratory by heating manganese(IV) oxide with concentrated hydrochloric acid.



Calculate the volume of $8.00 \text{ mol/dm}^3 \text{ HCl}(\text{aq})$ needed to react with 3.48 g of MnO_2 .

- moles of MnO_2 used

- moles of HCl needed

..... mol

- volume of HCl needed

..... mol

..... cm^3 [4]

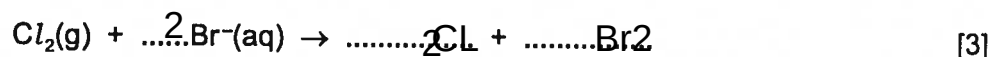
- (b) A student bubbled chlorine gas into a test-tube containing aqueous potassium bromide.

- (i) Describe the colour change seen in the test-tube.

from to [2]

- (ii) Complete the ionic equation for this reaction.

Include state symbols.



- (c) When one mole of bromine, Br_2 , reacts with one mole of propene, one organic product is formed.

- (i) Which part of the propene molecule reacts with bromine?

..... [1]

- (ii) What is the name of the type of reaction which takes place between bromine and propene?

..... [1]

8. O/N 16/P41/Q7

Calcium chloride can be made by reacting calcium carbonate with hydrochloric acid.



An excess of calcium carbonate was added to 50.0 cm^3 of 0.500 mol/dm^3 hydrochloric acid. The solution was filtered to remove the excess calcium carbonate.

(a) How many moles of HCl were used in this reaction?

..... mol [2]

(b) Deduce the number of moles of carbon dioxide gas made in this reaction.

..... mol [1]

(c) Calculate the mass of carbon dioxide made in this reaction.

..... g [2]

(d) Calculate the volume, in dm³, of carbon dioxide made in this reaction at room temperature and pressure (r.t.p.).

..... dm³ [1]

9. O/N 16/P43/Q3/a,b,d

When lead(II) nitrate is heated, two gases are given off and solid lead(II) oxide remains.
The equation for the reaction is shown.



(a) Calculate the M_r of lead(II) nitrate.

..... [1]

(b) 6.62 g of lead(II) nitrate are heated until there is no further change in mass.

(i) Calculate the mass of lead(II) oxide produced.

..... g [2]

(ii) Calculate the volume of oxygen, O₂, produced at room temperature and pressure (r.t.p.).

..... dm³ [2]

(c) Lead(II) oxide is insoluble. A student adds solid lead(II) oxide to dilute nitric acid until the lead(II) oxide is in excess. Aqueous lead(II) nitrate and water are produced.

(i) What is meant by the term *excess*?

.....[1]

(ii) How would the student know when the lead(II) oxide is in excess?

.....[1]

(iii) Write a chemical equation for the reaction.

.....[1]

10. M/J 16/P41/Q2/a

Period 3 contains the elements sodium to argon. This question asks about the chemistry of each of the Period 3 elements or their compounds.

(a) Sodium nitrate is a white crystalline solid. When heated it melts and the following reaction occurs.



A 3.40 g sample of sodium nitrate is heated.

Calculate the

- number of moles of NaNO_3 used,

..... mol

- number of moles of O_2 formed,

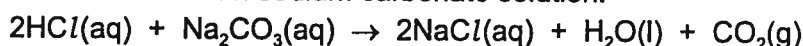
..... mol

- volume of O_2 formed, in dm^3 (measured at r.t.p.).

..... dm^3 [3]

11. M/J 16/P43/Q5

Dilute hydrochloric acid reacts with sodium carbonate solution.



(a) Explain why effervescence is seen during the reaction.

..... [1]

(b) Dilute hydrochloric acid was titrated with sodium carbonate solution.

- 10.0 cm^3 of 0.100 mol/dm^3 hydrochloric acid were placed in a conical flask.
- A few drops of methyl orange indicator were added to the dilute hydrochloric acid.
- The mixture was titrated with sodium carbonate solution.
- 16.2 cm^3 of sodium carbonate solution were required to react completely with the acid.

(i) What colour would the methyl orange indicator be in the hydrochloric acid?

..... [1]

(ii) Calculate how many moles of hydrochloric acid were used.

..... mol [1]

(iii) Use your answer to (b)(ii) and the equation for the reaction to calculate the number of moles of sodium carbonate that reacted.

..... mol [1]

(iv) Use your answer to (b)(iii) to calculate the concentration of the sodium carbonate solution in mol/dm^3 .

..... mol/dm^3 [2]

(c) In another experiment, 0.020 mol of sodium carbonate were reacted with excess hydrochloric acid.
Calculate the maximum volume (at r.t.p.) of carbon dioxide gas that could be made in this reaction.

..... dm^3 [3]

Answer Section

1. O/N 21/P42/Q2/b

- (a) (i) $\text{ZnCO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow \text{Zn}(\text{NO}_3)_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
 reactant states (1) product states (1) 2
- (ii) M1 125
 M2 $2.5 / 125 = 0.02(00)$
 M3 $0.02(00) \times 2 = 0.04(00)$
 M4 $0.04(00) \times 1000 / 20 = 2(00)$ 4

2. M/J 20/P41/Q1/b(ii,iii,iv),c,d

- (a) (i) ^{12}C 1
 (ii) a mole 1
 (iii) Avogadro constant 1
- (b) M1 $(3 \times 69) + (2 \times 71)$
 M2 $\frac{349}{5} = 69.8$
 M3 Y = Ga / gallium 3
- (c) (i) phosphorus / P 1
 (ii) gains electrons
 three electrons (when forming ion) 2

3. MAR 20/P42/Q4/d

- (a) $3+$
 $3+$ 2

4. MAR 19/P42/Q3/f(v)

- (a) (i) M1 $\text{CF}_2(1)$
 M2 $\text{CF}_2(1)$ 2

5. M/J 17/P42/Q5/b

- (a) (i) white 1
 (ii) 0.02 (mol) 1
 (iii) 0.02 (mol) 1
 (iv) 1:2 1
 VC l_2 1

6. M/J 17/P42/Q6/c

- (a) (i) colourless /decolourised 1
 bubbles /fizzing /effervescence 1
- (ii) addition 1
- $\begin{array}{c} \text{H} \quad \text{CO}_2\text{H} \\ | \quad | \\ \text{---C---C---} \\ | \quad | \\ \text{H} \quad \text{H} \end{array}$

repeat unit 1
 continuation bonds at both ends 1

7. O/N 16/P42/Q5/a,b,c

(a) 20 cm^3 **M1** M_r of MnO_2 : 87**M2** moles of MnO_2 used: $3.48/87 = 0.04$ **M3** moles of HCl needed: $0.04 \times 4 = 0.16$ **M4** volume of HCl needed: $(0.16/8.0) \times 1000$ **AND** 20 cm^3

4

(b) (i) from colourless
to yellow / orange / brown

2

(ii) $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{Cl}^-(\text{aq})$ **M1** (aq) as state symbols for the two products given**M2** correct products**M3** balancing

3

(c) (i) the (C=C) double bond

1

(ii) addition **OR** bromination

1

8. O/N 16/P41/Q7

(a) 0.025

M1 $50/1000 (=0.05)$ **M2** $(0.05 \times 0.5) = 0.025$

1

1

1

(b) 0.0125

(c) 0.55

M1 44**M2** 0.55

1

1

1

(d) 0.3

9. O/N 16/P43/Q3/a,b,d

(a) 331

1

(b) (i) **M1** $\text{mol} = 6.62/331$ **OR** 0.02

1

M2 $0.02 \times 223 = 4.46 \text{ (g)}$

1

(ii) **M1** $\text{mol O}_2 = 0.02 \div 2$ **OR** 0.01

1

M2 $\text{vol} = 0.01 \times 24 = 0.24 \text{ (dm}^3\text{)}$

1

(c) (i) more than enough to react (with all the acid)

OR

some lead oxide remains after the reaction

OR

(nitric) acid is limiting

1

(ii) solid stops dissolving

1

(iii) $\text{PbO} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$ **OR** $\text{PbO} + 2\text{H}^+ \rightarrow \text{Pb}^{2+} + \text{H}_2\text{O}$

1

10. M/J 16/P41/Q2/a

(a) number of moles of NaNO_3 used: $3.40/85 = 0.04(00) \text{ (mol)}$ **OR** $4.(00) \times 10^{-2} \text{ (mol)}$;number of moles of O_2 formed: $0.04/2 = 0.02(00) \text{ (mol)}$ **OR** $2.(00) \times 10^{-2} \text{ (mol)}$;volume of O_2 formed: $0.02 \times 24 = 0.48 \text{ (dm}^3\text{)}$;

3

11. M/J 16/P43/Q5

- (a) carbon dioxide / a gas is made; 1
- (b) (i) red; 1
- (ii) 0.001; 1
- (iii) 0.0005; 1
- (iv) 0.031 (2 marks) 1
- M1** (iii) / 0.0162; 2
- (c) 0.48 (dm³)
- M1** moles carbon dioxide = 0.02;
- M2** volume carbon dioxide = 0.02 × 24;
- M3** = 0.48 (dm³); 3

4.2: The mole concept**1. O/N 21/P41/Q2/d,e**

This question is about copper and its compounds.

- (a) When solid copper(II) nitrate is heated copper(II) oxide, nitrogen dioxide and oxygen are formed.



Calculate the volume of nitrogen dioxide formed at room temperature and pressure when 4.7 g of $\text{Cu}(\text{NO}_3)_2$ is heated.

Use the following steps:

- calculate the mass of one mole of $\text{Cu}(\text{NO}_3)_2$

..... g

- calculate the number of moles of $\text{Cu}(\text{NO}_3)_2$ used

..... moles

- determine the number of moles of nitrogen dioxide formed

..... moles

- calculate the volume of nitrogen dioxide formed at room temperature and pressure.

..... dm^3 [4]

- (b) Write the chemical equation to show the action of heat on sodium nitrate, NaNO_3 .

..... [2]

2. O/N 21/P43/Q3/a,b,c,d

Lead is a metallic element in Group IV. One of the ores of lead is galena, which is an impure form of lead(II) sulfide, PbS .

Lead also occurs in the ore cerussite, which contains lead(II) carbonate, PbCO_3 .

- (a) Calculate the relative formula mass, M_r , of PbCO_3 .

M_r of PbCO_3 = [1]

- (b) The M_r of PbS is 239.

Calculate the percentage of lead by mass in PbS .

percentage of lead by mass in PbS [1]

- (c) The percentage of lead by mass in PbCO_3 is 77.5%.

Use this information and your answer to (b) to suggest whether it would be better to extract lead from PbCO_3 or PbS .

Give a reason for your answer.

..... [1]

- (d) When lead(II) carbonate is heated it decomposes into lead(II) oxide, PbO , and carbon dioxide. Write a chemical equation for this reaction.

..... [1]

3. O/N 21/P43/Q4/c

- (a) The Avogadro constant is the number of particles in 1 mole.

The numerical value of the Avogadro constant is 6.02×10^{23} .

- (i) Calculate the number of molecules in 22.0 g of carbon dioxide, CO_2 .

..... molecules [2]

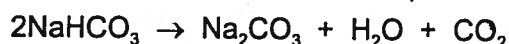
- (ii) Calculate the number of molecules in 6.00 dm^3 of carbon dioxide gas at room temperature and pressure.

..... molecules [1]

4. M/J 21/P42/Q3

Sodium hydrogencarbonate is found in baking powder.

When sodium hydrogencarbonate is heated it forms three products.



- (a) Name the type of reaction that takes place when sodium hydrogencarbonate reacts in this way.

..... [1]

- (b) Calculate the volume of carbon dioxide formed at room temperature and pressure when 12.6 g of NaHCO_3 is heated using the following steps:

- determine the mass of one mole of NaHCO_3

..... g

- calculate the number of moles of NaHCO_3 used

..... moles

- determine the number of moles of carbon dioxide formed

..... moles

- calculate the volume of carbon dioxide formed at room temperature and pressure.

..... dm³ [4]

- (c) Limewater is aqueous calcium hydroxide. Carbon dioxide turns limewater milky because a white precipitate forms.

Write the formula of:

- calcium hydroxide
- the white precipitate that forms when limewater turns milky..... [2]

5. M/J 21/P42/Q5/d

This question is about compounds of nitrogen.

- (a) Ammonium nitrate, NH_4NO_3 , is a compound of nitrogen.

- (i) Calculate the percentage by mass of nitrogen in ammonium nitrate.

percentage by mass of nitrogen = [2]

- (ii) State a use of ammonium nitrate in agriculture.

..... [1]

- (iii) State the name of a compound that will displace ammonia from ammonium nitrate.

..... [1]

6. M/J 21/P41/Q7

Many organic compounds contain carbon, hydrogen and oxygen only.

- (a) An organic compound **V** has the following composition by mass.

C, 48.65%; H, 8.11%; O, 43.24%

Calculate the empirical formula of compound **V**.

empirical formula = [3]

- (b) Compound **W** has the empirical formula CH_4O and a relative molecular mass of 32.
Calculate the molecular formula of compound **W**

molecular formula = [1]

- (c) Compounds **X** and **Y** have the same general formula.
X and **Y** are both carboxylic acids.

Compound **X** has the molecular formula $C_2H_4O_2$.

Compound **Y** has the molecular formula $C_4H_8O_2$.

(i) Deduce the general formula of compounds **X** and **Y**.

..... [1]

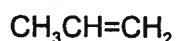
(ii) Draw the structure of compound **Y**. Show all of the atoms and all of the bonds.
Name compound **Y**.

name [3]

(iii) Give the name used to describe a 'family' of similar compounds with the same general formula, similar chemical properties and the same functional group.

..... [1]

(d) Propene is an unsaturated hydrocarbon. The formula of propene is shown.



(i) State the colour change observed when propene is added to aqueous bromine.

from to [1]

(ii) Propene can be produced by cracking long chain alkanes.

Pentadecane, $C_{15}H_{32}$, is cracked to produce an alkane and propene in a 1 : 2 molar ratio.
Complete the chemical equation for this reaction.



(iii) Propene can be converted into poly(propene).

Name the type of polymerisation that occurs when propene is converted into poly(propene).

..... [1]

(iv) Complete the diagram to show a section of poly(propene).



[2]

7. M/J 21/P43/Q7

Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **R** has the following composition by mass.

C, 69.77%; H, 11.63%; O, 18.60%

Calculate the empirical formula of compound **R**.

empirical formula = [2]

- (b) Compound **S** has the empirical formula CH_2O and a relative molecular mass of 60.
Calculate the molecular formula of compound **S**.

molecular formula = [2]

- (c) Compounds **T** and **V** have the same molecular formula, $\text{C}_3\text{H}_6\text{O}_2$.

- Compound **T** is an ester.
- Compound **V** contains a $-\text{COOH}$ functional group.

- (i) State the name given to compounds with the same molecular formula but different structures.

..... [1]

- (ii) Name the homologous series that **V** is a member of.

..... [1]

- (iii) Draw a structure of compound **T**. Show all of the atoms and all of the bonds.
Name compound **T**.

name [3]

- (iv) Draw the structure of compound **V**. Show all of the atoms and all of the bonds.
Name compound **V**.

name [2]

- (d) Ethanol can be produced from long chain alkanes such as decane, $\text{C}_{10}\text{H}_{22}$, in a two-step process.



For each of the two steps:

- name the type of chemical reaction that occurs
- write a chemical equation.

step 1: decane to ethene

type of reaction

chemical equation

step 2: ethene to ethanol

type of reaction

chemical equation [4]

8. O/N 20/P42/Q2/b(iii,iv),c,e

Soluble salts can be made by adding a metal carbonate to a dilute acid.

- (a) A student wanted to make hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, by adding excess iron(II) carbonate to dilute sulfuric acid. The student followed the procedure shown.

step 1 Add dilute sulfuric acid to a beaker.

step 2 Add small amounts of iron(II) carbonate to the dilute sulfuric acid in the beaker until the iron(II) carbonate is in excess.

step 3 Filter the mixture formed in **step 2**.

step 4 Heat the filtrate until it is a saturated solution. Allow to cool.

step 5 Once cold, pour away the remaining solution. Dry the crystals between filter papers.

- (i) Describe what should be done during **step 3** to ensure there is a maximum yield of crystals.

..... [1]

- (ii) A saturated solution is formed in **step 4**.

Describe what a saturated solution is.

.....
..... [2]

- (b) On analysing the crystals, the student found that one mole of the hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, had a mass of 278 g.

Determine the value of x using the following steps:

- calculate the mass of one mole of FeSO_4

mass = g

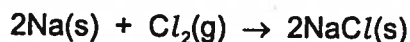
- calculate the mass of H_2O present in one mole of $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$

mass of H_2O = g

- determine the value of x .

x = [3]

- (c) Sodium chloride is an ionic salt. It can be made by reacting sodium with chlorine gas. The equation for this reaction is shown.



Calculate the volume of chlorine gas, in cm^3 , that reacts to form 2.34 g of NaCl .

The reaction takes place at room temperature and pressure.

volume of chlorine gas = cm^3 [3]

9. O/N 20/P41/Q3/b

- (a) A 2.8 g sample of impure ammonium sulfate is found to contain 0.7 g of impurities.
Calculate the percentage of ammonium sulfate in this sample.

percentage of ammonium sulfate = % [1]

10. O/N 20/P41/Q5/d

- (a) A sample of sulfuric acid, H_2SO_4 , has a concentration of 0.75 mol/dm^3 .
Calculate the concentration of sulfuric acid in g/dm^3 .

..... g/dm^3 [2]

11. O/N 20/P41/Q6/b,f

- (a) One mole of ethane, C_2H_6 , contains 6.02×10^{23} molecules.
Calculate how many molecules are in 15 g of ethane.

number of ethane molecules = [1]

- (b) A compound contains 85.7% carbon and 14.3% hydrogen by mass.

- (i) Calculate the empirical formula of this compound.
Show your working.

..... [2]

- (ii) The molecular mass of the compound is 112.
Calculate the molecular formula of this compound.

..... [1]

12. O/N 20/P43/Q4/d(ii)

Zinc is manufactured from zinc blende. Zinc blende is an ore which consists mainly of zinc sulfide, ZnS .

- (a) When hydrated magnesium sulfate crystals, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$, are heated they give off water.



A student carries out an experiment to determine the value of x in $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

Step 1 Hydrated magnesium sulfate crystals were weighed.

Step 2 Hydrated magnesium sulfate crystals were heated.

Step 3 The remaining solid was weighed.

- (i) In an experiment, all the water was removed from 1.23 g of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$. The mass of MgSO_4 remaining was 0.60 g.

M_r : $\text{MgSO}_4 = 120$; M_r : $\text{H}_2\text{O} = 18$

Determine the value of x using the following steps.

- Calculate the number of moles of MgSO_4 remaining.

moles of $\text{MgSO}_4 = \dots\dots\dots$

- Calculate the mass of H_2O given off.

mass of $\text{H}_2\text{O} = \dots\dots\dots$ g

- Calculate the moles of H_2O given off.

moles of $\text{H}_2\text{O} = \dots\dots\dots$

- Determine the value of x .

$x = \dots\dots\dots$ [4]

13. M/J 20/P42/Q4/e,f

Oxygen is produced by the decomposition of hydrogen peroxide. Manganese(IV) oxide is the catalyst for this reaction.

- (a) The equation for the decomposition of hydrogen peroxide is shown.



25.0 cm³ of aqueous hydrogen peroxide forms 48.0 cm³ of oxygen at room temperature and pressure (r.t.p.).

Calculate the concentration of aqueous hydrogen peroxide at the start of the experiment using the following steps.

- Calculate the number of moles of oxygen formed.

$\dots\dots\dots$ mol

- Deduce the number of moles of hydrogen peroxide that decomposed.

$\dots\dots\dots$ mol

- Calculate the concentration of hydrogen peroxide in mol/dm³.

$\dots\dots\dots$ mol/dm³ [3]

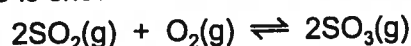
- (b) Oxygen can also be produced by the decomposition of potassium chlorate(V), KClO_3 .
The only products of this decomposition are potassium chloride and oxygen.
Write a chemical equation for this decomposition.

..... [2]

14. M/J 20/P41/Q3/b(vi)

Sulfur dioxide, SO_2 , is used in the manufacture of sulfuric acid.

- (a) The next stage of the process is a reaction which can reach equilibrium.
The equation for this stage is shown.



- (i) Calculate the percentage by mass of sulfur in sulfur trioxide, SO_3 .

percentage = [2]

15. M/J 20/P41/Q4/d

- (a) A student wanted to find the concentration of some dilute sulfuric acid by titration. The student found that 25.0 cm^3 of 0.0400 mol/dm^3 $\text{NaOH}(\text{aq})$ reacted exactly with 20.0 cm^3 of $\text{H}_2\text{SO}_4(\text{aq})$.
(i) Name a suitable indicator to use in this titration.

..... [1]

- (ii) Calculate the concentration of the $\text{H}_2\text{SO}_4(\text{aq})$ in mol/dm^3 using the following steps.
- Calculate the number of moles of NaOH in 25.0 cm^3 .

moles =

- Deduce the number of moles of H_2SO_4 that reacted with the 25.0 cm^3 of $\text{NaOH}(\text{aq})$.

moles =

- Calculate the concentration of $\text{H}_2\text{SO}_4(\text{aq})$ in mol/dm^3 .

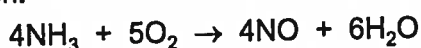
concentration = mol/dm^3 [3]

- (iii) Calculate the concentration of the 0.0400 mol/dm^3 $\text{NaOH}(\text{aq})$ in g/dm^3 .

concentration = g/dm^3 [2]

16. M/J 20/P43/Q2/d(ii,iii,iv)

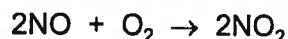
- (a) Ammonia, NH_3 , is used to produce nitric acid, HNO_3 . This happens in a three-stage process.
Stage 1 is a redox reaction.



- (i) In this reaction the predicted yield of NO is 512 g. The actual yield is 384 g.
Calculate the percentage yield of NO in this reaction.

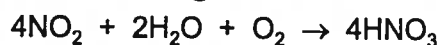
percentage yield of NO = [1]

- (ii) The equation for the reaction in **stage 2** is shown.



Which major environmental problem does NO₂ cause if it is released into the atmosphere?

- [1]
(iii) The equation for the reaction in **stage 3** is shown.



Calculate the volume of O₂ gas, at room temperature and pressure (r.t.p.), needed to produce 1260 g of HNO₃.

Use the following steps.

- Calculate the number of moles of HNO₃.

moles of HNO₃ =

- Deduce the number of moles of O₂ that reacted.

moles of O₂ =

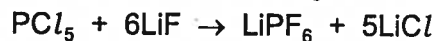
- Calculate the volume of O₂ gas that reacts at room temperature and pressure (r.t.p.).

volume of O₂ gas = dm³ [4]

17. MAR 20/P42/Q3/e

The Periodic Table is a method of classifying elements.

- (a) PCl₅ reacts with lithium fluoride, LiF, to form LiPF₆.



Calculate the mass of LiF needed to form 3.04 g of LiPF₆ using the following steps.

- Calculate the number of moles of LiPF₆ formed.

[M_r: LiPF₆, 152]

number of moles =

- Deduce the number of moles of LiF needed.

number of moles =

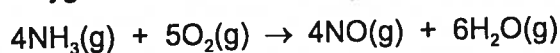
- Calculate the mass of LiF needed.

mass = g [3]

18. O/N 19/P42/Q3/d(i)

Ammonia is an important chemical.

(a) Ammonia reacts with oxygen as shown.



- (i) Calculate the volume of oxygen at room temperature and pressure, in dm^3 , that reacts with 4.80 dm^3 of ammonia.

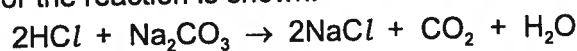
volume = dm^3 [3]**19. O/N 19/P41/Q2/e**

- (a) Sulfur dioxide reacts with aqueous sodium sulfite to produce a compound with the following composition by mass: 29.1% Na, 40.5% S and 30.4% O.
Calculate the empirical formula of this compound.

empirical formula = [3]

20. O/N 19/P41/Q6/a,b,c(i)Dilute hydrochloric acid, $\text{HCl}(\text{aq})$, reacts with aqueous sodium carbonate, $\text{Na}_2\text{CO}_3(\text{aq})$.

The chemical equation for the reaction is shown.



- (a) A 25.0 cm^3 portion of $\text{Na}_2\text{CO}_3(\text{aq})$ was placed in a conical flask with a few drops of a suitable indicator. It was titrated against $\text{HCl}(\text{aq})$ of concentration 0.180 mol/dm^3 .

 20.0 cm^3 of $\text{HCl}(\text{aq})$ was required to reach the end-point.Calculate the concentration of the $\text{Na}_2\text{CO}_3(\text{aq})$, in mol/dm^3 , using the following steps.

- Calculate the number of moles of HCl used in the titration.

..... mol

- Calculate the number of moles of Na_2CO_3 contained in the 25.0 cm^3 portion of $\text{Na}_2\text{CO}_3(\text{aq})$.

..... mol

- Calculate the concentration of the $\text{Na}_2\text{CO}_3(\text{aq})$ in mol/dm^3 .

..... mol/dm^3 [3]

- (b) In another experiment, the volume of carbon dioxide, CO_2 , produced was 48.0 cm^3 , measured at room temperature and pressure.

How many moles of CO_2 is this?

moles of CO_2 = mol [1]

- (c) A sample of concentrated hydrobromic acid, $\text{HBr}(\text{aq})$, was electrolysed using platinum electrodes.

The concentration of the hydrobromic acid was 8.89 mol/dm^3 .

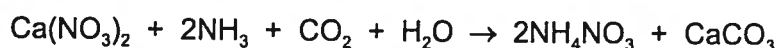
- (i) Calculate the concentration of the $\text{HBr}(\text{aq})$ in g/dm^3 .

concentration of $\text{HBr}(\text{aq})$ = g/dm^3 [1]

21. O/N 19/P43/Q5

Nitrates such as ammonium nitrate are used as fertilisers.

The final stage in the production of ammonium nitrate is shown in the equation.



Calculate the maximum mass of ammonium nitrate that can be produced from 820 g of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, using the following steps.

The relative formula mass, M_r , of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, = 164.

- Calculate the number of moles of $\text{Ca}(\text{NO}_3)_2$ in 820 g.

..... mol

- Deduce the number of moles of NH_4NO_3 produced.

..... mol

- Calculate the M_r of NH_4NO_3 .

M_r of $\text{NH}_4\text{NO}_3 = \dots\dots\dots$

- Calculate the maximum mass of ammonium nitrate produced.

 $\dots\dots\dots \text{ g [4]}$ **22. O/N 19/P43/Q6**

This question is about sulfuric acid and substances that can be made from sulfuric acid.

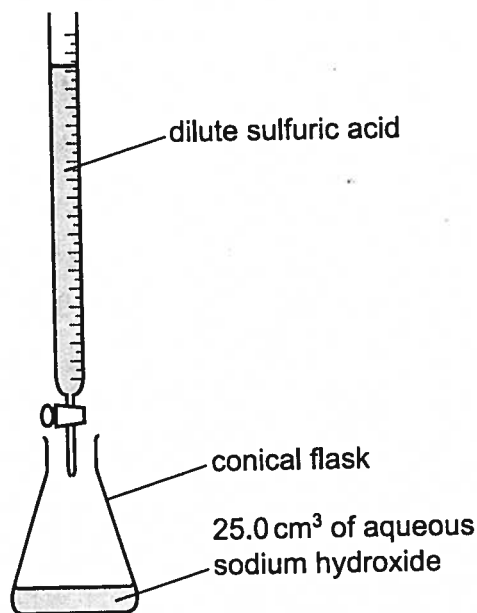
- (a) Sulfuric acid is a strong acid.

What is meant by the term *strong acid*?

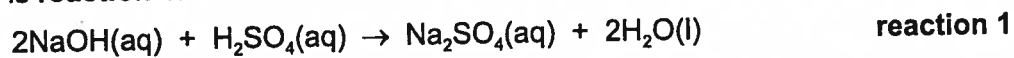
strong $\dots\dots\dots$

acid $\dots\dots\dots$ [2]

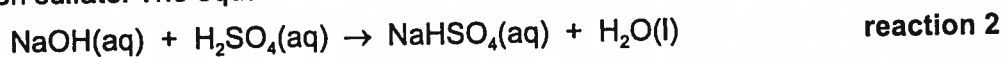
- (b) Dilute sulfuric acid and aqueous sodium hydroxide are used to make aqueous sodium sulfate, $\text{Na}_2\text{SO}_4(\text{aq})$, or aqueous sodium hydrogen sulfate, $\text{NaHSO}_4(\text{aq})$. The method includes use of the following apparatus.



25.0 cm^3 of aqueous sodium hydroxide of concentration 0.100 mol/dm^3 was neutralised by 25.0 cm^3 of dilute sulfuric acid of concentration 0.0500 mol/dm^3 . The equation for the reaction is shown. This is **reaction 1**.



The same technique and the same solutions can be used to make aqueous sodium hydrogen sulfate. The equation for the reaction is shown. This is **reaction 2**.



Complete the table to calculate the volume of dilute sulfuric acid that reacts with 25.0 cm³ of aqueous sodium hydroxide in **reaction 2**.

	volume of 0.0500 mol/dm ³ dilute sulfuric acid in cm ³	volume of 0.100 mol/dm ³ aqueous sodium hydroxide in cm ³
reaction 1	25.0	25.0
reaction 2		25.0

[1]

- (c) Aqueous sodium hydrogen sulfate, NaHSO₄(aq), contains the ions Na⁺(aq), H⁺(aq) and SO₄²⁻(aq).

Describe what you would **see** if the following experiments were done.

- (i) A flame test was done on aqueous sodium hydrogen sulfate.

..... [1]

- (ii) Solid copper(II) oxide was added to aqueous sodium hydrogen sulfate and the mixture was warmed.

.....
..... [2]

- (d) A test can be done to show the presence of SO₄²⁻(aq) by adding acidified aqueous barium chloride or acidified aqueous barium nitrate.

- (i) State the observation that would show that SO₄²⁻ is present.

..... [1]

- (ii) Write an ionic equation for the reaction that occurs if SO₄²⁻ is present. Include state symbols.

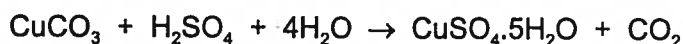
..... [2]

23. M/J 19/P42/Q5/a

Copper(II) sulfate crystals, CuSO₄·5H₂O, are hydrated.

Copper(II) sulfate crystals are made by reacting copper(II) carbonate with dilute sulfuric acid.

The equation for the overall process is shown.



step 1 Powdered solid copper(II) carbonate is added to 50.0 cm³ of 0.05 mol/dm³ sulfuric acid until the copper(II) carbonate is in excess.

step 2 The excess of copper(II) carbonate is separated from the aqueous copper(II) sulfate.

step 3 The aqueous copper(II) sulfate is heated until the solution is saturated.

step 4 The solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

- (a) Calculate the maximum mass of the copper(II) sulfate crystals, CuSO₄·5H₂O, that can form using the following steps.

- Calculate the number of moles of H_2SO_4 in 50.0 cm^3 of $0.05 \text{ mol/dm}^3 \text{ H}_2\text{SO}_4$.

..... mol

- Determine the number of moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ that can form.

..... mol

- The M_r of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 250.

Calculate the maximum mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ that can form.

..... g [3]

24. M/J 19/P41/Q4/c

Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Both ethanoic acid and hydrochloric acid dissociate in aqueous solution.

(a) Hydrochloric acid produces salts called chlorides.

Magnesium carbonate reacts with hydrochloric acid to produce magnesium chloride.



A student used 50.00 cm^3 of 2.00 mol/dm^3 hydrochloric acid in an experiment to produce magnesium chloride.

Calculate the mass, in g, of magnesium carbonate needed to react exactly with 50.00 cm^3 of 2.00 mol/dm^3 hydrochloric acid using the following steps.

- Calculate the number of moles of HCl present in 50.00 cm^3 of $2.00 \text{ mol/dm}^3 \text{ HCl}$.

..... mol

- Determine the number of moles of MgCO_3 which would react with 50.00 cm^3 of $2.00 \text{ mol/dm}^3 \text{ HCl}$.

..... mol

- Calculate the relative formula mass, M_r , of MgCO_3 .

 M_r of MgCO_3 =

- Calculate the mass of MgCO_3 needed to react exactly with 50.00 cm^3 of $2.00 \text{ mol/dm}^3 \text{ HCl}$.

mass = g [4]

25. M/J 19/P43/Q3/b,e

(a) Some car airbags contain sodium azide.

When a car airbag is used the sodium azide, NaN_3 , decomposes.

The products are nitrogen and sodium.

The equation for the decomposition of sodium azide is shown.



Calculate the mass, in g, of sodium azide needed to produce 144 dm^3 of nitrogen using the following steps.

- Calculate the number of moles in 144 dm^3 of N_2 measured at room temperature and pressure.

moles of N_2 = mol

- Determine the number of moles of NaN_3 needed to produce this number of moles of N_2 .

moles of NaN_3 = mol

- Calculate the relative formula mass, M_r , of NaN_3 .

M_r =

- Calculate the mass of NaN_3 needed to produce 144 dm^3 of N_2 .

..... g [4]

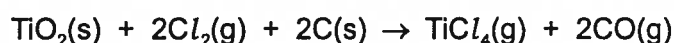
(b) An organic compound made from sodium azide has the composition by mass: 49.5% carbon, 7.2% hydrogen and 43.3% nitrogen.

Calculate the empirical formula of the organic compound.

[3]

26. MAR 19/P42/Q5/b

(a) Calculate the volume of chlorine gas, $\text{Cl}_2(\text{g})$, at room temperature and pressure, that reacts completely with 400 g of $\text{TiO}_2(\text{s})$ using the following steps.



- Calculate the relative formula mass, M_r , of TiO_2 .

M_r of TiO_2 =

- Calculate the number of moles in 400 g of TiO_2 .

..... mol

- Determine the number of moles of Cl_2 that react with 400 g of TiO_2 .

moles of Cl_2 = mol

- Calculate the volume of Cl_2 that reacts with 400 g of TiO_2 .

volume of Cl_2 = dm^3 [4]

27. O/N 18/P41/Q3/c

(a) A teacher heated 18.8 g of copper(II) nitrate.

- (i) Calculate the number of moles of copper(II) nitrate present in the 18.8 g.

..... mol [2]

- (ii) Calculate the maximum number of moles of oxygen that can be made by heating 18.8 g of copper(II) nitrate.

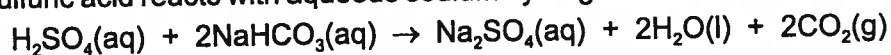
..... mol [1]

- (iii) Calculate the maximum volume of oxygen at room temperature and pressure, in cm^3 , that can be made by heating 18.8 g of copper(II) nitrate.

..... cm^3 [1]

28. O/N 18/P41/Q4/d

(a) Dilute sulfuric acid reacts with aqueous sodium hydrogencarbonate in a neutralisation reaction.



In a titration, 0.200 mol/dm^3 aqueous sodium hydrogencarbonate was used to neutralise 20.0 cm^3 of dilute sulfuric acid of concentration 0.150 mol/dm^3 .

- (i) Calculate the number of moles of dilute sulfuric acid used in the titration.

..... mol [1]

- (ii) Calculate the number of moles of sodium hydrogencarbonate needed to neutralise the dilute sulfuric acid.

..... mol [1]

- (iii) Calculate the volume, in cm^3 , of 0.200 mol/dm^3 aqueous sodium hydrogencarbonate needed to neutralise the dilute sulfuric acid.

..... cm^3 [1]

29. M/J 18/P42/Q6/a

- (a) All sodium salts are soluble in water. All nitrates are soluble in water. Barium carbonate is insoluble in water.

Describe how you would make a pure, dry sample of barium carbonate by precipitation.

Include:

- the names of the starting materials
- full practical details
- a chemical equation.

.....
.....
.....
.....
..... [5]

30. M/J 18/P42/Q7/a,b

Many organic compounds, such as alcohols, carboxylic acids and esters, contain the elements carbon, hydrogen and oxygen only.

- (a) Compound **R** has the following composition by mass: C, 60.00%; H, 13.33%; O, 26.67%.

Calculate the empirical formula of compound **R**.

empirical formula = [2]

- (b) Compound **S** has the empirical formula $\text{C}_2\text{H}_4\text{O}$ and a relative molecular mass of 88.

Calculate the molecular formula of compound **S**.

molecular formula = [2]

31. M/J 18/P41/Q4

This question is about masses, volumes and moles.

(a) Which term is defined by the following statement?

The average mass of naturally occurring atoms of an element on a scale where the ^{12}C atom has a mass of exactly 12 units.

..... [1]

(b) Butane, C_4H_{10} , has a relative **molecular** mass of 58.

Potassium fluoride, KF, has a relative **formula** mass of 58.

Explain why the term relative molecular mass can be used for butane but **cannot** be used for potassium fluoride.

..... [2]

(c) A 0.095 g sample of gaseous element Y occupies 60.0 cm^3 at room temperature and pressure.

- Determine the number of moles of element Y in 60.0 cm^3 .

moles of element Y mol

- Calculate the relative molecular mass of element Y and hence suggest the identity of element Y.

relative molecular mass

identity of element Y [3]

(d) A 1.68 g sample of phosphorus was burned and formed 3.87 g of an oxide of phosphorus.
Calculate the empirical formula of this oxide of phosphorus.

empirical formula [4]

(e) Another oxide of phosphorus has the empirical formula P_2O_3 .

One molecule of this oxide of phosphorus contains four atoms of phosphorus.

Calculate the mass of **one** mole of this oxide of phosphorus.

mass g [2]

32. M/J 18/P43/Q5/a

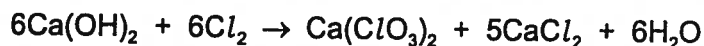
- (a) Nickel(II) iodide crystals are hydrated. A sample of hydrated nickel(II) iodide crystals has the following composition by mass: Ni, 14.01%; I, 60.33%; H, 2.85%; O, 22.81%.

Calculate the empirical formula of the hydrated nickel(II) iodide crystals.

empirical formula = [2]

33. M/J 18/P43/Q6/a

Calcium chlorate(V), $\text{Ca}(\text{ClO}_3)_2$, is made by reacting calcium hydroxide with chlorine gas.



- (a) 8.88 g of calcium hydroxide and 7200 cm^3 of chlorine gas are mixed together.

(i) How many moles is 8.88 g of calcium hydroxide?

(ii) How many moles of chlorine gas is 7200 cm^3 ?

..... mol [2]

- (iii) What is the maximum **number of moles** of calcium chlorate(V) that can be made from 8.88 g of calcium hydroxide and 7200 cm^3 of chlorine gas?

..... mol [1]

- (iv) What is the maximum **mass** of calcium chlorate(V) that can be made from 8.88 g of calcium hydroxide and 7200 cm^3 of chlorine gas?

..... mol [1]

..... g [2]

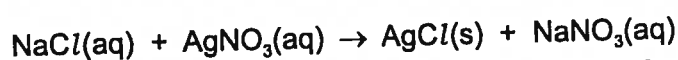
The experiment is repeated using different amounts of calcium hydroxide and chlorine gas. The maximum mass of calcium chlorate(V) that can be made in the experiment is 4.84 g.

- (v) The actual mass of calcium chlorate(V) made in the experiment is 3.63 g.
Calculate the percentage yield.

percentage yield = % [1]

34. MAR 18/P42/Q2/c(iv)

- (a) Silver chloride can be made by reacting aqueous sodium chloride with aqueous silver nitrate. The other product of the reaction is sodium nitrate. The chemical equation for the reaction is shown.



A student attempted to make the maximum amount of **sodium nitrate** crystals. The process involved three steps.

step 1 The student added aqueous sodium chloride to aqueous silver nitrate and stirred. Neither reagent was in excess.

step 2 The student filtered the mixture. The student then washed the residue and added the washings to the filtrate.

step 3 The student obtained sodium nitrate crystals from the filtrate.

- (i) The student started with 20 cm³ of 0.20 mol/dm³ NaCl(aq).
• Determine the amount of NaCl(aq) used.

amount of NaCl(aq) used = mol

The yield of NaNO₃ crystals was 90%.

- Calculate the mass of NaNO₃ crystals made.

mass of NaNO₃ crystals = g [4]

35. MAR 18/P42/Q3/c

- (a) Forsterite is another rock which contains a magnesium compound.

A sample of forsterite has the following composition by mass: Mg, 2.73 g; Si, 1.58 g; O, 3.60 g.

Calculate the empirical formula of forsterite.

empirical formula = [2]

36. MAR 18/P42/Q5/b

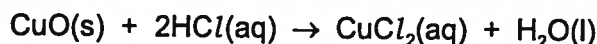
Alcohols are a 'family' of organic molecules which have the same general formula.

(a) Give the general formula of alcohols.

..... [1]

37. O/N 17/P41/Q7/b,c

Copper(II) oxide reacts with dilute hydrochloric acid.



6.00 g of copper(II) oxide were added to 50.0 cm³ of 1.00 mol/dm³ hydrochloric acid. This was an excess of copper(II) oxide.

(a) (i) Calculate the number of moles of copper(II) oxide added to the hydrochloric acid.

moles of copper(II) oxide mol [2]

(ii) Calculate the number of moles of hydrochloric acid used.

moles of hydrochloric acid mol [1]

(iii) Calculate the mass of copper(II) oxide that did not react.

mass of copper(II) oxide that did not react g [2]

(b) Crystals of hydrated copper(II) chloride were obtained from the solution at the end of the reaction.

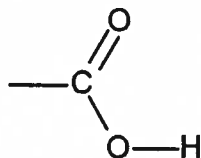
The crystals had the following composition by mass: Cl, 41.52%; Cu, 37.43%; H, 2.34%; O, 18.71%.

Calculate the empirical formula of the crystals.

empirical formula [2]

Answers Section

1. **O/N 21/P41/Q2/d,e**
 - (a) **M1** 188
 - M2** $4.7 / 188 = 0.025(0)$
 - M3** $0.025(0) \times 2 = 0.05(0)$
 - M4** $0.05(0) \times 24.0 = 1.2$ 4
 - (b) $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
 - NaNO_2 (1) 2
 - rest of equation (1)
2. **O/N 21/P43/Q3/a,b,c,d**
 - (a) 267 1
 - (b) $(207 / 239 \times 100 =)$ 86.6% 1
 - (c) PbS because the percentage of lead is larger in PbS or answer to (b) > 77.5% 1
 - (d) $\text{PbCO}_3 \rightarrow \text{PbO} + \text{CO}_2$ 1
3. **O/N 21/P43/Q4/c**
 - (a) (i) **M1** $(22 + 44 =)$ 0.5 (moles) (1) 2
 - M2** $3.01 \times 10^{23}(1)$ 1
 - (ii) 1.505×10^{23}
4. **M/J 21/P42/Q3**
 - (a) (thermal) decomposition 1
 - (b) 84 (1)
 - $12.6 / 84 = 0.15(00)$ (1)
 - $0.15(00) / 2 = 0.075(00)$ (1)
 - $0.075(0) \times 24.0 = 1.8$ (1) 4
 - (c) Ca(OH)_2 (1) 2
 - CaCO_3 (1)
5. **M/J 21/P42/Q5/d**
 - (a) (i) rfm of $\text{NH}_4\text{NO}_3 = 80$ (1) 2
 - mass of N = $2 \times 14 = 28$ and percentage N = $100 \times 28 / 80 = 35\%$ (1) 1
 - (ii) fertiliser 1
 - (iii) calcium hydroxide
6. **M/J 21/P41/Q7**
 - (a) $48.65 / 12 \ 8.11 / 1 \ 43.24 / 16$ (1)
 - OR evaluation**
 - $4.05 \ 8.11 \ 2.7(0)$
 - divide all by smallest
 - OR** $1.5 : 3 : 1$
 - OR** $6 : 3 : 2$ (1) 3
 - $\text{C}_3\text{H}_6\text{O}_2$ (1) **ALLOW** symbols in any order
 - (b) (M_r of $\text{CH}_4\text{O} = 32$) 1
 - CH_4O (1)
 - (c) (i) $\text{C}_n\text{H}_{2n}\text{O}_2$
 - OR**
 - $\text{C}_n\text{H}_{2n+1}\text{COOH}$ 1
 - (ii) butanoic acid (1)
 - fully displayed carboxylic acid group (1)



correct structure of butanoic acid showing all atoms and bonds (1)

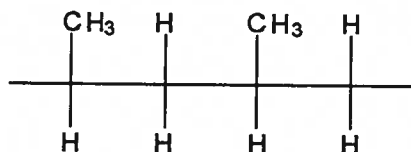
(iii) homologous series

(d) (i) brown to colourless

(ii) C_9H_{20} (1) $2C_3H_6$ (1)

(iii) addition

(iv)



any one repeat unit (1)

both repeat units fully correct (1)

7. M/J 21/P43/Q7

(a) 11.63 / 1 69.77 / 12 18.6 / 16

Or evaluation

11.63 5.81 1.16

Or ratio

10 : 5 : 1 (1)

 $C_5H_{10}O$ (1)(b) (M_r of CH_2Q) = 30 (1) $C_2H_4O_2$ (1)

(c) (i) structural isomers

(ii) carboxylic acids

(iii) name of ester, corresponding to $C_3H_6O_2$: ethyl methanoate or methyl ethanoate (1)

correctly displayed ester linkage (1)

fully correct displayed formula corresponding to $C_3H_6O_2$ and matching named ester (1) 3

(iv) displayed formula of propanoic acid (1)

propanoic acid (1)

(d) Step 1:

cracking (1)

 $C_{10}H_{22} \rightarrow C_2H_4 + C_8H_{18}$ (1)

Step 2:

(catalytic) addition (1)

 $C_2H_4 + H_2O \rightarrow C_2H_5OH$ (1)

8. O/N 20/P42/Q2/b(iii,iv),c,e

(a) (i) rinse the residue (with distilled water)

(ii) a solution that can dissolve no more solute (1)
at the specified temperature (1)(b) mass of $FeSO_4$ = 152 (1)mass of H_2O = $278 - 152 = 126$ (1)mass of H_2O = $126 / 18$ and $x = 7$ (1)(c) M1 mol of $NaCl$ = $2.34 / 58.5 = 0.04(00)$ M2 mol of Cl_2 = $M1/2 = 0.04(00)/2 = 0.02(00)$ M3 $0.02(00) \times 24000 = 480$ (cm^3)

9. O/N 20/P41/Q3/b 1
(a) 75(%)
10. O/N 20/P41/Q5/d 2
(a) ($M_r =$) 98
($0.75 \times 98 =$) 73.5
11. O/N 20/P41/Q6/b,f 1
(a) 3.01×10^{23} (molecules)
(b) (i) ($C = 85.7$, $H = 14.3$, M_r 112)
 $C = \frac{85.7}{12} = 7.14$ $H = \frac{14.3}{1} = 14.3$ (1)
(ratio = $7.13 : 14.3 = 1 : 2$)
 CH_2 (2) 2
(ii) C_8H_{16} 1
12. O/N 20/P43/Q4/d(ii) 4
(a) (i) (moles of $MgSO_4 =$) $0.005 / 5 \times 10^{-3}$ (1)
mass of water = 0.63 g (1)
moles of water = $0.63 \div 18 = 0.035 / 3.5 \times 10^{-2}$ (1)
($x = 0.035 + 0.005$) = 7 (1)
13. M/J 20/P42/Q4/e,f 3
(a) M1 moles of oxygen = $\frac{48.0}{24000}$ or moles of oxygen = 0.002 (1)
M2 moles of hydrogen peroxide = M1 $\times 2$ or moles of hydrogen peroxide = 0.004 (1)
M3 concentration = M2 $\times 40 = 0.16$ mol / dm³ (1)
allow ECF
(b) $2KClO_3 \rightarrow 2KCl + 3O_2$
ALL formulae correct (1) 2
balancing (1)
14. M/J 20/P41/Q3/b(vi) 1
(a) (i) M_r of $SO_3 = 80$ 1
 $100 \times \frac{32}{80} = 40\%$
15. M/J 20/P41/Q4/d 1
(a) (i) methyl orange
(ii) M1 mol of NaOH = $0.0400 \times \frac{25.0}{1000} = 0.001(00)$ mol
M2 mol of $H_2SO_4 = \frac{M1}{2} = \frac{0.001}{2} = 0.0005(00)$
M3 M2 $\times \frac{1000}{20.0} = 0.0005 \times \frac{1000}{20.0} = 0.025$ (mol / dm³) 3
allow ecf
(iii) M1 use of 40g/mol 2
M2 $40 \times 0.04 = 1.6$ (g/dm³)
16. M/J 20/P43/Q2/d(ii,iii,iv) 1
(a) (i) 75 1
(ii) (it could react with rain water to) form nitric acid / acid rain
(iii) (M_r of $HNO_3 =$) 63 (1)

- 20 (1)
5 (1)
120 (dm³) (1) 4
17. **MAR 20/P42/Q3/e**
(a) mol of LiPF₆ = 3.04 / 152 = 0.02(00) (1)
mol of LiF = 0.02(00) × 6 = 0.12(0) (1)
mass of LiF = 3.12 g (1) 3
18. **O/N 19/P42/Q3/d(i)**
(a) (i) mol of NH₃ = 4.8(0) / 24 = 0.2(0) (1)
mol of O₂ = 0.2 × 5 / 4 = 0.25 (1)
mol of O₂ = 0.25 × 24 = 6.(0) (1) 3
19. **O/N 19/P41/Q2/e**
(a) 29.1 / 23 40.5 / 32 30.4 / 16 or 1.2(65) 1.2(65) 1.9 (1)
1:1:1.5 (1)
Na₂S₂O₃ (1) 3
20. **O/N 19/P41/Q6/a,b,c(i)**
(a) correct final answer = 0.072(0)
M1 moles HCl = 0.0036(0)
M2 moles Na₂CO₃ = 0.0018(0) (M1 / 2)
M3 concentration Na₂CO₃ = 0.072 (M2 / 0.025) 3
(b) 0.002(00) 1
(c) (i) 720(.09) 1
21. **O/N 19/P43/Q5**
M1 5 moles of calcium nitrate (1)
M2 10 moles ammonium nitrate (1) or ecf M1 × 2
M3 M_r of ammonium nitrate = 80
M4 800 g or ecf M2 × M3 4
22. **O/N 19/P43/Q6**
(a) strong = exists entirely as ions in solution / fully dissociated 100% dissociated in solution (1)
acid = proton donor (1) 2
(b) 50.0 (cm³) 1
(c) (i) yellow flame 1
(ii) solid dissolves/ disappears (1)
blue solution (1) 2
(d) (i) white precipitate 1
(ii) Ba²⁺(aq) + SO₄²⁻(aq) → BaSO₄(s)
correct ionic equation (1)
state symbols (1) 2
23. **M/J 19/P42/Q5/a**
(a) M1 0.0025 / 2.5 × 10⁻³ (moles of H₂SO₄) (1)
M2 0.0025 / 2.5 × 10⁻³ (moles of CuSO₄•5H₂O) (1)
M3 0.625(g) (1) 3
24. **M/J 19/P41/Q4/c**
(a) M1 mol of HCl = 2.00 × $\frac{50.0}{1000}$ = 0.1(00) mol (1)

$$\text{M2 mol of MgCO}_3 = \frac{\text{M1}}{2} = 0.1(00) / 2 = 0.05(00) (1)$$

$$\text{M3 } M_r \text{ of MgCO}_3 = 84 (1)$$

$$\text{M4 mass of MgCO}_3 = \text{M3} \times \text{M2} = 84 \times 0.05(00) = 4.2(0)\text{g} (1)$$

4

25. M/J 19/P43/Q3/b,e

(a) mass sodium azide = 260 (g)

$$\infty \text{ Moles N}_2 = (144 / 24 =) 6$$

$$\infty \text{ Moles NaN}_3 = (6 \times 2 / 3 =) 4$$

$$\infty M_r \text{NaN}_3 = 65$$

$$\infty (4 \times 65 =) 260$$

4

(b) M1 49.5 / 12 7.2 / 1 43.3 / 14

OR

4.125 7.2 3.093....

M2 1.33 : 2.33 : 1 OR 4 : 7 : 3

M3 C₄H₇N₃

3

26. MAR 19/P42/Q5/b

(a) ∞ 80 ∞ 5 ∞ 10 ∞ 240

4

27. O/N 18/P41/Q3/c

(a) (i) M1 188

$$\text{M2 } (18.8 / 188) = 0.1(00)$$

[2]

(ii) 0.05

[1]

(iii) 1200

[1]

28. O/N 18/P41/Q4/d

(a) (i) 0.003

[1]

(ii) 0.006

[1]

(iii) 30

[1]

29. M/J 18/P42/Q6/a

(a) (mix) sodium carbonate **AND** barium nitrate / barium chloride

[1]

in solution / aqueous / dissolved (in water)

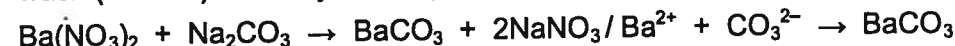
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filter / centrifuge (barium carbonate)

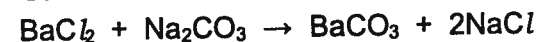
[1]

wash (residue) **AND** dry / description of washing and drying

[1]



OR



[1]

30. M/J 18/P42/Q7/a,b

(a) 60 / 12 : 13.33 / 1 : 26.67 / 16 or evaluation 5 : 13.33 : 1.67 or 3:8:1

[1]

C₃H₈O

[1]

(b) (C₂H₄O =) 44

[1]

C₄H₈O₂

[1]

31. M/J 18/P41/Q4

- (a) relative atomic mass [1]
 (b) C_4H_{10} is covalent [1]
 KF is ionic [1]
 (c) mol of Y = $0.060 / 24.0 = 2.5 \times 10^{-3}$ or 0.0025 [1]
 $M_r = 0.095 / 2.5 \times 10^{-3} = 38(.0)$ [1]
 fluorine [1]
 (d) mass of O = $3.87g - 1.68g = 2.19(g)$ [1]
 mol of P and mol of O [1]
 $1.68 / 31$ OR $0.054..$ $2.19 / 16$ OR $0.13...$
 ratio of P to O [1]
 P = $0.054... / 0.054$ O = $0.13... / 0.054...$
 = 1 = 2.5 [1]
 whole number ratio and P_2O_5 [1]
 = 2 = 5
 (e) the formula is P_4O_6 or (one mole of) $P_2O_3 = 110$ (g) [1]
 mass = 220 (g) [1]

32. M/J 18/P43/Q5/a

- (a) $14.01/59 : 60.33/127 : 2.85/1 : 22.81/16$
 OR
 $0.237 : 0.475 : 2.85 : 1.43$ [1]
 $NiI_2H_{12}O_6$ [1]

33. M/J 18/P43/Q6/a

- (a) (i) 74 [1]
 0.12 [1]
 (ii) 0.3 [1]
 (iii) 0.02 [1]
 (iv) 207 [1]
 4.14 [1]
 (v) 75% [1]

34. MAR 18/P42/Q2/c(iv)

- (a) (i) M1 (moles of NaCl = $0.20 \times 20 + 1000 = 4(.00) \times 10^{-3}$ or 0.004(00)
 M2 (M_r of $NaNO_3 = 85$
 M3 ($85 \times 4(.00) \times 10^{-3} = 0.34$ (g)
 M4 ($0.34 \times 90 / 100 = 0.306$ (g) OR 0.31 (g) [4]

35. MAR 18/P42/Q3/c

- (a)

	Mg	Si	O	
M1	2.73 / 24	1.58 / 28	3.60 / 16	
OR	0.11375	0.0564	0.23(0)	
M2	0.0.11375 / .0564	0.0564 / .0564	0.230 / .0564	leading to Mg_2SiO_4

 [2]

36. MAR 18/P42/Q5/b

- (a) $C_nH_{2n+2}O$ OR $C_nH_{2n+1}OH$ [1]

37. O/N 17/P41/Q7/b,c

- (a) (i) 0.075
 If full credit is not awarded, allow 1 mark for M_r of CuO = 80 [2]
 (ii) 0.05 [1]
 (iii) 4 (g)

M1 moles copper(II) oxide that reacted = $(0.05 / 2) = 0.025$ mol

M2 mass copper(II) oxide = $((0.075 - 0.025) \times 80) = 4$ g [2]

(b) $C_2H_4O_2$

M1 41.52 / 35.5; 37.43 / 64; 2.34 / 1; 18.71 / 16 **OR** 1.17 : 0.58 : 2.34 : 1.17

M2 appropriate scaling to give whole number ratios [2]



Unit 5

ELECTRICITY AND CHEMISTRY

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

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Unit 5: Electricity And Chemistry

1. M/J 22/P41/Q3

This question is about sodium and compounds of sodium.

- (a) (i) Describe the bonding in a metallic element such as sodium.
You may include a diagram as part of your answer.

.....
.....
..... [3]

- (ii) Describe how solid sodium conducts electricity.

..... [1]

- (b) Some properties of sodium chloride are shown:

- melting point of 801 °C
- non-conductor of electricity when solid
- conductor of electricity when molten
- soluble in water.

- (i) Name the type of bonding in sodium chloride.

..... [1]

- (ii) Explain why sodium chloride conducts electricity when molten.

..... [1]

- (c) A student determines the concentration of a solution of dilute sulfuric acid, H_2SO_4 , by titration with aqueous sodium hydroxide, NaOH .

step 1 25.0 cm³ of 0.200 mol/dm³ NaOH is transferred into a conical flask.

step 2 Three drops of methyl orange indicator are added to the conical flask.

step 3 A burette is filled with H_2SO_4 .

step 4 The acid in the burette is added to the conical flask until the indicator changes colour.
The volume of acid is recorded. This process is known as titration.

step 5 The titration is repeated several times until a suitable number of results is obtained.

- (i) Name the piece of apparatus used to measure exactly 25.0 cm³ of 0.200 mol/dm³ NaOH in **step 1**.

..... [1]

- (ii) State the colour change of the methyl orange indicator in **step 4**.

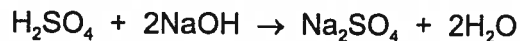
from to [1]

(iii) State how the student decides that a suitable number of results have been obtained.

..... [1]

(iv) 20.0 cm³ of H₂SO₄ reacts with 25.0 cm³ of 0.200 mol/dm³ NaOH.

The equation for the reaction is shown.



Calculate the concentration of H₂SO₄ using the following steps.

- Calculate the number of moles in 25.0 cm³ of 0.200 mol/dm³ NaOH.

..... mol

- Determine the number of moles of H₂SO₄ that react with the NaOH.

..... mol

- Calculate the concentration of H₂SO₄.

..... mol/dm³ [3]

2. M/J 22/P43/Q5

This question is about copper and its compounds.

(a) Describe the bonding in a metallic element such as copper.

You may include a diagram as part of your answer.

.....
.....
..... [3]

(b) A metal spoon is electroplated with copper.

State what is used as:

the positive electrode (anode)

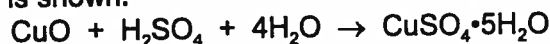
the negative electrode (cathode)

the electrolyte. [3]

(c) The formula for crystals of hydrated copper(II) sulfate is CuSO₄•5H₂O.

Hydrated copper(II) sulfate is made by reacting copper(II) oxide with dilute sulfuric acid.

The overall equation is shown.



The crystals are made using the following steps:

- step 1** 50.0 cm³ of 0.200 mol/dm³ dilute sulfuric acid is heated in a beaker. Powdered copper(II) oxide is added until the copper(II) oxide is in excess. Aqueous copper(II) sulfate is formed.
- step 2** The excess copper(II) oxide is separated from the aqueous copper(II) sulfate.
- step 3** The aqueous copper(II) sulfate is heated until a saturated solution is formed.
- step 4** The saturated solution is allowed to cool and crystallise.
- step 5** The crystals are removed and dried.

Calculate the maximum mass of copper(II) sulfate crystals, CuSO₄•5H₂O, that can form using the following steps.

- Calculate the number of moles of H₂SO₄ in 50.0 cm³ of 0.200 mol/dm³ H₂SO₄.

- Deduce the number of moles of CuSO₄•5H₂O that can form. mol

- The *M_r* of CuSO₄•5H₂O is 250. mol
Calculate the maximum mass of CuSO₄•5H₂O that can form.

..... g [3]

(d) State one observation that indicates the copper(II) oxide is in excess in **step 1**.

..... [1]

(e) **Step 1** is repeated without heating the dilute sulfuric acid.

All other conditions are kept the same.

The rate of reaction decreases.

Give a reason why the rate of reaction decreases. Explain your answer in terms of particles.

.....
.....
..... [3]

(f) Name a substance, other than copper(II) oxide, that can be added to dilute sulfuric acid to produce copper(II) sulfate in **step 1**.

..... [1]

(g) Name the process used to separate excess copper(II) oxide from aqueous copper(II) sulfate in **step 2**.

..... [1]

(h) Suggest what is meant by the term *saturated solution* in **step 3**.

.....
..... [2]

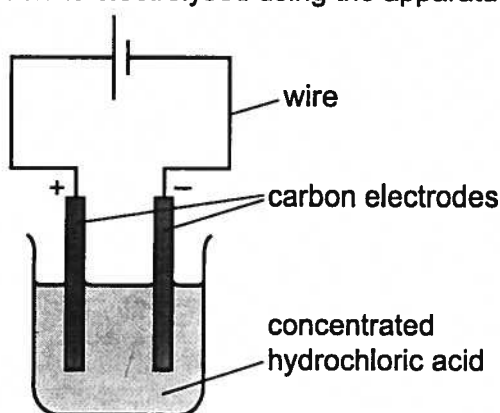
(i) The phrase 'heating to dryness' means heating until no more water is given off.
Explain why aqueous copper(II) sulfate is **not** heated to dryness in **step 3**.

..... [1]

3. O/N 21/P41/Q3

This question is about electrolysis.

Concentrated hydrochloric acid is electrolysed using the apparatus shown.



(a) Chloride ions are discharged at the anode.

(i) Complete the ionic half-equation for this reaction.



(ii) State whether oxidation or reduction takes place. Explain your answer.

..... [1]

(b) Describe what is seen at the cathode.

..... [1]

(c) Write the ionic half-equation for the reaction at the cathode.

..... [2]

(d) The pH of the electrolyte is measured throughout the experiment.

(i) Suggest the pH of the electrolyte at the beginning of the experiment.

..... [1]

(ii) State how the pH changes, if at all, during the experiment.

Explain your answer.

.....
..... [2]

- (e) The electrolysis is repeated using molten lead(II) bromide.
Describe what is seen at the:

- cathode
- anode. [2]

- (f) State **two** properties of graphite (carbon) which make it suitable for use as an electrode.

- 1
- 2 [2]

4. O/N 21/P43/Q2

This question is about electrolysis.

- (a) State the meaning of the term *electrolyte*.

.....

..... [2]

- (b) The table gives information about the electrolysis of two electrolytes. Carbon (graphite) electrodes are used in each experiment.

- (i) Complete the table to show the observations and products of electrolysis.

electrolyte	positive electrode (anode)		negative electrode (cathode)	
	observations	name of product	observations	name of product
aqueous copper(II) sulfate	colourless bubbles			
concentrated aqueous sodium bromide			colourless bubbles	hydrogen

[5]

- (ii) Hydrogen is produced at the negative electrode (cathode) during the electrolysis of concentrated aqueous sodium bromide.

Write the ionic half-equation for this reaction.

..... [2]

- (iii) State **two** reasons why carbon (graphite) is suitable to use as an electrode.

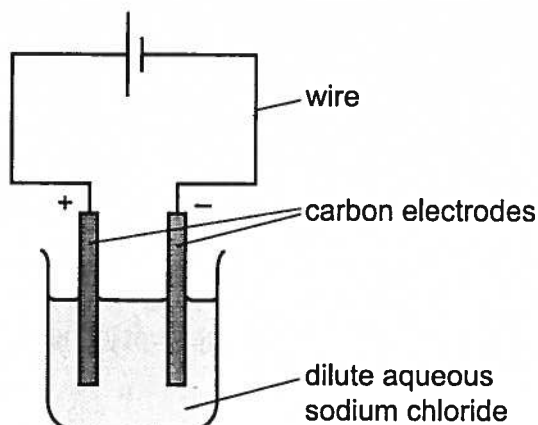
- 1
- 2 [2]

- (iv) Name the particle responsible for the conduction of electricity in the metal wires used in a circuit.

..... [1]

5. M/J 21/P42/Q4

A student carries out an electrolysis experiment using the apparatus shown.



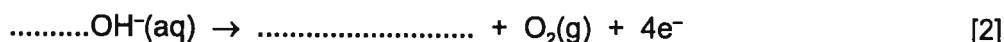
The student uses dilute aqueous sodium chloride.

(a) State the name given to any solution which undergoes electrolysis.

..... [1]

(b) Hydroxide ions are discharged at the anode.

(i) Complete the ionic half-equation for this reaction.



(ii) Explain how the ionic half-equation shows the hydroxide ions are being oxidised.

..... [1]

(c) Describe what the student observes at the cathode.

..... [1]

(d) Write the ionic half-equation for the reaction at the cathode.

..... [2]

(e) The student repeats the experiment using concentrated aqueous sodium chloride.

(i) Describe what the student observes at:

- the cathode
- the anode. [2]

(ii) The student added litmus to the solution after the electrolysis of concentrated aqueous sodium chloride.

State the colour seen in the solution. Give a reason for your answer.

colour of solution

reason [2]

- (f) Carbon electrodes are used because they are inert.
State another element that can be used instead of carbon.

..... [1]

6. M/J 21/P41/Q3

Potassium reacts with chlorine to form potassium chloride, KCl .

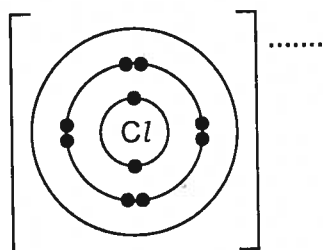
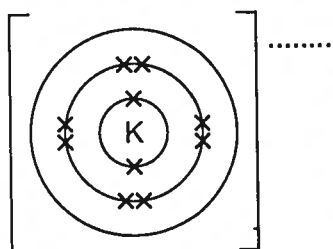
- (a) Write a chemical equation for this reaction.

..... [2]

- (b) Potassium chloride is an ionic compound.

Complete the diagram to show the electron arrangement in the outer shells of the ions present in potassium chloride.

Give the charges on both ions.



[3]

- (c) Molten potassium chloride undergoes electrolysis.

- (i) State what is meant by the term *electrolysis*.

..... [2]

- (ii) Name the products formed at the positive electrode (anode) and negative electrode (cathode) when molten potassium chloride undergoes electrolysis.

anode

cathode [2]

- (d) Concentrated aqueous potassium chloride undergoes electrolysis.

- (i) Write an ionic half-equation for the reaction at the negative electrode (cathode).

..... [2]

- (ii) Name the product formed at the positive electrode (anode).

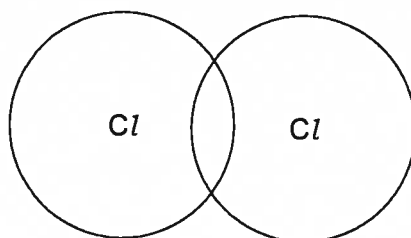
..... [1]

- (iii) Name the potassium compound that remains in the solution after electrolysis.

..... [1]

- (e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of chlorine, Cl_2 .

Show the outer electrons only.



[1]

- (f) The melting points and boiling points of chlorine and potassium chloride are shown.

	melting point /°C	boiling point /°C
chlorine	-101	-35
potassium chloride	770	1500

- (i) Deduce the physical state of chlorine at -75°C . Use the data in the table to explain your answer.

physical state

explanation [2]

- (ii) Explain, in terms of structure and bonding, why potassium chloride has a much higher melting point than chlorine.

Your answer should refer to the:

- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

.....

 [3]

7. M/J 21/P43/Q3

Sodium reacts with fluorine to form sodium fluoride, NaF.

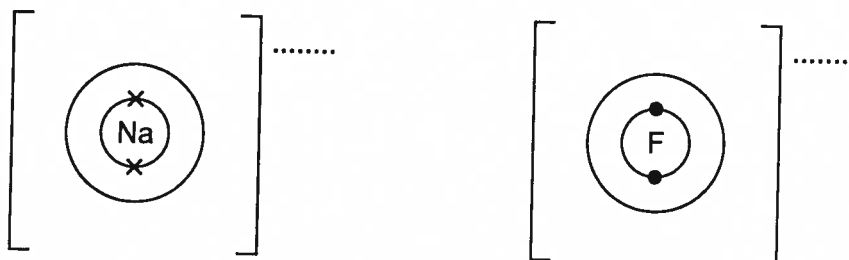
- (a) Write a chemical equation for this reaction.

..... [2]

- (b) Sodium fluoride is an ionic compound.

Complete the diagram to show the electron arrangement in the outer shells of the ions present in sodium fluoride.

Give the charges on both ions.



[3]

(c) Aqueous sodium fluoride undergoes electrolysis.

(i) State what is meant by the term *electrolysis*.

..... [2]

(ii) Name the products formed at the positive electrode (anode) and the negative electrode (cathode) when dilute aqueous sodium fluoride undergoes electrolysis.

anode

cathode [2]

(d) Molten sodium fluoride undergoes electrolysis.

(i) Name the products formed at the positive electrode (anode) and the negative electrode (cathode) when molten sodium fluoride undergoes electrolysis.

anode

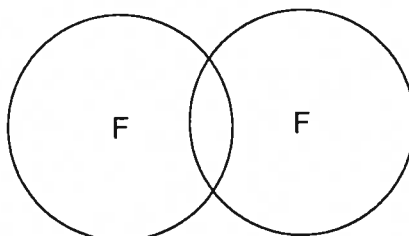
cathode [2]

(ii) Write the ionic half-equation for the reaction at the negative electrode (cathode).

..... [1]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of fluorine, F_2 .

Show the outer electrons only.



[1]

(f) The melting points and boiling points of fluorine and sodium fluoride are shown.

	melting point /°C	boiling point /°C
fluorine	-220	-188
sodium fluoride	993	1695

(i) Deduce the physical state of fluorine at -195°C . Use the data in the table to explain your answer.

physical state

explanation [2]

- (ii) Explain, in terms of structure and bonding, why sodium fluoride has a much higher melting point than fluorine.

Your answer should refer to the:

- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

.....
.....
..... [3]

8. O/N 20/P42/Q2/f(ii,iii,iv)

- (a) Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

- (i) Name the product formed at the positive electrode when electricity is passed through molten sodium chloride.

..... [1]

- (ii) State the type of change that occurs at the positive electrode in (ii).

Explain your answer in terms of electron transfer.

type of change

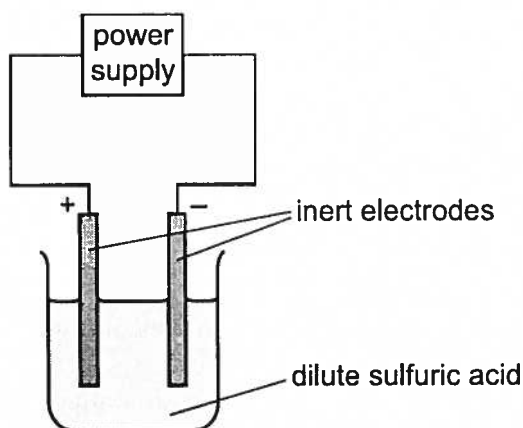
explanation [2]

- (iii) Describe what else can be done to sodium chloride to allow it to conduct electricity.

..... [1]

9. O/N 20/P41/Q5/a

- (a) Dilute sulfuric acid is electrolysed using the apparatus shown in the diagram.



- (i) State what is meant by the term *electrolysis*.

.....
..... [2]

(ii) Explain why inert electrodes are used.

..... [1]

(iii) Name the products formed at each electrode.

negative electrode

positive electrode [2]

(iv) Write an ionic half-equation for the reaction at the negative electrode.

..... [2]

10. O/N 20/P43/Q5/d

(a) Sodium is extracted from sodium chloride by electrolysis.

(i) State the meaning of the term *electrolysis*.

.....

..... [2]

(ii) State what must be done to sodium chloride before it can be electrolysed to produce sodium.

..... [1]

(iii) Write an ionic half-equation for the change that occurs at the cathode during this electrolysis.

..... [1]

11. M/J 20/P42/Q5

Electrolysis of concentrated aqueous sodium chloride using inert electrodes forms chlorine, hydrogen and sodium hydroxide.

(a) What is meant by the term *electrolysis*?

.....

..... [2]

(b) Name a substance that can be used as the inert electrodes.

..... [1]

(c) Write an ionic half-equation for the formation of hydrogen during this electrolysis.

..... [1]

(d) Give the formulae of the **four** ions present in concentrated aqueous sodium chloride.

..... [2]

(e) Explain how sodium hydroxide is formed during this electrolysis.

.....

..... [2]

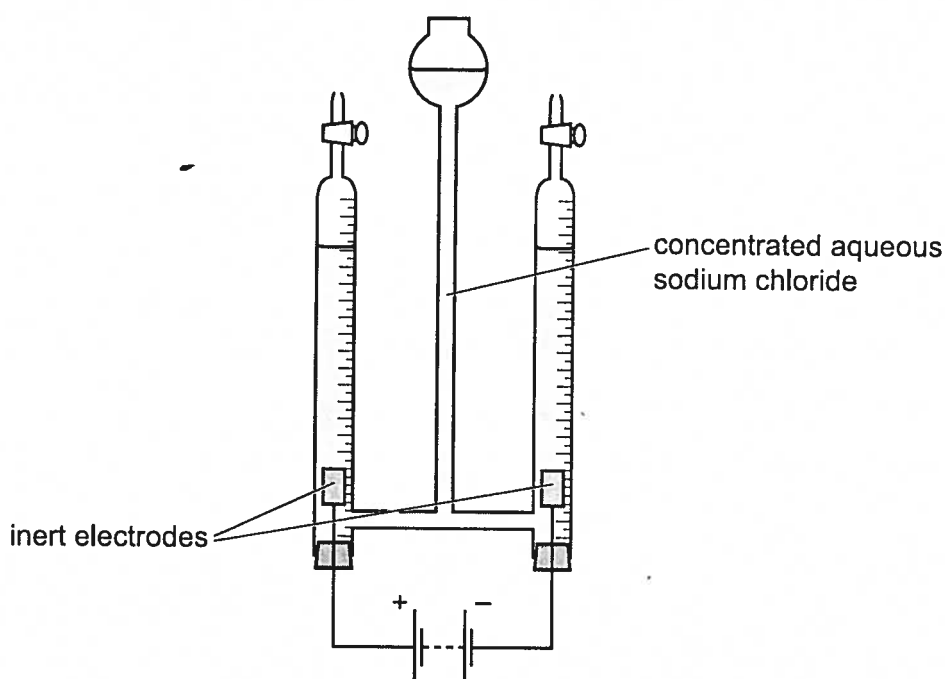
12. O/N 19/P42/Q4

Many substances conduct electricity.

(a) Identify all the particles responsible for the passage of electricity in:

- graphite.....
- magnesium ribbon
- molten copper(II) bromide. [4]

(b) A student used the following apparatus to electrolyse concentrated aqueous sodium chloride using inert electrodes.



(i) Suggest the name of a metal which could be used as the inert electrodes.

..... [1]

(ii) Name the gas formed at the positive electrode.

..... [1]

(iii) Write an ionic half-equation for the reaction occurring at the negative electrode. Include state symbols.

..... [3]

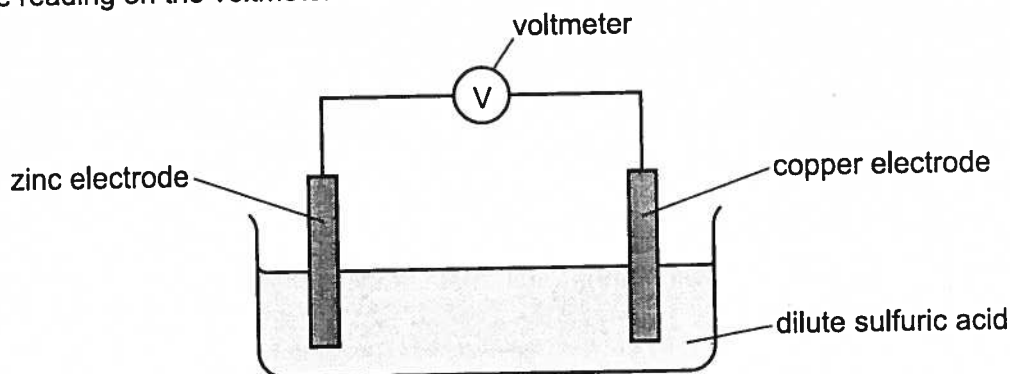
(iv) How, if at all, does the pH of the solution change during the electrolysis? Explain your answer.

.....

.....

..... [3]

- (c) A student used the following electrochemical cell.
The reading on the voltmeter was +1.10 V.



- (i) Draw an arrow on the diagram to show the direction of electron flow. [1]
- (ii) Suggest the change, if any, in the voltmeter reading if the zinc electrode was replaced with an iron electrode. Explain your answer. [2]
-
-
- (iii) The zinc electrode was replaced with a silver electrode. The reading on the voltmeter was -0.46 V. Suggest why the sign of the voltmeter reading became negative. [1]
-

13. O/N 19/P41/Q6/c(ii,iii,iv,v)

- (a) A sample of concentrated hydrobromic acid, HBr(aq) , was electrolysed using platinum electrodes.

The concentration of the hydrobromic acid was 8.89 mol/dm^3 .

- (i) Explain why concentrated HBr(aq) can conduct electricity.

..... [2]

- (ii) Magnesium is **not** a suitable material from which to make the electrodes. Explain why.

..... [1]

- (iii) Predict the product formed at the anode when concentrated HBr(aq) is electrolysed.

..... [1]

- (iv) Write the ionic half-equation for the reaction occurring at the cathode.

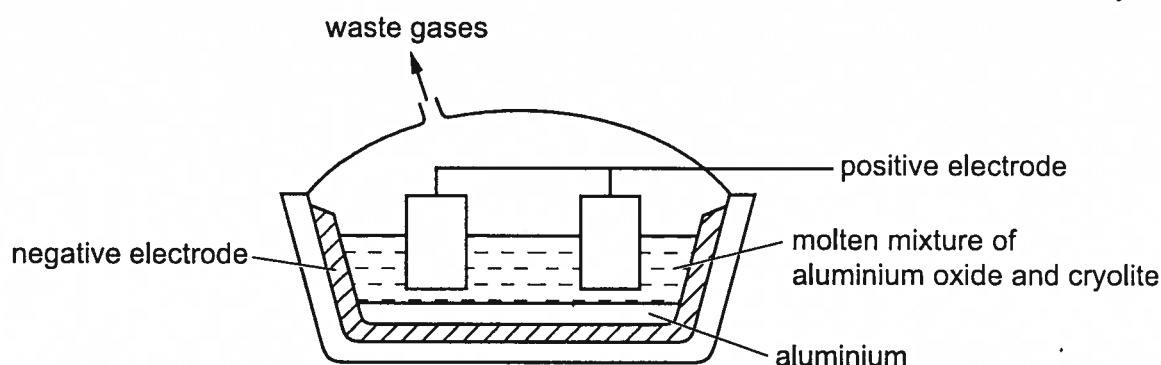
..... [2]

14. O/N 19/P43/Q3/a,b

- (a) Name the ore of aluminium which mainly consists of aluminium oxide.

..... [1]

- (b) Aluminium is produced by the electrolysis of aluminium oxide dissolved in molten cryolite.



- (i) Give **two** reasons why the electrolysis is done using a molten mixture of aluminium oxide and cryolite instead of molten aluminium oxide only.

1

2 [2]

- (ii) Write ionic half-equations for the reactions occurring at the electrodes.

positive electrode

negative electrode [2]

- (iii) The anodes are made of carbon and have to be replaced regularly.
Explain why the carbon anodes have to be replaced regularly.

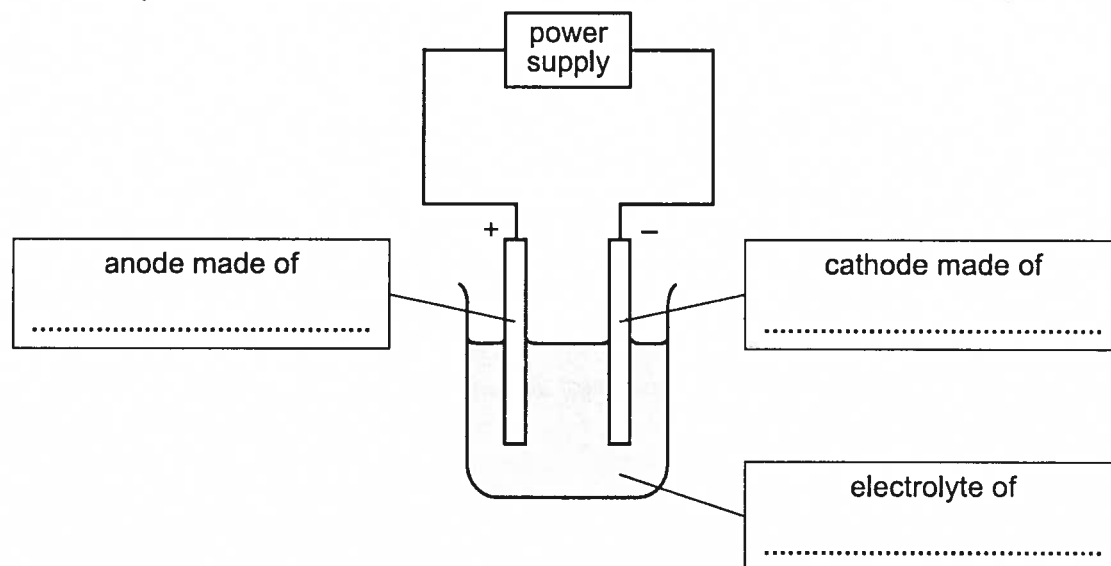
.....

..... [2]

15. M/J 19/P42/Q7/f

- (a) Copper is refined (purified) by electrolysis. Nickel can be refined using a similar method.

- (i) The diagram shows the refining of nickel by electrolysis.
Complete the labels in the boxes.



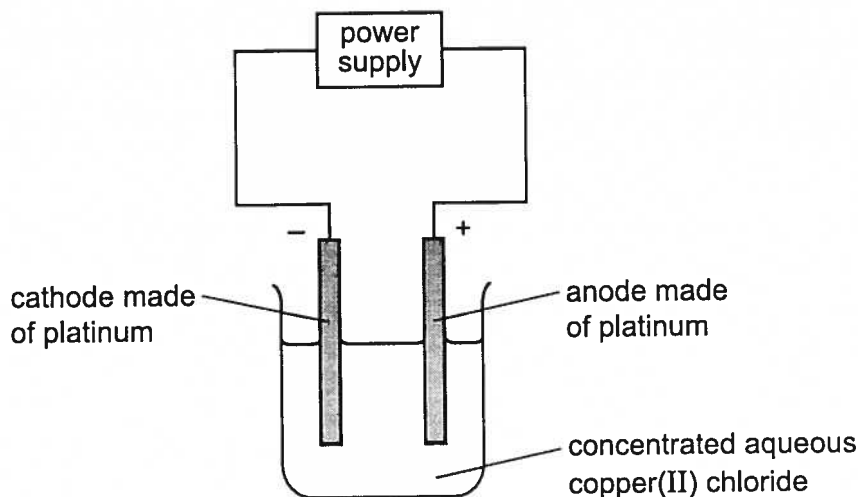
[3]

(ii) Indicate, by writing **N** on the diagram, where nickel is produced. [1]

16. M/J 19/P43/Q4

Solutions of ionic compounds can be broken down by electrolysis.

(a) Concentrated aqueous copper(II) chloride was electrolysed using the apparatus shown.



The ionic half-equations for the reactions at the electrodes are shown.

negative electrode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$

positive electrode: $2\text{Cl}^{-}(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^{-}$

(i) Platinum is a solid which is a good conductor of electricity.

State **one** other property of platinum which makes it suitable for use as electrodes.

..... [1]

(ii) State what would be **seen** at the positive electrode during this electrolysis.

..... [1]

(iii) State and explain what would happen to the mass of the negative electrode during this electrolysis.

.....

..... [2]

(iv) The concentrated aqueous copper(II) chloride electrolyte is green.

Suggest what would happen to the colour of the electrolyte during this electrolysis.
Explain your answer.

.....

..... [2]

(v) Identify the species that is oxidised during this electrolysis.

Explain your answer.

species that is oxidised

explanation [2]

(b) Metal objects can be electroplated with silver.

(i) Describe how a metal spoon can be electroplated with silver.

Include:

- what to use as the positive electrode and as the negative electrode
- what to use as the electrolyte
- an ionic half-equation to show the formation of silver.

You may include a diagram in your answer.

.....
ionic half-equation [4]

(ii) Give **one** reason why metal spoons are electroplated with silver.

..... [1]

17. MAR 19/P42/Q5/e

(a) Magnesium cannot be produced by electrolysis of aqueous magnesium chloride using inert electrodes.

(i) Name the product formed at the negative electrode (cathode) during the electrolysis of aqueous magnesium chloride.

..... [1]

(ii) Suggest how magnesium can be produced from magnesium chloride by electrolysis.

..... [1]

18. M/J 18/P42/Q4/a,b,c,d,e

Potassium reacts with bromine at room temperature to form potassium bromide.

(a) Write a chemical equation for this reaction. Include state symbols.

..... [3]

(b) Potassium bromide exists as an ionic lattice.

Potassium bromide does **not** conduct electricity when solid but does conduct electricity when molten.

(i) What is meant by the term *ionic lattice*?

.....
..... [2]

(ii) Explain why potassium bromide does **not** conduct electricity when solid but does conduct electricity when molten.

.....
..... [2]

(c) Concentrated aqueous potassium bromide is an electrolyte.

(i) What is meant by the term *electrolyte*?

.....
..... [2]

(ii) Describe the electrolysis of concentrated aqueous potassium bromide.

Include:

- an ionic half-equation for the reaction at the cathode
- the name of the product at the anode
- the name of the potassium compound formed.

.....
.....
.....
..... [4]

(iii) When molten potassium bromide is electrolysed, the product at the cathode is different.

Name the product at the cathode when molten potassium bromide is electrolysed.

..... [1]

(d) Iodine reacts with chlorine to form iodine monochloride, ICl , as the only product.

(i) Write a chemical equation for this reaction.

..... [2]

(ii) Draw a dot-and-cross diagram to show the electron arrangement in a molecule of iodine monochloride. Show outer shell electrons only.

[2]

(e) Potassium bromide has a melting point of 734°C .

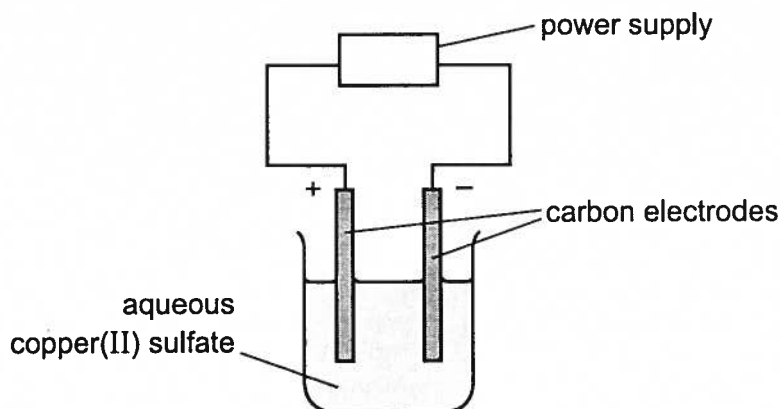
Iodine monochloride has a melting point of 27°C .

In terms of attractive forces, explain why there is a large difference between these melting points.

.....
.....
..... [3]

19. M/J 18/P41/Q5/e

(a) A student electrolyses aqueous copper(II) sulfate using the apparatus shown.



Oxygen gas forms at the positive electrode (anode).

(i) Write an ionic half-equation for the reaction at the negative electrode (cathode). Include state symbols.

..... [3]

(ii) Describe what the student observes at the negative electrode.

..... [1]

(iii) Give **two other** observations which the student makes during the electrolysis.

1

2 [2]

(iv) What difference would the student observe at the positive electrode if the aqueous copper(II) sulfate were replaced by concentrated aqueous copper(II) chloride?

..... [1]

20. M/J 18/P43/Q1/c

The following are the symbols and formulae of some elements and compounds.

Ar Ca(OH)₂ Cl₂ CO₂ Cu Fe SO₂ V₂O₅

Answer the following questions using only the elements or compounds in the list.

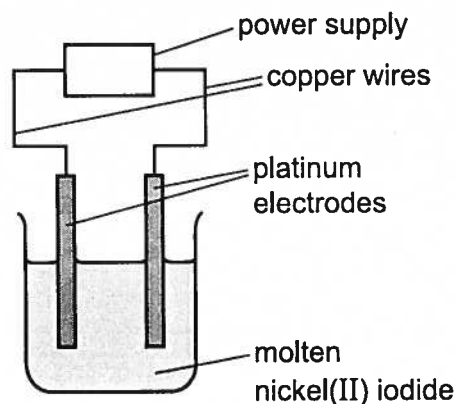
Each element or compound may be used once, more than once or not at all.

State which element or compound is used:

(a) as an electrical conductor in cables [1]

21. M/J 18/P43/Q5/b,c

(a) Molten nickel(II) iodide can be electrolysed using the apparatus shown.



During electrolysis, charge is transferred through the copper wires and through the molten nickel(II) iodide.

(i) Name the type of particles which transfer charge through the copper wires.

..... [1]

(ii) Name the type of particles which transfer charge through the molten nickel(II) iodide.

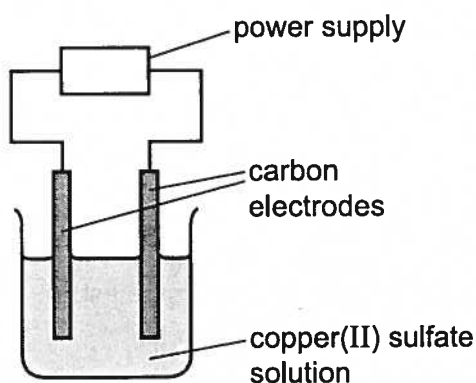
..... [1]

(iii) Predict the products of the electrolysis of molten nickel(II) iodide. Write an ionic half-equation for the formation of **one** of these products.

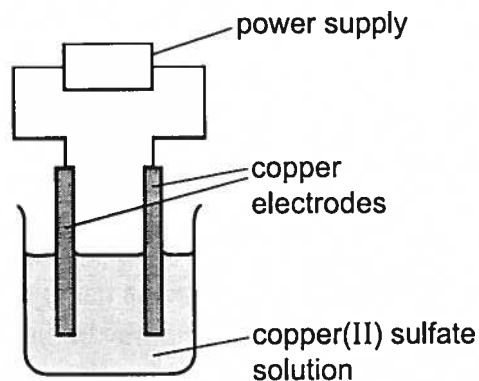
products.....

ionic half-equation [3]

(b) A student electrolysed copper(II) sulfate solution using the two sets of apparatus shown.



apparatus A



apparatus B

In apparatus A the student used carbon electrodes.

In apparatus B the student used copper electrodes.

The student made the following observations.

apparatus A	apparatus B
The mass of the negative electrode increased.	The mass of the negative electrode increased.
The mass of the positive electrode stayed the same.	The mass of the positive electrode decreased.
Bubbles were seen at the positive electrode.	No bubbles were seen at the positive electrode.

(i) Explain why the mass of the negative electrode increased in **both** sets of apparatus.

..... [1]

(ii) Name the gas that formed the bubbles seen in apparatus A.

..... [1]

(iii) Explain why the mass of the positive electrode decreased in apparatus B.

..... [1]

(iv) Suggest what happens to the colour of the solution in apparatus A and apparatus B as the electrolysis progresses.

Explain your answer.

colour of the solution in apparatus A

colour of the solution in apparatus B

explanation [3]

22. MAR 18/P42/Q2/b

Sodium chloride is a typical ionic compound.

(a) Electrolysis of concentrated aqueous sodium chloride is an important industrial process.

(i) What is meant by the term *electrolysis*?

..... [2]

(ii) Name the products of the electrolysis of concentrated aqueous sodium chloride.

1

2

3 [3]

(iii) Write an ionic half-equation for the reaction at the cathode.

Include state symbols.

..... [2]

23. O/N 17/P41/Q6/a

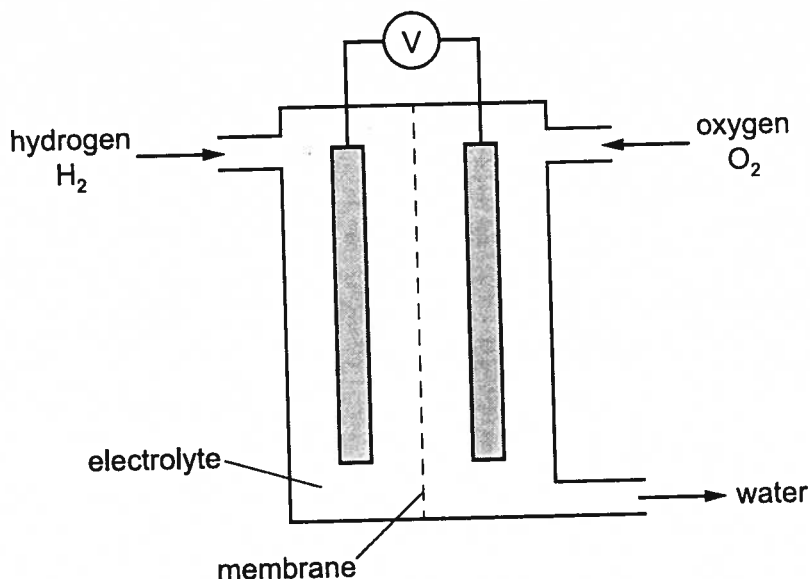
Aluminium is extracted from aluminium oxide by electrolysis.

(a) Why is aluminium **not** extracted by heating aluminium oxide with carbon?

..... [1]

24. O/N 17/P43/Q4/a,b,c,d,f

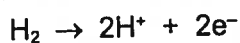
Hydrogen and oxygen react together in a hydrogen fuel cell. A hydrogen fuel cell is shown in the diagram.



(a) Name the process by which oxygen is obtained from air.

..... [1]

(b) (i) In a hydrogen fuel cell, the hydrogen molecules are converted into hydrogen ions, H^+ , according to the ionic half-equation shown.



What type of reaction does this ionic half-equation represent?

..... [1]

(ii) What **type** of substance reacts by donating hydrogen ions, H^+ ?

..... [1]

(c) Write a chemical equation for the overall reaction that occurs in a hydrogen fuel cell.

..... [1]

(d) Hydrogen fuel cells are being developed as alternatives to petrol engines in cars.

(i) Give **one** advantage of hydrogen fuel cells compared to petrol engines.

..... [1]

(ii) Give **one** disadvantage of hydrogen fuel cells compared to petrol engines.

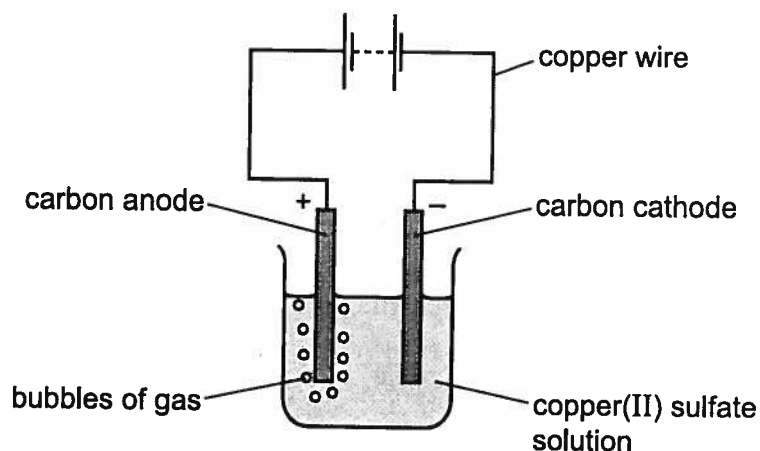
..... [1]

(e) Name the process occurring when electrical energy is used to break down an ionic compound.

..... [1]

25. O/N 16/P41/Q5

Copper(II) sulfate solution was electrolysed using the apparatus shown.



- (a) A gas was formed at the anode.

Identify this gas and give the test for this gas.

gas

test

result of test [3]

- (b) During electrolysis, electricity passes through the copper(II) sulfate solution.

Solid copper(II) sulfate does not conduct electricity.

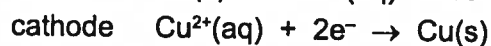
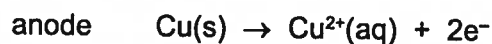
Explain **both** of these statements.

.....

.....

..... [3]

- (c) The electrolysis was repeated using copper electrodes in place of carbon electrodes. The ionic half-equations for the reactions at the two electrodes are shown.



- (i) Which species is reduced during the electrolysis? Explain your answer.

.....

..... [2]

- (ii) The masses of the copper electrodes changed during the electrolysis.

State how **and** explain why the masses of the **two** copper electrodes changed.

Use the ionic half-equations to help you.

.....

.....

- [3]
- (iii) Explain why, during the electrolysis, the colour of the copper(II) sulfate solution does **not** change.
- [1]

26. O/N 16/P43/Q4/e,f,g

Silicon(IV) oxide and sodium chloride have different types of bonding and structure.

- (a) A **concentrated** aqueous solution of sodium chloride is electrolysed using carbon electrodes.

- (i) Name the products formed at the electrodes.

product at the positive electrode (anode)

product at the negative electrode (cathode) [2]

- (ii) Write an ionic half-equation for the reaction occurring at the negative electrode.

..... [1]

- (b) A **dilute** aqueous solution of sodium chloride is electrolysed using carbon electrodes.
Name the main product formed at the positive electrode.

..... [1]

- (c) Molten sodium chloride is electrolysed using carbon electrodes.

- (i) Name the product formed at the negative electrode.

..... [1]

- (ii) Write an ionic half-equation for the reaction occurring at the negative electrode.

..... [1]

- (iii) Chlorine is produced at the positive electrode.

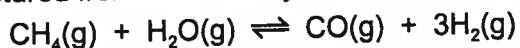
Give the test for chlorine.

test

result [2]

27. M/J 16/P42/Q4/e,f

Hydrogen can be manufactured from methane by steam reforming.



The reaction is carried out using a nickel catalyst at temperatures between 700 °C and 1100 °C and using a pressure of one atmosphere.

The forward reaction is endothermic.

- (a) Hydrogen can also be manufactured by electrolysis. The electrolyte is concentrated aqueous sodium chloride. The electrodes are inert.

The products of electrolysis are hydrogen, chlorine and sodium hydroxide.

- (i) Define the term *electrolysis*.

.....

- [2]
- (ii) Name a substance that can be used as the inert electrodes. [2]
- [1]
- (iii) Write an ionic half-equation for the reaction in which hydrogen is produced. [1]
- [1]
- (iv) Where is hydrogen produced in the electrolytic cell? [1]
- [1]
- (v) Describe a test for chlorine. [1]
- test
- result [2]
- (b) The electrolysis of concentrated aqueous sodium chloride can be represented by the following word equation.
- sodium chloride + water $\rightarrow$ sodium hydroxide + hydrogen + chlorine
- Construct a chemical equation to represent this reaction. Do not include state symbols.
- [2]

Answers Section

1. M/J 22/P41/Q3

- | | | |
|---------|--|--------|
| (a) (i) | positive ions / cations (1)
sea of electrons / mobile electrons / delocalised electrons (1)
attraction between positive ions and electrons (1) | 3 |
| (ii) | electrons move /
electrons mobile /
electrons flow | 1
1 |
| (b) (i) | ionic | 1 |
| (ii) | ions move /
ions are mobile /
ions flow | 1
1 |
| (c) (i) | pipette | 1 |
| (ii) | yellow to orange | 1 |
| (iii) | at least two results are within 0.2 cm ³ or less | 1 |
| (iv) | 0.005 / 5 × 10 ⁻³ (1)
0.0025 / 2.5 × 10 ⁻³ (1)
0.125 (1) | 3 |

2. M/J 22/P43/Q5

- | | | |
|-----|--|--------|
| (a) | (lattice of) positive ions (1)
sea of / delocalised / mobile electrons (1)
attraction between positive ions and electrons (1) | 3 |
| (b) | copper (1)
spoon (1)
(aqueous or solution) of named copper salt (1) | 3 |
| (c) | (0.2 × 50 / 1000) = 0.01 (1)
0.01 (1)
(250 × 0.01) = 2.5 (1) | 3
1 |
| (d) | solid undissolved | 1 |
| (e) | particles have less energy
fewer collisions (between particles) occur per second / per unit time
a smaller percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation energy / have energy equal to activation energy | 3
1 |
| (f) | copper(II) carbonate OR copper(II) hydroxide | 1 |
| (g) | filtration | 1 |
| (h) | (a solution that) can dissolve no more solute
at a given temperature | 2 |
| (i) | forms anhydrous (copper sulfate)
OR forms (white) powder | 1 |

3. O/N 21/P41/Q3

- | | | |
|---------|---|--------|
| (a) (i) | $2Cl^- \rightarrow Cl_2 + 2e^-$
Cl_2 (1)
rest of equation (1) | 2
1 |
| (ii) | Oxidation AND lose electrons | 1 |
| (b) | effervescence (of colourless gas) | 1 |

- (c) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 $\text{H}^+ + \text{e}^-$ as only species on LHS (1)
 rest of equation fully correct (1) 2
- (d) (i) 1 1
 (ii) **M1** increase (1)
M2 H^+ ions being removed (1) 2
- (e) **M1** cathode: silver / grey solid (1)
M2 anode: bubbles of orange / brown gas (1) 2
- (f) **M1** inert (1)
M2 conducts electricity (1) 2
4. **O/N 21/P43/Q2**
- (a) **M1** ionic compound
AND either molten or aqueous(or both)(1)
M2 conducts electricity / undergoes electrolysis(1) 2
- (b) (i) **M1** oxygen (1)
M2 pink / brown solid (1)
M3 copper (1)
M4 orange / brown / yellow liquid (1)
M5 bromine (1) 5
 (ii) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (2) 2
 (iii) **M1** inert (1)
M2 good conductor of electricity (1) 2
 (iv) electron 1
5. **M/J 21/P42/Q4**
- (a) electrolyte 1
- (b) (i) $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
 balance of charge (1)
 rest of equation (1) 2
 (ii) $(\text{OH}^-(\text{aq}))$ ions lose electrons 1
- (c) fizzing 1
- (d) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 species correct (1)
 fully correct equation (1) 2
- (e) (i) fizzing (1)
 green gas (1) 2
 (ii) (litmus turns) blue **and** alkali / base forms (1)
 Sodium hydroxide / NaOH (forming) (1) 2
- (f) platinum 1
6. **M/J 21/P41/Q3**
- (a) $2\text{K} + \text{Cl}_2 \rightarrow 2\text{KCl}$
 Cl_2 on left hand side (1)
 equation fully correct (1) 2
- (b) K outer shell with **8 crosses** (1)
 Cl outer shell with **7 dots and 1 cross** (1)
 + and - (1) 3

- (c) (i) **breakdown** by (the passage of) **electricity** (1)
of an **ionic compound** in **molten or aqueous** (state) (1) 2
- (ii) (anode) chlorine 2
(cathode)potassium
- (d) (i) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 H^+ and e^- on left hand side (1) 2
equation fully correct (1) 1
- (ii) chlorine 1
- (iii) potassium hydroxide (1) 1
- (e) one shared pair of electrons and 6 non-bonding electrons on each chlorine atom 1
- (f) (i) liquid (1)
BOTH melting point is below -75°C **AND** boiling point is above -75°C
OR
BOTH -75°C is higher than -101°C / melting point **AND** lower than -35°C / boiling point
OR
 -75°C is **between** melting point or -101°C **and** boiling point or -35°C 2
- (ii) **ionic bonds** in KCl (1)
attraction **between molecules** in Cl_2 (1)
weaker attraction (between particles) in Cl_2 **ORA** (1) 3

7. M/J 21/P43/Q3

- (a) $2\text{Na} + \text{F}_2 \rightarrow 2\text{NaF}$
 F_2 (1) 2
equation fully correct (1)
- (b) Na outer shell with 8 crosses (1)
F outer shell with 7 dots and 1 cross (1) 3
 Na^+ and F^- (1)
- (c) (i) **breakdown** by (the passage of) **electricity** (1)
of an **ionic compound** in **molten / aqueous** (state) (1) 2
- (ii) oxygen 2
hydrogen
- (d) (i) (anode) fluorine (1) 2
(cathode) sodium (1)
- (ii) $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$ 1
- (e) one shared pair of electrons and 6 non-bonding electrons on each fluorine atom 1
- (f) (i) liquid (1)
BOTH melting point is below -195°C **and** boiling point is above -195°C
OR -195°C is in between melting point and boiling point / -220°C and -188°C (1) **OR**
BOTH -195°C is higher than -220°C / melting point **AND** lower than -188°C /
boiling point 2
- (ii) **ionic bonds** in NaF (1)
attraction **between molecules** in F_2 (1)
weaker attraction (between particles) in F_2 **ORA** (1) 3

8. O/N 20/P42/Q2/f(ii,iii,iv)

- (a) (i) chlorine 1
- (ii) oxidation (1) 2
electrons are lost (1)
- (iii) dissolve it (in water) 1

9. O/N 20/P41/Q5/a

- (a) (i) breakdown by (the passage of) electricity (1)
of an ionic compound in molten/aqueous (state) (1) 2
- (ii) they do not react 1
- (iii) negative electrode: hydrogen (gas) (1)
positive electrode: oxygen (gas) (1) 2
- (iv) $\text{H}^+ + \text{e}^-$ as the only species on the left (1)
equation fully correct (1)
 $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (scores 2) 2

10. O/N 20/P43/Q5/d

- (a) (i) **breakdown** by (the passage of) **electricity** (1)
of an **ionic compound** in **molten or aqueous** (state) (1) 2
- (ii) heat until it melts / heat to or above melting point 1
- (iii) $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$ 1

11. M/J 20/P42/Q5

- (a) **breakdown** of an ionic **compound** when molten or in aqueous solution (1)
(using) electricity / electric current / electrical energy (1) 2
- (b) platinum / graphite 1
- (c) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ 1
- (d) $\text{Na}^+ \text{H}^+ \text{Cl}^- \text{OH}^-$ 2
all four (2) 3 or 2 (1)
- (e) H^+ and Cl^- are discharged / removed (1)
 Na^+ and OH^- remain (1) 2

12. O/N 19/P42/Q4

- (a) electrons (1)
electrons (1)
 Cu^{2+} (ions) (1)
 Br^- (ions) (1) 4
- (b) (i) platinum 1
(ii) chlorine 1
- (iii) $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$
 $\text{H}^+ + \text{e}^-$ on left hand side (1)
rest of equation (1)
state symbols of (aq) $\rightarrow$ (g) (1) 3
- (iv) increases
(sodium) hydroxide is formed
(sodium) hydroxide is an alkali 3
- (c) (i) arrow (anywhere) going from $\text{Zn} \rightarrow \text{Cu}$ 1
- (ii) reading would decrease (1)
Fe less reactive than Zn (1)
OR
difference in reactivity (between Fe and Cu) is smaller 2
- (iii) Ag less reactive than Cu 1

13. O/N 19/P41/Q6/c(ii,iii,iv,v)
- (a) (i) (it contains) ions (1)
(ions) are able to move (1) 2
 - (ii) magnesium is not inert 1
 - (iii) bromine / Br₂ 1
 - (iv) H⁺ and e⁽⁻⁾ on LHS (1)
fully correct, i.e.: 2H⁺ + 2e⁻ → H₂ (1) 2
14. O/N 19/P43/Q3/a,b
- (a) bauxite 1
 - (b) (i) improves conductivity / better conductor (1)
lower (operating) temperature (1) 2
 - (ii) positive: 2O²⁻ → O₂ + 4e⁻ (1)
negative: Al³⁺ + 3e⁻ → Al (1) 2
 - (iii) anodes or carbon react with oxygen (1)
(form) carbon dioxide (1) 2
15. M/J 19/P42/Q7/f
- (a) (i) M1 electrolyte aqueous or solution of named nickel salt (1)
M2 anode impure nickel (1)
M3 cathode pure nickel (1) 3
 - (ii) nickel produced at cathode under the liquid surface (1) 1
16. M/J 19/P43/Q4
- (a) (i) inert / unreactive / does not react with chlorine 1
 - (ii) bubbles / fizzing / effervescence 1
 - (iii) M1 increases
M2 (solid) copper deposited 2
 - (iv) M1 colour fades / becomes pale(r) / becomes colourless / becomes lighter
M2 copper (ions) removed (from solution) 2
 - (v) M1 species oxidised: chloride (ions) / Cl⁻
M2 explanation: loss of electrons / increase in oxidation state 2
 - (b) (i) M1 spoon as cathode
M2 (pure)silver as anode
M3 aqueous silver nitrate as electrolyte
M4 Ag⁺ + e⁻ → Ag 4
 - (ii) any one from:
∞ Improves appearance
∞ prevent / resist corrosion / oxidation
∞ antibacterial 1
17. MAR 19/P42/Q5/e
- (a) (i) hydrogen 1
 - (ii) Heat until magnesium chloride is molten and electrolyse 1
18. M/J 18/P42/Q4/a,b,c,d,e
- (a) 2K(s) + Br₂(l) → 2KBr(s)
1 mark for formulae all correct
1 mark for balancing
1 mark for state symbols 3

- (b) (i) (ionic): made of, positive and negative ions / anions and cations / oppositely charged ions / unlike charged ions / different charged ions (lattice): regular / sequence / pattern / alternating / repeated / framework / ordered / organised / network / uniform 1
- (ii) (in solid) ions don't move 1
(when molten) ions move / ions mobile 1
- (c) (i) substance that conducts electricity / (undergoes) electrolysis 1
decomposed / chemically changed 1
- OR**
molten or liquid or solution or aqueous **AND** containing ions/or ionic 1
- (ii) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
1 mark for $\text{H}^+ + \text{e}^-$ as the only species on the left
1 mark for equation fully correct
1 mark for bromine at the anode
1 mark for potassium hydroxide 4
- (iii) potassium 1
- (d) (i) $\text{I}_2 + \text{Cl}_2 \rightarrow 2\text{ICl}$
1 mark for formulae all correct
1 mark for correct balancing 2
- (ii) one bonding pair 1
6 non-bonding electrons on each atom 1
- (e) (potassium bromide): ionic bonds / attraction between ions 1
(iodine monochloride): intermolecular forces / forces between molecules / named intermolecular forces, e.g. van der Waals / London forces / dispersion forces / dipole- dipole 1
bonds in KBr are stronger / need more energy to break bonds / ORA 1
- 19. M/J 18/P41/Q5/e**
- (a) (i) $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$
1 mark for any equation which has Cu as the product or Cu^{2+} ions on left
1 mark for correct species
1 mark for correct state symbols 3
- (ii) (a pink / brown) solid / deposit forms 1
- (iii) bubbles / fizzing (at the anode) 1
solution becomes paler / less blue / colourless 1
- (iv) a green gas would be seen (on the anode) 1
- 20. M/J 18/P43/Q1/c**
- (a) Cu / copper 1
- 21. M/J 18/P43/Q5/b,c**
- (a) (i) electrons 1
(ii) (positive and negative) ions 1
(iii) nickel 1
iodine 1
 $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$ **OR** $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ 1
- (b) (i) copper formed/copper deposited 1
(ii) oxygen 1
(iii) copper removed or copper lost **or** copper forms ions 1

- (iv) any **three** from:
 (apparatus **A**): solution becomes paler/fades in **A**
 (apparatus **B**): solution stays the same colour in **B**
 (explanation):
 copper ions removed (but not added) copper ions not replaced in **A**
OR
 copper ions both removed and added (at the same rate) copper ions are being replaced (continually)

3

22. MAR 18/P42/Q2/b

- (a) (i) M1 breakdown of an ionic compound when molten or in aqueous solution
 M2 (by the passage of) electricity / electric current / electrical energy
- (ii) hydrogen
 chlorine
 sodium hydroxide
- (iii) $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$
 M1 H^+ on left hand side with e^- added
 M2 fully correct equation

2

3

2

23. O/N 17/P41/Q6/a

- (a) aluminium is more reactive than carbon

1

24. O/N 17/P43/Q4/a,b,c,d,f

- (a) fractional distillation
- (b) (i) oxidation
 (ii) acid(ic)
- (c) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- (d) (i) no carbon dioxide produced / more efficient
 (ii) storage of hydrogen is difficult / takes more space to store (hydrogen) /
 high likelihood of (hydrogen) leaks / lack of availability of hydrogen
- (e) electrolysis

1

1

1

1

1

1

1

1

25. O/N 16/P41/Q5

- (a) (gas) oxygen
 (test) glowing splint
 (result of test) relights
- (b) reference to ions / ionic
 ions cannot move in solid **OR** are in fixed positions in solid
 ions can move when in solution
- (c) (i) copper ions / Cu^{2+}
 gain of electrons / oxidation number decreases
- (ii) any 3 from:
 anode decreases (in mass)
 copper removed (from anode) / solid (copper from anode) becomes aqueous
 cathode increases (in mass)
 copper deposited / added / Cu^{2+} deposited as Cu (on cathode)
- (iii) copper is both added and removed (at same rate)
OR
 the concentration (of copper ions) does not change

1

1

1

1

1

1

1

1

3

1

26. O/N 16/P43/Q4/e,f,g

- (a) (i) *product at the positive electrode:* chlorine 1
product at the negative electrode: hydrogen 1
- (ii) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
OR
 $2\text{H}_3\text{O}^+ + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{H}_2\text{O}$ 1
- (b) oxygen 1
- (c) (i) sodium 1
(ii) $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$ 1
(iii) *test:* (damp blue) litmus 1
result: bleached / removes colour / (turns) white 1

27. M/J 16/P42/Q4/e,f

- (a) (i) **M1** breakdown of an ionic compound when molten or in aqueous solution;
M2 (using) electricity / electric current / electrical energy; 2
- (ii) carbon / graphite / platinum; 1
- (iii) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$;
OR
 $2\text{H}_3\text{O}^+ + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{H}_2\text{O}$; 1
- (iv) cathode / negative electrode; 1
- (v) **M1** damp blue litmus paper;
M2 bleaches / loses colour / turns white / turns colourless; 2
- (b) $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2 + \text{Cl}_2$
all formulae correct; balancing; 2

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Unit 6

CHEMICAL ENERGETICS

IGCSE Chemistry P-4
Topical Workbook
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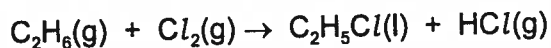
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Unit 6: Chemical Energetics**1. O/N 22/P42/Q5**

Ethane is an alkane which undergoes a photochemical reaction with chlorine as shown.



(a) Write the general formula of alkanes.

..... [1]

(b) State why this reaction is described as a photochemical reaction.

..... [1]

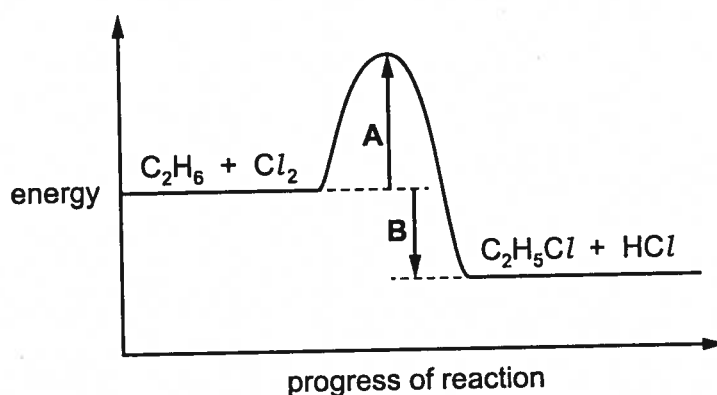
(c) In this reaction, an atom of hydrogen is replaced with a chlorine atom.
State the name of this type of organic reaction.

..... [1]

(d) In this reaction, one of the products is chloroethane.
Name the other product.

..... [1]

(e) The energy profile diagram of this reaction is shown.



(i) Name the energy change labelled A.

..... [1]

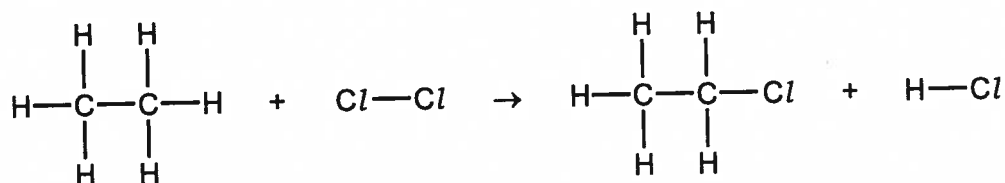
(ii) Name the energy change labelled B.

..... [1]

(iii) State how the energy profile diagram shows this is an exothermic reaction.

..... [1]

(f) The equation for the reaction can be represented as shown.



Some bond energies are given.

bond	bond energy / kJ mol
C-H	410
C-C	350
Cl-Cl	240
C-Cl	340
H-Cl	430

Use the bond energies in the table to calculate the energy change in this reaction.
Use the following steps.

- Calculate the energy needed to break bonds.

energy kJ

- Calculate the energy released in making bonds.

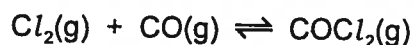
energy kJ

- Determine the energy change in this reaction.

energy change in this reaction kJ/mol [3]

2. O/N 21/P42/Q4

Chlorine reacts with carbon monoxide to produce phosgene gas, $\text{COCl}_2(\text{g})$. A catalyst is used.



The reaction is exothermic.

- (a) Explain why the reaction is exothermic in terms of the energy changes of bond breaking and bond making.

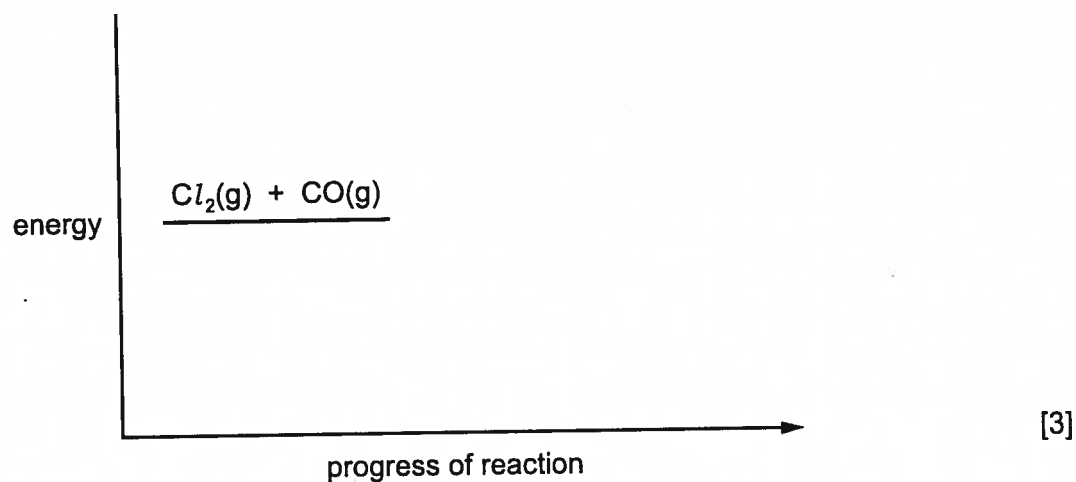
.....

 [3]

- (b) (i) Complete the energy level diagram for this reaction.

On your diagram show:

- the product of the reaction
- an arrow representing the energy change, labelled ΔH
- an arrow representing the activation energy, labelled A.



(ii) State why a catalyst is used.

..... [1]

(c) Describe and explain the effect, if any, on the position of equilibrium when:

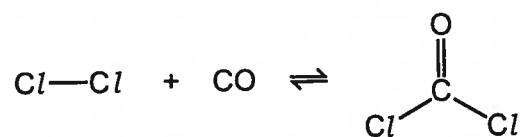
(i) the pressure is increased

.....
 [2]

(ii) the temperature is increased.

.....
 [2]

(d) The reaction between chlorine and carbon monoxide can be represented as shown.



When one mole of chlorine reacts with one mole of carbon monoxide, 230 kJ of energy is released.

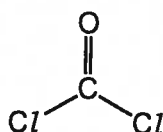
Some bond energies are shown in the table.

bond	bond energy in kJ/mol
Cl-Cl	240
C=O	745
C-Cl	400

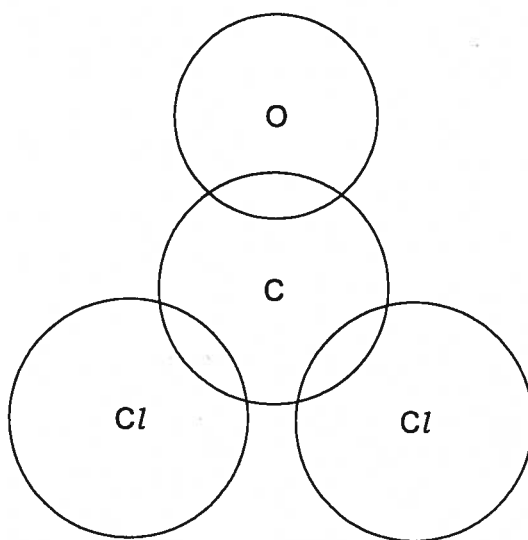
Use the information to calculate the energy of the bond between the C and the O in carbon monoxide, CO.

bond energy in carbon monoxide, CO = kJ/mol [3]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of COCl_2 .



Show outer electrons only.

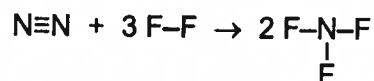


[3]

3. M/J 21/P42/Q5/b

(a) Nitrogen reacts with fluorine to form nitrogen trifluoride, NF_3 .

(i) The chemical equation can be represented as shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
$\text{N}\equiv\text{N}$	945
$\text{F}-\text{F}$	160
$\text{N}-\text{F}$	300

Calculate the energy change for the reaction between nitrogen and fluorine, using the following steps:

- energy taken in to break bonds

..... kJ

- energy released when bonds are formed

..... kJ

- energy change during the reaction.

..... kJ/mol [3]

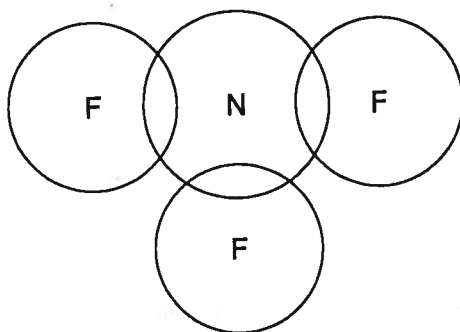
- (ii) Use your answer to (i) to deduce whether this reaction is endothermic or exothermic. Explain your answer.

..... [1]

- (iii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of NF_3 .

Use dots for nitrogen electrons and crosses for fluorine electrons.

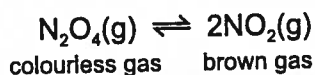
Show outer electrons only.



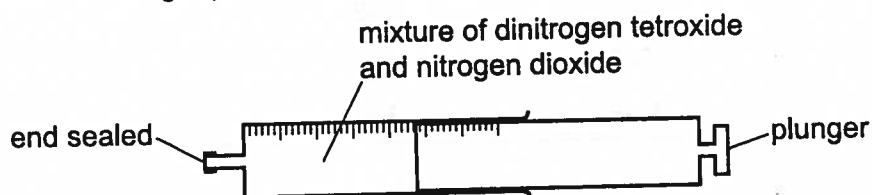
[3]

4. M/J 21/P41/Q4

Dinitrogen tetroxide, N_2O_4 , decomposes into nitrogen dioxide, NO_2 . The reaction is reversible.



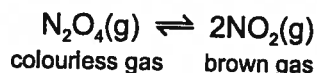
A gas syringe containing a mixture of dinitrogen tetroxide and nitrogen dioxide gases was sealed and heated. After reaching equilibrium the mixture was a pale brown colour.



- (a) State what is meant by the term *equilibrium*.

.....
 [2]

- (b) The plunger of the gas syringe is pushed in. The temperature does not change. The mixture initially turns darker brown. After a few seconds the mixture turns lighter brown because the equilibrium shifts to the left.



- (i) Explain why the mixture initially turns darker brown.

..... [1]

- (ii) Explain why the position of equilibrium shifts to the left.

..... [1]

- (c) The forward reaction is endothermic.

- (i) State what happens to the position of equilibrium when the temperature of the mixture is increased.

..... [1]

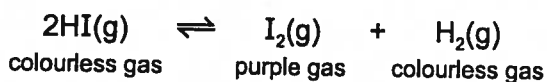
- (ii) State what happens to the rate of the forward reaction and the rate of the backward reaction when the temperature of the mixture is increased.

rate of the forward reaction

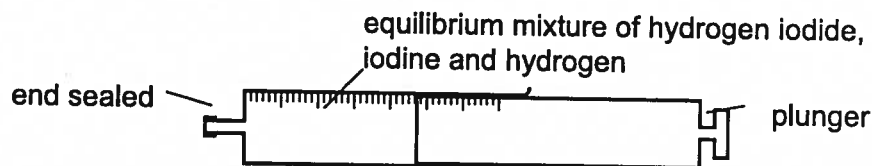
rate of the backward reaction [2]

5. M/J 21/P43/Q4

Hydrogen iodide, HI, decomposes into iodine and hydrogen. The reaction is reversible.



A gas syringe containing a mixture of hydrogen iodide, iodine and hydrogen gases was sealed. After reaching equilibrium the mixture was a pale purple colour.

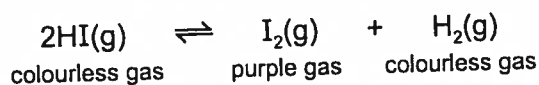


- (a) State what is meant by the term *equilibrium*.

.....
 [2]

- (b) The plunger of the gas syringe is pushed in. The position of equilibrium does not change. The colour of the gaseous mixture turns darker purple.

The temperature remains constant.



(i) Explain why the position of equilibrium does **not** change.

..... [1]

(ii) Suggest why the colour of the gaseous mixture turns darker purple even though the position of equilibrium does not change.

..... [1]

(c) The forward reaction is endothermic.

(i) State what happens to the position of equilibrium when the temperature is decreased.

..... [1]

(ii) State what happens to the rate of the forward reaction and the rate of the backward reaction when the temperature of the mixture is decreased.

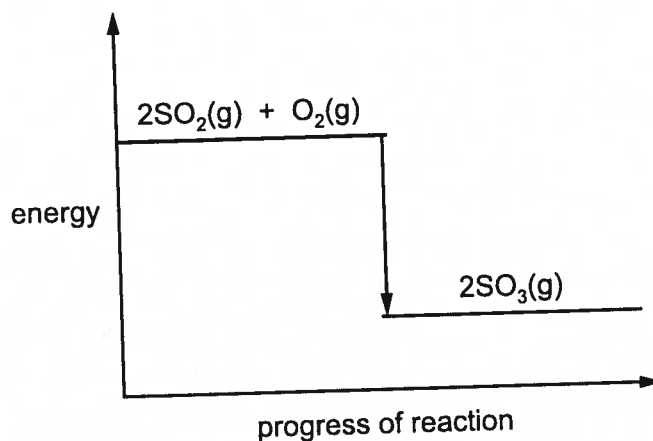
rate of the forward reaction

rate of the backward reaction [2]

6. O/N 20/P41/Q5/b(iii)

(a) Sulfuric acid is manufactured using the Contact process. This manufacture involves four stages.

(i) The energy level diagram for the forward reaction in **stage 2** is shown.

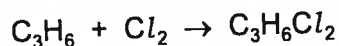


Explain what the diagram shows about the energy changes in the forward reaction.

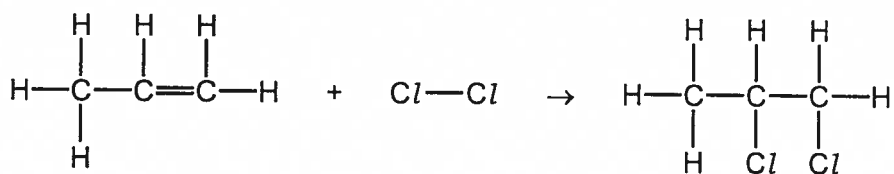
..... [2]

7. M/J 20/P42/Q6/b(ii)

(a) Propene reacts with chlorine in an addition reaction as shown.



(i) The structures of the reactants and products of this reaction are shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
C–C	347
C=C	612
C–H	413
C–Cl	339
Cl–Cl	242

Calculate the energy change for the reaction between propene and chlorine using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

- Calculate the energy released when bonds are formed.

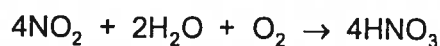
..... kJ

- Calculate the energy change for the reaction between propene and chlorine.

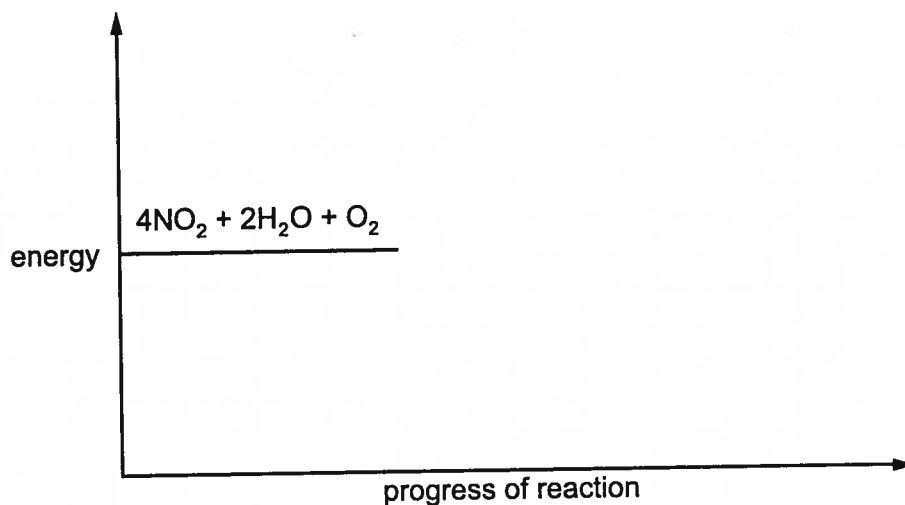
..... kJ/mol [3]

8. M/J 20/P43/Q2/e

(a) The reaction in **stage 3** is exothermic.



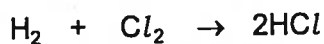
Complete the energy level diagram for this reaction. Include an arrow that clearly shows the energy change during the reaction.



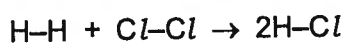
[3]

9. M/J 20/P43/Q3/d

(a) Hydrogen and chlorine react to form hydrogen chloride gas, as shown in the equation.



This equation can be represented as shown.



Some bond energies are shown in the table.

bond	bond energy in kJ/mol
H-H	436
Cl-Cl	243
H-Cl	432

Calculate the energy change for the reaction between hydrogen and chlorine, using the following steps.

- Calculate the energy needed to break the bonds.

..... kJ

- Calculate the energy released when bonds are formed.

..... kJ

- Calculate the energy change for the reaction.

..... kJ/mol [3]

10. M/J 20/P43/Q6/a

(a) An endothermic reaction occurs when calcium nitrate is heated.

(i) Balance the equation for this reaction.



[1]

(ii) State the type of reaction shown by the equation.

..... [1]

11. MAR 20/P42/Q1/e

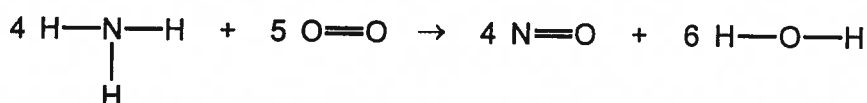
(a) Hydrogen fuel cells can be used to power vehicles.

Write the word equation for the overall reaction that takes place in a hydrogen fuel cell.

..... [1]

12. O/N 19/P42/Q3/d(ii)

(a) (i) The chemical equation for the reaction can be represented as shown.



Use the bond energies in the table to calculate the energy change, in kJ/mol, which occurs when **one** mole of NH_3 reacts.

bond	N-H	O=O	N=O	O-H
bond energy in kJ/mol	391	498	587	464

- Energy needed to break bonds.

..... kJ

- Energy released when bonds are formed.

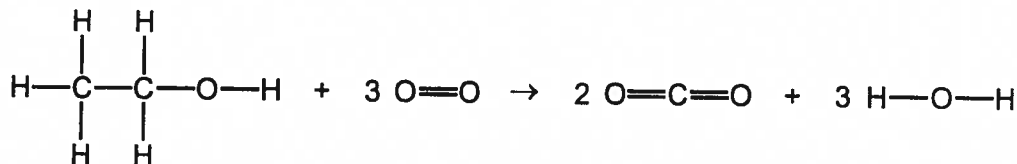
..... kJ

- Energy change when **one** mole of NH_3 reacts.

energy change = kJ/mol [4]

13. O/N 19/P41/Q7/b

(a) The equation for the complete combustion of ethanol is shown.



Use the bond energies in the table to calculate the energy change, in kJ/mol, for the complete combustion of ethanol.

bond	bond energy in kJ/mol
C—C	347
C—H	413
C—O	358
C=O	805
O—H	464
O=O	498

- Energy needed to break bonds.

..... kJ

- Energy released when bonds are formed.

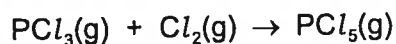
..... kJ

- Energy change for the complete combustion of ethanol.

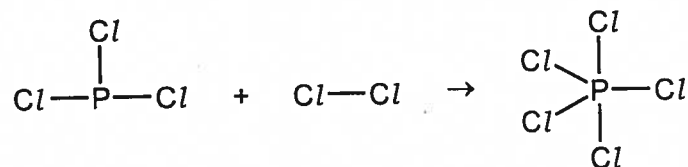
energy change = kJ/mol [3]

14. O/N 19/P43/Q4/c,d,e,f,g,h

- (a) Gaseous phosphorus(III) chloride, PCl_3 , reacts with gaseous chlorine to form gaseous phosphorus(V) chloride, PCl_5 .



The chemical equation for this reaction can be represented as shown.



- (i) Use the bond energies in the table to calculate the energy change, in kJ/mol, of the reaction.

bond	bond energy in kJ/mol
P—Cl	326
Cl—Cl	243

- Energy needed to break bonds.

..... kJ

- Energy released when bonds are formed.

..... kJ

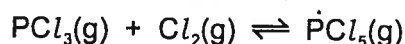
- Energy change of reaction.

energy change = kJ/mol [3]

- (ii) Deduce whether the energy change for this reaction is exothermic or endothermic. Explain your answer.

..... [1]

- (b) Under certain conditions the reaction reaches equilibrium.

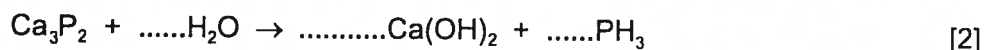


State and explain the effect, if any, on the **position of equilibrium** if the pressure is increased. All other conditions are unchanged.

.....

..... [2]

- (c) Phosphine, PH_3 , is produced by the reaction between water and calcium phosphide, Ca_3P_2 . Balance the chemical equation for this reaction.



[2]

- (d) The phosphonium ion, PH_4^+ , is similar to the ammonium ion.

- (i) State the formula of the ammonium ion. [1]

- (ii) Suggest the formula of phosphonium iodide. [1]

- (e) Calcium phosphate contains the phosphate ion, PO_4^{3-} .

What is the formula of calcium phosphate?

..... [1]

- (f) Phosphorus forms another compound with hydrogen with the following composition by mass: P, 93.94%; H, 6.06%.

- (i) Calculate the empirical formula of the compound.

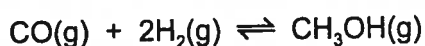
empirical formula = [2]

- (ii) The compound has a relative molecular mass of 66.
Deduce the molecular formula of the compound.

molecular formula = [1]

15. M/J 19/P42/Q4/a,b

Methanol is made industrially by reacting carbon monoxide with hydrogen. The gases react at a temperature of 250°C and a pressure of 75 atmospheres.



The forward reaction is exothermic.

- (a) Suggest a source of hydrogen for this industrial process.

..... [1]

- (b) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the reverse reaction	effect on the equilibrium yield of CH ₃ OH(g)
adding a catalyst		no change
increasing the temperature	increases	
decreasing the pressure		

[4]

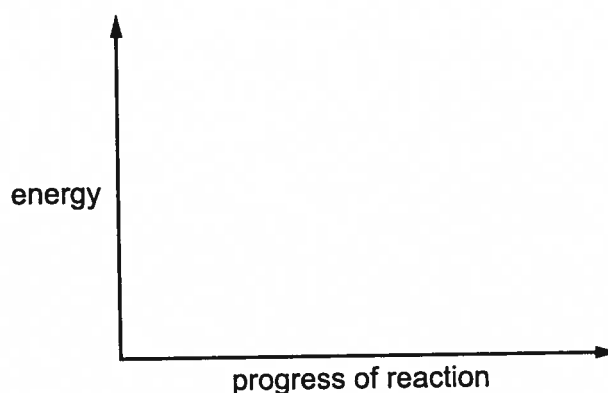
16. M/J 19/P43/Q5/b

- (a) The reaction between ethanoic acid and ethanol is exothermic.

Draw an energy level diagram for this reaction.

On your diagram label:

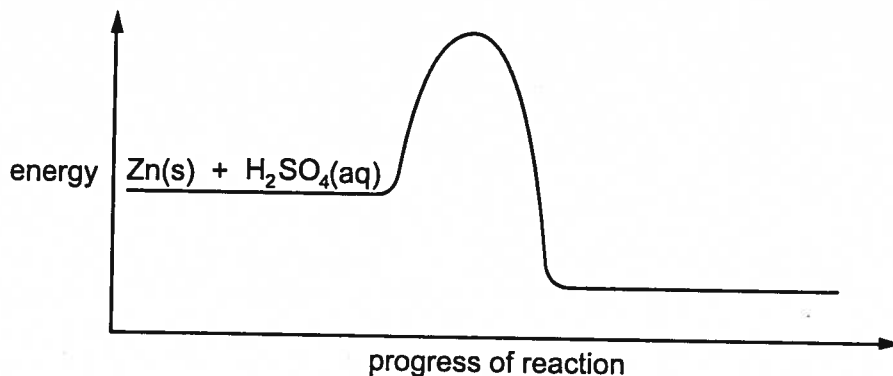
- the reactants and products
- the energy change of the reaction, ΔH .



[3]

17. M/J 18/P41/Q5/c,d

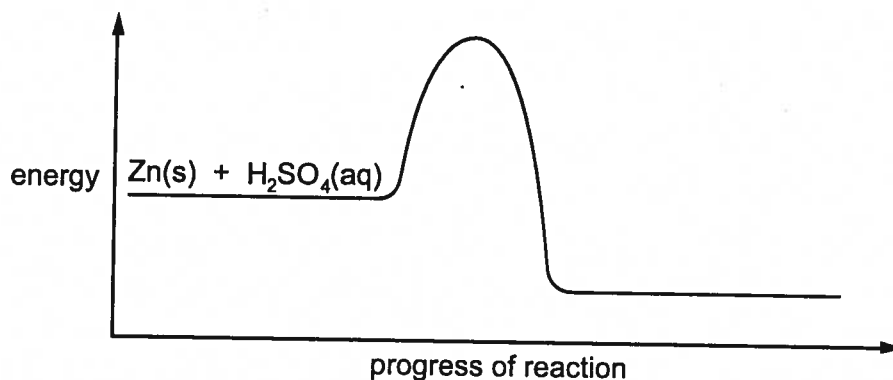
- (a) The energy level diagram shows the energy profile for the reaction between zinc and dilute sulfuric acid.



- (i) Complete the diagram by adding the formulae of the products. Include state symbols. [3]
(ii) Draw an arrow on the diagram to represent the activation energy. [1]
(iii) Is the reaction endothermic or exothermic? Explain your answer.

- [1]
(b) The reaction between zinc and dilute sulfuric acid can be catalysed by the addition of aqueous copper(II) sulfate.

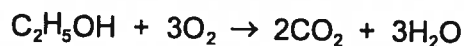
On the diagram, add the energy profile for the catalysed reaction.



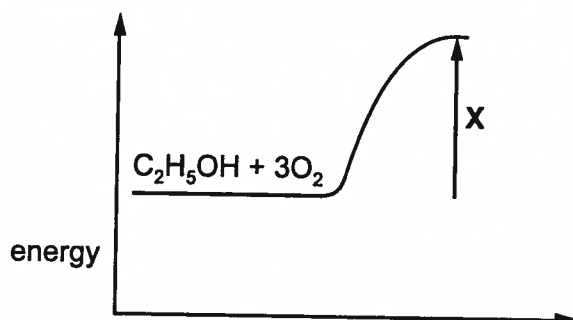
[1]

18. O/N 17/P42/Q3/a,b,c

The chemical equation for the complete combustion of ethanol, C_2H_5OH , is shown.



The energy released when one mole of ethanol undergoes complete combustion is 1280 kJ.
Part of the energy level diagram for this reaction is shown.



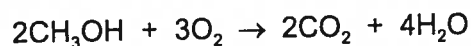
- (a) Complete the energy level diagram to show
- the products of the reaction,
 - the overall energy change of the reaction.

[3]

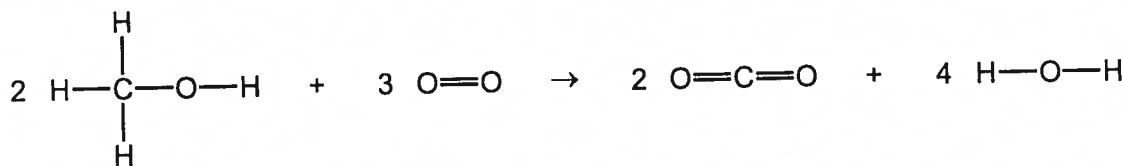
- (b) What does **X** represent?

..... [1]

- (c) The chemical equation for the complete combustion of methanol, CH_3OH , is shown.



The equation can be represented as shown.



Use the bond energies in the table to determine the energy change, ΔH , for the complete combustion of **one** mole of methanol.

bond	bond energy in kJ/mol
C-H	410
C-O	360
O-H	460
O=O	500
C=O	805

- energy needed to break bonds

..... kJ

- energy released when bonds are formed

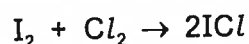
..... kJ

- energy change, ΔH , for the complete combustion of **one** mole of methanol

..... kJ/mol [4]

19. M/J 17/P42/Q5/d

- (a) Iodine reacts with chlorine. The chemical equation is shown.



Use the bond energies to answer the questions.

bond	bond energy in kJ/mol
I-I	151
Cl-Cl	242
I-Cl	208

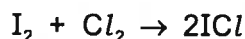
- (i) Calculate the total amount of energy required to break the bonds in 1 mole of I_2 and 1 mole of Cl_2 .

..... kJ [1]

- (ii) Calculate the total amount of energy given out when the bonds in 2 moles of ICl are formed.

..... kJ [1]

- (iii) Use your answers to (d)(i) and (d)(ii) to calculate the overall energy change for the reaction.

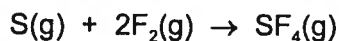


..... kJ/mol [1]

20. M/J 16/P41/Q2/f

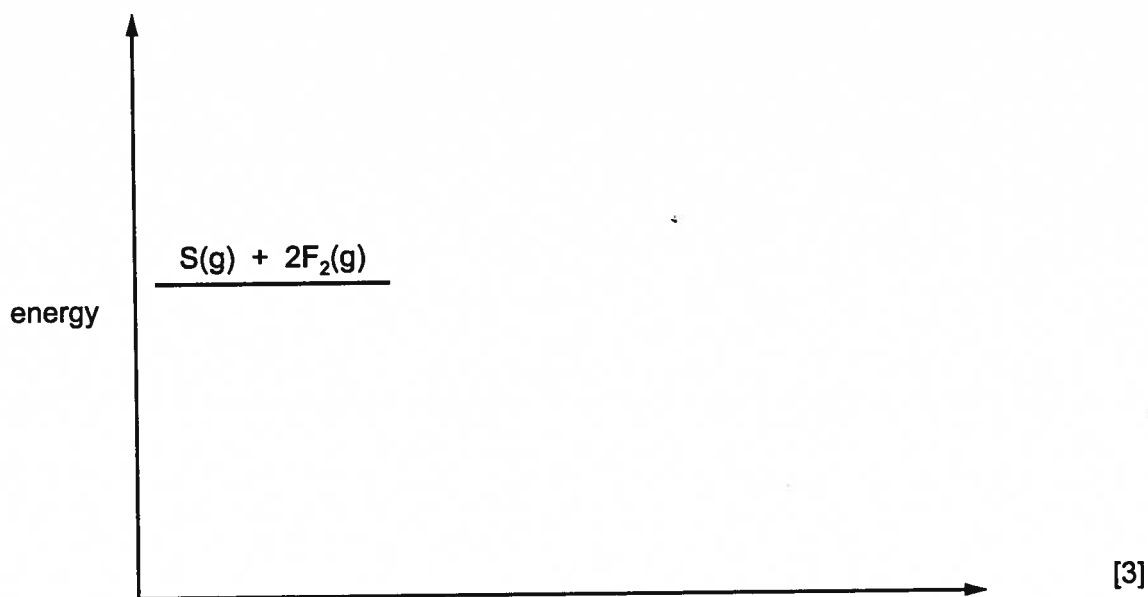
Period 3 contains the elements sodium to argon. This question asks about the chemistry of each of the Period 3 elements or their compounds.

- (a) Sulfur tetrafluoride, SF_4 , can be made by combining gaseous sulfur with fluorine.



The reaction is exothermic.

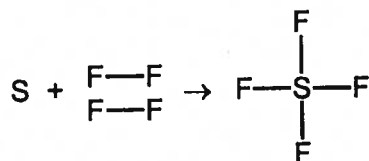
- (i) Complete the energy level diagram for this reaction. Include an arrow which clearly shows the energy change during the reaction.



(ii) During the reaction the amount of energy given out is 780 kJ/mol.

The F–F bond energy is 160 kJ/mol.

Use this information to determine the bond energy, in kJ/mol, of one S–F bond in SF_4 .



..... kJ/mol [3]

Answers Section**1. O/N 22/P42/Q5**

- (a) C_nH_{2n+2} 1
 (b) needs ultraviolet (light) 1
 (c) substitution 1
 (d) hydrogen chloride 1
 (e) (i) activation energy 1
 (ii) energy (change) of reaction 1
 (iii) energy of products is lower than energy of reactants 1
 (f) M1 energy needed to break bonds
 $6 \times C-H + C-C + C-Cl = 6 \times 410 + 350 + 240 = 3050 \text{ (kJ / mol) (1)}$
 M2 energy released in making bonds
 $5 \times C-H + C-C + C-Cl + H-Cl = 5 \times 410 + 350 + 340 + 430 = 3170 \text{ (kJ / mol) (1)}$
 M3 energy change $M1 - M2 = -120 \text{ (kJ / mol) (1)}$ 3

2. O/N 21/P42/Q4

- (a) M1 E of making bonds > breaking bonds
 M2 bond making releases energy
 M3 bond breaking requires energy 3
- (b) (i) M1 exothermic mark
 horizontal line below energy level to R.H.S. of reactants line
 and
 labelled $COCl_2(g)$ (1)
 M2 activation E mark
 activation energy 'hump' with upward arrow labelled A / activation energy (1)
 M3 energy change mark
 one downward arrow labelled ΔH
 and
 energy change starting from E level of reactants and finishing at E level
 of products (1) 3
 (ii) increases rate of reaction 1
- (c) (i) equilibrium shifts to right hand side (1)
 fewer moles (of gas) on right hand side (1) 2
 (ii) equilibrium shifts to left hand side (1)
 (forward) reaction is exothermic (1) 2
- (d) M1 bond energy in making bonds
 $= [(2 \times 400) + 745] = 1545 \text{ (kJ mol}^{-1}\text{)}$
 M2 use of total E change
 $-230 = [240 + E(C \equiv O)] - 1545$
 OR
 $[240 + E(C \equiv O)] = -230 + 1545 = (+1315)$
 M3 $E(C \equiv O)$
 $= [-230 + 1545] - 240 = 1075 \text{ (kJ mol}^{-1}\text{)}$ 3
- (e) M1 all single bonding dot and cross pairs correct
 M2 double $C=O$ bond dot and cross pairs are correct
 M3 complete diagram is correct 3

3. M/J 21/P42/Q5/b

- (a) (i) bonds broken $[945 + (3 \times 160)] = 1425$ (1)
 bonds formed $(2 \times 3 \times 300) = 1800$ (1)
 energy change = $M1 - M2 = 1425 - 1800 = -375$ (1) 3
- (ii) *Answer must reflect answer in 5(b)(i)*
 exothermic
 and
 more energy released (in bond formation) than used/taken in (in bond breaking) 1
- (iii) N with 1 bonding pair with each F (1)
 2 non-bonding dots for N (1)
 6 non-bonding crosses for F (1) 3

4. M/J 21/P41/Q4

- (a) the **rate** of forward reaction equals the rate of the reverse reaction (1)
concentrations of reactants and products are constant (1) 2
- (b) (i) (increased pressure) nitrogen dioxide **particles** or **molecules** (forced) closer together
OR
 same number of nitrogen dioxide **particles** or **molecules** in a smaller volume 1
- (ii) fewer number of gas moles or molecules on left hand side or reactant side (of the equation) **ORA** 1
- (c) (i) shifts to the right 1
- (ii) increase / faster (1)
 increase / faster (1) 2

5. M/J 21/P43/Q4

- (a) the **rate** of forward reaction equals (the rate of the) reverse reaction (1)
 concentrations of reactants and products are constant (1) 2
- (b) (i) same number of gas moles on both sides of the equilibrium / same number of gas molecule on both sides of the equilibrium 1
- (ii) iodine **particles** or **molecules** (forced) closer together / same number of iodine **particles** or **molecules** in a smaller volume 1
- (c) (i) shifted to the left 1
- (ii) decrease / slower (1)
 decrease / slower (1) 2

6. O/N 20/P41/Q5/b(iii)

- (a) (i) exothermic / heat / energy is released / surroundings warm up
 products have lower energy than reactants / **ORA** 2

7. M/J 20/P42/Q6/b(ii)

- (a) (i) (energy required to break bonds =) 854 (1)
 (energy given out when bonds form =) 1025 (1)
 overall energy change $854 - 1025 = -171$ (1) 3

8. M/J 20/P43/Q2/e

- (a) • horizontal product energy line at lower energy level than reactant
 • label of product
 • correct direction of vertical arrow – arrow must start level with reactant energy and finish level with product level and one arrow head **ONLY** 3

9. M/J 20/P43/Q3/d
 (a) energy needed to break bonds = $436 + 243 = 679$
 energy released when bonds formed = $2 \times 432 = 864$
 energy change = $679 - 864 = -$ AND 185 3
10. M/J 20/P43/Q6/a
 (a) (i) 2, 2, 4, 1 1
 (ii) thermal decomposition 1
11. MAR 20/P42/Q1/e
 (a) hydrogen + oxygen $\rightarrow$ water 1
12. O/N 19/P42/Q3/d(ii)
 (a) (i) M1 Bonds broken
 $[4 \times 3 \times 391] + [5 \times 498]$
 $= 4692 + 2490 = 7182$
 M2 Bonds formed
 $[4 \times 587] + [12 \times 464]$
 $= 2348 + 5568 = 7916$
 M3 Energy change = $7182 - 7916 = -734$
 M4 = M3 / 4 = $-734 / 4 = -183.5$ 4
13. O/N 19/P41/Q7/b
 (a) (energy to break bonds) = 4728 (1)
 (energy released by making bonds) = 6004 (1)
 -1276 (1) 3
14. O/N 19/P43/Q4/c,d,e,f,g,h
 (a) (i) method 1
 ∞ (bond breaking) = 1221 or $(326 \times 3) + 243$ (1)
 ∞ (bond forming) = 1630 or (326×5) (1)
 ∞ energy change = -409 kJ (1) negative sign essential
 OR
 method 2 (ignoring 3 P-Cl bonds on both sides)
 ∞ bond breaking = 243 (1)
 ∞ bond forming = 652 or 326×2 (1)
 ∞ energy change = -409 kJ (1) negative sign essential 3
 (ii) exothermic AND energy released when bonds form is greater than energy absorbed to break bonds
 OR exothermic AND overall energy change has a negative sign 1
- (b) fewer OR less molecules OR moles + on right OR in product (1) ORA
 equilibrium shifts to the right (1) 2
- (c) any two numbers correct (1)
 equation fully balanced (1)
 $\text{Ca}_3\text{P}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Ca}(\text{OH})_2 + 2\text{PH}_3$ 2
- (d) (i) NH_4^+ 1
 (ii) PH_4I 1
- (e) $\text{Ca}_3(\text{PO}_4)_2$ 1

- (f) (i) 93.94 / 31 and 6.06 / 1

OR 3.03 and 6.06

OR 1 : 2 ratio (1)

PH₂ (1)

2

- (ii) P
- ₂
- H
- ₄

1

15. M/J 19/P42/Q4/a,b

1

- (a) water / natural gas / hydrocarbons

- (b)

effect on the rate of the reverse reaction	effect on the percentage of methanol in the equilibrium mixture
M1 increases(1)	no change
increases	M3 decreases(1)
M2 decreases(1)	M4 decreases(1)

4

16. M/J 19/P43/Q5/b

- (a)
- M1**
- right hand energy level lower than left hand side energy level

M2 reactants and product positions identified**M3** energy change shown as approximately vertical line indicating gap between reactants and product with arrow head pointing from reactant to products. Arrow needs to be labelled

3

17. M/J 18/P41/Q5/c,d

1

- (a) (i) ZnSO
- ₄

1

H₂ written on product line

1

states (aq) **AND** (g)

- (ii) (labelled) arrow pointing upwards starting level with reactants and finishing level with top of the hump.

1

(iii) exothermic **AND** products are at lower energy (than reactants)

1

- (b) lower hump starting from reactants line

1

18. O/N 17/P42/Q3/a,b,c

- (a)
- exothermic mark:*
- horizontal line representing the energy of the products below the energy of the reactants

label of products mark: product line labelled with 2CO₂ + 3H₂O*correct direction of vertical heat of reaction arrow:* arrow starts level with reactant energy and finishes level with product energy **AND** has (only) **one** arrow head

3

- (b) activation energy / E
- _a

- (c) -650 kJ/mol

M1 bonds broken

$$2 \times ((3 \times 410) + 360 + 460) + (3 \times 500)$$

$$2 \times (1230 + 360 + 460) + 1500$$

$$2 \times 2050 + 1500$$

$$4100 + 1500 = 5600$$

1

M2 bonds formed

$$(2 \times (2 \times 805)) + (4 \times (2 \times 460))$$

$$2 \times 1610 + 4 \times 920$$

$$3220 + 3680 = 6900$$

$$\mathbf{M3} = \mathbf{M1} - \mathbf{M2}$$

$$\text{energy change of reaction} = 5600 - 6900 = -1300$$

$$\mathbf{M4} = \mathbf{M3} / 2$$

4

19. M/J 17/P42/Q5/d

(a) (i) 393 (kJ)

(ii) 416 (kJ)

(iii) -23 (kJ/mol)

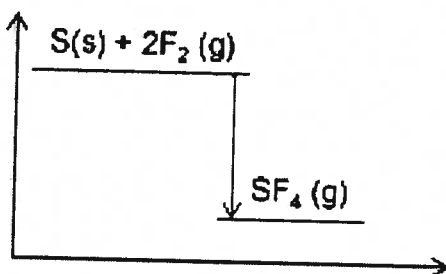
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1

20. M/J 16/P41/Q2/f

(a) (i)

**M1** exothermic mark: horizontal product energy line at lower energy than that of reactant energy line;**M2** label of product mark: SF_4 ;**M3** correct direction of vertical heat of reaction arrow: arrow must start level with reactant energy and finish level with product energy **and** must have only **one** (correct) arrow-head;

3

(ii) **M1** bond energy of 2F_2 : $2 \times \text{F-F} = 2 \times 160 = 320$ (kJ/mol);**M2** bond energy of all bonds in SF_4 : $780 + 320 = 1100$ (kJ/mol);**M3** calculated bond energy of SF_4 divided by 4: $1100/4 = 275$ (kJ/mol);

3

23. O/N 15/P23/Q2/a,b,c,d 1
- (a) (i) plastics; 1
- (b) third box ticked / exothermic; 1
- (c) 4; 1
- (d) (i) speed up the reaction / increase reaction rate; 1
- (ii) carbon dioxide; 1
- water; 1
24. M/J 15/P21/Q2/a 1
- (a) temperature rises;
25. O/N 14/P21/Q4/a,b,c,d,e [1]
- (a) thermometer
- (b) Any **two** from:
- same volume of water in can
 - same height of burner (from can)
 - wick same height
 - same rate / amount of stirring of water
 - **allow:** same temperature of water at start
 - **allow:** same amount of fuels burnt / same temperature rise
 - **allow:** same type of can
- [2]
- (c) so same temperature throughout the water / to stop differences in temperature in the different parts of the water / otherwise the temperature will be higher at the bottom (of the water) / so not hotter in one place [1]
- ignore:** to mix the water / so there are no convection currents [1]
- (d) decreases / goes down
- idea of liquid or fuel turning to vapour / gas;
- allow:** gases formed
- ignore:** fuels evaporate [1]
- note:** 2nd mark dependent on first [1]
- (e) F
26. O/N 14/P21/Q5/b [1]
- (a) exothermic
27. O/N 13/P21/Q6/c [1]
- (a) endothermic
28. M/J 12/P22/Q5/c(i) [1]
- (a) (i) exothermic;



Unit 7

CHEMICAL REACTIONS

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 7.1 Rate (speed) of reaction
- 7.2 Reversible reactions
- 7.3 Redox

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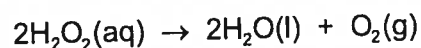
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Unit 7: Chemical Reactions

7.1: Rate (speed) of reaction

1. O/N 21/P43/Q6

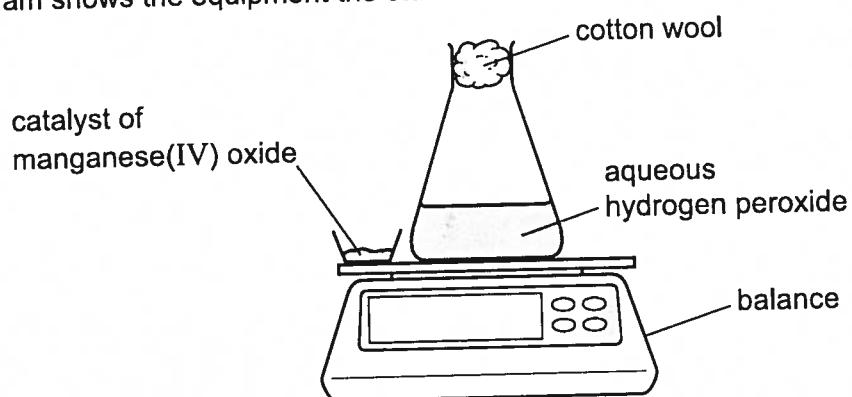
A student investigates the decomposition of hydrogen peroxide in the presence of a catalyst of manganese(IV) oxide.



(a) State the meaning of the term *catalyst*.

.....
..... [2]

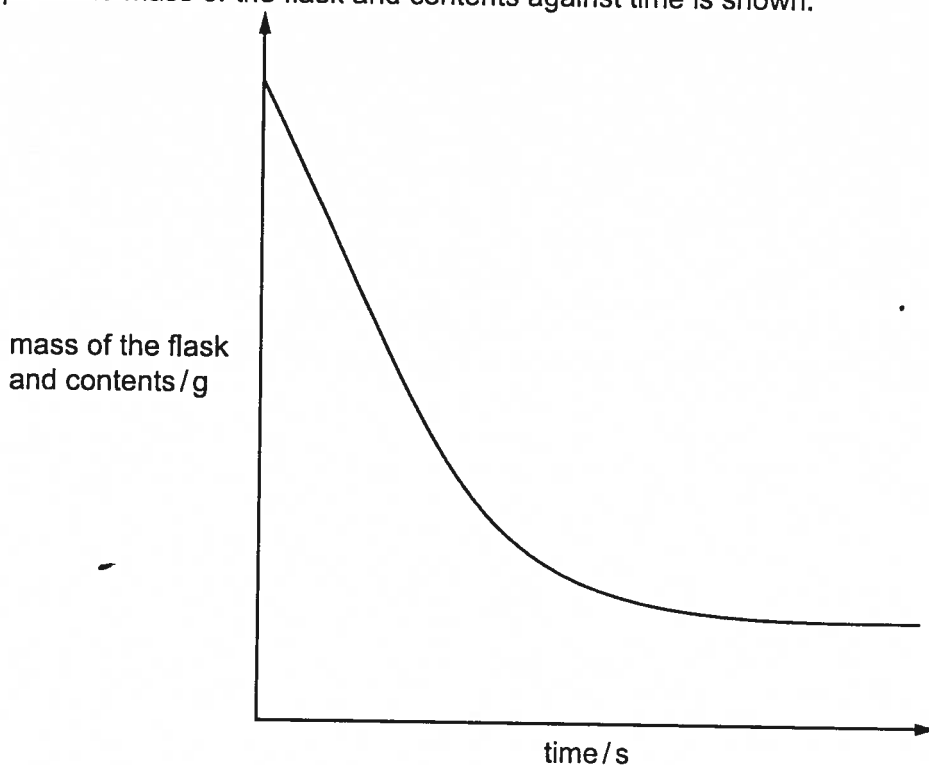
(b) The diagram shows the equipment the student uses.



The student uses this method:

- the catalyst is added to the aqueous hydrogen peroxide
- the stop-clock is started
- the mass of the flask and contents is recorded at regular time intervals.

A graph of the mass of the flask and contents against time is shown.



- (i) Suggest why the mass of the flask and contents decreases as time increases.

..... [1]

- (ii) Describe what happens to the rate of the reaction as time increases.

.....
..... [2]

- (c) The student repeats the experiment at a higher temperature. All other conditions stay the same. The rate of reaction increases.

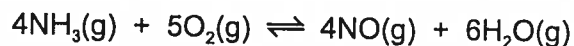
- (i) Explain, in terms of collisions between particles, why the rate of reaction increases at a higher temperature.

.....
.....
..... [3]

- (ii) Draw a line on the graph in (b) for the experiment at a higher temperature. [2]

2. O/N 20/P43/Q3/c(i,iii)

- (a) Some of the ammonia made by the Haber process is converted into nitric acid.
The first stage of this process is the oxidation of ammonia to make nitrogen monoxide.



The process is carried out at 900 °C and a pressure of 5 atmospheres using an alloy of platinum and rhodium as a catalyst.

The forward reaction is exothermic.

- (i) State the meaning of the term *catalyst*.

.....

..... [2]

- (ii) Complete the table using the words **increase**, **decrease** or **no change**.

	effect on the rate of the forward reaction	effect on the equilibrium yield of NO(g)
increasing the temperature		
increasing the pressure		

[4]

3. M/J 20/P42/Q3/a(i,ii)

- (a) Sulfuric acid is made from sulfur in a four-stage process.

stage 1 Sulfur is converted into sulfur dioxide.

stage 2 Sulfur dioxide is converted into sulfur trioxide.

stage 3 Sulfur trioxide is converted into oleum.

stage 4 Oleum is converted into sulfuric acid.

- (i) How is sulfur converted into sulfur dioxide in **stage 1**?

..... [1]

- (ii) Describe how sulfur dioxide is converted into sulfur trioxide in **stage 2**.

Your answer should include:

- an equation for the reaction
- the temperature used
- the name of the catalyst used.

.....

.....

..... [3]

4. M/J 20/P42/Q4/a,b,c,d

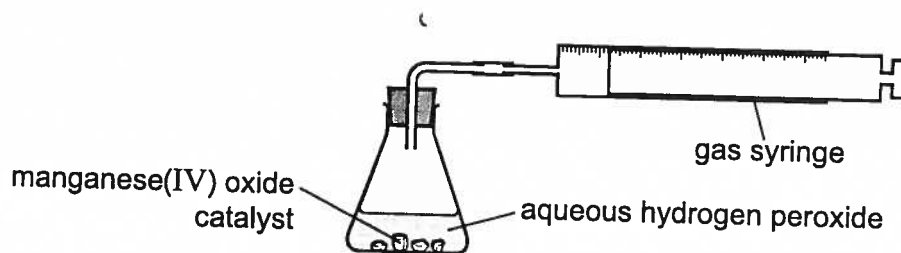
Oxygen is produced by the decomposition of hydrogen peroxide. Manganese(IV) oxide is the catalyst for this reaction.

- (a) What is meant by the term *catalyst*?

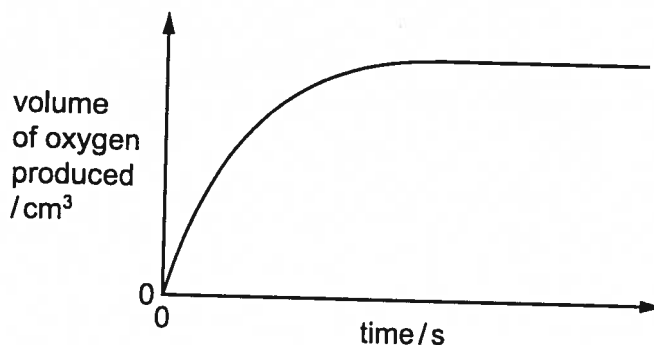
.....

..... [2]

- (b) A student measures the volume of oxygen produced at regular time intervals using the apparatus shown. Large lumps of manganese(IV) oxide are used.



A graph of the results is shown.



What happens to the **rate** of this reaction as time increases?
In your answer, explain why the rate changes in this way.

..... [4]

(c) The experiment is repeated using the same mass of manganese(IV) oxide. Powdered manganese(IV) oxide is used instead of large lumps. All other conditions stay the same.

Sketch a graph on the axes in (b) to show how the volume of oxygen changes with time. [2]

(d) In terms of particles, explain what happens to the rate of this reaction when the temperature is increased.

..... [3]

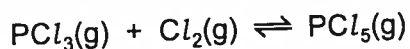
5. M/J 20/P43/Q2/c

(a) Explain, in terms of particles, what happens to the rate of this reaction when the temperature is increased.

..... [3]

6. MAR 20/P42/Q3/d(ii)

- (a) PCl_3 reacts with chlorine, Cl_2 , to form PCl_5 . This reaction is exothermic and reaches an equilibrium.



- (i) State the effect, if any, on the position of this equilibrium when the following changes are made.
Explain your answers.

temperature is increased

.....

pressure is increased

..... [4]

7. MAR 20/P42/Q4/a

Iron is a typical transition element.

Iron:

- acts as a catalyst
- forms coloured compounds
- has more than one oxidation state.

- (a) Name **one** major industrial process that uses iron as a catalyst and name the product made in this process.

process

product made [2]

8. O/N 19/P41/Q7/c(i),d(ii)

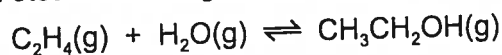
- (a) Ethanol can be oxidised by hydrogen peroxide to form ethanal, CH_3CHO . A catalyst for this reaction is Fe^{3+} .

- (i) What is meant by the term *catalyst*?

.....

..... [2]

- (b) Ethene gas reacts with steam to form gaseous ethanol.



The reaction can reach a position of equilibrium. The forward reaction is exothermic.

- (i) Increasing the pressure of a gas increases its concentration.

State and explain the effect of increasing the pressure on the **rate** of the reaction.
All other conditions are unchanged.

.....

..... [2]

9. M/J 19/P43/Q5/c,d

- (a) Concentrated sulfuric acid is a catalyst for this reaction.
What is meant by the term *catalyst*?

.....
 [2]

- (b) The rate of reaction can be increased by increasing the temperature.
Explain why increasing the temperature increases the rate of reaction.

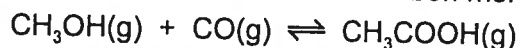
.....

 [4]

10. MAR 19/P42/Q4/a

This question is about ethanoic acid, CH_3COOH .

- (a) Ethanoic acid is manufactured from methanol and carbon monoxide.



The process is done at 200°C and 30 atmospheres pressure.

The forward reaction is exothermic.

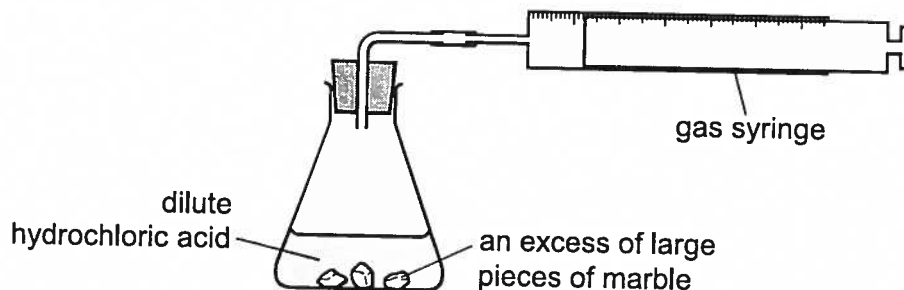
Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the forward reaction	effect on the equilibrium yield of $\text{CH}_3\text{COOH}(\text{g})$
adding a catalyst		no change
increasing the temperature		
decreasing the pressure	decreases	

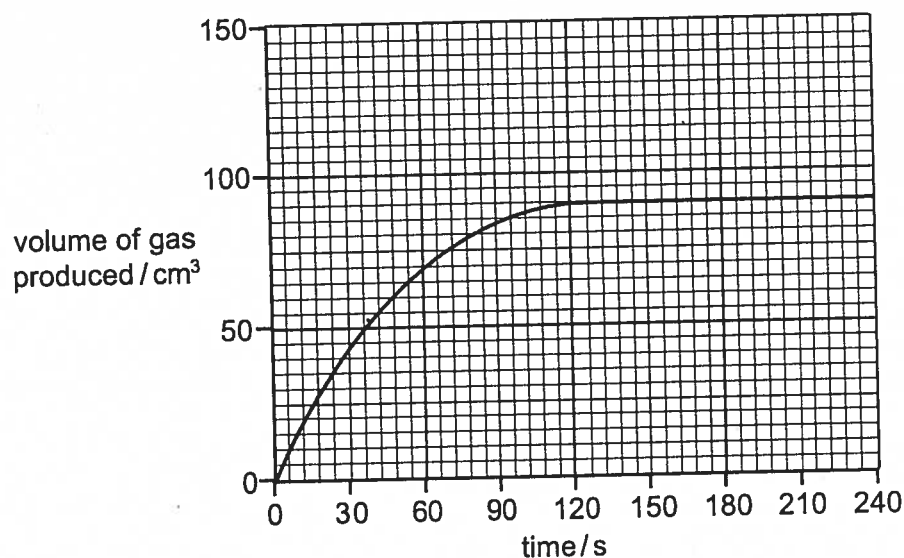
[4]

11. O/N 18/P42/Q4

A student investigated the progress of the reaction between dilute hydrochloric acid, HCl , and an excess of large pieces of marble, CaCO_3 , using the apparatus shown.



- (a) A graph of the volume of gas produced against time is shown.



(i) How does the shape of the graph show that the rate of reaction decreased as the reaction progressed?

..... [1]

(ii) Why did the rate of reaction decrease as the reaction progressed?

..... [1]

(iii) After how many seconds did the reaction finish?

..... s [1]

(b) The experiment was repeated using the same mass of smaller pieces of marble. All other conditions were kept the same.

Draw a graph **on the grid** to show the progress of the reaction using the smaller pieces of marble. [2]

(c) The original experiment was repeated at a higher temperature. All other conditions were kept the same.

Describe and explain, in terms of collisions between particles, the effect of using a higher temperature on the time taken for the reaction to finish.

.....
.....
.....
.....
..... [5]

12. M/J 18/P42/Q3

Complete the following table.

particle	number of protons	number of electrons	number of neutrons	number of nucleons
$^{23}_{11}\text{Na}$	11	11		23
$^{37}_{17}\text{Cl}^-$			20	
$^{56}_{26}\text{.....}$	26	24	30	56

[6]

13. M/J 18/P43/Q3/d(ii),(iii)

(a) Cobalt reacts with dilute hydrochloric acid to make the salt cobalt(II) chloride. Bubbles of hydrogen gas are produced.

- (i) The rate of reaction of cobalt with dilute hydrochloric acid can be made faster by heating the acid or by increasing its concentration.

State **one** other way to make the rate of reaction faster.

..... [1]

- (ii) Use collision theory to explain how heating the dilute hydrochloric acid makes the rate of reaction faster.

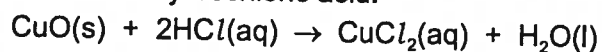
.....

.....

..... [3]

14. O/N 17/P41/Q7/a

Copper(II) oxide reacts with dilute hydrochloric acid.

6.00 g of copper(II) oxide were added to 50.0 cm³ of 1.00 mol/dm³ hydrochloric acid. This was an excess of copper(II) oxide.

- (a) The rate of the reaction can be increased by increasing the concentration of the hydrochloric acid or by heating it.

- (i) In terms of collisions, explain why increasing the concentration of the hydrochloric acid increases the rate of the reaction.

.....

..... [2]

- (ii) In terms of collisions, explain why heating the hydrochloric acid increases the rate of the reaction.

.....

..... [2]

15. O/N 17/P43/Q5

- (a) (i) Name the products formed when sodium nitrate is heated.

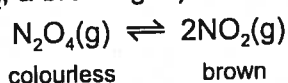
..... [2]

- (ii) When copper(II) nitrate, $\text{Cu}(\text{NO}_3)_2$, undergoes thermal decomposition, three products are formed. One of the products is nitrogen dioxide, NO_2 .

Write a chemical equation for the thermal decomposition of copper(II) nitrate.

..... [2]

- (b) The chemical equation shows the equilibrium between dinitrogen tetroxide (N_2O_4 , a colourless gas) and nitrogen dioxide (NO_2 , a brown gas).



A mixture of dinitrogen tetroxide and nitrogen dioxide is allowed to reach equilibrium in a closed gas syringe.

- (i) In chemistry, what is meant by the term *equilibrium*?

.....

..... [2]

- (ii) If the equilibrium mixture is heated at constant pressure, a darker brown colour is seen inside the gas syringe.

What does this information indicate about the decomposition of dinitrogen tetroxide?

Explain your answer in terms of the position of the equilibrium.

.....

..... [2]

- (iii) Suggest what you would see if the pressure on the equilibrium mixture were increased at constant temperature.

Explain your answer in terms of the position of the equilibrium.

.....

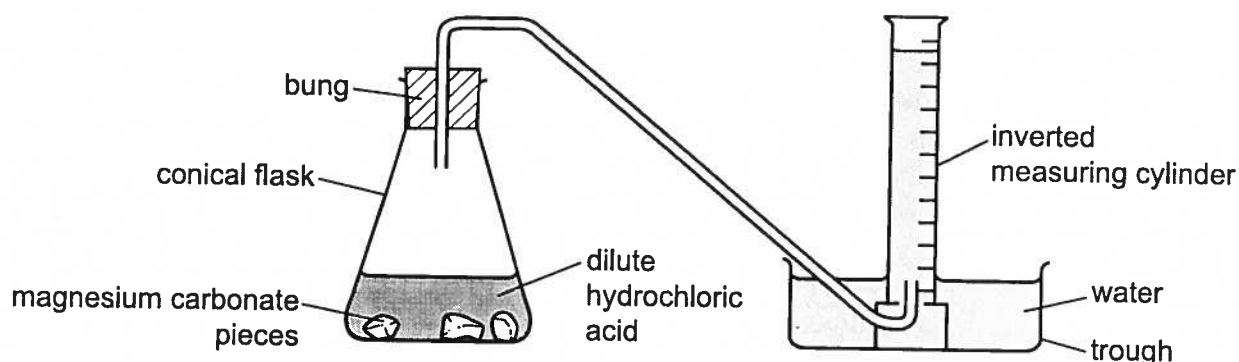
..... [2]

16. O/N 16/P41/Q8

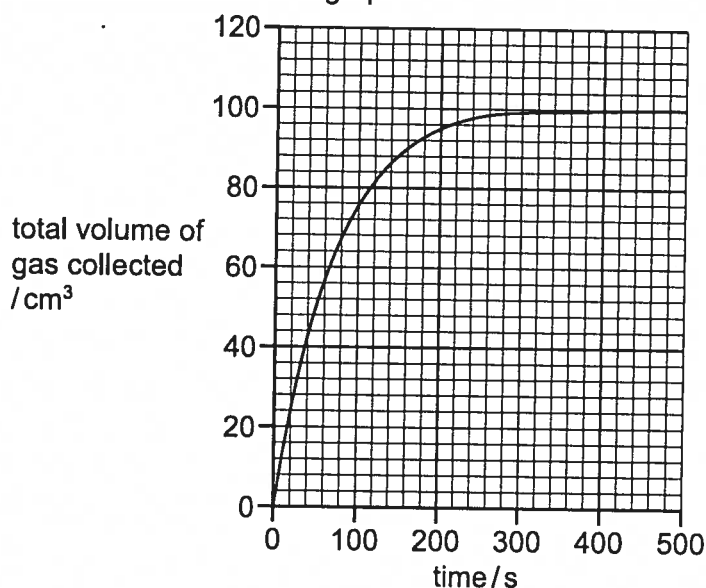
Magnesium carbonate reacts with dilute hydrochloric acid.



An excess of magnesium carbonate pieces was added to dilute hydrochloric acid. The apparatus in the diagram was used to measure the volume of gas produced. The total volume of gas collected was recorded every 20 seconds.



(a) The results obtained are shown on the graph.



- (i) Describe how the rate of this reaction changed during the reaction. Explain why the rate changed in this way.

.....

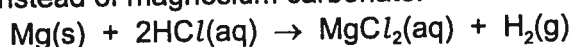
 [4]

- (ii) The experiment was repeated using the same mass of **powdered** magnesium carbonate with the same volume and concentration of dilute hydrochloric acid. Explain how the initial rate of reaction and total volume of gas collected would compare to the first experiment.

initial rate of reaction

 total volume of gas
 [4]

- (b) A piece of magnesium ribbon was cleaned. The experiment was repeated using this clean magnesium ribbon instead of magnesium carbonate.



This reaction is exothermic.

The rate of the reaction gradually increased over the first 2 minutes.

Explain why the rate of the reaction increased.

.....

.....

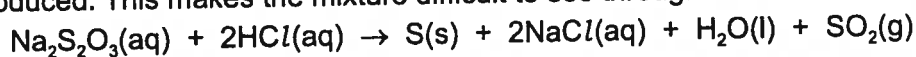
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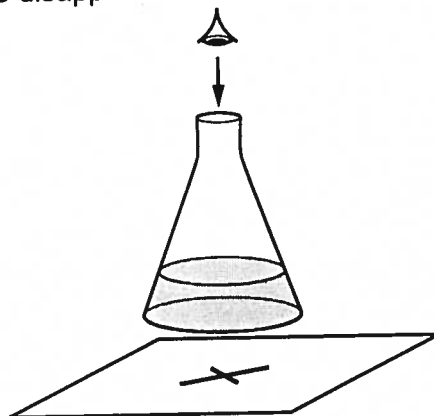
..... [5]

17. M/J 16/P41/Q3

When aqueous sodium thiosulfate and dilute hydrochloric acid are mixed, a precipitate of insoluble sulfur is produced. This makes the mixture difficult to see through.



The time taken for the cross to disappear from view is measured.



A student adds the following volumes of aqueous sodium thiosulfate, dilute hydrochloric acid and distilled water to the conical flask.

The time taken for the formation of the precipitate of sulfur to make the cross disappear from view is recorded.

experiment number	volume of sodium thiosulfate /cm ³	volume of hydrochloric acid /cm ³	volume of distilled water /cm ³	time taken for cross to disappear from view/s
1	10	10	40	56
2	20	10	30	28
3				

- (a) State the order in which the aqueous sodium thiosulfate, hydrochloric acid and distilled water should be added to the flask.

..... [1]

(b) In experiment 3 the student wanted the sodium thiosulfate to be double the concentration used in experiment 2.

(i) Complete the table to show the **volumes** which should be used and the **expected** time taken for the cross to disappear from view in experiment 3. [2]

(ii) Use collision theory to explain why increasing the concentration of sodium thiosulfate would change the rate of reaction.

.....

..... [2]

(c) The student repeated experiment 1 at a higher temperature.

Use collision theory to explain why the rate of reaction would increase.

.....

.....

..... [3]

Answers Section

1. O/N 21/P43/Q6

- (a) **M1** (a substance that) speeds up a reaction / increases the rate of a reaction (1) 2
M2 unchanged chemically at the end (1) 1
- (b) (i) oxygen gas escapes 1
(ii) **M1** rate decreases
OR
rate / reaction is fastest at the start (1)
M2 reaction stops or
rate = zero (1) 2
- (c) (i) **M1** (particles) have more kinetic energy / particles move faster(1)
M2 more collisions per unit time (1)
M3 more particles have energy greater than or equal to activation energy
OR
more particles have sufficient energy to react
OR
A greater percentage or greater proportion or greater fraction of collisions have sufficient greater energy to react **OR**
A greater percentage or greater proportion or greater fraction of collisions have energy greater than or equal to activation energy
OR
A greater percentage or greater proportion or greater fraction of collisions are successful 3
- (ii) **M1** steeper gradient from start (1)
M2 levels off at the same mass (1) 2

2. O/N 20/P43/Q3/c(i,iii)

- (a) (i) substance that speeds up a reaction / increases rate (1)
unchanged (chemically) at the end
OR not used up 2
OR lowers activation energy (1)

(ii)

effect on the rate of the forward reaction	effect on the equilibrium yield of NO(g)
increase (1)	decrease (1)
increase (1)	decrease (1)

4

3. M/J 20/P42/Q3/a(i,ii)

- (a) (i) heat in air 1
(ii) $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$ (1)
450°C (1)
vanadium(V) oxide (1) 3

4. M/J 20/P42/Q4/a,b,c,d

- (a) substance that speeds up a reaction / increases rate (1)
unchanged (chemically) at the end **OR** not used up **OR** lowers activation energy **OR** provides alternative pathway (1) 2
- (b) rate decreases (1)

- particles further apart / less particles per unit volume (1)
fewer collisions per unit time / lower collision frequency (1)
reaction stops because all hydrogen peroxide is used up 4
- (c) steeper gradient (1)
reaches same volume of oxygen (1) 2
- (d) particles gain kinetic energy / particles move faster (1)
greater number of collisions with activation energy (or more) / greater number of particles with activation energy (or more) / greater number of particles with energy required for reaction (1)
more collisions are successful / more collisions are fruitful / more collisions lead to reaction (1) 3
5. **M/J 20/P43/Q2/c**
(a) (particles) have more energy OR (particles) move faster
more collisions per second OR greater collision rate
more (of the) particles OR collisions have sufficient energy / activation energy to react OR
a greater percentage / proportion / fraction of collisions are successful 3
6. **MAR 20/P42/Q3/d(ii)**
(a) (i) *increased temperature:*
(equilibrium) shifts to LHS (1)
(forward) reaction is exothermic (1)
increased pressure:
(equilibrium) shifts to RHS (1)
fewer moles (of gas) on RHS (1) 4
7. **MAR 20/P42/Q4/a**
(a) Haber (process) (1)
ammonia (1) 2
8. **O/N 19/P41/Q7/c(i),d(ii)**
(a) (i) speeds up a (chemical) reaction (1)
not used up or unchanged (at end) (1) 2
(b) (i) (reaction is faster) because more collisions per second (1)
particles / molecules closer together or more particles / molecules per unit volume (1) 2
9. **M/J 19/P43/Q5/c,d**
(a) **M1** (a substance which) increases the rate of a reaction
M2 without being used up (at the end) OR unchanged (chemically) at the end OR without changing mass 2
(b) **M1** particles / molecules in explanation
M2 (particles) move faster / more energy
M3 more collisions per second or greater collision rate
M4 more of the (colliding) molecules / particles have sufficient energy (activation energy) to react / more of the collisions have sufficient energy (activation energy) to react 4
10. **MAR 19/P42/Q4/a**
(a)
- | | |
|------------------------|------------------------|
| M1 increases(1) | No change |
| M2 increases(1) | M3 decreases(1) |
| decreases | M4 decreases(1) |
- 4
11. **O/N 18/P42/Q4**
(a) (i) Gradient gets less 1
(ii) Concentration of HCl is decreasing 1
(iii) 120 seconds 1

- (b) M1 New line steeper than printed line and starts at origin
M2 New line reaches same final volume as printed line 2
- (c) M1 Time taken is less
M2 (particles) have more energy
M3 (particles) move faster
M4 More collisions (of particles) occur per second / per unit time
M5 More (of the) particles / collisions have energy greater than activation energy
or
More (of the) particles / collisions have sufficient energy to react
or
A greater percentage / proportion / fraction of collisions (of particles) are successful 5

12. M/J 18/P42/Q3

particles	number of protons	number of electrons	number of neutrons	number of nucleons
			12 (1)	
	17 (1)	18 (1)		37 (1)
Fe (1) 2+ (1)				

6

13. M/J 18/P43/Q3/d(ii),(iii)

- (a) (i) any one from:
increase surface area (of cobalt)
powder the metal
add a catalyst 1
- (ii) (particles) have more energy / (particles) move faster 1
more collisions per second / greater collision rate 1
more of the colliding molecules have sufficient energy (activation energy) to react 1

14. O/N 17/P41/Q7/a

- (a) (i) more particles (of acid) in a given volume / dm^3 / cm^3 1
more collisions per second / unit time **OR** greater collision rate 1
- (ii) particles have more energy / particles move faster / more collisions per second /
more collisions per unit time / greater collision rate 1
more (of the) particles / collisions have energy greater than the activation energy /
more particles have sufficient energy to react / more collisions have sufficient energy
to react / a greater percentage of collisions are successful 1

15. O/N 17/P43/Q5

- (a) (i) oxygen / O_2 1
sodium nitrite / sodium nitrate(III) / NaNO_2 1
- (ii) $2\text{Cu}(\text{NO}_3)_2 \rightarrow 2\text{CuO} + \text{O}_2 + 4\text{NO}_2$
M1 CuO 2
M2 rest of equation fully correct
- (b) (i) reversible reaction in which the **rate** of the forward reaction **equals** the **rate**
of the backward reaction 1
concentration of all reactants and products becomes constant / does not change 1
- (ii) forward reaction is endothermic 1
(increased temperature) causes equilibrium to shift to the right /
to shift in the endothermic direction / to form more nitrogen dioxide /
to form more product(s) 1
- (iii) less brown / lighter / paler / colour fades 1
more molecules / moles / volume on the right **ORA** 1

ORequilibrium shifts in the direction of **fewer** molecules / moles / lower volume

1

16. O/N 16/P41/Q8**(a) (i)** any 4 from:

slowed down

acid became less concentrated **OR** fewer particles per unit volumefewer collisions per second **OR** lower collision rate

(then the reaction) stopped

all the hydrochloric acid reacted

4

(ii) any 4 from:

faster (reaction)

(powder has) larger surface area

more collisions per second **OR** higher collision rate

same volume of gas

amount / moles hydrochloric acid is not changed

4

(b) any 5 from:

temperature increased

particles have more energy

(particles) move faster

more collisions per second **OR** higher collision rate

more particles have sufficient energy to react / activation energy

more of the collisions are successful

5

17. M/J 16/P41/Q3**(a)** 1 Na₂S₂O₃ 2 H₂O 3 HCl **OR**1 HCl 2 H₂O 3 Na₂S₂O₃ **OR**1 H₂O 2 Na₂S₂O₃ 3 HCl **OR**1 H₂O 2 HCl 3 Na₂S₂O₃ ;

1

(b) (i) **M1** volumes 40 : 10 : 10;**M2** time = 14;

2

(ii) **M1** more particles per unit volume / particles are closer together;**M2** increases the rate of collisions / there are more collisions per unit time;

2

(c) **M1** particles gain more energy **and** move faster;**M2** increasing rate of collisions / more collisions per unit time;**M3** higher proportion of particles have sufficient energy to react /

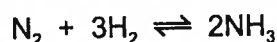
collisions have sufficient energy to react / are above the activation energy;

3

7.2: Reversible reactions

1. O/N 22/P41/Q3

Ammonia is made in an industrial process starting with nitrogen. The equation for the reaction is shown.



- (a) Name the industrial process used to make ammonia.

..... [1]

- (b) State the raw material from which nitrogen is obtained.

..... [1]

- (c) State what is meant by the symbol $\rightleftharpoons$.

..... [1]

- (d) State the temperature and pressure used in this industrial process.

temperature = °C

pressure = atm [2]

- (e) Name the catalyst used in this industrial process.

..... [1]

- (f) The forward reaction is exothermic.

State the effect, if any, on the position of the equilibrium when the following changes are made.
Explain your answers.

temperature is reduced

.....

.....

pressure is reduced

.....

..... [4]

- (g) Explain, in terms of particles, what happens to the rate of reaction when the temperature is reduced.

.....

.....

..... [3]

(h) Give the formula of the compound formed when sulfuric acid reacts with ammonia.

..... [1]

2. O/N 22/P43/Q4

This question is about compounds of phosphorus.

(a) Gaseous phosphorus(V) chloride decomposes into gaseous phosphorus(III) chloride and gaseous chlorine.

When the three gases are present in a closed container the system reaches equilibrium.



(i) Complete the table using only the words *increases*, *decreases* or *no change*.

	effect on the rate of the forward reaction	effect on the equilibrium yield ($\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$)
increasing the temperature		increases
decreasing the pressure		
adding a catalyst		no change

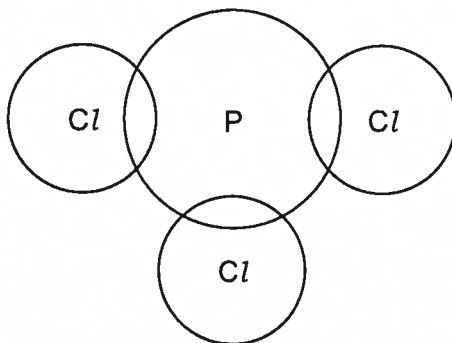
[4]

(ii) The table shows that when the temperature increases, the equilibrium yields of $\text{PCl}_3(\text{g})$ and $\text{Cl}_2(\text{g})$ increase.

State what conclusion can be made from this.

..... [1]

(b) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of phosphorus(III) chloride, PCl_3 . Show outer shell electrons only.



[2]

(c) Phosphorus oxychloride has the formula POCl_3 .

Phosphorus oxychloride is the only product of the reaction between phosphorus(V) chloride, PCl_5 , and phosphorus(V) oxide, P_4O_{10} .

Write a chemical equation for the reaction between phosphorus(V) chloride and phosphorus(V) oxide.

..... [2]

- (d) Compound X has the following composition by mass.

H, 3.66%; P, 37.80%; O, 58.54%

Calculate the empirical formula of compound X.

empirical formula = [2]

- (e) Compound Y has the empirical formula H_3PO_4 and a relative molecular mass of 98.

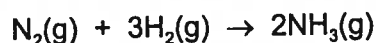
Deduce the molecular formula of compound Y.

molecular formula = [1]

3. O/N 20/P41/Q3/e,f

- (a) Ammonia is produced in the Haber process.

The equation for the reaction is shown.



- (i) In the Haber process, a temperature of 450°C and a pressure of 200 atmospheres are used in the presence of finely-divided iron.

A larger equilibrium yield of ammonia would be produced if a lower temperature and a higher pressure are used.

Explain why a lower temperature and a higher pressure are **not** used.

lower temperature [2]

higher pressure [2]

- (ii) State the role of iron in the Haber process.

..... [1]

- (b) Ammonia is a weak base.

- (i) Explain the meaning of the term *base*.

..... [1]

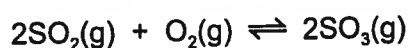
- (ii) Suggest the pH of aqueous ammonia.

..... [1]

4. O/N 20/P41/Q5/b(ii)

- (a) Sulfuric acid is manufactured using the Contact process. This manufacture involves four stages.

- (i) The equation for **stage 2** is shown.



The reaction can reach equilibrium.

Explain what is meant by the term *equilibrium*.

.....
..... [2]

5. O/N 20/P41/Q6/c(iii)

(a) Propane reacts with chlorine.

(i) State the name of this type of reaction.

..... [1]

6. O/N 20/P43/Q3/b

(a) Nitrogen can be converted into ammonia by the Haber process.

(i) Describe how nitrogen is obtained for the Haber process.

.....
..... [2]

(ii) Give the essential reaction conditions and write a chemical equation for the reaction occurring in the Haber process.

chemical equation:

reaction conditions:

.....
.....
..... [5]

7. M/J 20/P42/Q3/a(iii)

(a) Sulfuric acid is made from sulfur in a four-stage process.

stage 1 Sulfur is converted into sulfur dioxide.

stage 2 Sulfur dioxide is converted into sulfur trioxide.

stage 3 Sulfur trioxide is converted into oleum.

stage 4 Oleum is converted into sulfuric acid.

(i) The reaction in **stage 2** can reach equilibrium.

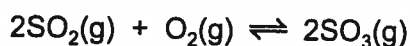
What is meant by the term *equilibrium*?

.....
..... [2]

8. M/J 20/P41/Q3/b(i,ii,iii,iv,v)

(a) The next stage of the process is a reaction which can reach equilibrium.

The equation for this stage is shown.



- (i) Describe
- two**
- features of an equilibrium.

.....
 [2]

- (ii) Name the catalyst used in this stage.

..... [1]

- (iii) Why is a catalyst used?

..... [1]

- (iv) Explain, in terms of particles, why a high temperature increases the rate of this reaction.

.....

 [3]

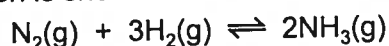
- (v) In this stage, only a moderate temperature of 450 °C is used.
-
- What does this suggest about the forward reaction?

..... [1]

9. M/J 20/P43/Q2/a,b

Ammonia is manufactured by the Haber process.

- (a) The equation for the reaction is shown.



- (i) State what is meant by the symbol
- $\rightleftharpoons$
- .

..... [1]

- (ii) State
- one**
- source of hydrogen used in the manufacture of ammonia.

..... [1]

- (b) The table shows some data for the production of ammonia.

pressure / atm	temperature / °C	percentage yield of ammonia
250	350	58
100	450	28
400	450	42
250	550	20

Deduce the effect on the percentage yield of ammonia of:

- increasing the pressure of the reaction

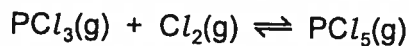
.....

- increasing the temperature of the reaction.

..... [2]

10. MAR 20/P42/Q3/d(i,iii)

- (a) PCl_3 reacts with chlorine, Cl_2 , to form PCl_5 . This reaction is exothermic and reaches an equilibrium.



- (i) Describe **two** features of an equilibrium.

.....
..... [2]

- (ii) Explain, in terms of particles, what happens to the rate of the forward reaction when the reaction mixture is heated.

.....
.....
..... [3]

11. O/N 19/P42/Q3/a

Ammonia is an important chemical.

- (a) Ammonia is manufactured by the Haber process. The reaction is reversible.

- (i) What is the sign for a reversible reaction?

..... [1]

- (ii) State the essential conditions for the manufacture of ammonia by the Haber process starting from hydrogen and nitrogen. Include a chemical equation to show the reaction which occurs.

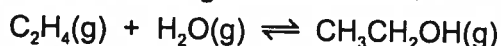
.....
.....
.....
.....
..... [5]

- (iii) Name **one** raw material which is a source of the hydrogen used in the Haber process.

..... [1]

12. O/N 19/P41/Q7/d(i,iii)

- (a) Ethene gas reacts with steam to form gaseous ethanol.



The reaction can reach a position of equilibrium. The forward reaction is exothermic.

- (i) State and explain the effect of increasing the pressure on the **position of equilibrium**. All other conditions are unchanged.

.....
..... [2]

- (ii) State and explain the effect of increasing the temperature on the **position of equilibrium**. All other conditions are unchanged.

.....
..... [2]

13. M/J 19/P43/Q5/e

- (a) The reaction between ethanoic acid and ethanol reaches equilibrium.

- (i) The reaction between ethanoic acid and ethanol is exothermic.

State and explain the effect, if any, of increasing the temperature on the amount of ester at equilibrium.

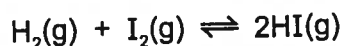
.....
..... [2]

- (ii) State and explain the effect, if any, of removing water from the mixture on the amount of ester at equilibrium.

.....
..... [2]

14. O/N 18/P41/Q5

Hydrogen gas reacts with iodine gas. The equation is shown.

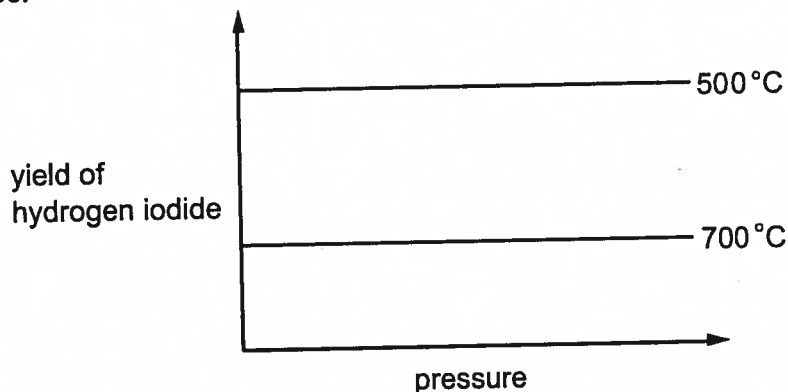


The reaction is reversible and can reach equilibrium.

- (a) What is meant by the term *equilibrium*?

.....
..... [2]

- (b) The graphs show how pressure affects the yield of hydrogen iodide, HI, at two different temperatures.



- (i) Explain why the yield at 500 °C does **not** change as the pressure is increased.

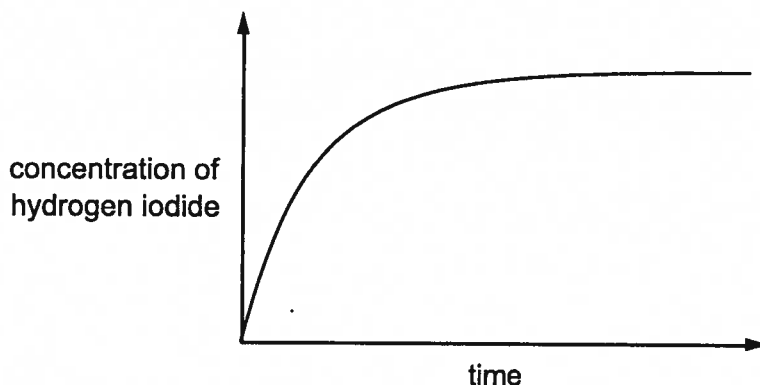
..... [1]

- (ii) What can you conclude from the difference in the yield of hydrogen iodide at the two temperatures shown? Explain your answer.

.....

..... [2]

- (c) The graph shows how the concentration of hydrogen iodide, HI, changes after hydrogen gas and iodine gas are mixed together in a sealed container.



- (i) When is the rate of reaction fastest?

..... [1]

- (ii) The reaction was repeated at the same temperature and pressure but in the presence of a catalyst.

Draw a graph on the same axes to show how the concentration of hydrogen iodide changes with time in the presence of a catalyst. [2]

- (d) A mixture of hydrogen gas and iodine gas is allowed to reach equilibrium.

- (i) Increasing the pressure of a gas increases its concentration.

State and explain the effect of increasing the pressure on the **rate** of the forward reaction.

.....

..... [2]

- (ii) State and explain the effect of increasing the temperature on the **rate** of the reverse reaction.

.....

.....

..... [3]

15. M/J 18/P42/Q3

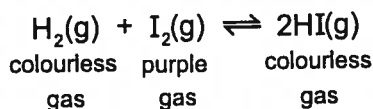
Complete the following table.

particle	number of protons	number of electrons	number of neutrons	number of nucleons
$^{23}_{11}\text{Na}$	11	11		23
$^{37}_{17}\text{Cl}^-$			20	
$^{56}_{26}\text{.....}$	26	24	30	56

[6]

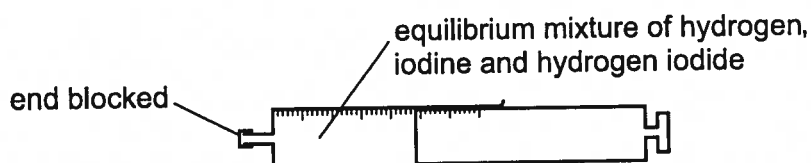
16. M/J 18/P42/Q5

Hydrogen and iodine react together in a reversible reaction. Hydrogen iodide is formed.



The forward reaction is exothermic.

A gas syringe containing an equilibrium mixture of hydrogen, iodine and hydrogen iodide gases was sealed and heated to 250 °C. The equilibrium mixture was a pale purple colour.

(a) What is meant by the term *equilibrium*?

.....

..... [2]

(b) The plunger of the gas syringe was pressed in while the end of the gas syringe was blocked. This increased the pressure. The position of the equilibrium did **not** change. The colour of the gaseous mixture turned darker purple.

(i) Give a reason why the position of the equilibrium did **not** change.

..... [1]

(ii) Suggest why the gaseous mixture turned darker purple, even though the position of the equilibrium did **not** change.

..... [1]

(c) The temperature of the gas syringe was increased to 300 °C.

(i) What happened to the **position** of the equilibrium when the temperature of the gas syringe was increased from 250 °C to 300 °C?

..... [1]

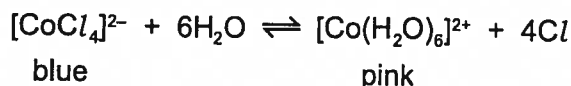
(ii) What happened to the **rate** of the forward reaction and the **rate** of the backward reaction when the temperature of the gas syringe was increased from 250 °C to 300 °C?

rate of the forward reaction

rate of the backward reaction [2]

17. M/J 18/P43/Q3/e

- (a) When cobalt(II) chloride is added to water an equilibrium is established.



- (i) A student adds water to a blue solution containing $[\text{CoCl}_4]^{2-}$ ions.
Describe what the student observes. Give a reason for your answer in terms of the position of the equilibrium.

..... [2]

- (ii) Another student cools a blue solution containing $[\text{CoCl}_4]^{2-}$. The blue solution turns pink.
What does this information indicate about the forward reaction?

..... [1]

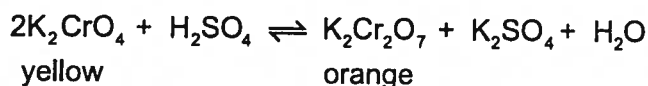
18. O/N 17/P42/Q5

Some chemical reactions are reversible.

- (a) Aqueous potassium chromate(VI), K_2CrO_4 , is a yellow solution.

Aqueous potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7$, is an orange solution.

The two compounds interconvert when the pH of the solution changes.



Solution Y is a mixture of aqueous potassium chromate(VI) and aqueous potassium dichromate(VI) at equilibrium.

- Explain, in terms of the position of the equilibrium, what you would see if sulfuric acid were added to solution Y.

.....

.....

.....

- Explain, in terms of the position of the equilibrium, what you would see if sodium hydroxide were added to solution Y.

.....

.....

..... [5]

- (b) Hydrogen can be manufactured using a reversible reaction between methane and steam.



At 900°C, in the presence of a nickel catalyst, the yield of hydrogen is 70%.

- (i) What volume of hydrogen is produced from 100 cm³ of methane under these conditions?

..... cm³ [2]

Under different conditions, different yields of hydrogen are obtained.

- (ii) If the pressure is increased, the yield of hydrogen becomes less than 70%.

Explain why, in terms of the position of the equilibrium.

..... [1]

- (iii) If the temperature is decreased, the yield of hydrogen decreases.

What does this information indicate about the reaction between methane and steam?

..... [1]

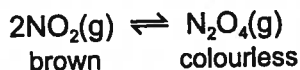
- (iv) Why is a catalyst used in this reaction?

..... [1]

19. O/N 17/P41/Q5/c

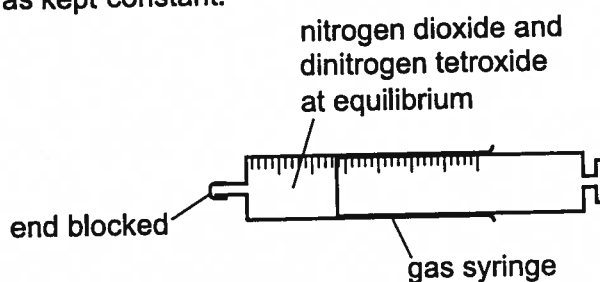
- (a) Nitrogen dioxide, NO₂, exists in equilibrium with dinitrogen tetroxide, N₂O₄.

Nitrogen dioxide is brown and dinitrogen tetroxide is colourless.



- (i) A sample of nitrogen dioxide and dinitrogen tetroxide at equilibrium was placed in a closed gas syringe.

The syringe plunger was pushed in. This increased the pressure in the gas syringe. The temperature was kept constant.



State how the colour of the gas in the syringe changed. Explain your answer in terms of the position of the equilibrium.

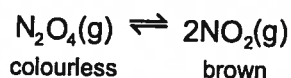
..... [3]

- (ii) A sealed tube containing nitrogen dioxide and dinitrogen tetroxide at equilibrium was cooled in an ice bath at constant pressure. The contents of the tube became paler. Suggest an explanation for this observation in terms of the position of the equilibrium.

.....
 [2]

20. O/N 17/P43/Q5/b

- (a) The chemical equation shows the equilibrium between dinitrogen tetroxide (N_2O_4 , a colourless gas) and nitrogen dioxide (NO_2 , a brown gas).



A mixture of dinitrogen tetroxide and nitrogen dioxide is allowed to reach equilibrium in a closed gas syringe.

- (i) In chemistry, what is meant by the term *equilibrium*?

.....
 [2]

- (ii) If the equilibrium mixture is heated at constant pressure, a darker brown colour is seen inside the gas syringe.

What does this information indicate about the decomposition of dinitrogen tetroxide? Explain your answer in terms of the position of the equilibrium.

.....
 [2]

- (iii) Suggest what you would see if the pressure on the equilibrium mixture were increased at constant temperature.

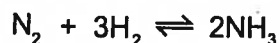
Explain your answer in terms of the position of the equilibrium.

.....
 [2]

21. M/J 17/P42/Q3

This question is about nitrogen and some of its compounds.

- (a) Nitrogen in the air can be converted into ammonia by the Haber process. The chemical equation for the reaction is shown.



- (i) State the temperature and pressure used in the Haber process.

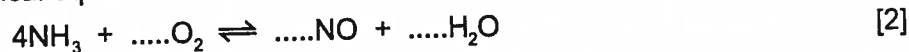
temperature
 pressure [2]

- (ii) Name the catalyst used in the Haber process.

..... [1]

- (b) The ammonia produced in the Haber process can be oxidised to nitrogen(II) oxide at 900 °C. The reaction is exothermic.

(i) Balance the chemical equation for this reaction.



(ii) Suggest a reason, other than cost, why a temperature greater than 900 °C is **not** used.

..... [1]

(iii) Suggest a reason why a temperature less than 900 °C is **not** used.

..... [1]

- (c) Nitrogen(II) oxide can be reacted with oxygen and water to produce nitric acid as the only product.

Write a chemical equation for this reaction.

..... [2]

- (d) Describe how you would prepare a pure dry sample of copper(II) nitrate crystals in the laboratory using dilute nitric acid and solid copper(II) carbonate.

Include a series of key steps in your answer.

You should include a chemical equation for the reaction.

.....
.....
.....
.....
.....
.....
..... [6]

22. O/N 16/P41/Q4

- (a) Ammonia, NH_3 , is made by reacting nitrogen with hydrogen in the Haber process.

(i) Write a chemical equation for the formation of ammonia in the Haber process.

..... [2]

(ii) Name the raw materials from which nitrogen and hydrogen are obtained.

nitrogen

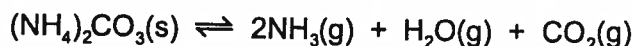
hydrogen [2]

(iii) State the temperature and pressure used in the Haber process. Include the units.

temperature

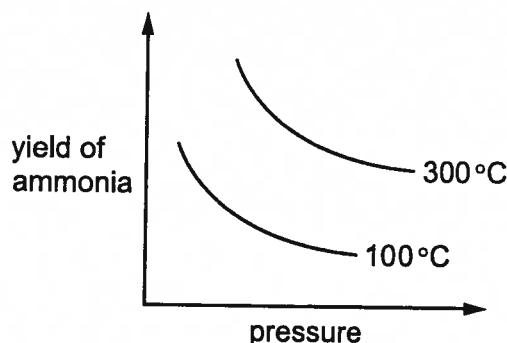
pressure [2]

- (b) Ammonia is also made when ammonium carbonate decomposes.



The reaction is reversible and can reach a position of equilibrium.

The graph shows how the yield of ammonia at equilibrium changes with temperature and pressure.



- (i) What is meant by the term *equilibrium* for a reversible reaction?

.....
..... [2]

- (ii) Using information from the graph, explain whether the reaction is endothermic or exothermic.

..... [1]

- (iii) State and explain the effect of increasing the pressure on the yield of ammonia in this reaction.

.....
.....
..... [3]

23. M/J 16/P42/Q4/a,b,c,d

Hydrogen can be manufactured from methane by steam reforming.



The reaction is carried out using a nickel catalyst at temperatures between 700 °C and 1100 °C and using a pressure of one atmosphere. The forward reaction is endothermic.

- (a) What is meant by the term *catalyst*?

.....
..... [2]

- (b) Suggest **two** reasons why a temperature lower than 700 °C is not used.

.....
..... [2]

(c) Suggest **one** advantage of using a pressure greater than one atmosphere.

..... [1]

(d) Suggest **one** disadvantage of using a pressure greater than one atmosphere.

..... [1]

Answers Section

1. O/N 22/P41/Q3

- (a) Haber (process) 1
- (b) air 1
- (c) reversible 1
- (d) 450 (1) 2
200 (1)
- (e) Iron / Fe 1
- (f) reduced temperature:
M1 (position of) equilibrium moves to right-hand side (1)
M2 reaction is exothermic (1)
reduced pressure:
M3 (position of) equilibrium moves to left-hand side (1)
M4 more (gaseous) moles on left hand side (1) 4
- (g) M1 rate decreases **and** particles have less energy (1)
M2 less collisions (between particles) occur per second / per unit time (1)
M3 less of the particles/collisions have energy equal to or above the activation energy (1)
or
less of the particles / collisions have sufficient energy to react
or
a lower percentage / proportion / fraction of collisions (of particles)
• are successful or
• have energy equal to or above activation energy 3
- (h) $(\text{NH}_4)_2\text{SO}_4$ 1

2. O/N 22/P43/Q4

- (a) (i)
- | | |
|------------------|------------------|
| M1 increases (1) | |
| M2 decreases (1) | M3 increases (1) |
| M4 increases (1) | |
- 4
- (ii) forward reaction is endothermic 1
- (b) M1 3 dot and cross bonding pairs (1)
M2 All lone pairs correct.
All octets should be complete AND dots and crosses should be different on P and all Cl (1) 2
- (c) $\text{P}_4\text{O}_{10} + 6\text{PCl}_5 \rightarrow 10\text{POCl}_3$
M1 all formula correct and on the correct sides of the arrow (1)
M2 equation fully correct (1) 2
- (d) M1 H 3.66 / 1
P 37.80 / 31
O 58.54 / 16
OR H = 3.66 P = 1.22 O = 3.66
OR H:P:O 3:1:3 (1)
M2 H_3PO_3 (1) 2

- (e) H_3PO_4 (1) 1
3. O/N 20/P41/Q3/e,f
- (a) (i) lower temperature: (rate of reaction) slower (1) 2
higher pressure: expensive/specialist equipment 1
(ii) catalyst 1
- (b) (i) proton acceptor 1
(ii) any value greater than 7 up to 12 1
4. O/N 20/P41/Q5/b(ii)
- (a) rate of forward reaction is equal to rate of reverse reaction (1) 2
constant concentration (of reactants and products) (1)
5. O/N 20/P41/Q6/c(iii)
- (a) substitution 1
6. O/N 20/P43/Q3/b
- (a) (i) METHOD 1
liquid air (1)
fractional distillation (1)
METHOD 2
hydrogen
burns in air (to remove the oxygen and then scrub out the carbon dioxide) 2
(ii) (pressure) 200 atmospheres (1)
(temperature) 450°C (1)
iron catalyst (1)
 $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (1)
equilibrium / reversible (1) 5
7. M/J 20/P42/Q3/a(iii)
- (a) rate of forward reaction and rate of backward reaction are equal (1) 2
concentrations of reactants and products are constant (1)
8. M/J 20/P41/Q3/b(i,ii,iii,iv,v)
- (a) (i) reaction is reversible 2
rate of forward reaction = rate of reverse reaction 1
(ii) vanadium(V) oxide 1
(iii) increases the rate of reaction 1
(iv) particles have more energy (E) 1
rate of collisions increase 1
a higher proportion of particles have energy greater than activation energy ($E > E_A$) 1
(v) exothermic 1
9. M/J 20/P43/Q2/a,b
- (a) (i) reversible reaction 1
(ii) hydrocarbons (reacting with steam) 1
(b) [increasing pressure] increases yield 2
[increasing temperature] decreases yield
10. MAR 20/P42/Q3/d(i,iii)
- (a) (i) constant concentrations (1) 2
rate of forward reaction = rate of reverse reaction (1)
(ii) rate increases and particles have more energy (1)

more collisions (between particles) occur per second / per unit time
 more (of the) particles / collisions have energy greater than activation energy
 or
 more (of the) particles / collisions have sufficient energy to react
 or
 a greater percentage / proportion / fraction of collisions (of particles) are successful 3

11. O/N 19/P42/Q3/a

- (a) (i) $\rightleftharpoons$ 1
- (ii) pressure 100–300 atmospheres/atm (1)
 temperature in range 330 to 500 °C (1)
 iron (catalyst) (1)
 species: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (1)
 fully correctly equation (1) 5
- (iii) water / steam or methane / natural gas 1

12. O/N 19/P41/Q7/d(i,iii)

- (a) (i) moves right (1)
 fewer moles / molecules (of gas) on right (1) 2
- (ii) moves left (1)
 (forward) reaction is exothermic or backward reaction is endothermic (1) 2

13. M/J 19/P43/Q5/e

- (a) (i) M1 less ester
 M2 equilibrium moves left and because forward reaction is exothermic 2
- (ii) M1 more ester
 M2 (equilibrium moves right) to replace water 2

14. O/N 18/P41/Q5

- (a) M1 forward and back reactions occur at equal rates 1
 M2 concentration (of substances) remains constant 1
- (b) (i) equal / same number of moles on each side
 or
 amount / molecules (of gas) on each side is the same 1
- (ii) M1 (forward) reaction exothermic or reverse reaction endothermic
 M2 yield lower at higher temperature or (position of) equilibrium moves left at
 higher temperature ORA 2
- (c) (i) at the start / beginning 1
- (ii) M1 new line is steeper than printed line and starts at origin 1
 M2 new line reaches same final volume as printed line 1
- (d) (i) M1 Faster
 and
 More particles per unit volume / $\text{dm}^3 / \text{cm}^3$
 M2 More collisions per second / unit time or greater collision rate 2
- (ii) Reaction faster and (particles) have more energy or (particles) move faster 1
 more collisions per second or greater collision rate 1
 more (of the) particles / collisions have energy greater than the activation energy
 or more particles / collisions have sufficient energy to react or a greater percentage /
 proportion / fraction of collisions are successful 1

15. M/J 18/P42/Q3

particle	number of protons	number of electrons	number of neutrons	number of nucleons
			12(1)	
	17(1)	18(1)		37(1)
Fe(1) 2+ (1)				

6

16. M/J 18/P42/Q5

- (a) the rate of forward reaction equals (the rate of the) reverse reaction
concentrations of reactants and products are constant
- (b) (i) same number of gas moles on both sides of the equilibrium /
same number of gas molecules on both sides of the equilibrium
(ii) (increased pressure) particles or molecules (forced) closer together /
same number of particles or molecules in a smaller volume
- (c) (i) to left / towards reactants / in reverse direction
(ii) increase / faster
increase / faster

1
1
1
1
1
1
1

17. M/J 18/P43/Q3/e

- (a) (i) becomes pink / becomes purple
equilibrium moves right
(ii) (forward reaction is) exothermic

1
1
1

18. O/N 17/P42/Q5

- (a) both colours referred to correctly as observations in both parts of the answer
(if sulfuric acid is added to solution Y,) equilibrium moves to the right-hand side
because the concentration of acid has increased
(if sodium hydroxide is added to solution Y,) equilibrium moves to the left-hand side
because sodium hydroxide reacts with / neutralises sulfuric acid
- (b) (i) 210 cm³
M1 expected volume of hydrogen = 300 cm³
M2 70% of M1
(ii) fewer moles / molecules / particles (of gas) on the left-hand side
(iii) endothermic
(iv) increases rate (of reaction)

1
1
1
1
1
2
1
1
1

19. O/N 17/P41/Q5/c

- (a) (i) becomes paler
equilibrium moves right
(because) fewer moles (of gas) on right
(ii) equilibrium moved right / more N₂O₄ / less NO₂
(forward) reaction exothermic

1
1
1
1
1

20. O/N 17/P43/Q5/b

- (a) (i) reversible reaction in which the rate of the forward reaction equals the
rate of the backward reaction
concentration of all reactants and products becomes constant / does not change
(ii) forward reaction is endothermic
(increased temperature) causes equilibrium to shift to the right /

1
1
1
1

- to shift in the endothermic direction / to form more
nitrogen dioxide / to form more product(s) 1
- (iii) less brown / lighter / paler / colour fades 1
- more molecules / moles / volume on the right **ORA OR**
- equilibrium shifts in the direction of fewer molecules / moles / lower volume 1

21. M/J 17/P42/Q3

- (a) (i) 450 °C 1
- 200 atmospheres 1
- (ii) iron 1
- (b) (i) 4(NO) 1
- 5(O₂) **AND** 6(H₂O) 1
- (ii) lower yield of NO / lower yield of nitric acid / lower yield of product /
equilibrium shifts to left (at higher temperatures) /
backward reaction favoured (at higher temperatures) **ORA** 1
- (iii) too slow / rate decreases **ORA** 1
- (c) 4NO + 3O₂ + 2H₂O → 4HNO₃
- M1** all formulae correct
- M2** balancing 2
- (d) add copper(II) carbonate (to acid) until it stops dissolving or no more effervescence /
bubbling / fizzing
- filter (to remove copper(II) carbonate)
- evaporate / heat / warm / boil / leave in sun
- AND**
- until most of the water has gone / some water is left / evaporate **some** of the water / until it is
concentrated / saturation (point) / crystallisation point / crystals form on glass rod or
microscope slide / crystals **start** to form
- (for any solution) leave / allow to cool / allow to crystallise
- OR**
- (for any crystals) filter / wash / dry with filter paper / dry in warm place / dry in a (**low**) oven /
leave to dry formula of Cu(NO₃)₂
- equation: CuCO₃ + 2HNO₃ → Cu(NO₃)₂ + CO₂ + H₂O 6

22. O/N 16/P41/Q4

- (a) (i) N₂ + 3H₂ ⇌ 2NH₃
- M1** formulae
- M2** balancing 2
- (ii) (nitrogen) air / atmosphere 1
- (hydrogen) steam / water / hydrocarbons / natural gas 1
- (iii) (temperature) answer in range 370–470 °C 1
- (pressure) answer in range 150–300 atm 1
- (b) (i) **M1** forward and reverse reactions (occur) 1
- M2** amounts / moles / concentrations (of reagents and products) constant
- OR**
- M2** rate of forward and reverse reactions equal 1
- (ii) endothermic **AND** yield increases as temperature increases 1
- (iii) **M1** yield decreases (as pressure increases) 1
- M2** because more moles / molecules (of gas) on the right 1
- M3** so position of equilibrium moves left 1

23. M/J 16/P42/Q4/a,b,c,d

- (a) **M1** substance that speeds up a reaction/increases rate;
M2 unchanged (chemically) at the end /not used up/lowers activation energy/
provides alternative pathway; 2
- (b) **M1** too slow / slower;
M2 lower yield /less product(s) /equilibrium shifts to left/
equilibrium shifts in direction of reactants /backward
reaction favoured /reverse reaction favoured; 2
1
- (c) faster /increase rate;
- (d) lower yield /less product(s) /equilibrium shifts to left/
equilibrium shifts in direction of reactants /backward reaction
favoured /reverse reaction favoured;
- OR**
- higher cost /expensive;
- OR**
- safety risks; 1

7.3: Redox**1. O/N 20/P41/Q1/a(iv)**

This question is about elements.

aluminium
carbon
iron
hydrogen
oxygen
silicon
sodium
sulfur

- (a) (i) Name the element that has oxidation states of +2 and +3.

..... [1]

2. O/N 20/P43/Q3/c(ii)

- (a) Some of the ammonia made by the Haber process is converted into nitric acid.

The first stage of this process is the oxidation of ammonia to make nitrogen monoxide.



The process is carried out at 900 °C and a pressure of 5 atmospheres using an alloy of platinum and rhodium as a catalyst.

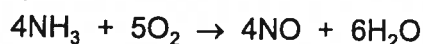
The forward reaction is exothermic.

- (i) State the meaning of the term *oxidation*.

..... [1]

3. M/J 20/P43/Q2/d(i)

- (a) Ammonia, NH_3 , is used to produce nitric acid, HNO_3 . This happens in a three-stage process. **Stage 1** is a redox reaction.



- (i) Identify what is oxidised in **stage 1**.

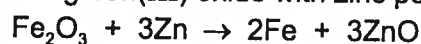
Give a reason for your answer.

substance oxidised

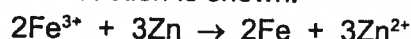
reason [2]

4. O/N 19/P41/Q3/c(ii)

- (a) Iron can be obtained by heating iron(III) oxide with zinc powder.



- (i) The ionic equation for this reaction is shown.



Identify the oxidising agent in this reaction. Explain your answer in terms of electron transfer.

oxidising agent

explanation [2]

5. M/J 19/P42/Q3/b(ii)

(a) Phosphorus, P_4 , reacts with air to produce phosphorus(V) oxide, P_4O_{10} .

(i) What type of chemical reaction is this?

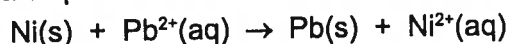
..... [1]

6. M/J 19/P42/Q7/a(ii)

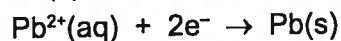
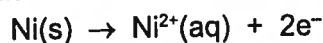
(a) Displacement reactions occur between metals and metal ions.

Displacement reactions can be used to determine the order of reactivity of metals such as lead (Pb), nickel (Ni), and silver (Ag).

The ionic equation for a displacement reaction is shown.



The ionic half-equations for this reaction are shown.



The ionic half-equations show that electrons are donated by nickel atoms and accepted by lead ions.

(i) What is the general term given to the type of reaction in which electrons are transferred from one species to another?

..... [1]

7. O/N 18/P41/Q4/c

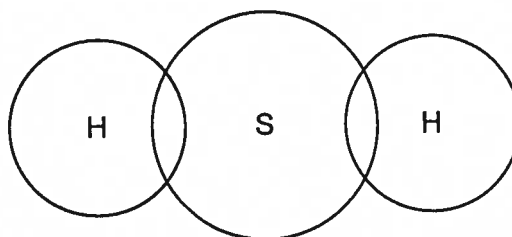
(a) The gas hydrogen sulfide, H_2S , is produced when concentrated sulfuric acid is added to solid potassium iodide.

The reaction involves oxidation.

(i) Define the term *oxidation* in terms of electron transfer.

..... [1]

(ii) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of hydrogen sulfide. Show outer shell electrons only.



[2]

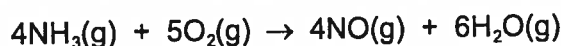
(iii) Hydrogen sulfide has a simple molecular structure.

Explain why hydrogen sulfide has a low boiling point.

.....
..... [2]

8. MAR 18/P42/Q4/c

(a) Ammonia reacts with oxygen at high temperatures in the presence of a suitable catalyst to form nitric oxide, NO.



(i) Explain how this chemical equation shows ammonia acting as a reducing agent.

..... [1]

(ii) Suggest a suitable catalyst for the reaction from the list of metals. Give a reason for your answer.

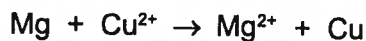
aluminium calcium platinum potassium sodium

suitable catalyst

reason [2]

9. O/N 17/P41/Q3/a

(a) When magnesium is added to aqueous copper(II) sulfate a reaction occurs.
The ionic equation for the reaction is shown.



(i) Give **one** change you would observe during this reaction.

..... [1]

(ii) Explain why this is a redox reaction.

..... [1]

(iii) Identify the oxidising agent in this reaction. Give a reason for your answer.

.....

..... [2]

(iv) A redox reaction occurs when magnesium is heated with iron(III) oxide.

Write a chemical equation for the reaction between magnesium and iron(III) oxide.

..... [2]

Answers Section

1. **O/N 20/P41/Q1/a(iv)**
(a) (i) iron 1

2. **O/N 20/P43/Q3/c(ii)**
(a) gain of oxygen / loss of hydrogen / electron loss / increase in oxidation state (oxidation number) 1

3. **M/J 20/P43/Q2/d(i)**
(a) (i) N / NH₃
change in oxidation state of N from -3 to +2 / increase in oxidation number / gain in oxygen / loss of electrons 2

4. **O/N 19/P41/Q3/c(ii)**
(a) (i) Fe³⁺ (1)
accept / take / gain electrons (1) 2

5. **M/J 19/P42/Q3/b(ii)**
(a) (i) redox / combustion 1

6. **M/J 19/P42/Q7/a(ii)**
(a) (i) redox 1

7. **O/N 18/P41/Q4/c**
(a) (i) (oxidation is) loss of electrons 1
(ii) M1 one shared pair between each H and S 1
M2 four unpaired electrons on S giving S a total of 8 outer shell electrons and no other unpaired electrons 1
(iii) M1 weak (attractive) forces OR (attractive) forces need little energy to overcome 1
M2 forces between molecules / intermolecular 1

8. **MAR 18/P42/Q4/c**
(a) (i) ammonia / it is oxidised / oxygen is reduced 1
(ii) M1 platinum 2
M2 transition metal / element

9. **O/N 17/P41/Q3/a**
(a) (i) brown / orange solid (forms / is made) OR solution becomes paler / colourless 1
(ii) magnesium is oxidised AND copper ions are reduced
OR
magnesium loses electrons AND copper ions gain electrons
OR
magnesium increases in oxidation number AND copper decreases in oxidation number 1
(iii) Cu²⁺ OR copper(II) ions OR copper ions 1
gains electrons 1
(iv) $3\text{Mg} + \text{Fe}_2\text{O}_3 \rightarrow 3\text{MgO} + 2\text{Fe}$
M1 Fe₂O₃ AND MgO 2
M2 fully correct



Unit 8

ACIDS, BASES AND SALTS

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 8.1 The characteristic properties of acids and bases
- 8.2 Types of oxides
- 8.3 preparation of salts
- 8.4 Identification of ions and gases

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Unit 8: Acids, Bases And Salts

8.1: The characteristic properties of acids and bases

1. M/J 22/P42/Q2

Calcium hydroxide, $\text{Ca}(\text{OH})_2$, is slightly soluble in water.

(a) Calcium hydroxide can be made by the reaction of calcium with water.

(i) Write the chemical equation for this reaction.

..... [2]

(ii) Name another substance that reacts with water to form calcium hydroxide.

..... [1]

(b) When calcium hydroxide dissolves in water, it dissociates into ions and forms a weakly alkaline solution.

(i) Suggest the pH of aqueous calcium hydroxide.

..... [1]

(ii) Give the formula of the ion responsible for making the solution alkaline.

..... [1]

(c) Limewater is a saturated solution of calcium hydroxide, $\text{Ca}(\text{OH})_2(\text{aq})$.

(i) Name the gas limewater is used to test for.

..... [1]

(ii) Suggest what is meant by the term *saturated solution*.

.....

..... [2]

(iii) Describe how you would make a sample of limewater starting with solid calcium hydroxide.

.....

..... [2]

(iv) Describe how you would test for the presence of calcium ions in a sample of limewater.

test

observations

..... [3]

(d) A 25.0 cm^3 sample of limewater is placed in a conical flask. The concentration of $\text{Ca}(\text{OH})_2$ in the limewater is determined by titration with dilute hydrochloric acid, HCl .

(i) Name the item of apparatus used to measure the volume of acid in this titration.

..... [1]

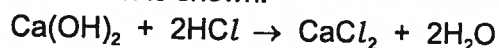
(ii) State the type of reaction which takes place.

..... [1]

(iii) As well as limewater and dilute hydrochloric acid, state what other type of substance must be added to the conical flask.

..... [1]

(iv) The equation for the reaction is shown.



20.0 cm³ of 0.0500 mol/dm³ HCl reacts with the 25.0 cm³ of Ca(OH)₂.

Determine the concentration of Ca(OH)₂ in g/dm³. Use the following steps.

- Calculate the number of moles in 20.0 cm³ of 0.0500 mol/dm³ HCl

..... mol

- Determine the number of moles of Ca(OH)₂ in 25.0 cm³ of the limewater.

..... mol

- Calculate the concentration of Ca(OH)₂ in mol/dm³.

..... mol/dm³

- Determine the concentration of Ca(OH)₂ in g/dm³.

..... g/dm³ [5]

2. O/N 21/P42/Q2/a

Acids are important laboratory chemicals.

(a) Some acids completely dissociate in water to form ions.

(i) State the term applied to acids that completely dissociate in water.

..... [1]

(ii) Complete the equation to show the complete dissociation of sulfuric acid in water.



(iii) State the colour of methyl orange in sulfuric acid.

..... [1]

3. O/N 21/P43/Q3/e

Lead is a metallic element in Group IV. One of the ores of lead is galena, which is an impure form of lead(II) sulfide, PbS.

Lead also occurs in the ore cerussite, which contains lead(II) carbonate, PbCO₃.

- (a) Lead(II) carbonate reacts with dilute nitric acid. One of the products is aqueous lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$.
Write a chemical equation for this reaction.

..... [2]

4. M/J 21/P42/Q5/e

- (a) Ammonia is a base which forms a weakly alkaline solution when dissolved in water.

(i) Define the term *base*.

..... [1]

(ii) Suggest the pH of aqueous ammonia.

..... [1]

5. O/N 20/P42/Q5/b

- (a) Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Complete the table to show the similarities and differences in the properties of samples of these two acids of equal concentration.

	dilute ethanoic acid	dilute hydrochloric acid
extent of dissociation		
colour after adding universal indicator solution		
observation when magnesium ribbon is added		

[6]

6. O/N 20/P41/Q3/a

- (a) Aqueous ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, is warmed with aqueous sodium hydroxide. The pungent-smelling gas ammonia, NH_3 , is produced.

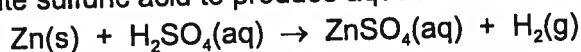
Balance the equation for this reaction.



[1]

7. O/N 20/P43/Q4/c(iii)

- (a) Zinc reacts with dilute sulfuric acid to produce aqueous zinc sulfate.



Hydrated zinc sulfate crystals are made from aqueous zinc sulfate.

Step 1 Solid zinc is added to dilute sulfuric acid until zinc is in excess.

Step 2 Excess zinc is separated from aqueous zinc sulfate by filtration.

Step 3 Aqueous zinc sulfate is heated until the solution is saturated.

Step 4 The saturated solution is allowed to cool and crystallise.

Step 5 The crystals are removed and dried.

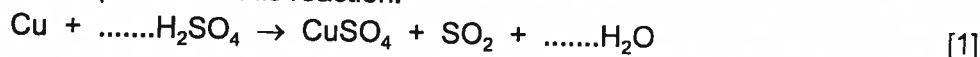
- (i) Name **two** compounds each of which react with dilute sulfuric acid to produce aqueous zinc sulfate.

1

2 [2]

8. M/J 20/P42/Q3/d

- (a) When copper is reacted with hot concentrated sulfuric acid, sulfur dioxide gas is formed.
Balance the chemical equation for this reaction.



9. M/J 20/P41/Q3/c

- (a) Concentrated sulfuric acid is a dehydrating agent which can chemically remove water from substances.

Both hydrated copper(II) sulfate crystals and sucrose (a sugar), $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, can be completely dehydrated by concentrated sulfuric acid.

Name the solid product formed in each case.

hydrated copper(II) sulfate crystals

sucrose [2]

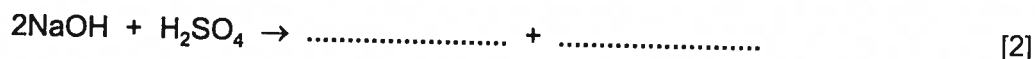
10. M/J 20/P41/Q4/c

- (a) Aqueous sodium hydroxide, NaOH(aq) , is a strong alkali that reacts with dilute sulfuric acid exothermically.

- (i) What type of reaction is this?

..... [1]

- (ii) Complete the equation for the reaction between aqueous sodium hydroxide and dilute sulfuric acid.



11. M/J 20/P43/Q1/b

- (a) Some acids are described as weak acids.

State the meaning of the term *weak acid*.

weak

acid [2]

12. MAR 20/P42/Q2/c(ii,iii,iv,v)

- (a) Aluminium oxide reacts with acids and with alkalis.

- (i) Aluminium oxide reacts with dilute sulfuric acid to form a salt.

State the name and write the formula of the salt formed.

name

formula [2]

- (ii) Aluminium oxide reacts with dilute sodium hydroxide to form a salt and one other product.
Name the other product.

..... [1]

- (iii) Aluminium hydroxide, Al(OH)_3 , decomposes when heated to form aluminium oxide and water.

Write the chemical equation for this reaction.

..... [2]

- (iv) Suggest the names of **two** other aluminium compounds that decompose when heated to form aluminium oxide.

.....
 [2]

13. O/N 19/P41/Q4/d

Insoluble salts can be made by precipitation reactions.

A student mixed solutions of some soluble salts.

The results the student obtained are shown in the table.

		second salt solution		
		$\text{Co}(\text{NO}_3)_2(\text{aq})$	$\text{AgNO}_3(\text{aq})$	$\text{Pb}(\text{NO}_3)_2(\text{aq})$
first salt solution	$\text{NaI}(\text{aq})$	no change	yellow precipitate	yellow precipitate
	$\text{Na}_2\text{CO}_3(\text{aq})$	purple precipitate	yellow precipitate	white precipitate
	$\text{Na}_2\text{SO}_4(\text{aq})$	no change	white precipitate	white precipitate

All sodium salts are soluble in water.

Use only results from the table to answer the following questions.

- (a) Aqueous silver nitrate produces a yellow precipitate with both iodide ions and carbonate ions. When testing an unknown solution for iodide ions, the aqueous silver nitrate is acidified. Explain why the aqueous silver nitrate is acidified.

..... [1]

14. M/J 19/P42/Q3/c,e

- (a) Phosphorus(V) oxide, P_4O_{10} , is an acidic oxide.

Phosphorus(V) oxide, P_4O_{10} , reacts with aqueous sodium hydroxide to form a salt containing the phosphate ion, PO_4^{3-} . Water is the only other product.

Write a chemical equation for the reaction between phosphorus(V) oxide and aqueous sodium hydroxide.

..... [2]

- (b) Phosphine, PH_3 , has a similar chemical structure to ammonia, NH_3 . Ammonia acts as a base when it reacts with sulfuric acid.

- (i) What is meant by the term *base*?

..... [1]

- (ii) Write a chemical equation for the reaction between ammonia and sulfuric acid.

..... [2]

15. M/J 19/P41/Q4/a,b

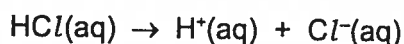
Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Both ethanoic acid and hydrochloric acid dissociate in aqueous solution.

(a) (i) Define the term *acid*.

..... [1]

(ii) The chemical equation shows the changes which occur when the **strong** acid, hydrochloric acid, is added to water.



Complete the chemical equation to show the changes which occur when the **weak** acid, ethanoic acid, is added to water.

$\text{CH}_3\text{COOH(aq)}$ [2]

(b) A student does experiments to show that hydrochloric acid is a strong acid and ethanoic acid is a weak acid. The student adds an excess of hydrochloric acid and an excess of ethanoic acid to separate samples of lumps of calcium carbonate.

Only the identity of the acid is changed between the experiments. All other conditions are kept the same.

(i) State **two** observations which would show that hydrochloric acid is a stronger acid than ethanoic acid.

1

2 [2]

(ii) The student uses the same size container and checks that the pressure is the same for each experiment.

State **three** other conditions which must be kept the same to ensure fair testing.

1

2

3 [3]

16. MAR 19/P42/Q4/c

(a) Ethanoic acid is a weak acid.

(i) What is meant by the term *acid*?

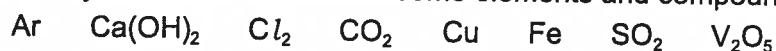
..... [1]

(ii) Why is ethanoic acid described as *weak*?

..... [1]

17. M/J 18/P43/Q1/e

The following are the symbols and formulae of some elements and compounds.



Answer the following questions using only the elements or compounds in the list.

Each element or compound may be used once, more than once or not at all.

State which element or compound is used:

(a) to neutralise excess acidity in soil [1]

18. M/J 18/P43/Q3/d(i)

Cobalt is a transition element. Potassium is in Group I of the Periodic Table.

(a) Cobalt reacts with dilute hydrochloric acid to make the salt cobalt(II) chloride. Bubbles of hydrogen gas are produced.

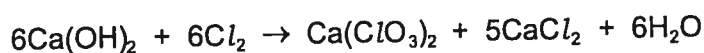
(i) Describe a test for hydrogen.

test

result [2]

19. M/J 18/P43/Q6/c

Calcium chlorate(V), $\text{Ca}(\text{ClO}_3)_2$, is made by reacting calcium hydroxide with chlorine gas.



(a) Chloric(V) acid, HClO_3 , is a strong acid. It can be made from calcium chlorate(V).

(i) What colour is methyl orange indicator in chloric(V) acid?

..... [1]

(ii) Define the term *acid* in terms of proton transfer.

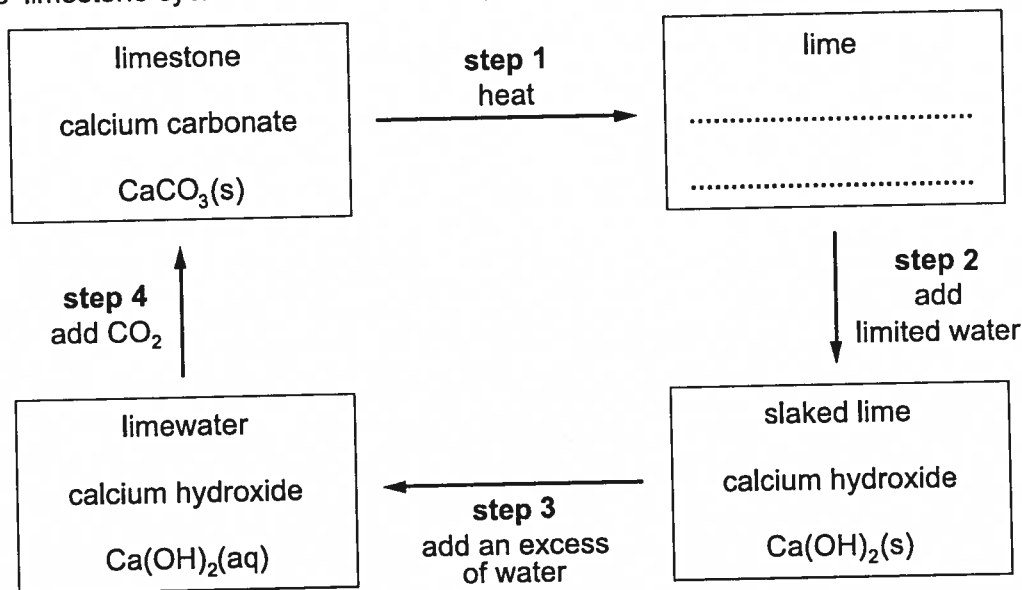
..... [1]

(iii) Complete the chemical equation to show HClO_3 behaving as an acid in water.

**20. MAR 18/P42/Q3/a,b**

Limestone rock is mainly calcium carbonate, CaCO_3 .

(a) The 'limestone cycle' is shown. Each step is numbered.



(i) Complete the box to give the chemical name and formula of lime.

[2]

(ii) Which step involves a physical change?

..... [1]

(iii) What type of reaction is **step 1**?

..... [1]

(iv) Suggest how **step 2** could be reversed.

..... [1]

(v) Write a chemical equation for **step 4**.

..... [1]

(vi) Explain why **step 4** is a neutralisation reaction. Refer to the substances reacting in your answer.

.....

..... [2]

(b) Dolomite is a similar rock to limestone. Dolomite contains magnesium carbonate, MgCO_3 . Write a chemical equation for the reaction between magnesium carbonate and dilute nitric acid.

..... [2]

21. O/N 16/P42/Q4

Dilute nitric acid behaves as a typical acid in some reactions but **not** in other reactions.

(a) Dilute nitric acid behaves as a typical acid when reacted with copper(II) oxide and with copper(II) carbonate.

Describe what you would **see** if excess dilute nitric acid is added separately to solid samples of copper(II) carbonate and copper(II) oxide followed by warming the mixtures.

copper(II) carbonate

.....

.....

copper(II) oxide

.....

..... [4]

(b) When dilute nitric acid is added to pieces of copper and heated, a reaction takes place and copper(II) nitrate is formed.

(i) Part of the chemical equation for the reaction between copper and dilute nitric acid is shown.

Complete the chemical equation by inserting the formula of copper(II) nitrate and balancing the equation.

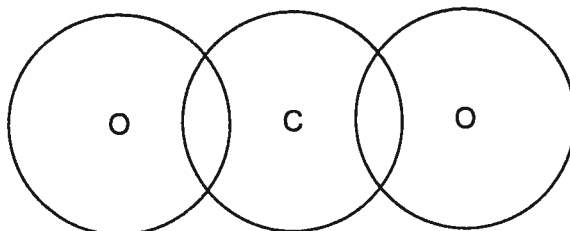


(ii) How is the reaction of dilute nitric acid with copper different from that of a typical metal with a typical acid?

..... [1]

22. O/N 16/P43/Q1/d

- (a) Complete the diagram to show the electron arrangement in a carbon dioxide molecule. Show the outer shell electrons only.



[2]

23. O/N 16/P43/Q5/f

Sulfuric acid can be manufactured from the raw materials sulfur, air and water. The process can be divided into four stages.

stage 1 converting sulfur into sulfur dioxide

stage 2 converting sulfur dioxide into sulfur trioxide

stage 3 converting sulfur trioxide into oleum, $\text{H}_2\text{S}_2\text{O}_7$

stage 4 converting oleum into sulfuric acid

- (a) Sulfuric acid reacts with a hydrocarbon called benzene to produce benzenesulfonic acid, $\text{C}_6\text{H}_5\text{SO}_3\text{H}$. Benzenesulfonic acid is a strong acid which ionises to produce hydrogen ions, H^+ , and benzenesulfonate ions, $\text{C}_6\text{H}_5\text{SO}_3^-$.

- (i) What is meant by the term *strong acid*?

..... [1]

- (ii) Describe how to show that a 1 mol/dm^3 solution of benzenesulfonic acid is a strong acid.

..... [2]

- (iii) Write a chemical equation for the reaction between benzenesulfonic acid and sodium carbonate, Na_2CO_3 .

..... [2]

24. M/J 16/P41/Q2/b

Period 3 contains the elements sodium to argon. This question asks about the chemistry of each of the Period 3 elements or their compounds.

- (a) Magnesium reacts slowly with warm water to form a base, magnesium hydroxide.

- (i) Explain what is meant by the term *base*.

..... [1]

- (ii) Write a chemical equation for the reaction between magnesium and warm water.

..... [2]

Answers Section

- 1. M/J 22/P42/Q2**
- (a) (i) $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$
M1 H_2 as product
M2 Fully correct equation 2
- (ii) calcium oxide 1
- (b) (i) $7 < \text{pH} \leq 12$ 1
- (ii) OH^- 1
- (c) (i) carbon dioxide 1
- (ii) (a solution that) can dissolve no more solute
at a given temperature 2
- (iii) add excess (solid) calcium hydroxide (to water)
filter 2
- (iv) (aqueous) sodium hydroxide
white ppt 1
- insoluble / remains in excess 2
- (d) (i) burette 1
- (ii) neutralisation 1
- (iii) indicator 1
- (iv) M1 $\text{Mol HCl} = 0.0500 \times 20.0 / 1000 = 0.001(00)$
M2 $\text{Mol Ca(OH)}_2 = \text{M1} / 2 = 0.00100 / 2 = 0.0005(00)$
M3 $\text{M2} \times 1000 / 25 = 0.0005(00) \times 40 = 0.02(00)$
M4 $M_r \text{ Ca(OH)}_2 = 74$
M5 $\text{M4} \times \text{M3} = 74 \times 0.02 = 1.48 \text{ (g / dm}^3\text{)}$ 5
- 2. O/N 21/P42/Q2/a**
- (a) (i) strong 1
- (ii) $2\text{H}^+ + \text{SO}_4^{2-}$
 H^+ (1) correct equation (1) 2
- (iii) pink / red 1
- 3. O/N 21/P43/Q3/e**
- (a) $\text{PbCO}_3 + 2\text{HNO}_3 \rightarrow \text{Pb(NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$ 2
- 4. M/J 21/P42/Q5/e**
- (a) (i) proton acceptor 1
- (ii) $7 < x \leq 11$ 1
- 5. O/N 20/P42/Q5/b**
- (a) partial (dissociation) (1)
full / 100% (dissociation) (1)
both acid colours (1)
 HCl indicating a lower pH acid colour than CH_3COOH (1)
fizzing OR dissolving / disappearing in both (1)
either observation happens quicker with HCl (1) 6

6. O/N 20/P41/Q3/a 1
(a) (i) $2 \rightarrow 2 + 2$
7. O/N 20/P43/Q4/c(iii) 2
(a) (i) one mark for each of any two from:
 - zinc oxide
 - zinc hydroxide
 - zinc carbonate
8. M/J 20/P42/Q3/d 1
(a) $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
9. M/J 20/P41/Q3/c 2
(a) anhydrous copper(II) sulfate
carbon
10. M/J 20/P41/Q4/c 1
(a) (i) neutralisation
(ii) Na_2SO_4 2
 $2\text{H}_2\text{O}$
11. M/J 20/P43/Q1/b 2
(a) donate or lose protons
does not fully ionise / partially ionises / forms an equilibrium mixture
12. MAR 20/P42/Q2/c(ii,iii,iv,v) 2
(a) (i) aluminium sulfate (1) $\text{Al}_2(\text{SO}_4)_3$ (1) 1
(ii) water
(iii) $2\text{Al}(\text{OH})_3 \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$ 2
species (1) balance (1)
(iv) aluminium carbonate (1) 2
aluminium nitrate (1)
13. O/N 19/P41/Q4/d 1
(a) (nitric) acid reacts with / removes carbonate ions
14. M/J 19/P42/Q3/c,e 2
(a) $\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$
M1 Na_3PO_4 (1) 1
M2 equation completely correct (1)
(b) (i) proton / H^+ / hydrogen ion acceptor
(ii) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$ 2
M1 $(\text{NH}_4)_2\text{SO}_4$ (1)
M2 equation completely correct (1)
15. M/J 19/P41/Q4/a,b 1
(a) (i) proton donor 2
(ii) $(\text{CH}_3\text{COOH}) \rightleftharpoons \text{CH}_3\text{COO}^-$ (1) + H^+ (1)

- (b) (i) any **two** from:
 ∞ faster rate of fizzing
 ∞ solid dissolves quicker / disappears quicker / gets smaller quicker
 ∞ fizzing stops quicker
 ∞ dissolving stops quicker 2
- (ii) any **three** from:
 ∞ temperature
 ∞ volume (of acid)
 ∞ concentration (of acid)
 ∞ mass / amount (of CaCO_3)
 ∞ particle size / surface area (of CaCO_3) 3
16. MAR 19/P42/Q4/c
 (a) (i) proton donor 1
 (ii) ionises / dissociates partially or incompletely 1
17. M/J 18/P43/Q1/e
 (a) Ca(OH)_2 / calcium hydroxide [1]
18. M/J 18/P43/Q3/d(i)
 (a) (i) test: lighted splint / flame
 result: (squeaky) pop [2]
19. M/J 18/P43/Q6/c
 (a) (i) red [1]
 (ii) proton donor [1]
 (iii) $\rightarrow \text{ClO}_3^- + \text{H}_3\text{O}^+$ [1]
20. MAR 18/P42/Q3/a,b
 (a) (i) M1 calcium oxide
 M2 CaO 2
 (ii) (step) 3 1
 (iii) thermal decomposition 1
 (iv) heating 1
 (v) $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$ 1
 (vi) M1 CO_2 is acidic
 M2 Ca(OH)_2 is a base / alkali 2
 (b) $\text{MgCO}_3 + 2\text{HNO}_3 \rightarrow \text{Mg(NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$
 M1 $\text{Mg(NO}_3)_2$
 M2 rest of equation 2
21. O/N 16/P42/Q4
 (a) *copper(II) carbonate*
 fizzes/bubbles/effervescence
 dissolves/disappears [2]
copper(II) oxide
 dissolves/disappears
 blue (solution formed) [2]
 (b) (i) $\text{Cu(NO}_3)_2$
 3Cu AND $3\text{Cu(NO}_3)_2$ [2]

- (ii) hydrogen (gas) is not produced (when copper reacts with nitric acid) [1]
22. O/N 16/P43/Q1/d [1]
(a) two double bonds with no extra electrons on the carbon atoms [1]
both oxygen atoms with four non-bonding electrons
23. O/N 16/P43/Q5/f
(a) (i) exists completely as ions (in solution)/completely dissociates (in solution)/completely ionises (in solution) [1]
(ii) Universal Indicator/pH paper/pH indicator/pH meter [1]
Universal Indicator or pH paper or pH indicator turns red/pH 0–1 [1]
(iii) $\text{Na}_2\text{CO}_3 + 2\text{C}_6\text{H}_5\text{SO}_3\text{H} \rightarrow 2\text{C}_6\text{H}_5\text{SO}_3\text{Na} + \text{CO}_2 + \text{H}_2\text{O}$
formula of $\text{C}_6\text{H}_5\text{SO}_3\text{Na}$ [2]
all formulae correct and balancing correct
24. M/J 16/P41/Q2/b [1]
(a) (i) (a substance which is) a proton/ H^+ /hydrogen ion acceptor;
(ii) $\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$ [2]
 Mg(OH)_2 ; rest of equation;

8.2: Types of oxides

1. MAR 21/P42/Q4

A student wanted to make some zinc chloride crystals.

The student followed the procedure shown.

step 1 Add excess zinc powder to dilute hydrochloric acid to form aqueous zinc chloride.

step 2 Remove unreacted zinc powder from the aqueous zinc chloride.

step 3 Heat the solution until it is saturated.

step 4 Allow the saturated solution to cool and remove the crystals that form.

(a) Write the equation for the reaction in **step 1**. Include state symbols.

.....[3]

(b) Explain why **excess** zinc powder is added in **step 1**.

.....[1]

(c) Suggest how unreacted zinc powder is removed in **step 2**.

.....[1]

(d) A saturated solution is formed in **step 3**.

Suggest what is meant by the term *saturated solution*.

.....
.....[2]

(e) Explain why crystals form as the solution cools in **step 4**.

.....[1]

(f) Name **two** zinc compounds which react with dilute hydrochloric acid to form zinc chloride.

.....
.....[2]

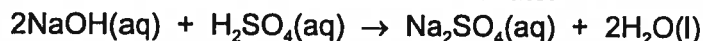
(g) If excess calcium metal is used instead of excess zinc powder in **step 1**, pure calcium chloride crystals do **not** form.

Explain why.

.....[1]

(h) Some salts can be made by titration.

In a titration experiment, 20.0 cm³ of aqueous sodium hydroxide reacts exactly with 25.0 cm³ of 0.100 mol/dm³ dilute sulfuric acid to make sodium sulfate.



(i) Circle the name of the type of reaction that takes place.

decomposition **neutralisation** **precipitation** **reduction** [1]

(ii) Calculate the concentration of the aqueous sodium hydroxide in g/dm³ using the following steps.

- Calculate the number of moles of dilute sulfuric acid used.

..... mol

- Determine the number of moles of sodium hydroxide which react with the dilute sulfuric acid.

..... mol

- Calculate the concentration of the aqueous sodium hydroxide in mol/dm³.

..... mol/dm³

- Calculate the concentration of the aqueous sodium hydroxide in g/dm³.

..... g/dm³ [5]

2. O/N 20/P41/Q1/a(iii)

- (a) This question is about elements.

aluminium
carbon
iron
hydrogen
oxygen
silicon
sodium
sulfur

- (i) Name the element that forms an amphoteric oxide.

.....[1]

3. MAR 20/P42/Q2/c(i)

- (a) Aluminium oxide reacts with acids and with alkalis.

- (i) What term is used to describe an oxide that reacts with acids and with alkalis?

.....[1]

4. O/N 19/P41/Q3/d

- (a) Zinc oxide is amphoteric.

Describe **two** simple experiments to show that zinc oxide is amphoteric.

Name the reagents you would use and describe the observations you would make.

reagent 1

observation.....

reagent 2

observation..... [3]

5. M/J 19/P42/Q3/b(i)

(a) Phosphorus, P_4 , reacts with air to produce phosphorus(V) oxide, P_4O_{10} .

(i) Write a chemical equation for this reaction.

..... [2]

6. M/J 19/P43/Q3/c

(a) Some airbags contain silicon(IV) oxide.

When the airbag is used sodium oxide is formed.

Oxides can be classified as acidic, amphoteric, basic or neutral.

Classify each of these oxides:

sodium oxide

silicon(IV) oxide. [2]

7. MAR 19/P42/Q1/b,d,f

Period 3 of the Periodic Table is shown.

sodium	magnesium	aluminium	silicon	phosphorus	sulfur	chlorine	argon
--------	-----------	-----------	---------	------------	--------	----------	-------

Answer the following questions using only these elements.

Each element may be used once, more than once or not at all.

State which element:

(a) forms a basic oxide with a formula of the form X_2O

..... [1]

(b) forms an oxide which causes acid rain

..... [1]

(c) forms an oxide which has a macromolecular structure

..... [1]

8. O/N 18/P41/Q6/b

(a) Ethane can react with chlorine in a substitution reaction.

(i) State **one** essential reaction condition.

..... [1]

(ii) Draw the structure of the organic product formed by substitution of **one** of the hydrogen atoms in ethane with chlorine. Show all of the atoms and all of the bonds.

- (iii) Name the product of the substitution reaction between ethane and chlorine that does **not** contain carbon.

..... [1]

9. O/N 17/P43/Q2

- (a) (i) Define the term *molecule*.

.....
.....[2]

- (ii) Define the term *element*.

.....[1]

- (b) The table shows the composition of four atoms or ions, **A**, **B**, **C** and **D**.

	number of protons	number of neutrons	number of electrons
A	10	10	10
B	10	12	10
C	12	10	10
D	13	14	10

- (i) What is the atomic number of **A**?

.....[1]

- (ii) What is the nucleon number of **B**?

.....[1]

- (iii) Which of **A**, **B**, **C** and **D** are isotopes of each other?

.....[1]

- (iv) Which of **A**, **B**, **C** and **D** are atoms?

.....[1]

- (v) Which of **A**, **B**, **C** and **D** are positive ions?

.....[1]

- (c) Complete the table.

	number of protons	number of electrons
Na		
S ²⁻		
Cl ₂		

[3]

10. M/J 16/P42/Q3/c

Gallium is a metallic element in Group III. It has similar properties to aluminium.

(a) Gallium(III) oxide, Ga_2O_3 , is amphoteric.

(i) Write the chemical equation for the reaction between gallium(III) oxide and dilute nitric acid to form a salt and water only.

.....[2]

(ii) The reaction between gallium(III) oxide and sodium hydroxide solution forms only water and a salt containing the negative ion $\text{Ga}_2\text{O}_4^{2-}$.

Write the chemical equation for this reaction.

.....[2]

11. M/J 16/P41/Q2/c

Period 3 contains the elements sodium to argon. This question asks about the chemistry of each of the Period 3 elements or their compounds.

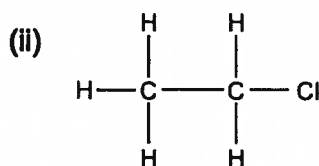
(a) Aluminium oxide is amphoteric. It is insoluble in water.

Describe experiments to show that aluminium oxide is amphoteric.

.....
.....
.....
.....[3]

Answers Section

- 1. MAR 21/P42/Q4**
- (a) $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
 ZnCl_2 or H_2 as a product (1)
 correct equation (1)
 states (1) 3
- (b) to make sure all the (hydrochloric) acid reacts 1
- (c) filtration 1
- (d) a solution that can dissolve no more solute (1)
 at a given temperature (1) 2
- (e) solubility (of ZnCl_2 /solids) decreases (as temperature decreases) 1
- (f) zinc oxide 2
 zinc carbonate 1
- (g) Ca will also react with water 1
- (h) (i) neutralisation
- (ii) $0.100 \times 25 / 1000 = 0.0025(0)$ (1)
 $0.0025 \times 2 = 0.005(00)$ (1)
 $0.005 \times 1000 / 20 = 0.25(0)$ (1)
 $M_r = 40$ (1) 5
 $0.25 \times 40 = 10.(0)$ (1)
- 2. O/N 20/P41/Q1/a(iii)** 1
- (a) aluminium
- 3. MAR 20/P42/Q2/c(i)** 1
- (a) amphoteric
- 4. O/N 19/P41/Q3/d**
- (a) (add a) named acid (1)
 (add a) named alkali (1)
 disappears / dissolves in both (1) 3
- 5. M/J 19/P42/Q3/b(i)**
- (a) (i) $\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$
M1 all formulae correct (1) 2
M2 equation correctly balanced (1)
- 6. M/J 19/P43/Q3/c**
- (a) **M1** (sodium oxide) basic 2
M2 (silicon dioxide) acidic
- 7. MAR 19/P42/Q1/b,d,f** 1
- (a) sodium 1
- (b) sulfur 1
- (c) silicon
- 8. O/N 18/P41/Q6/b** 1
- (a) (i) ultraviolet light / sunlight



1

(iii) hydrogen chloride

1

9. O/N 17/P43/Q2

- (a) (i) (two or more) **atoms** [1]
combined/joined/sharing electrons (by a covalent bond)/bonded [1]
- (ii) substance that cannot be split up/broken down/decomposed (into anything simpler)
OR
(substance) made of **atoms** with the same atomic number/number of protons/
proton number [1]
- (b) (i) 10 [1]
(ii) 22 [1]
(iii) A AND B [1]
(iv) A AND B [1]
(v) C AND D [1]

(c)

	number of protons	number of electrons
Na	11	11
S ²⁻	16	18
Cl ₂	34	34

[3]

10. M/J 16/P42/Q3/c

- (a) (i) $\text{Ga}_2\text{O}_3 + 6\text{HNO}_3 \rightarrow 2\text{Ga}(\text{NO}_3)_3 + 3\text{H}_2\text{O}$
formula of $\text{Ga}(\text{NO}_3)_3$; all formulae and balancing correct; [2]
- (ii) $\text{Ga}_2\text{O}_3 + 2\text{NaOH} \rightarrow \text{Na}_2\text{Ga}_2\text{O}_4 + \text{H}_2\text{O}$;
formula of $\text{Na}_2\text{Ga}_2\text{O}_4$; all formulae and balancing correct; [2]

11. M/J 16/P41/Q2/c

- (a) M1 add a *named* acid, e.g. HCl and a named alkali, e.g. NaOH ; [1]
M2 Al_2O_3 will react with/neutralises both reagents; [1]
M3 and so it will dissolve into the reagent/form a solution; [1]

8.3: Preparation of salts

1. O/N 22/P42/Q4

A student prepares magnesium sulfate crystals, MgSO_4 , by adding excess magnesium to dilute sulfuric acid.

- (a) Write the chemical equation for this reaction.

.....[1]

- (b) Describe **two** observations which show the reaction has finished.

1

2[2]

- (c) The excess magnesium is removed by filtration.

State the general name given to a solid separated from a solution by filtration.

.....[1]

- (d) The aqueous magnesium sulfate is heated until crystals begin to appear.

- (i) Suggest the name for a solution in which no more solute can dissolve.

.....[1]

- (ii) Suggest why more crystals of magnesium sulfate appear on cooling.

.....[1]

- (e) Magnesium sulfate crystals have the formula, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$, where x is a whole number of molecules of water.

The student heats the crystals to remove the molecules of water.



- (i) Name the term given to crystals containing molecules of water.

.....[1]

- (ii) The student heats a sample of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ and finds it has lost 0.140 moles of H_2O and has 2.40 g of MgSO_4 remaining.

Determine the value of x . Use the following steps.

- Calculate the M_r of MgSO_4 .

$M_r =$

- Determine the number of moles of MgSO_4 formed.

moles of MgSO_4 formed =

- Determine the value of x in $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

$x = \dots\dots\dots$ [3]

- (f) If the student uses dilute nitric acid instead of dilute sulfuric acid, the salt formed is magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$.

Write the chemical equation for the reaction when solid magnesium nitrate is heated.

..... [2]

2. O/N 22/P41/Q4

A student prepares calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, by adding calcium carbonate to dilute nitric acid.

- (a) Write the chemical equation for this reaction.

..... [2]

- (b) Describe **two** observations during this reaction.

1

2 [2]

- (c) The student continues to add calcium carbonate until it is in excess. The student then removes the excess calcium carbonate by filtration and collects the aqueous calcium nitrate.

State the general term given to a solution collected from filtration.

..... [1]

- (d) The student gently heats the aqueous calcium nitrate until the solution is saturated.

- (i) Suggest what is meant by the term *saturated solution*.

.....

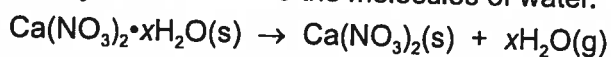
..... [2]

- (ii) Describe how crystals are produced from a hot saturated solution.

..... [1]

- (e) Calcium nitrate crystals are hydrated and have the formula $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$ where x is a whole number of molecules of water.

The student heats the crystals to remove the molecules of water.



- (i) State the term used to describe the calcium nitrate after the molecules of water have been removed.

..... [1]

- (ii) The student heats a sample of $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$ and forms 2.46 g of $\text{Ca}(\text{NO}_3)_2$ and 0.0600 moles of H_2O .

Determine the value of x . Use the following steps.

- Calculate the M_r of $\text{Ca}(\text{NO}_3)_2$.

$M_r =$

- Determine the number of moles of $\text{Ca}(\text{NO}_3)_2$ formed.

moles of $\text{Ca}(\text{NO}_3)_2$ formed =

- Determine the value of x in $\text{Ca}(\text{NO}_3)_2 \cdot x\text{H}_2\text{O}$.

$x =$ [3]

- (f) Nitrates decompose on heating.
Write the chemical equation for the reaction when solid sodium nitrate is heated.

..... [2]

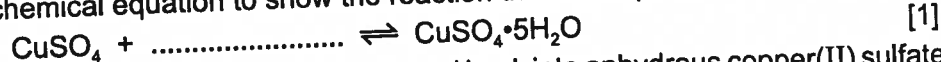
3. O/N 21/P41/Q2/b,c

- (a) Anhydrous copper(II) sulfate is used to test for the presence of water. When this test is positive, hydrated copper(II) sulfate is formed.

- (i) State the colour change seen during this test.

from to [2]

- (ii) Complete the chemical equation to show the reaction that takes place.



- (iii) State how hydrated copper(II) sulfate can be turned back into anhydrous copper(II) sulfate.

..... [1]

- (iv) Describe a test for pure water.

.....
..... [2]

(b) Aqueous copper(II) sulfate contains $\text{Cu}^{2+}(\text{aq})$ ions.

(i) Describe what is seen when aqueous copper(II) sulfate is added to aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$.

..... [1]

(ii) Write the ionic equation for the reaction between aqueous copper(II) sulfate and aqueous sodium hydroxide.
Include state symbols.

..... [3]

4. O/N 21/P43/Q3/h

(a) When lead(II) nitrate is heated it decomposes to produce the same gaseous products as when copper(II) nitrate is heated.

(i) One of the gaseous products is oxygen.
Describe a test for oxygen.

test

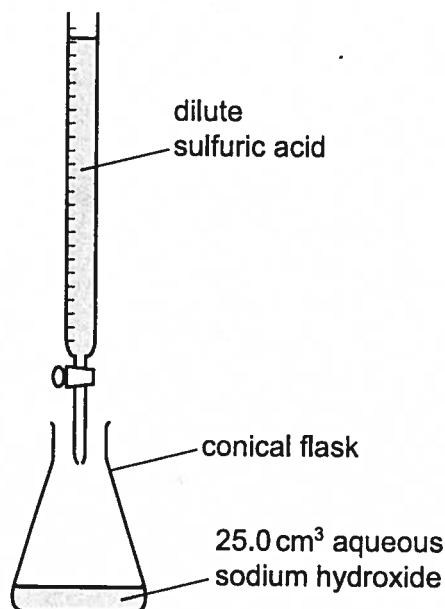
observations [2]

(ii) Name the other gaseous product.

..... [1]

5. O/N 21/P43/Q5

(a) Dilute sulfuric acid and aqueous sodium hydroxide can be used to prepare sodium sulfate crystals using a method that involves titration.



(i) Suggest why universal indicator is **not** suitable for this titration.

..... [1]

(ii) Name an indicator that can be used in this titration.

..... [1]

20.0 cm³ of dilute sulfuric acid neutralises 25.0 cm³ of 1.00 mol/dm³ aqueous sodium hydroxide. At the end of the titration the conical flask contains aqueous sodium sulfate with the dissolved indicator as an impurity.

- (b) Describe how to prepare a **pure** sample of sodium sulfate crystals from the original solutions of dilute sulfuric acid and aqueous sodium hydroxide of the same concentrations.

You are not required to give details of how to carry out the titration.

.....

.....

.....

.....

- [5]
- (c) Sodium hydrogensulfate, NaHSO₄, dissolves in water to produce an aqueous solution, X, containing Na⁺, H⁺ and SO₄²⁻ ions.

State the observations when the following tests are done.

- (i) A flame test is carried out on X.

..... [1]

- (ii) Copper(II) oxide is warmed with an excess of X.

.....

..... [2]

- (iii) Acidified aqueous barium nitrate is added to X.

..... [1]

6. M/J 21/P41/Q5

This question is about salts.

- (a) Salts that are insoluble in water are made by precipitation.

- Lead(II) iodide, PbI₂, is insoluble in water.
- All nitrates are soluble in water.
- All sodium salts are soluble in water.

You are provided with solid lead(II) nitrate, Pb(NO₃)₂, and solid sodium iodide, NaI.

Describe how you would make a pure sample of lead(II) iodide by precipitation.

Your answer should include:

- practical details
- a chemical equation for the precipitation reaction.

.....

.....

.....

.....

..... [5]

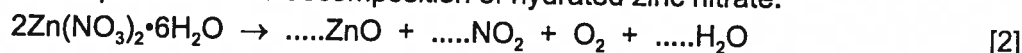
(b) Nitrates decompose when heated.

- (i) When hydrated zinc nitrate is heated, oxygen gas is given off.
Describe a test for oxygen.

test

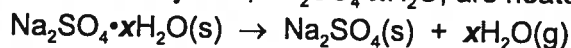
observations [2]

(ii) Complete the equation for the decomposition of hydrated zinc nitrate.



(c) Some sulfates are hydrated.

When hydrated sodium sulfate crystals, $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$, are heated, they give off water.



A student carries out an experiment to determine the value of x in $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$.

step 1 Hydrated sodium sulfate crystals are weighed.

step 2 The hydrated sodium sulfate crystals are then heated.

step 3 The remaining solid is weighed.

- (i) Describe how the student can check that all the water has been given off.

.....
..... [2]

(ii) In an experiment, 1.61 g of $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$ is heated until all the water is given off. The mass of Na_2SO_4 remaining is 0.71 g.

[M_r : Na_2SO_4 , 142; H_2O , 18]

Determine the value of x using the following steps.

- Calculate the number of moles of Na_2SO_4 remaining.

..... mol

- Calculate the mass of H_2O given off.

..... g

- Calculate the number of moles of H_2O given off.

..... mol

- Determine the value of x .

$x = \dots\dots\dots$ [4]

7. M/J 21/P43/Q5

This question is about salts.

- (a) Salts that are soluble in water can be made by the reaction between insoluble carbonates and dilute acids. Zinc sulfate is soluble in water.

You are provided with solid zinc carbonate, ZnCO_3 , and dilute sulfuric acid, H_2SO_4 .

Describe how you would make a pure sample of zinc sulfate crystals.

Your answer should include:

- practical details
- how you would make sure that all the dilute sulfuric acid has reacted
- a chemical equation for the reaction.

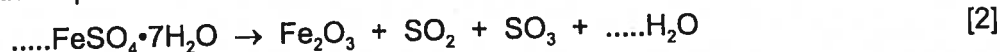
.....
.....
.....
..... [5]

- (b) Some sulfates decompose when heated.
When hydrated iron(II) sulfate is heated strongly, sulfur dioxide gas is given off.

- (i) Describe a test for sulfur dioxide.

test
observations [2]

- (ii) Complete the equation for the decomposition of hydrated iron(II) sulfate.



- (c) Some chlorides are hydrated.

When hydrated barium chloride crystals, $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$, are heated they give off water.



A student carries out an experiment to determine the value of x in $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$.

- step 1** Hydrated barium chloride crystals are weighed.
step 2 The hydrated barium chloride crystals are then heated.
step 3 The remaining solid is weighed.

- (i) Describe how the student can be sure that all the water is given off.

.....
..... [2]

- (ii) In an experiment, 4.88 g of $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ is heated until all the water is given off. The mass of BaCl_2 remaining is 4.16 g.

[M_r : BaCl_2 , 208; H_2O , 18]

Determine the value of x using the following steps.

- Calculate the number of moles of BaCl_2 remaining.

..... mol

- Calculate the mass of H_2O given off.

..... g

- Calculate the number of moles of H_2O given off.

..... mol

- Determine the value of x .

$x =$ [4]

8. O/N 20/P42/Q2/a,b(i,ii,v),d

Soluble salts can be made by adding a metal carbonate to a dilute acid.

- (a) Give the formula of the dilute acid which reacts with a metal carbonate to form a nitrate salt.

..... [1]

- (b) A student wanted to make hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, by adding excess iron(II) carbonate to dilute sulfuric acid. The student followed the procedure shown.

step 1 Add dilute sulfuric acid to a beaker.

step 2 Add small amounts of iron(II) carbonate to the dilute sulfuric acid in the beaker until the iron(II) carbonate is in excess.

step 3 Filter the mixture formed in **step 2**.

step 4 Heat the filtrate until it is a saturated solution. Allow to cool.

step 5 Once cold, pour away the remaining solution. Dry the crystals between filter papers.

- (i) Why must the iron(II) carbonate be added in excess in **step 2**?

..... [1]

- (ii) State **two** observations in **step 2** that would show that iron(II) carbonate was in excess.

1

2 [2]

- (iii) Name a different compound that could be used instead of iron(II) carbonate to produce hydrated iron(II) sulfate crystals from dilute sulfuric acid.

..... [1]

- (c) Insoluble salts can be made by mixing solutions of two soluble salts.

A student followed the procedure shown to make silver bromide, an insoluble salt.

step 1 Add aqueous silver nitrate to a beaker. Then add aqueous potassium bromide and stir.

step 2 Filter the mixture formed in **step 1**.

step 3 Dry the residue.

- (i) State the term used to describe this method of making salts.

..... [1]

- (ii) Give the observation the student would make during **step 1**.

..... [1]

- (iii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous potassium bromide.

Include state symbols.

..... [3]

9. O/N 20/P43/Q4/d(i)

- (a) When hydrated magnesium sulfate crystals, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$, are heated they give off water.



A student carries out an experiment to determine the value of x in $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

Step 1 Hydrated magnesium sulfate crystals were weighed.

Step 2 Hydrated magnesium sulfate crystals were heated.

Step 3 The remaining solid was weighed.

- (i) Describe how the student can ensure that all the water is given off.

.....

..... [2]

10. M/J 19/P41/Q4/e(ii)

- (a) Silver chloride, AgCl , is insoluble. It can be made by a precipitation reaction between aqueous barium chloride and a suitable aqueous silver salt.

- (i) Name a suitable silver salt to use to prepare silver chloride.

Complete the chemical equation to show the formation of insoluble silver chloride from aqueous barium chloride and the silver salt you have named.

name of a suitable silver salt



11. M/J 19/P43/Q3/d(i,ii)

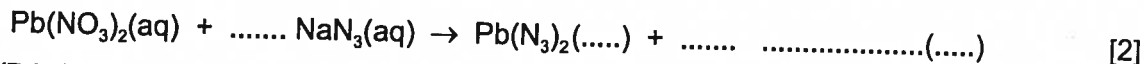
- (a) Lead(II) azide is insoluble in water. Solid lead(II) azide can be made in a precipitation reaction between aqueous lead(II) nitrate and aqueous sodium azide.

Lead(II) azide has the formula $\text{Pb}(\text{N}_3)_2$.

- (i) Deduce the formula of the azide ion.

..... [1]

- (ii) Complete the chemical equation for the reaction between aqueous lead(II) nitrate and aqueous sodium azide to form solid lead(II) azide and aqueous sodium nitrate. Include state symbols.



12. MAR 19/P42/Q4/b

- (a) How would you show that an aqueous solution of ethanoic acid is an acid **without** using an indicator or measuring the pH?

State the reagent you would use and give the expected observations. Write a chemical equation for the reaction that you describe.

- reagent

.....

- expected observations

.....

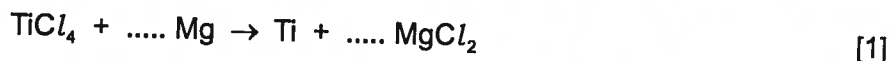
- chemical equation

..... [3]

13. MAR 19/P42/Q5/c(i,ii)

- (a) Titanium(IV) chloride, TiCl_4 , is heated with an excess of magnesium, in an atmosphere of argon.

- (i) Balance the chemical equation for the reaction.



- (ii) Titanium(IV) chloride can be reacted with sodium instead of magnesium.

The reaction between titanium(IV) chloride and sodium is similar to the reaction between titanium(IV) chloride and magnesium.

Write a chemical equation for the reaction between titanium(IV) chloride and sodium.

..... [1]

14. O/N 18/P42/Q3/c,d(i)

Sulfur is an important element.

- (a) Dilute sulfuric acid is used to make salts known as sulfates.

A method consisting of three steps is used to make zinc sulfate from zinc carbonate.

step 1 Add an excess of zinc carbonate to 20 cm^3 of 0.4 mol/dm^3 dilute sulfuric acid until the reaction is complete.

step 2 Filter the mixture.

step 3 Heat the filtrate until a saturated solution forms and then allow it to crystallise.

- (i) Name a suitable piece of apparatus for measuring 20 cm^3 of dilute sulfuric acid in **step 1**.

..... [1]

- (ii) State **two** observations which would show that the reaction is complete in **step 1**.

1

2 [2]

(iii) Why is it important to add an excess of zinc carbonate in **step 1**?

.....[1]

(iv) What is meant by the term *saturated solution* in **step 3**?

.....
.....[2]

(v) The equation for the reaction is shown.



Complete the equation by inserting the state symbol for zinc sulfate. [1]

(vi) Name another zinc compound which could be used to make zinc sulfate from dilute sulfuric acid using this method.

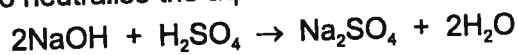
.....[1]

(vii) Suggest why this method would **not** work to make barium sulfate from barium carbonate and dilute sulfuric acid.

.....[1]

(b) In a titration, a student added 25.0 cm³ of 0.200 mol/dm³ aqueous sodium hydroxide to a conical flask. The student then added a few drops of methyl orange to the solution in the conical flask.

Dilute sulfuric acid was then added from a burette to the conical flask. The volume of dilute sulfuric acid needed to neutralise the aqueous sodium hydroxide was 20.0 cm³.



(i) What was the colour of the methyl orange in the aqueous sodium hydroxide?

.....[1]

15. M/J 18/P42/Q6

(a) All sodium salts are soluble in water. All nitrates are soluble in water. Barium carbonate is insoluble in water.

Describe how you would make a pure, dry sample of barium carbonate by precipitation.

Include:

- the names of the starting materials
- full practical details
- a chemical equation.

.....
.....
.....
.....
.....[5]

(b) Nitrates decompose when heated.

(i). Write a chemical equation for the decomposition of sodium nitrate when it is heated.

.....[2]

(ii) The unbalanced chemical equation for the decomposition of hydrated copper(II) nitrate crystals is shown.

Balance the chemical equation for this reaction.



(iii) When the hydrated copper(II) nitrate crystals are heated, steam is produced. When the steam condenses on a cool surface, it turns into a colourless liquid.

Anhydrous cobalt(II) chloride is used to show that the colourless liquid contains water.

How does the colour of the anhydrous cobalt(II) chloride change?

from to [2]

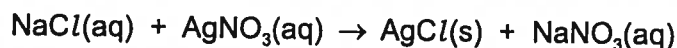
(iv) How would the student test to determine if the water produced in (b)(iii) is pure?

..... [1]

16. MAR 18/P42/Q2/c(i,ii,iii,v)

Sodium chloride is a typical ionic compound.

(a) Silver chloride can be made by reacting aqueous sodium chloride with aqueous silver nitrate. The other product of the reaction is sodium nitrate. The chemical equation for the reaction is shown.



A student attempted to make the maximum amount of **sodium nitrate** crystals. The process involved three steps.

step 1 The student added aqueous sodium chloride to aqueous silver nitrate and stirred. Neither reagent was in excess.

step 2 The student filtered the mixture. The student then washed the residue and added the washings to the filtrate.

step 3 The student obtained sodium nitrate crystals from the filtrate.

(i) Describe what the student observed in **step 1**.

..... [1]

(ii) Why was the residue washed in **step 2**?

..... [1]

(iii) Give the names of the **two** processes which occurred in **step 3**:

1

2 [2]

(v) Write a chemical equation for the action of heat on sodium nitrate crystals.

..... [2]

17. M/J 17/P42/Q3/d

This question is about nitrogen and some of its compounds.

- (a) Describe how you would prepare a pure dry sample of copper(II) nitrate crystals in the laboratory using dilute nitric acid and solid copper(II) carbonate.

Include a series of key steps in your answer.

You should include a chemical equation for the reaction.

.....

.....

.....

.....

.....

..... [6]

18. M/J 17/P42/Q4/a,b

Nickel, copper and zinc are three consecutive elements in the Periodic Table.

- (a) Nickel and copper are transition elements.

State **three** chemical properties of transition elements.

.....

.....

..... [3]

- (b) Copper(II) oxide is a basic oxide but zinc oxide is an amphoteric oxide. Both oxides are insoluble in water.

You are provided with a mixture of solid copper(II) oxide and solid zinc oxide. Describe how you would obtain a sample of copper(II) oxide from this mixture.

.....

.....

..... [3]

19. M/J 17/P42/Q6/b

- (a) Propan-1-ol is a member of the homologous series of alcohols. It reacts in the same way as ethanol with acidified potassium manganate(VII) and with carboxylic acids.

Name the **type** of compound that is formed when propan-1-ol is heated with

acidified potassium manganate(VII),

ethanoic acid and a suitable catalyst. [2]

20. M/J 16/P42/Q3/c

Gallium is a metallic element in Group III. It has similar properties to aluminium.

(a) Gallium(III) oxide, Ga_2O_3 , is amphoteric.

- (i) Write the chemical equation for the reaction between gallium(III) oxide and dilute nitric acid to form a salt and water only.

..... [2]

- (ii) The reaction between gallium(III) oxide and sodium hydroxide solution forms only water and a salt containing the negative ion $\text{Ga}_2\text{O}_4^{2-}$.

Write the chemical equation for this reaction.

..... [2]

21. M/J 16/P43/Q4/b

(a) Potassium iodide and lead nitrate are both soluble. Lead iodide is insoluble.

- (i) Describe how a pure dry sample of lead iodide could be made from solid potassium iodide and solid lead nitrate.

.....

.....

.....

..... [4]

- (ii) Write an ionic equation for the formation of lead iodide, PbI_2 , when potassium iodide and lead nitrate react with each other.

State symbols are **not** required.

..... [2]

Answers Section

1. O/N 22/P42/Q4

- (a) $\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$ 1
- (b) **M1** no (more) fizzing (1) 2
M2 solid stops dissolving or a solid remains / is visible (in the mixture) (1) 1
- (c) residue 1
- (d) (i) saturated 1
(ii) solubility (of MgSO_4 / solid) decreases (as temperature decreases) 1
- (e) (i) hydrated 1
(ii) **M1** $M_r \text{MgSO}_4 = 120$ (1)
M2 $\text{mol MgSO}_4 = 2.40 / M1 = 2.40 / 120 = 0.02(00)$ (1)
M3 $0.02(00) / 0.02(00) = 1$
 $0.140 / 0.02(00) = 7$ and $x = 7$ (1) 3
- (f) $2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$
M1 $\text{NO}_2 + \text{O}_2$ (1) 2
M2 equation completely correct (1)

2. O/N 22/P41/Q4

- (a) $\text{CaCO}_3 + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$
M1 H_2O and CO_2 as product (1) 2
M2 rest of equation correct (1)
- (b) **M1** fizzing / effervescence (1) 2
M2 solid disappears / dissolves (1) 1
- (c) filtrate 1
- (d) (i) **M1** a solution that can contain no more solute (1) 2
M2 at a given temperature (1) 1
(ii) cool the solution 1
- (e) (i) anhydrous 1
(ii) **M1** $M_r \text{Ca}(\text{NO}_3)_2 = 164$ (1)
M2 $\text{mol Ca}(\text{NO}_3)_2 = 2.46 / 164 = 0.015(00)$ (1)
M3 $0.015(00) / 0.015(00) = 1$
 $0.0600 / 0.015(00) = 4$ and $x = 4$ (1) 3
- (f) $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
M1 NaNO_2 on the right-hand side 2
M2 equation completely correct

3. O/N 21/P41/Q2/b,c

- (a) (i) **M1** white (1) 2
M2 to (light) blue (1) 1
- (ii) $5\text{H}_2\text{O}$ 1
- (iii) heating 1
- (iv) **M1** boiling point (1)
M2 is 100°C
OR
M1 freezing point (1) 2
M2 is 0°C (1) 1
- (b) (i) blue precipitate

- (ii) Alternative suggestion:
M1 $\text{Cu}(\text{OH})_2$ (as only product) (1)
M2 Cu^{2+} and 2OH^- (as reactants) (1)
M3 state symbols (1) 3
4. **O/N 21/P43/Q3/h**
- (a) (i) **M1** glowing splint (1)
M2 relights (1) 2
(ii) nitrogen dioxide/
nitrogen(IV) oxide 1
5. **O/N 21/P43/Q5**
- (a) (i) too many colour changes 1
(ii) Any acid-base indicator, e.g. methyl orange or phenolphthalein 1
- (b) **M1** repeat without indicator using same volumes **OR** remove indicator by adding charcoal or carbon and filtering (1)
M2 evaporate / heat / warm/
boil/leave in hot place (1)
M3 until most of the water is gone / some water left /
saturation(point) /
crystallisation (point) /
evaporate **some** of the water (1)
M4 cool /
leave to crystallise(1)
M5 description of drying (1) 5
- (c) (i) yellow flame 1
(ii) **M1** solid dissolves / disappears(1)
M2 blue solution(1) 2
(iii) white precipitate 1
6. **M/J 21/P41/Q5**
- (a) (add) **water** (to both salts) (1)
dissolve both salts / make **solutions** (1)
filter (lead(II) iodide)(1)
wash (residue of lead(II) iodide) with **water AND dry** e.g. with filter paper / description of
washing and drying (1)
 $\text{Pb}(\text{NO}_3)_2 + 2\text{NaI} \rightarrow 2\text{NaNO}_3 + \text{PbI}_2$
OR $\text{Pb}^{2+} + 2\text{I}^- \rightarrow \text{PbI}_2$ (1) 5
- (b) (i) glowing splint (1)
relights / rekindles (1) 2
(ii) **2ZnO(s) and 4NO₂(g)** (1)
12H₂O(g) (1) 2
- (c) (i) **heat again and weigh** again/ repeat steps 2 and 3 (1)
until mass is constant (1) 2
(ii) 0.005 (1)
0.9 (1)
 $(0.9 + 18 =) 0.05$ (1)
 $(0.05 + 0.005 =) 10$ (1) 4
7. **M/J 21/P43/Q5**
- (a) add zinc carbonate to sulfuric acid until
• it stops dissolving
OR

- no more effervescence (1)
- filter (zinc carbonate) (1)
- evaporation of filtrate to form dry crystals (1)
- $\text{ZnCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$
- CO_2 product in equation (1)
- rest of equation correct (1) 5
- (b) (i) acidified aqueous potassium manganate(VII) (1) 2
- purple to colourless (1)
- (ii) $2 \text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (1) 2
- $14 \text{H}_2\text{O}$ (1)
- (c) (i) heat + weigh + repeat (1) 2
- until mass is constant (1)
- (ii) M1 0.02 (1)
- M2 0.72 (1)
- M3 $M2 + 18 = 0.72/18 = 0.04$ (1) 4
- M4 $M3 + M1 = 0.04 + 0.02 = 2$ (1)
- 8. O/N 20/P42/Q2a,b(i,ii,v),d 1
- (a) HNO_3 1
- (b) (i) to make sure all the (sulfuric) acid reacts
- (ii) no (more) fizzing (1) 2
- (FeCO_3) stops dissolving or a solid remains / is visible (in the mixture) (1) 1
- (iii) iron(II) oxide / iron(II) hydroxide 1
- (c) (i) precipitation 1
- (ii) cream precipitate
- (iii) $\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$
- AgBr (as only product) (1)
- Ag^+ and Br^- (as reactants)(1) 3
- state symbols(1)
- 9. O/N 20/P43/Q4/d(i) 2
- (a) (i) heat again and weigh again / repeat steps 2 and 3 (1)
- until mass is constant (1)
- 10. M/J 19/P41/Q4/e(ii) 3
- (a) (i) (silver) nitrate (1)
- $\text{BaCl}_2 + 2\text{AgNO}_3 \rightarrow 2\text{AgCl} + \text{Ba}(\text{NO}_3)_2$
- formulae (1)
- balance(1)
- 11. M/J 19/P43/Q3/d(i,ii) 1
- (a) (i) N_3^-
- (ii) M1 state symbols on right correct (s) then (aq) 2
- M2 $(\text{Pb}(\text{NO}_3)_2 +) 2 (\text{NaN}_3) \rightarrow (\text{Pb}(\text{N}_3)_2 +) 2\text{NaNO}_3$
- 12. MAR 19/P42/Q4/b 3
- (a) M1 Suitable metal e.g. magnesium / any carbonate / any base(1)
- M2 suitable observation e.g. insoluble base / insoluble carbonate / metal dissolve or disappear or metal / carbonate bubbles(1)
- M3 balanced equation fully correct(1)

13. MAR 19/P42/Q5/c(i,ii)

- (a) (i) $\text{TiCl}_4 + 2\text{Mg} \rightarrow \text{Ti} + 2\text{MgCl}_2$ [1]
 (ii) $\text{TiCl}_4 + 4\text{Na} \rightarrow \text{Ti} + 4\text{NaCl}$ [1]

14. O/N 18/P42/Q3/c,d(i)

- (a) (i) Measuring cylinder [1]
 (ii) M1 No more fizzing;
 M2 (ZnCO_3) stops dissolving or a (white) solid remains / is visible [2]
 (iii) To use up all the acid / H^+ ions [1]
 (iv) M1 A solution that can hold no more solute
 M2 at the specified temperature [2]
 (v) (aq) [1]
 (vi) Zinc oxide or zinc hydroxide [1]
 (vii) Barium sulfate is insoluble [1]
 (b) (i) yellow [1]

15. M/J 18/P42/Q6

- (a) (mix) sodium carbonate **AND** barium nitrate / barium chloride [1]
 in solution / aqueous / dissolved (in water) [1]
 filter / centrifuge (barium carbonate) [1]
 wash (residue) **AND** dry / description of washing and drying [1]
 $\text{Ba}(\text{NO}_3)_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{BaCO}_3 + 2\text{NaNO}_3$ / $\text{Ba}^{2+} + \text{CO}_3^{2-} \rightarrow \text{BaCO}_3$
OR
 $\text{BaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{BaCO}_3 + 2\text{NaCl}$ [1]
 (b) (i) $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
 1 mark for either NaNO_2 or O_2 on the right-hand side
 1 mark for fully correct equation [2]
 (ii) $2\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}(\text{s}) \rightarrow 2\text{CuO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
 all 3 numbers = 2 marks
 any 2 numbers = 1 mark [2]
 (iii) blue [1]
 pink [1]
 (iv) boiling point sharp / melting point sharp / freezing point sharp / boiling point $100\text{ }(^{\circ}\text{C})$ /
 freezing point or melting point $0(^{\circ}\text{C})$ [1]

16. MAR 18/P42/Q2/c(i,ii,iii,v)

- (a) (i) white precipitate [1]
 (ii) to ensure all sodium nitrate / NaNO_3 was collected [1]
 (iii) M1 evaporation
 M2 crystallisation [2]
 (iv) $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
 M1 = NaNO_2
 M2 = rest of equation [2]

17. M/J 17/P42/Q3/d

(a) add copper(II) carbonate (to acid) until it stops dissolving or no more effervescence /bubbling /fizzing [1]

filter (to remove copper(II) carbonate)

evaporate / heat / warm / boil / leave in sun [1]

AND

until most of the water has gone / some water is left / evaporate **some** of the water /

until it is concentrated / saturation [1]

(point) / crystallisation point / crystals form on glass rod or microscope slide / crystals **start** to form (for any solution) leave / allow to cool / allow to crystallise

OR

(for any crystals) filter / wash / dry with filter paper / dry in warm place / dry in a (**low**) oven /

leave to dry [1]

formula of $\text{Cu}(\text{NO}_3)_2$ [1]

equation: $\text{CuCO}_3 + 2\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$ [1]

18. M/J 17/P42/Q4/a,b

(a) any 3 from:

- catalyst
- more than one / variable oxidation state / oxidation number / valency
- form coloured **compounds** / coloured **ions**
- forms complex ions / complexes [3]

(b) add sodium hydroxide (solution) / NaOH / potassium hydroxide (solution) / KOH [1]

zinc oxide dissolves / reacts **OR** copper(II) oxide does not dissolve / react [1]

filter / decant / centrifuge (copper(II) oxide) [1]

19. M/J 17/P42/Q6/b

(a) carboxylic acid / aldehyde [1]

ester [1]

20. M/J 16/P42/Q3/c

(a) (i) $\text{Ga}_2\text{O}_3 + 6\text{HNO}_3 \rightarrow 2\text{Ga}(\text{NO}_3)_3 + 3\text{H}_2\text{O}$
formula of $\text{Ga}(\text{NO}_3)_3$; all formulae and balancing correct; [2]

(ii) $\text{Ga}_2\text{O}_3 + 2\text{NaOH} \rightarrow \text{Na}_2\text{Ga}_2\text{O}_4 + \text{H}_2\text{O}$;
formula of $\text{Na}_2\text{Ga}_2\text{O}_4$; all formulae and balancing correct; [2]

21. M/J 16/P43/Q4/b

(a) (i) **M1** dissolve solids (in water) and mix / combine / add; [1]

M2 filter; [1]

M3 wash the residue (with water); [1]

M4 leave to dry / place in oven / dry between filter papers; [1]

(ii) $\text{Pb}^{2+} + 2\text{I}^- \rightarrow \text{PbI}_2$
formulae of ions correct; rest correct; [2]

8.4: Identification of ions and gases

1. O/N 21/P43/Q1

A list of substances is shown.

ammonia
bauxite
carbon dioxide
carbon monoxide
ethanol
hematite
oxygen
sodium chloride
sulfur dioxide

Answer the questions using the list of substances.

Each substance may be used once, more than once or not at all.

State which substance is:

- (a) an element [1]
- (b) an ore of aluminium [1]
- (c) a gas that causes acid rain [1]
- (d) used as a fuel [1]
- (e) an ionic compound [1]
- (f) produced in the Haber process [1]
- (g) a product of respiration [1]
- (h) a toxic product of the incomplete combustion of hydrocarbons
..... [1]
- (i) a gas produced in the test for nitrate ions. [1]

2. M/J 21/P42/Q2/c,d,e

(a) Aqueous silver nitrate is a colourless solution containing $\text{Ag}^+(\text{aq})$ ions.

(i) Describe what is seen when aqueous silver nitrate is added to aqueous sodium iodide, $\text{NaI}(\text{aq})$.

..... [1]

(ii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous sodium iodide.

Include state symbols.

..... [3]

- (b) In the positive test for aqueous nitrate ions, aqueous sodium hydroxide and one other substance are warmed with the nitrate ions.

Name this other substance and the gas formed.

name of substance

name of gas [2]

- (c) When silver nitrate is exposed to sunlight, silver is formed.
Name the type of reaction which needs light to make it happen.

..... [1]

3. M/J 20/P42/Q3/g(i)

- (a) Barium sulfate is an insoluble salt.

Barium sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

- (i) Name a solution that can be added to aqueous ammonium sulfate to produce a precipitate of barium sulfate.

..... [1]

4. M/J 20/P43/Q6/b

- (a) Describe the test for a nitrate ion.

test.....

.....

result [3]

5. MAR 20/P42/Q4/b,c

- (a) When aqueous sodium hydroxide is added to aqueous iron(II) sulfate, a precipitate forms.

- (i) What colour is this precipitate?

..... [1]

- (ii) Write the ionic equation for this reaction. Include state symbols.

..... [3]

- (b) Iron(II) sulfate can be converted to iron(III) sulfate by potassium manganate(VII) at room temperature.

- (i) What is the role of potassium manganate(VII) in this reaction?

..... [1]

- (ii) What condition must be used for this reaction to occur?

..... [1]

- (iii) In terms of electron transfer, what happens to the iron(II) ions in this reaction?

..... [1]

- (iv) State the colour change seen during this reaction.

from purple to [1]

6. O/N 19/P42/Q3/c

(a) When aqueous ammonia is added to aqueous iron(II) sulfate a green precipitate is seen. This green precipitate turns red-brown at the surface.

(i) Name the green precipitate.

..... [1]

(ii) Suggest why the green precipitate turns red-brown at the surface.

.....
..... [2]

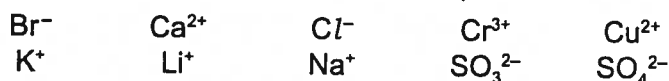
(iii) State what happens when an excess of aqueous ammonia is added to the green precipitate.

..... [1]

7. O/N 19/P41/Q1

This question is about ions and ionic compounds.

(a) Choose from the following list of ions to answer the questions.



Each ion may be used once, more than once or not at all.

State which ion:

(i) gives a lilac colour in a flame test [1]

(ii) forms a grey-green precipitate with aqueous ammonia [1]

(iii) forms a white precipitate with aqueous sodium hydroxide [1]

(iv) forms a cream precipitate with acidified aqueous silver nitrate [1]

(v) forms a white precipitate with acidified aqueous barium nitrate. [1]

(b) Describe how to do a flame test on a sample of a salt.

.....
..... [2]

(c) Magnesium phosphate contains magnesium ions, Mg^{2+} , and phosphate ions, PO_4^{3-} .
Deduce the formula of magnesium phosphate.

..... [1]

8. O/N 19/P41/Q4/a,b,c

Insoluble salts can be made by precipitation reactions.

A student mixed solutions of some soluble salts.

The results the student obtained are shown in the table.

		second salt solution		
		$\text{Co}(\text{NO}_3)_2(\text{aq})$	$\text{AgNO}_3(\text{aq})$	$\text{Pb}(\text{NO}_3)_2(\text{aq})$
first salt solution	$\text{NaI}(\text{aq})$	no change	yellow precipitate	yellow precipitate
	$\text{Na}_2\text{CO}_3(\text{aq})$	purple precipitate	yellow precipitate	white precipitate
	$\text{Na}_2\text{SO}_4(\text{aq})$	no change	white precipitate	white precipitate

All sodium salts are soluble in water.

Use only results from the table to answer the following questions.

(a) Name:

(i) an insoluble cobalt salt [1]

(ii) an insoluble yellow lead salt. [1]

(b) Write the chemical equation for the reaction in which silver carbonate is formed.

..... [2]

(c) Write the ionic equation for the reaction in which lead(II) iodide is formed.

..... [2]

9. M/J 19/P43/Q2/b(ii,iii)

(a) All isotopes of magnesium react with dilute hydrochloric acid to make hydrogen and a salt.

(i) Write a chemical equation for the reaction between magnesium and dilute hydrochloric acid.

..... [2]

(ii) Describe a test for hydrogen.

test

result [2]

10. MAR 19/P42/Q5/d(i)

(a) After titanium(IV) chloride is heated with magnesium, the unreacted magnesium is removed by adding an excess of dilute hydrochloric acid to the mixture.

The dilute hydrochloric acid also dissolves the magnesium chloride.

The dilute hydrochloric acid does **not** react with the titanium or dissolve it.

(i) Give **two** observations and write a chemical equation for the reaction that occurs when dilute hydrochloric acid reacts with magnesium.

1

2

chemical equation [3]

11. O/N 18/P42/Q2/e

Magnesium, calcium and strontium are Group II elements.

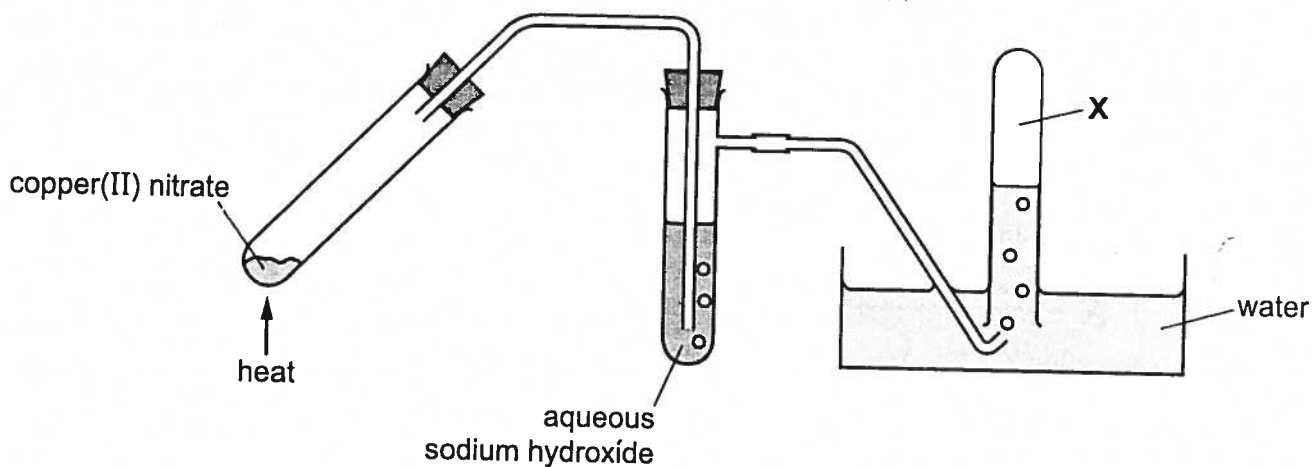
- (a) Aqueous magnesium chloride is added to aqueous silver nitrate. A white precipitate forms.
Write an **ionic** equation for this reaction. Include state symbols.

..... [2]

12. O/N 18/P41/Q3/a,d

- (a) Copper(II) nitrate decomposes when heated. Two gases, oxygen and nitrogen dioxide, and a solid are made in the reaction.

A sample of copper(II) nitrate was decomposed using the apparatus shown.



- (i) Complete the chemical equation for the reaction.



- (ii) Only oxygen gas is collected at X.

Explain why.

..... [1]

- (b) A sample of copper(II) nitrate was dissolved in water to form an aqueous solution.

The aqueous solution was split into three portions. A separate test was done on each portion as shown.

test	reagent added	result
1	aqueous sodium hydroxide	light blue precipitate forms
2	zinc powder	solution changes from blue to colourless and a brown solid forms
3		ammonia gas is produced

- (i) Give the formula of the light blue precipitate formed in **test 1**.

..... [1]

- (ii) Explain the changes seen in **test 2**.

.....
.....
..... [3]

- (iii) Identify the **two** reagents that must be added to the aqueous copper(II) nitrate in **test 3**.

1
2 [2]

13. O/N 17/P41/Q5/b

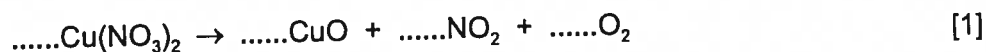
- (a) Copper(II) carbonate reacts with dilute nitric acid. One of the products of the reaction is a solution of copper(II) nitrate.

- (i) Describe tests for copper(II) ions and nitrate ions. Include the results of the tests.

copper(II) ions
.....
nitrate ions
..... [4]

- (ii) Copper(II) nitrate undergoes thermal decomposition.

Balance the chemical equation for the thermal decomposition of copper(II) nitrate.



14. O/N 16/P43/Q3/c

When lead(II) nitrate is heated, two gases are given off and solid lead(II) oxide remains. The equation for the reaction is shown.



- (a) Describe a test for oxygen.

test
result [2]

15. M/J 16/P42/Q6/d

Zinc is extracted from an ore called zinc blende, which consists mainly of zinc sulfide, ZnS.

- (a) When a sample of steel is added to dilute hydrochloric acid, an aqueous solution of iron(II) chloride, FeCl_2 , is formed.

When a sample of rust is added to dilute hydrochloric acid, an aqueous solution of iron(III) chloride, FeCl_3 , is formed.

- (i) Aqueous sodium hydroxide is added to the solutions of iron(II) chloride and iron(III) chloride.

Complete the table below, showing the observations you would expect to make.

	iron(II) chloride solution	iron(III) chloride solution
aqueous sodium hydroxide		

[2]

Solutions of iron(II) chloride and iron(III) chloride were added to solutions of potassium iodide and acidified potassium manganate(VII). The results are shown in the table.

	iron(II) chloride solution	iron(III) chloride solution
potassium iodide solution	no change	solution turns from colourless to brown
acidified potassium manganate(VII) solution	solution turns from purple to colourless	no change

- (ii) What **types** of substance cause potassium iodide solution to turn from colourless to brown?

..... [1]

- (iii) What **types** of substance cause acidified potassium manganate(VII) solution to turn from purple to colourless?

..... [1]

- (iv) Which **ion** in iron(III) chloride solution causes potassium iodide solution to turn from colourless to brown?

..... [1]

- (v) Which **ion** in iron(II) chloride solution causes acidified potassium manganate(VII) solution to turn from purple to colourless?

..... [1]

Answers Section

- | | |
|---|---|
| 1. O/N 21/P43/Q1 | 1 |
| (a) oxygen | 1 |
| (b) bauxite | 1 |
| (c) sulfur dioxide | 1 |
| (d) ethanol | 1 |
| (e) sodium chloride | 1 |
| (f) ammonia | 1 |
| (g) carbon dioxide | 1 |
| (h) carbon monoxide | 1 |
| (i) ammonia | 1 |
| 2. M/J 21/P42/Q2/c,d,e | 1 |
| (a) (i) yellow precipitate | 1 |
| (ii) $\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$ | |
| AgI (as only product) (1) | |
| Ag ⁺ and I ⁻ (as <i>only</i> reactants) (1) | |
| state symbols (1) | 3 |
| (b) aluminium (1) | 2 |
| ammonia (1) | 1 |
| (c) photochemical | 1 |
| 3. M/J 20/P42/Q3/g(i) | 1 |
| (a) (i) barium nitrate / barium chloride | 1 |
| 4. M/J 20/P43/Q6/b | 3 |
| (a) • add aqueous sodium hydroxide | |
| • then (reduction with) aluminium (foil) (and warm) | |
| • (ammonia gas produced which) turns damp red litmus blue | 3 |
| 5. MAR 20/P42/Q4/b,c | 1 |
| (a) (i) green | 1 |
| (ii) $\text{Fe}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Fe}(\text{OH})_2(\text{s})$ | |
| Fe(OH) ₂ (as only product) (1) | |
| Fe ²⁺ and 2OH ⁻ (as reactants) (1) | |
| state symbols (1) | 3 |
| (a) (i) oxidising agent | 1 |
| (ii) presence of an acid | 1 |
| (iii) lose an electron | 1 |
| (iv) colourless | 1 |
| 6. O/N 19/P42/Q3/c | 1 |
| (a) (i) iron(II) hydroxide | 1 |
| (ii) any two from: | |
| ∞ it (iron(II) hydroxide) is oxidised | |
| ∞ to form iron(III) (hydroxide) / (oxide) | |
| ∞ by (iron(II) hydroxide reacting with) air / oxygen | 2 |
| (iii) (green ppt) Remains | 1 |
| 7. O/N 19/P41/Q1 | 1 |
| (a) (i) K ⁺ | 1 |
| (ii) Cr ³⁺ | 1 |

- (iii) Ca^{2+} 1
 (iv) Br 1
 (v) SO_4^{2-} 1
- (b) (compound / salt) on wooden splint or (nichrome / platinum) wire (1)
 into (roaring) Bunsen flame (1) 2
- (c) $\text{Mg}_3(\text{PO}_4)_2$ 1
8. O/N 19/P41/Q4/a,b,c
- (a) (i) cobalt carbonate 1
 (ii) lead iodide 1
- (b) $2\text{AgNO}_3 + \text{Na}_2\text{CO}_3 \rightarrow \text{Ag}_2\text{CO}_3 + 2\text{NaNO}_3$
 formula of silver carbonate correct (1)
 fully correct equation (1) 2
- (c) $\text{Pb}^{2+} + 2\text{I}^- \rightarrow \text{PbI}_2$
 Pb^{2+} and I^- on left of equation (1)
 fully correct equation (1) 2
9. M/J 19/P43/Q2/b(ii,iii)
- (a) (i) $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$
 M1 MgCl_2 as product
 M2 fully correct equation 2
- (ii) M1 Test: lighted / burning splint
 M2 Result: (squeaky) pop 2
10. MAR 19/P42/Q5/d(i)
- (a) M1 / 2
 bubbles / fizzing / effervescence(1)
 M1 / 2 (magnesium or solid) dissolves / disappears / forms solution(1)
 M3
 $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2(1)$ 3
11. O/N 18/P42/Q2/e
- (a) $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
 M1 Species
 M2 States [2]
12. O/N 18/P41/Q3/a,d
- (a) (i) 4NO_2 2CuO
 M1 CuO as a product (1)
 M2 rest fully correct (1) 2
- (ii) nitrogen dioxide is acidic
 OR
 nitrogen dioxide reacts with sodium hydroxide 1
- (b) (i) $\text{Cu}(\text{OH})_2$ [1]
 (ii) Any three from:
 1 zinc more reactive than copper
 2 displacement / redox reaction OR zinc displaces copper OR zinc reacts with copper ions
 3 copper is solid / copper is brown
 4 zinc nitrate is colourless (solution)
 OR

- blue colour disappears because Cu^{2+} ions removed (from solution) [3]
- (iii) M1 sodium hydroxide / NaOH [2]
M2 aluminium / Al
13. O/N 17/P41/Q5/b [1]
(a) (i) (copper(II) ions) add sodium hydroxide (solution) [1]
(copper(II) ions) blue ppt. [1]
(nitrate ions) add aluminium AND aqueous sodium hydroxide AND warm [1]
ammonia given off / gas turns damp (red) litmus blue [1]
(ii) 2 / 2 / 4 / 1
14. O/N 16/P43/Q3/c
(a) test: glowing splint [2]
result: relights / rekindles
15. M/J 16/P42/Q6/d [1]
(a) (i) green precipitate; [1]
red-brown / brown / orange precipitate; [1]
(ii) oxidising agent / oxidant; [1]
(iii) reducing agent / reductant; [1]
(iv) iron(III) / Fe^{3+} ; [1]
(v) iron(II) / Fe^{2+} ; [1]



Unit 9

THE PERIODIC TABLE

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 9.1 The periodic table
- 9.2 Periodic trends
- 9.3 Group properties
- 9.4 Transition elements
- 9.5 Nobel gases

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Unit 9: The Periodic Table

9.1: The periodic table

1. O/N 22/P41/Q1

The names of the elements of Period 2 of the Periodic Table are shown.

lithium beryllium boron carbon nitrogen oxygen fluorine neon

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

Identify the element which:

(a) is a product of photosynthesis

..... [1]

(b) has an oxide found in clean, dry air

..... [1]

(c) forms a basic oxide with the formula X_2O

..... [1]

(d) is a main component of fertilisers used to improve crop growth

..... [1]

(e) has the highest rate of diffusion at room temperature

..... [1]

(f) produces a red flame in a flame test

..... [1]

(g) has only 5 electrons in each of its atoms

..... [1]

(h) has an oxide responsible for acid rain.

..... [1]

2. M/J 22/P42/Q1

The symbols of the elements of Period 3 of the Periodic Table are shown.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

Answer the following questions about these elements.

Each symbol may be used once, more than once or not at all.

Write the symbol of the element which:

(a) forms a stable ion with a 2+ charge [1]

(b) is the least reactive in the period [1]

(c) is used in water treatment [1]

(d) forms an oxide which is the main impurity in iron ore [1]

(e) is an important component of fertilisers [1]

(f) is stored under oil [1]

(g) is used in food containers [1]

(h) is found in the ore zinc blende. [1]

3. **M/J 21/P42/Q1**

The symbols of the elements of Period 3 of the Periodic Table are shown.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

Write the symbol of an element which:

(a) is malleable

..... [1]

(b) has only two electrons in its outermost shell

..... [1]

(c) forms an oxide which leads to acid rain

..... [1]

(d) forms an ion with a 2- charge

..... [1]

(e) is extracted from an ore called bauxite

..... [1]

(f) does not form an oxide

..... [1]

(g) forms an oxide with a macromolecular structure

..... [1]

(h) forms an amphoteric oxide

..... [1]

(i) exists as diatomic molecules

..... [1]

(j) forms a binary compound with hydrogen that is a strong acid.

..... [1]

4. O/N 20/P42/Q3/c

(a) When Group I metals are added to water they fizz and an alkaline solution forms.

(i) Name the gas given off.

..... [1]

(ii) Identify the ion present in the solution which makes the solution alkaline.

..... [1]

(iii) Write the chemical equation for the reaction between sodium and water.

..... [2]

5. O/N 20/P43/Q5/a,b,c

Group I elements, Group VII elements and transition elements are found in different parts of the Periodic Table.

(a) Describe the trend in the reactivity of Group I elements.

..... [1]

(b) When potassium is added to water a chemical reaction occurs.

(i) State **two** observations that can be made when potassium is added to water.

..... [1]

..... [2]

(ii) Write a chemical equation for the reaction of potassium with water.

..... [2]

(c) Excess aqueous potassium iodide is added to chlorine.

(i) Write a chemical equation for the reaction that occurs when aqueous potassium iodide is added to chlorine.

..... [2]

(ii) State the final colour of the reaction mixture.

..... [1]

6. MAR 20/P42/Q3/a,b

The Periodic Table is a method of classifying elements.

(a) Identify the element which is in Group VI and Period 4.

..... [1]

(b) Calcium is in Group II and chlorine is in Group VII of the Periodic Table.

Explain, in terms of number of outer shell electrons and electron transfer, how calcium atoms and chlorine atoms form ions. Give the formulae of the ions formed.

.....

.....

.....

.....

..... [5]

7. M/J 19/P42/Q6/a

The halogens are the elements in Group VII of the Periodic Table.

(a) Predict the physical state and colour of astatine at room temperature and pressure.

physical state

colour [2]

8. MAR 19/P42/Q1

Period 3 of the Periodic Table is shown.

sodium	magnesium	aluminium	silicon	phosphorus	sulfur	chlorine	argon
--------	-----------	-----------	---------	------------	--------	----------	-------

Answer the following questions using only these elements.

Each element may be used once, more than once or not at all.

State which element:

(a) is a gas at room temperature and pressure

..... [1]

(b) forms a basic oxide with a formula of the form X_2O

..... [1]

(c) is made of atoms which have a full outer shell of electrons

..... [1]

(d) forms an oxide which causes acid rain

..... [1]

(e) is extracted from bauxite

..... [1]

(f) forms an oxide which has a macromolecular structure

..... [1]

(g) consists of diatomic molecules.

..... [1]

9. MAR 19/P42/Q6/a,b,c,d

This question is about transition elements.

(a) Transition elements are harder and stronger than Group I elements.

Describe **two** other differences in **physical** properties between transition elements and Group I elements.

1

2 [2]

(b) State **one** physical property of transition elements that is similar to Group I elements.

..... [1]

(c) State **two** chemical properties of transition elements.

1

2 [2]

(d) Cobalt is a transition element. Anhydrous cobalt(II) chloride is used to test for water.

State the colour change that occurs when water is added to anhydrous cobalt(II) chloride.

from to [2]

10. MAR 18/P42/Q4/a

Ammonia is an important chemical.

(a) Ammonia is a base.

(i) In chemistry, what is meant by the term *base*?

..... [1]

(ii) Write a word equation to show ammonia behaving as a base.

.....

..... [2]

11. M/J 18/P42/Q2

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

(a) forms an oxide with a macromolecular structure

..... [1]

(b) is extracted from the ore bauxite

..... [1]

(c) is soft, metallic and stored in oil

..... [1]

(d) is a green gas at room temperature and pressure

..... [1]

(e) provides an inert atmosphere in lamps

..... [1]

(f) forms **two** different oxides during the Contact process

..... [1]

(g) is non-metallic and an important component of fertilisers.

..... [1]

12. M/J 18/P41/Q2/a,b,c,e

Flerovium, Fl, atomic number 114, was first made in research laboratories in 1998.

(a) Flerovium was made by bombarding atoms of plutonium, Pu, atomic number 94, with atoms of element Z.

- The nucleus of **one** atom of plutonium combined with the nucleus of **one** atom of element Z.
- This formed the nucleus of **one** atom of flerovium.

Suggest the identity of element Z.

..... [1]

(b) In which period of the Periodic Table is flerovium?

..... [1]

(c) Predict the number of outer shell electrons in an atom of flerovium.

..... [1]

(d) Only a relatively small number of atoms of flerovium have been made in the laboratory and the properties of flerovium have not yet been investigated.

It has been suggested that flerovium is a typical metal.

(i) Suggest **two** physical properties of flerovium.

1

2 [2]

(ii) Suggest **one** chemical property of flerovium oxide.

..... [1]

13. M/J 17/P42/Q5

(a) The elements in Group VII are known as the halogens. Some halogens react with aqueous solutions of halides.

(i) Complete the table by adding a ✓ to indicate when a reaction occurs and a ✗ to indicate when no reaction occurs.

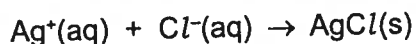
	aqueous potassium chloride	aqueous potassium bromide	aqueous potassium iodide
chlorine	x	✓	
bromine		x	
iodine			x

[3]

(ii) Write a chemical equation for the reaction between chlorine and aqueous potassium bromide.

..... [1]

- (b) A sample of vanadium chloride was weighed and dissolved in water. An excess of aqueous silver nitrate, acidified with dilute nitric acid, was added. A precipitate of silver chloride was formed. The ionic equation for this reaction is shown.



The mass of silver chloride formed was 2.87 g.

- (i) State the colour of the precipitate of silver chloride.

..... [1]

- (ii) The relative formula mass of silver chloride, AgCl , is 143.5.
Calculate the number of moles in 2.87 g of AgCl .

moles of AgCl = mol [1]

- (iii) Use your answer to (b)(ii) and the ionic equation to deduce the number of moles of chloride ions, Cl^- , that produced 2.87 g of AgCl .

moles of Cl^- = mol [1]

- (iv) The amount of vanadium chloride in the sample was 0.01 moles.

Use this and your answer to (b)(iii) to deduce the **whole number** ratio of moles of vanadium chloride : moles of chloride ions.

Deduce the formula of vanadium chloride.

moles of vanadium chloride : moles of chloride ions :

formula of vanadium chloride [2]

- (c) Astatine is at the bottom of Group VII. Use your knowledge of the properties of the halogens to

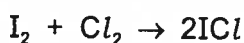
- (i) predict the physical state of astatine at room temperature and pressure,

..... [1]

- (ii) write a chemical equation for the reaction between sodium and astatine.

..... [2]

(d) Iodine reacts with chlorine. The chemical equation is shown.



Use the bond energies to answer the questions.

bond	bond energy in kJ/mol
I–I	151
Cl–Cl	242
I–Cl	208

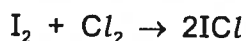
(i) Calculate the total amount of energy required to break the bonds in 1 mole of I_2 and 1 mole of Cl_2 .

..... kJ [1]

(ii) Calculate the total amount of energy given out when the bonds in 2 moles of ICl are formed.

..... kJ [1]

(iii) Use your answers to (d)(i) and (d)(ii) to calculate the overall energy change for the reaction.



..... kJ/mol [1]

14. M/J 16/P42/Q1

(a) For each of the following, give the name of an element from Period 2 (lithium to neon), which matches the description.

Elements may be used once, more than once or not at all.

(i) an element which is gaseous at room temperature and pressure

..... [1]

(ii) an element which forms an oxide that is a reactant in photosynthesis

..... [1]

(iii) an element that is a product of photosynthesis

..... [1]

(iv) an element that makes up approximately 78% by volume of the air

..... [1]

- (v) an element which has atoms with a full outer shell of electrons
..... [1]
- (vi) an element which exists as both diamond and graphite
..... [1]
- (vii) an element that reacts vigorously with cold water
..... [1]
- (viii) a soft metallic element which is stored in oil
..... [1]
- (b) Give the formula of a compound that contains
- (i) only boron and oxygen, [1]
- (ii) only lithium and nitrogen. [1]

Answers Section

- | | |
|--|---|
| 1. O/N 22/P41/Q1 | |
| (a) oxygen | 1 |
| (b) carbon | 1 |
| (c) lithium | 1 |
| (d) nitrogen | 1 |
| (e) neon | 1 |
| (f) lithium | 1 |
| (g) boron | 1 |
| (h) nitrogen | 1 |
| 2. M/J 22/P42/Q1 | |
| (a) Mg | 1 |
| (b) Ar | 1 |
| (c) Cl | 1 |
| (d) Si | 1 |
| (e) P | 1 |
| (f) Na | 1 |
| (g) Al | 1 |
| (h) S | 1 |
| 3. M/J 21/P42/Q1 | |
| (a) Na or Mg or Al | 1 |
| (b) Mg | 1 |
| (c) S | 1 |
| (d) S | 1 |
| (e) Al | 1 |
| (f) Ar | 1 |
| (g) Si | 1 |
| (h) Al | 1 |
| (i) Cl | 1 |
| (j) Cl | 1 |
| 4. O/N 20/P42/Q3/c | |
| (c) (i) hydrogen | 1 |
| (ii) hydroxide / OH | 1 |
| (iii) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ | |
| NaOH (as a product) (1) | |
| rest of equation (1) | 2 |
| 5. O/N 20/P43/Q5/a,b,c | |
| (a) become more reactive down the group ORA (1) | 1 |
| (b) (i) one mark each for any two of: | |
| • floats | |
| • dissolves / disappears / melts | |
| • moves | |
| • bubbles / fizzes / effervesces | |

- lilac flame 2
- (ii) $2K + 2H_2O \rightarrow 2KOH + H_2$
all formulae (1)
equation fully correct (1) 2
- (c) (i) $Cl_2 + 2KI \rightarrow 2KCl + I_2$
OR $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$
all formulae (1)
equation fully correct (1) 2
- (ii) brown / black 1
- 6. **MAR 20/P42/Q3/a,b** 1
 - (a) selenium / Se
 - (b) Ca has 2 and Cl has outer electrons 7 (1)
Ca (atoms) lose electrons (1)
Cl (atoms) gain electrons (1)
Ca²⁺ (ions) (1)
Cl⁻ (ions) (1) 5
- 7. **M/J 19/P42/Q6/a** 2
 - (a) M1 solid (1)
M2 black (1)
- 8. **MAR 19/P42/Q1**
 - (a) chlorine / argon 1
 - (b) sodium 1
 - (c) argon 1
 - (d) sulfur 1
 - (e) aluminium 1
 - (f) silicon 1
 - (g) chlorine 1
- 9. **MAR 19/P42/Q6/a,b,c,d**
 - (a) M1 Transition element has higher melting point / high boiling point **ORA**(1) 2
M2 Transition element has higher density **ORA**(1) 1
 - (b) good conduction of heat or electricity **OR** malleability **OR** ductility 1
 - (c) 1 mark each for any two of:
 - catalyst
 - more than one or different or variable oxidation state / oxidation number / valency 2
 - coloured **compounds** / coloured **ions** 2
 - (d) from blue(1) to pink(1)
- 10. **MAR 18/P42/Q4/a** 1
 - (a) (i) proton acceptor
 - (ii) ammonia + named acid $\rightarrow$ correct ammonium salt
M1 ammonium product (from ammonia / ammonium hydroxide + acid) 2
M2 fully correct equation
- 11. **M/J 18/P42/Q2**
 - (a) silicon / Si 1
 - (b) aluminium / Al 1
 - (c) sodium / Na 1
 - (d) chlorine / Cl₂ / Cl 1
 - (e) argon / Ar 1

- (f) sulfur / S 1
(g) phosphorus / P 1
12. M/J 18/P41/Q2/a,b,c,e
- (a) calcium / Ca 1
(b) 7 1
(c) 4 1
(d) (i) any **two** from:
high melting point / boiling point
hard
dense
conduct electricity
conduct heat
ductile / malleable
sonorous
lustrous / shiny 2
(ii) basic (oxide) 1
13. M/J 17/P42/Q5
- (a) (i)
- | | aqueous potassium chloride | aqueous potassium bromide | aqueous potassium iodide |
|----------|----------------------------|---------------------------|--------------------------|
| chlorine | | | ✓ |
| bromine | ✗ | | ✓ |
| iodine | ✗ | ✗ | |
- 5 cells completed correctly = [3]
3 or 4 cells completed correctly = [2]
2 cells completed correctly = [1] 3
- (ii) $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$
OR
 $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$ 1
- (b) (i) white 1
(ii) 0.02 (mol) 1
(iii) 0.02 (mol) 1
(iv) 1:2
 VCl_2 2
- (c) (i) solid 1
(ii) $2\text{Na} + \text{At}_2 \rightarrow 2\text{NaAt}$
M1 formula of NaAt
M2 equation fully correct 2
- (d) (i) 393 (kJ) 1
(ii) 416 (kJ) 1
(iii) -23 (kJ/mol) 1
14. M/J 16/P42/Q1
- (a) (i) nitrogen/oxygen/fluorine/neon; 1
(ii) carbon; 1

(iii) oxygen;	1
(iv) nitrogen;	1
(v) neon;	1
(vi) carbon;	1
(vii) lithium/fluorine;	1
(viii) lithium;	1
(b) (i) B_2O_3 ;	1
(ii) Li_3N ;	1

9.2: Periodic trends

1. O/N 20/P43/Q5/e

(a) Chromium is a transition element.

- Chromium has a high melting point.
- Chromium is a good conductor of electricity.
- Many chromium compounds are soluble in water.
- Hydrated chromium(III) sulfate is green.
- Chromium forms the chlorides CrCl_2 and CrCl_3 .
- Oxides of chromium act as catalysts in the manufacture of poly(ethene).

(i) Use this information to give **two** properties of chromium which are different from properties of Group I elements such as sodium.

1

2 [2]

(ii) Use this information to give **two** properties of chromium which are similar to properties of Group I elements such as sodium.

1

2 [2]

2. M/J 20/P43/Q7/e(ii)

(a) Iron is a transition element.

(i) Give **two** ways in which the properties of transition elements differ from the properties of Group I metals.

1

2 [2]

3. M/J 18/P42/Q2/a,c,e

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

(a) forms an oxide with a macromolecular structure

..... [1]

(b) is soft, metallic and stored in oil

..... [1]

(c) provides an inert atmosphere in lamps

..... [1]

Answers Section**1. O/N 20/P43/Q5/e**

(a) (i) one mark for each of any two from:

- (chromium has) high melting point **ORA**
- (chromium forms) coloured ions / coloured compounds **ORA**
- (chromium has) variable valency / variable oxidation state / variable oxidation number **ORA**
- catalytic behaviour **ORA**

ORA ALLOW group 1 **or** sodium **if stated**

- no colour **or** white **or** colourless ions or compounds
- fixed valency / +1 charge **only or one** oxidation state / forms **one** chloride
- low melting point
- doesn't behave as a catalyst

2

(ii) one mark for each of any two from:

- (chromium / sodium) **conducts** electricity
- (chromium / sodium) compounds are **soluble** (in water)
- (chromium / sodium) form hydrated salts / form hydrated compounds

2

2. M/J 20/P43/Q7/e(ii)

(a) (i) any **two** from:

- act as catalysts
- variable oxidation numbers
- form coloured compounds / coloured ions
- higher melting point
- higher density
- harder

2

3. M/J 18/P42/Q2/a,c,e

(a) silicon / Si

1

(b) sodium / Na

1

(c) argon / Ar

1

9.3: Group properties

1. M/J 22/P41/Q7

The Periodic Table can be used to classify elements.

(a) Group I elements react with cold water to form alkaline solutions.

(i) Place the Group I elements caesium, lithium, potassium, rubidium and sodium in their order of reactivity with water.

Put the most reactive element first.

most reactive
→
 least reactive

--	--	--	--	--

(ii) Name the alkaline solution formed when caesium reacts with cold water. [1]

..... [1]

(b) Group I elements have lower melting points than transition elements.

Describe one **other** difference in the **physical** properties of Group I elements and transition elements.

..... [1]

(c) Group VII elements are known as the halogens.

Astatine is below iodine in Group VII.

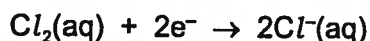
Predict the physical state of astatine at room temperature and pressure.

..... [1]

(d) Some Group VII elements react with aqueous solutions containing halide ions.

When aqueous chlorine is added to aqueous potassium bromide a reaction occurs.

The ionic half-equations for the reaction are shown.



(i) Describe the colour change of the solution.

original colour of potassium bromide solution

final colour of reaction mixture [2]

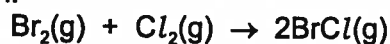
(ii) Identify the species that is oxidised.

Explain your decision.

species oxidised

explanation [2]

(e) Bromine monochloride, BrCl , is made by the reaction between bromine and chlorine. The chemical equation is shown.



bond	bond energy in kJ/mol
Br–Br	190
Cl–Cl	242
Br–Cl	218

Calculate the overall energy change for the reaction using bond energies.

Use the following steps.

- Calculate the total amount of energy required to break the bonds in 1 mole of $\text{Br}_2(\text{g})$ and 1 mole of $\text{Cl}_2(\text{g})$.

..... kJ

- Calculate the total amount of energy released when the bonds in 2 moles of $\text{BrCl}(\text{g})$ are formed.

..... kJ

- Calculate the overall energy change for the reaction.

..... kJ/mol [3]

2. M/J 22/P43/Q6

The Periodic Table can be used to classify elements.

(a) The Group I metals react with cold water. Transition elements do not react with cold water.

- (i) Describe two **other** differences in the **chemical** properties between Group I metals and transition elements.

1

2 [2]

- (ii) Describe the observations when potassium is added to cold water. Write a balanced equation for the reaction. Include state symbols.

observations

.....

.....

.....

equation [5]

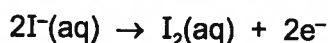
- (b) Transition elements are stronger than Group I metals.

Describe two **other** differences in the **physical** properties of Group I metals and transition elements.

1

2 [2]

- (c) Some Group VII elements react with aqueous solutions containing halide ions.
When aqueous bromine is added to aqueous potassium iodide a reaction occurs.
The ionic half-equations for the reaction are shown.



- (i) Describe the colour change of the solution.

original colour of potassium iodide solution

final colour of reaction mixture [2]

- (ii) State the name of the general term given to the type of reaction in which electrons are transferred from one species to another.

..... [1]

- (iii) Identify the oxidising agent in this reaction. Give a reason for your answer.

oxidising agent

reason [2]

- (d) Use the key to complete the table to show the results of adding aqueous halogens to aqueous solutions of halides. One has been completed for you.

		halides		
		KCl(aq)	KBr(aq)	KI(aq)
halogens	Cl ₂ (aq)			
	Br ₂ (aq)			✓
	I ₂ (aq)			

key
✓ = reaction
x = no reaction

[2]

3. O/N 21/P42/Q5

Iron is a transition element. Potassium is a Group I element.

- (a) Iron and potassium have the same type of bonding.

Name and describe the type of bonding in these two elements.

name

description

.....

..... [4]

(b) Transition elements and Group I elements have some similar physical properties.

They can both:

- be hammered into a shape
- conduct electricity
- be stretched into wires.

(i) Name the term used to describe the ability of elements to be hammered into a shape.

..... [1]

(ii) Describe what happens to the particles in iron when it is hammered into a shape.

..... [1]

(iii) Suggest why copper, rather than other transition elements, is used for wires which conduct electricity.

..... [1]

(c) Transition elements are harder and stronger than Group I elements.

Describe how **two** other **physical** properties of transition elements are different from those of Group I elements.

1

2 [2]

(d) Chemical properties of some Group I elements are shown in the table.

element	reaction with cold water	reaction with oxygen	flame test colour
lithium	• steadily effervesces • forms a colourless solution	very slowly forms an oxide layer	red
sodium	• strongly effervesces • forms a colourless solution	slowly forms an oxide layer	
potassium	• very strongly effervesces • forms a colourless solution	quickly forms an oxide layer	
rubidium			ruby red

(i) Add to the table:

- the flame test colours for sodium and potassium
- the predicted reactions of rubidium with water and with oxygen.

[4]

(ii) Name the gas produced when Group I elements react with water.

..... [1]

(iii) Name the solution formed when potassium reacts with water.

..... [1]

(iv) Predict the pH of the colourless solution formed when potassium reacts with water.

..... [1]

(v) Write the chemical equation for the reaction of sodium with oxygen.

..... [2]

(e) Iron is a typical transition element. It is the catalyst used in the Haber process.

(i) Write the equation for the reaction that occurs in the Haber process.

..... [2]

(ii) State the temperature and pressure used in the Haber process. Include units.

temperature

pressure [2]

4. MAR 19/P42/Q6/a,b,c

This question is about transition elements.

(a) Transition elements are harder and stronger than Group I elements.

Describe **two** other differences in **physical** properties between transition elements and Group I elements.

1

2 [2]

(b) State **one** physical property of transition elements that is similar to Group I elements.

..... [1]

(c) State **two** chemical properties of transition elements.

1

2 [2]

5. MAR 19/P42/Q3/a,b,c,d

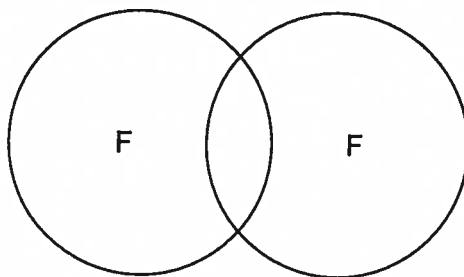
Fluorine is a Group VII element. Fluorine forms compounds with metals and non-metals.

(a) Predict the physical state of fluorine at room temperature and pressure.

..... [1]

(b) Fluorine exists as diatomic molecules.

Complete the dot-and-cross diagram to show the electron arrangement in a molecule of fluorine. Show outer shell electrons only.



[2]

(c) Write a chemical equation for the reaction between sodium and fluorine.

..... [2]

(d) Explain why chlorine does **not** react with aqueous sodium fluoride.

..... [1]

6. M/J 18/P42/Q4/a,b,d,e,f

Potassium reacts with bromine at room temperature to form potassium bromide.

(a) Write a chemical equation for this reaction. Include state symbols.

..... [3]

(b) Potassium bromide exists as an ionic lattice.

Potassium bromide does **not** conduct electricity when solid but does conduct electricity when molten.

(i) What is meant by the term *ionic lattice*?

.....

..... [2]

(ii) Explain why potassium bromide does **not** conduct electricity when solid but does conduct electricity when molten.

.....

..... [2]

(c) Iodine reacts with chlorine to form iodine monochloride, ICl , as the only product.

(i) Write a chemical equation for this reaction.

..... [2]

(ii) Draw a dot-and-cross diagram to show the electron arrangement in a molecule of iodine monochloride. Show outer shell electrons only.

[2]

(d) Potassium bromide has a melting point of 734°C .

Iodine monochloride has a melting point of 27°C .

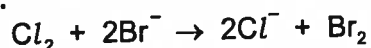
In terms of attractive forces, explain why there is a large difference between these melting points.

.....

.....

..... [3]

(e) When chlorine gas is passed through aqueous potassium bromide, a redox reaction occurs. The ionic equation is shown.



(i) Write an ionic half-equation showing what happens to the chlorine molecules, Cl_2 , in this reaction.

..... [1]

(ii) Explain why the bromide ions, Br^- , act as reducing agents in this reaction.

..... [1]

7. M/J 18/P42/Q7/d,f

Many organic compounds, such as alcohols, carboxylic acids and esters, contain the elements carbon, hydrogen and oxygen only.

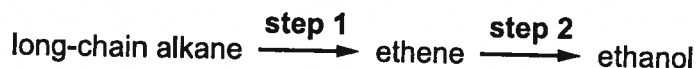
(a) Compound **W** has the molecular formula $\text{C}_2\text{H}_6\text{O}$. Compound **W** reacts when heated with ethanoic acid and a catalyst to produce a sweet-smelling liquid.

(i) Give the name of the homologous series to which compound **W** belongs.

..... [1]

(ii) Draw the structure of compound **W**. Show all of the atoms and all of the bonds.

(b) Ethanol can be produced from long-chain alkanes as shown. [1]



Describe the **two-stage** manufacture of ethanol from the long-chain alkane octane, C_8H_{18} . Include:

- the names of the types of chemical reactions that occur
- reaction equations
- reaction conditions.

step 1

.....

step 2

.....

..... [5]

8. M/J 18/P43/Q3/c

Cobalt is a transition element. Potassium is in Group I of the Periodic Table.

(a) When a small piece of potassium is added to cold water, the potassium floats and disappears as it reacts.

Give **two** other observations that would be made when a small piece of potassium is added to cold water.

1

2 [2]

9. M/J 17/P42/Q2/c

Carbon and silicon are elements in Group IV of the Periodic Table. Both carbon and silicon exist as more than one isotope.

(a) Silicon has a giant structure which is similar to the structure of diamond.

(i) Name the type of bond which is present between silicon atoms in silicon.

..... [1]

(ii) Suggest **two** physical properties of silicon.

Use your knowledge of structure and bonding to explain why silicon has these physical properties.

property 1

reason 1

property 2

reason 2 [4]

10. M/J 17/P42/Q5/a,c

(a) The elements in Group VII are known as the halogens. Some halogens react with aqueous solutions of halides.

(i) Complete the table by adding a ✓ to indicate when a reaction occurs and a X to indicate when no reaction occurs.

	aqueous potassium chloride	aqueous potassium bromide	aqueous potassium iodide
chlorine	X	✓	
bromine		X	
iodine			X

[3]

(ii) Write a chemical equation for the reaction between chlorine and aqueous potassium bromide.

..... [1]

(b) Astatine is at the bottom of Group VII. Use your knowledge of the properties of the halogens to

(i) predict the physical state of astatine at room temperature and pressure,

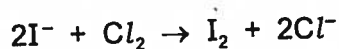
..... [1]

(ii) write a chemical equation for the reaction between sodium and astatine.

..... [2]

11. M/J 16/P43/Q4/c

(a) When chlorine gas is bubbled through an aqueous solution of potassium iodide, a redox reaction takes place.



(i) State the colour change expected in this reaction.

start colour

end colour [2]

(ii) Identify the reducing agent in this reaction. Explain your answer.

.....

..... [2]

Answers Section

1. M/J 22/P41/Q7

- (a) (i) from left to right
caesium → rubidium → potassium → sodium → lithium 1
- (ii) caesium hydroxide 1
- (b) Group I element is less strong / not strong
ORA
OR
Group I element has low(er) density ORA
OR
Group I element is soft(er) ORA 1
- (c) solid 1
- (d) (i) colourless (1)
orange / brown / yellow (1) 2
- (ii) Br⁻ (1)
loses electron(s) (1) 2
- (e) 432(1)
436(1)
- 4(1) 3

2. M/J 22/P43/Q6

- (a) (i) Group 1 metals do not show catalytic behaviour
Group 1 have fixed oxidation states 2
- (ii) any 2 observations from:
- moves / floats
 - dissolves / disappears
 - bubbles / effervescence / fizzes
 - lilac flame
 - explodes
 - melts / forms a spherical shape
- $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
KOH or H₂ product (1) in equation
fully correct equation (1)
state symbols (1) 5
- (b) transition elements have high(er) melting point (1)
transition elements have high(er) density (1) 2
- (c) (i) colourless (1)
brown (1) 2
- (ii) redox 1
- (iii) Br₂ (1)
(bromine) is reduced (1) 2
- (d) two ticks in first row (1)
three crosses in the other three boxes (1) 2

3. O/N 21/P42/Q5

- (a) metallic (1)
(lattice of) positive ions (1)
sea of / delocalised / mobile electrons (1)
attraction between positive ions and electrons (1) 4
- (b) (i) malleability / malleable 1
(ii) (particles) slide (over each other) 1
(iii) unreactive 1
- (c) high(er) density
high(er) melting points 2
- (d) (i) Na – yellow (1)
K – lilac (1)
colourless solution (1)
clear idea of increased reactivity in **both** reactions (1) 4
(ii) hydrogen 1
(iii) potassium hydroxide 1
(iv) any number in the range $7 < \text{pH} \leq 14$ 1
(v) $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$
 Na_2O (1) correct equation (1) 2
- (e) (i) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
species (1) correct equation (1) 2
(ii) 450°C (1)
200 atm (1) 2

4. MAR 19/P42/Q6/a,b,c

- (a) M1 Transition element has higher melting point / high boiling point **ORA**(1)
M2 Transition element has higher density **ORA**(1) 2
- (b) good conduction of heat or electricity **OR** malleability **OR** ductility 1
- (c) 1 mark each for any two of:
 ∞ catalyst
 ∞ more than one or different or variable oxidation state / oxidation number / valency
 ∞ coloured **compounds** / coloured **ions** 2

5. MAR 19/P42/Q3/a,b,c,d

- (a) gas / gaseous 1
- (b) M1 1 shared pair of electrons(1)
M2 6 non-bonding electrons on each atom to complete an octet(1) 2
- (c) $2\text{Na} + \text{F}_2 \rightarrow 2\text{NaF}$
M1 NaF anywhere(1) M2 equation fully correct(1) 2
- (d) chlorine less reactive than fluorine **ORA** 1

6. M/J 18/P42/Q4/a,b,c,d,e,f

- (a) $2\text{K(s)} + \text{Br}_2(\text{l}) \rightarrow 2\text{KBr(s)}$
1 mark for formulae all correct
1 mark for balancing
1 mark for state symbols 3
- (b) (i) (ionic): made of, positive and negative ions / anions and cations /
oppositely charged ions / unlike charged ions / different charged ions 1
(lattice): regular / sequence / pattern / alternating / repeated / framework /
ordered / organised / network / uniform 1

- (ii) (in solid) ions don't move 1
(when molten) ions move / ions mobile 1
- (c) (i) substance that conducts electricity / (undergoes) electrolysis 1
decomposed / chemically changed 1
OR 1
molten or liquid or solution or aqueous **AND** containing ions/or ionic 1
- (ii) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
1 mark for $\text{H}^+ + \text{e}^-$ as the only species on the left
1 mark for equation fully correct
1 mark for bromine at the anode
1 mark for potassium hydroxide 4
- (iii) potassium 1
- (d) (i) $\text{I}_2 + \text{Cl}_2 \rightarrow 2\text{ICl}$
1 mark for formulae all correct
1 mark for correct balancing 2
- (ii) one bonding pair 1
6 non-bonding electrons on each atom 1
- (e) (potassium bromide): ionic bonds / attraction between ions 1
(iodine monochloride): intermolecular forces / forces between molecules /
named intermolecular forces, e.g. van der Waals / London forces /
dispersion forces / dipole- dipole 1
bonds in KBr are stronger / need more energy to break bonds / ORA 1
- (f) (i) $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ 1
(ii) (bromide ions) lose electrons / donate electrons / are oxidised 1
7. M/J 18/P42/Q7/d,f 1
- (a) (i) alcohol / alkanol 1
- (ii)
- $$\begin{array}{c}
 \text{H} \quad \text{H} \\
 | \quad | \\
 \text{H} - \text{C} - \text{C} - \text{O} - \text{H} \\
 | \quad | \\
 \text{H} \quad \text{H}
 \end{array}$$
- 1
- (b) (step 1) crack/cracking (of octane) 1
(step 1) equation with only C_8H_{18} on left hand side and C_2H_4
+ other correct product(s) on right hand side e.g. $\text{C}_8\text{H}_{18} \rightarrow \text{C}_2\text{H}_4 + \text{C}_6\text{H}_{14}$ 1
- (step 2) hydration / addition 1
(step 2) one correct condition for either process required 1
(cracking): 450 (°) C to 800 (°)C / zeolites / aluminosilicates / silica / SiO_2 /
aluminium oxide / Al_2O_3 / alumina / china / broken
pot / chromium oxide / Cr_2O_3 / up to 70 atmospheres
(hydration): phosphoric acid / H_3PO_4 / 300 (°)C / 60 atmospheres
 $\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{OH} / \text{CH}_3\text{CH}_2\text{OH}$ 1

8. M/J 18/P43/Q3/c

(a) any two from:

potassium melts / potassium forms a ball
fizzes / bubbles
potassium moves
(lilac) flame

2

9. M/J 17/P42/Q2/c

(a) (i) covalent

1

(ii) award 1 mark for each correct property and one mark for each correct matching reason.

property: high melting point / high boiling point

reason: bonds between atoms are strong OR covalent bonds are strong /
bonds need large amount of energy to break

property: non-conductor / poor conductor (of electricity) / insulator

reason: no moving charged particles / no moving ions / no moving electrons /
all (outer shell) electrons used in bonding

property: hard

reason: bonds between atoms are strong OR covalent bonds are strong

property: brittle

reason: bonds between atoms are strong OR covalent bonds are strong /
bonds are directional

property: insoluble

reason: does not form hydrogen bonds with water / no ions that can be hydrated

4

10. M/J 17/P42/Q5/a,c

(a) (i)

	aqueous potassium chloride	aqueous potassium bromide	aqueous potassium iodide
chlorine			✓
bromine	x		✓
iodine	x	x	

5 cells completed correctly = [3]

3 or 4 cells completed correctly = [2]

2 cells completed correctly = [1]

3

(ii) $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$

OR

 $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$

1

(b) (i) solid

1

(ii) $2\text{Na} + \text{At}_2 \rightarrow 2\text{NaAt}$

M1 formula of NaAt

M2 equation fully correct

2

11. M/J 16/P43/Q4/c

(a) (i) start colour: colourless;
end colour: brown;

2

(ii) M1 iodide / I ;

M2 it is oxidised OR it loses electrons / it increases oxidation number /
it reduces the chlorine;

2

9.4: Transition elements

1. M/J 22/P42/Q3

Transition elements are found in the middle block of the Periodic Table.

(a) Chromium has several isotopes. Manganese has only one isotope.

(i) State what is meant by the term *isotopes*.

.....
..... [2]

(ii) State the nucleon number of manganese.

..... [1]

(iii) Complete the table to show the number of protons, neutrons and electrons in a $^{52}_{24}\text{Cr}^{3+}$ ion.

protons	neutrons	electrons

[3]

(b) One chemical property of transition elements is that they form coloured compounds.

(i) Give the colours of the following hydrated salts.

- hydrated copper(II) sulfate
- hydrated cobalt(II) chloride [2]

(ii) State two **other** chemical properties of transition elements.

- 1
- 2 [2]

(c) Transition elements and Group I elements are metals. They share many physical properties including the ability to:

- conduct electricity
- be hammered into shape.

(i) Explain why transition elements and Group I elements conduct electricity.

..... [1]

(ii) State the property that describes a material which can be hammered into shape.

..... [1]

(d) Transition elements and Group I elements differ in other physical properties. Transition elements are harder and stronger than Group I elements.

Describe two **other** ways in which the physical properties of transition elements differ from Group I elements.

- 1
- 2 [2]

2. MAR 19/P42/Q6/d

This question is about transition elements.

- (a) Cobalt is a transition element. Anhydrous cobalt(II) chloride is used to test for water.

State the colour change that occurs when water is added to anhydrous cobalt(II) chloride.

from to [2]

3. M/J 18/P41/Q2/e

Flerovium, Fl, atomic number 114, was first made in research laboratories in 1998.

- (a) Only a relatively small number of atoms of flerovium have been made in the laboratory and the properties of flerovium have not yet been investigated.

It has been suggested that flerovium is a typical metal.

- (i) Suggest **two** physical properties of flerovium.

1

2 [2]

- (ii) Suggest **one** chemical property of flerovium oxide.

..... [1]

4. M/J 18/P41/Q3/a,b

This question is about iron.

- (a) Three of the raw materials added to a blast furnace used to extract iron from hematite are coke, hematite and limestone.

Name **one** other raw material added to the blast furnace.

..... [1]

- (b) A series of reactions occurs in a blast furnace during the extraction of iron from hematite. Describe these reactions.

Include:

- **one** chemical equation for the reduction of hematite
- **one** chemical equation for the formation of slag.

.....

.....

.....

..... [5]

5. M/J 18/P41/Q5/a,b

- (a) The table gives some chemical properties of transition elements and their compounds, and of Group I elements and their compounds.

chemical property	transition elements	Group I elements
ability to act as catalysts	yes	no
exist as coloured compounds	yes	no

(i) What is meant by the term *catalyst*?

..... [2]

(ii) Give **one** other chemical property shown by transition elements which is **not** shown by Group I elements.

..... [1]

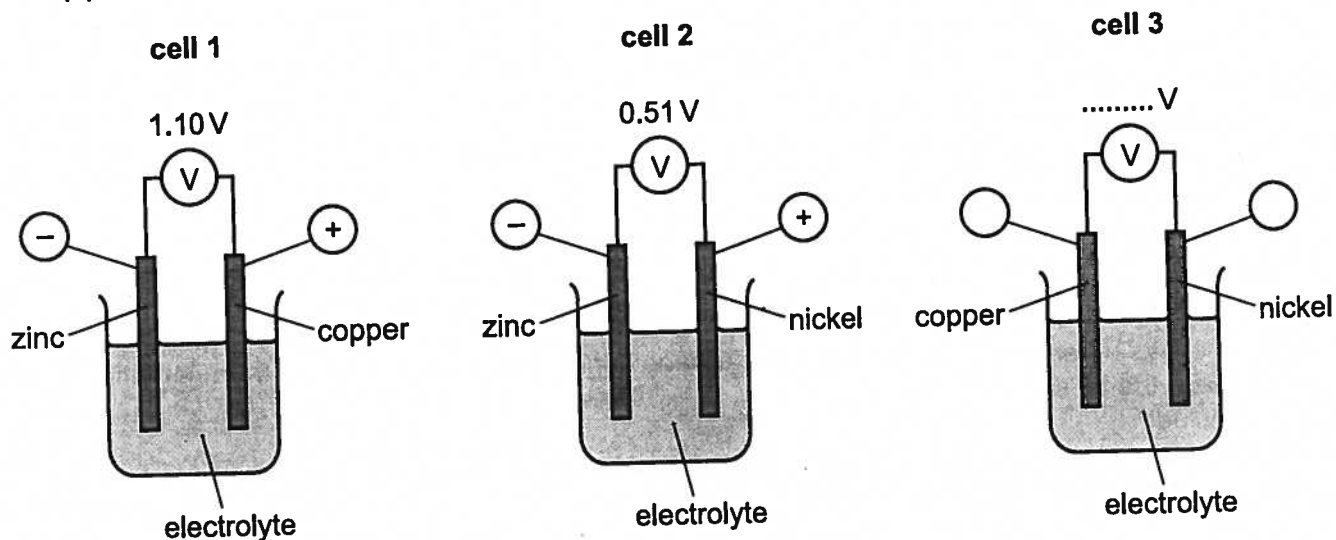
(b) Give **two** physical properties shown by transition elements which are **not** shown by Group I elements.

1 [2]

2 [2]

6. M/J 17/P42/Q4/c

(a) Three cells are set up each using two metals.



(i) Write the ionic half-equation for the reaction occurring at the zinc electrode in **cell 1**.

..... [2]

(ii) Put the **three** metals, copper, nickel and zinc, in order of reactivity.

most reactive

↓

least reactive

[1]

(iii) Complete the labelling in **cell 3** by writing the polarity (+/-) of each electrode in the circles and calculating the reading on the voltmeter. [2]

Answers Section

1. **M/J 22/P42/Q3**
 - (a) (i) **atom(s)** of the same element
with different number of neutrons 2
 - (ii) 55 1
 - (iii) 24 protons; 28 neutrons; 21 electrons 3
 - (b) (i) blue
pink 2
 - (ii) (act as) catalysts
variable oxidation state 2
 - (c) (i) mobile electrons 1
 - (ii) malleability / malleable 1
 - (d) high(er) density
high(er) melting points 2
2. **MAR 19/P42/Q6/d**
 - (a) from blue(1) to pink(1) 2
3. **M/J 18/P41/Q2/e**
 - (a) (i) any **two** from:
high melting point / boiling point
hard
dense
conduct electricity
conduct heat
ductile / malleable
sonorous
lustrous / shiny 2
 - (ii) basic (oxide) 1
4. **M/J 18/P41/Q3/a,b**
 - (a) (hot) air 1
 - (b) coke is burned (to form carbon dioxide)
OR
 $C + O_2 \rightarrow CO_2$ 1
carbon dioxide is reduced by (more) coke to form carbon monoxide or CO
OR
 $C + CO_2 \rightarrow 2CO$ 1
 $3CO + Fe_2O_3 \rightarrow 2Fe + 3CO_2$ 1
limestone (decomposes to) form lime / CaO / calcium oxide (and carbon dioxide)
OR
 $CaCO_3 \rightarrow CaO + CO_2$ 1
 $CaO + SiO_2 \rightarrow CaSiO_3$ 1
5. **M/J 18/P41/Q5/a,b**
 - (a) (i) (a substance which) increases the rate of a reaction
without being used up (at the end) / remains unchanged or unaffected or without
changing mass 1
 - (ii) variable oxidation states 1

- (b) any **two** from:
high(er) melting point / boiling point
(very) hard(er)
(very) strong(er)
dense(r)

2

6. M/J 17/P42/Q4/c

- (a) (i) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
M1 formula of Zn^{2+} on the right-hand side
M2 equation fully correct
- (ii) zinc/Zn
nickel/Ni
copper/Cu
- (iii) copper (+) and nickel (–)
0.59 V

2

1

1

1

9.5: Noble gases

1. M/J 18/P43/Q1/d

The following are the symbols and formulae of some elements and compounds.

Ar Ca(OH)₂ Cl₂ CO₂ Cu Fe SO₂ V₂O₅

Answer the following questions using only the elements or compounds in the list.

Each element or compound may be used once, more than once or not at all.

State which element or compound is used:

(a) as an inert atmosphere in lamps [1]

2. M/J 16/P41/Q2/h

Period 3 contains the elements sodium to argon. This question asks about the chemistry of each of the Period 3 elements or their compounds.

(a) Argon is an unreactive noble gas.

(i) Explain why argon is unreactive.

..... [1]

(ii) Give one use of argon.

..... [1]

Answers Section

- | | |
|--|---|
| 1. M/J 18/P43/Q1/d | 1 |
| (a) Ar / argon | |
| 2. M/J 16/P41/Q2/h | 1 |
| (a) (i) it has a complete outer shell / a full outer shell / 8 electrons in the outer shell; | |
| (ii) (in) lamps; | 1 |



Unit 10

METALS

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 10.1 Properties of metals
- 10.2 Reactivity series
- 10.3 Extraction of metals
- 10.4 Uses of metals

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Unit 10: Metals

10.1: Properties of metals

1. O/N 20/P42/Q3/a,b

Group I metals are very reactive. Transition elements are also metals but are less reactive than Group I metals.

(a) State **two** physical properties of Group I metals which are similar to those of transition metals.

1

2 [2]

(b) Describe **two** ways in which the physical properties of Group I metals are different from those of transition metals.

1

2 [2]

2. M/J 20/P43/Q7/e(i)

(a) Iron is a transition element.

(i) Iron forms hydrated iron(III) oxide when it rusts.

Write a word equation to represent the formation of rust.

..... [2]

3. O/N 19/P41/Q3/a

This question is about metals and metal oxides.

(a) Most metals have a high melting point.

State **one** other physical property that all metals have.

..... [1]

4. M/J 19/P42/Q7/d,e

(a) Nickel is a transition element. Nickel is stronger than sodium.

Describe **two** other differences in the physical properties of nickel and sodium.

1

2 [2]

(b) Predict **one** difference in the appearance of aqueous solutions of nickel compounds compared to aqueous solutions of sodium compounds.

..... [1]

5. MAR 19/P42/Q3/d

(a) Explain why chlorine does **not** react with aqueous sodium fluoride.

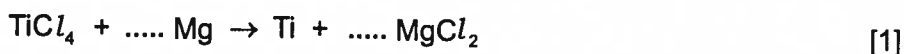
..... [1]

6. MAR 19/P42/Q5/c,d

Titanium is extracted from an ore called rutile. Rutile is an impure form of titanium(IV) oxide, TiO_2 .

(a) Titanium(IV) chloride, TiCl_4 , is heated with an excess of magnesium, in an atmosphere of argon.

(i) Balance the chemical equation for the reaction.



(ii) Titanium(IV) chloride can be reacted with sodium instead of magnesium.

The reaction between titanium(IV) chloride and sodium is similar to the reaction between titanium(IV) chloride and magnesium.

Write a chemical equation for the reaction between titanium(IV) chloride and sodium.

..... [1]

(iii) Suggest why the reaction between titanium(IV) chloride and magnesium is done in an atmosphere of argon and **not** in air.

..... [1]

(b) After titanium(IV) chloride is heated with magnesium, the unreacted magnesium is removed by adding an excess of dilute hydrochloric acid to the mixture.

The dilute hydrochloric acid also dissolves the magnesium chloride.

The dilute hydrochloric acid does **not** react with the titanium or dissolve it.

(i) Give **two** observations and write a chemical equation for the reaction that occurs when dilute hydrochloric acid reacts with magnesium.

1

2

chemical equation [3]

(ii) Name the process that is used to separate the titanium from the mixture after all the magnesium has been removed.

..... [1]

(iii) Titanium does not react with the dilute hydrochloric acid or dissolve in it.

Suggest why titanium does **not** react with dilute hydrochloric acid.

..... [1]

7. M/J 16/P42/Q3/d

Gallium is a metallic element in Group III. It has similar properties to aluminium.

(a) Alloys of gallium and other elements are often more useful than the metallic element itself.

Suggest **two** reasons why alloys of gallium are more useful than the metallic element.

.....

..... [2]

Answers Section

1. **O/N 20/P42/Q3/a,b**
 - (a) any **two** from:
 - shiny / lustrous
 - conduct electricity
 - conduct heat
 - (b) low(er) density (1)
 - low(er) melting points (1)
2. **M/J 20/P43/Q7/e(i)**
 - (a) (i) iron + water + oxygen → (hydrated) iron oxide
3. **O/N 19/P41/Q3/a**
 - (a) malleable / conduct electricity / conduct heat
4. **M/J 19/P42/Q7/d,e**
 - (a) **nickel** has
 - M1** higher density (1) **ORA**
 - nickel** has
 - M2** higher melting point / boiling point (1) **ORA**
 - (b) solutions of nickel compounds are coloured **ORA**
5. **MAR 19/P42/Q3/d**
 - (a) chlorine less reactive than fluorine **ORA**
6. **MAR 19/P42/Q5/c,d**
 - (a) (i) $\text{TiCl}_4 + 2\text{Mg} \rightarrow \text{Ti} + 2\text{MgCl}_2$
 - (ii) $\text{TiCl}_4 + 4\text{Na} \rightarrow \text{Ti} + 4\text{NaCl}$
 - (iii) magnesium burns in air or oxygen **OR** reacts with air or oxygen / argon is unreactive or inert
 - (b) (i) **M1 / 2**
 - bubbles / fizzing / effervescence(1)
 - M1 / 2** (magnesium or solid) dissolves / disappears / **forms** solution(1)
 - M3**
 - $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2(1)$
 - (ii) filtration
 - (iii) titanium is below hydrogen in the reactivity series **ORA**
 - OR** titanium less reactive than hydrogen **ORA**
 - OR** titanium coated with an oxide layer
7. **M/J 16/P42/Q3/d**
 - (a) any 2 from:
 - (do not) corrode;
 - strong;
 - hard;
 - (improved) appearance;

10.2: Reactivity series

1. M/J 22/P41/Q1

A list of substances is shown.

aluminium oxide	carbon dioxide	chlorine	diamond	ethanol
glucose	iron(III) oxide	limestone	nitrogen	oxygen

Answer the questions using the list of substances.

Each substance may be used once, more than once or not at all.

State which of the substances:

(a) is a reactant in photosynthesis

..... [1]

(b) is the main constituent of bauxite

..... [1]

(c) are **two** products of fermentation

..... and [2]

(d) is used as a fuel

..... [1]

(e) is a gas used to convert iron into steel

..... [1]

(f) is a greenhouse gas

..... [1]

(g) is a gas that is approximately 78% of clean, dry air

..... [1]

(h) is a form of carbon.

..... [1]

2. O/N 21/P41/Q1

Some elements are shown in the order they appear in the reactivity series. The most reactive element is at the top.

sodium
calcium
magnesium
aluminium
zinc
iron
hydrogen
copper

(a) Answer the questions using the list of elements. Each element may be used once, more than once or not at all.

Identify:

- (i) a non-metal [1]
- (ii) a metal which is stored under oil [1]
- (iii) the main component of steel [1]
- (iv) a metal with three electrons in the outer shell of its atoms [1]
- (v) a metal found in brass [1]
- (vi) a metal that forms chlorides of the type XCl_2 and XCl_3 [1]
- (b) Name the main ores of:
- (i) zinc [1]
- (ii) aluminium. [1]
- (c) In an experiment, a sample of aluminium appeared less reactive than expected.
Explain why.
..... [1]
- (d) Name **two** metals from the list which are extracted by reduction of their ores using carbon.
1
2 [2]
- (e) When zinc granules are added to aqueous copper(II) sulfate, a reaction occurs. During the reaction, a red-pink solid is formed and the solution becomes colourless.
- (i) Name the red-pink solid.
..... [1]
- (ii) Name the colourless solution.
..... [1]
- (iii) Explain, in terms of particles, why the rate of this reaction increases when the temperature is increased.
.....
.....

..... [1]

- (iii) Zinc has a melting point of 420 °C and a boiling point of 907 °C. The temperature inside the furnace is 1200 °C.

Explain how this information shows that the zinc produced inside the furnace is a gas.

..... [1]

- (iv) The gaseous zinc is converted to molten zinc.
Name this change of state.

..... [1]

5. O/N 19/P42/Q1/b

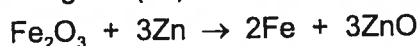
- (a) State which metal in the first 36 elements:

(i) is the Group I element which reacts most vigorously with water [1]

(ii) reacts with air to form lime. [1]

6. O/N 19/P41/Q3/c(i)

- (a) Iron can be obtained by heating iron(III) oxide with zinc powder.

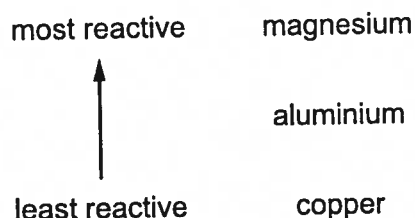


- (i) What can be deduced about the reactivity of zinc from this reaction?

..... [1]

7. O/N 19/P43/Q3/c,d

- (a) The positions of some common metals in the reactivity series are shown.



- (i) When magnesium is placed in aqueous copper(II) sulfate a displacement reaction occurs immediately.

Write an ionic equation for the reaction. Include state symbols.

..... [2]

- (ii) State **two** observations you would make when magnesium is placed in aqueous copper(II) sulfate.

1

2 [2]

- (iii) When aluminium foil is added to aqueous copper(II) sulfate no immediate reaction takes place.

Explain why.

..... [1]

- (b) Aluminium powder reacts with iron(III) oxide to produce aluminium oxide and iron.
Write a chemical equation for this reaction.

..... [2]

8. M/J 19/P42/Q6/b(i),c

- (a) When chlorine reacts with aqueous potassium bromide a displacement reaction occurs.
(i) Describe the colour change of the solution.

from to

[2]

- (b) Reactions occur when some aqueous solutions of halogens are added to aqueous solutions of halides.

Use the key to complete the table to show the results of adding halogens to halides.

key

✓ = reaction

x = no reaction

		halides		
		KCl(aq)	KBr(aq)	KI(aq)
halogens	Cl ₂ (aq)		✓	
	Br ₂ (aq)			
	I ₂ (aq)			

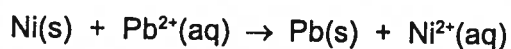
[2]

9. M/J 19/P42/Q7/a(i),c

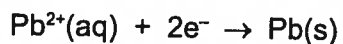
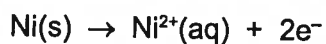
- (a) Displacement reactions occur between metals and metal ions.

Displacement reactions can be used to determine the order of reactivity of metals such as lead (Pb), nickel (Ni), and silver (Ag).

The ionic equation for a displacement reaction is shown.



The ionic half-equations for this reaction are shown.



The ionic half-equations show that electrons are donated by nickel atoms and accepted by lead ions.

- (i) Identify the reducing agent in the displacement reaction. Give a reason for your answer.

reducing agent.....

reason.....

[2]

- (b) Use the information in (a) and (b) to put the **three** metals lead, nickel and silver in order of reactivity.

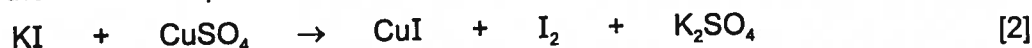
	most reactive
	↑
	least reactive

[1]

10. M/J 19/P41/Q3/e

- (a) Aqueous potassium iodide reacts with aqueous copper(II) sulfate to produce iodine.

- (i) Balance the chemical equation for this reaction.



- (ii) Deduce the charge on the copper ion in CuI.

..... [1]

- (iii) In terms of electron transfer, explain why copper is reduced in this reaction.

..... [1]

- (iv) Identify the reducing agent.

..... [1]

11. M/J 19/P43/Q3/a

- (a) (i) Sodium is in Group I of the Periodic Table.

Describe **two** physical properties of sodium which are different from the physical properties of transition elements such as copper.

1

2 [2]

- (ii) Sodium reacts rapidly with water.

Give **one** observation made when sodium is added to water.

..... [1]

12. MAR 19/P42/Q1/e

Period 3 of the Periodic Table is shown.

sodium	magnesium	aluminium	silicon	phosphorus	sulfur	chlorine	argon
--------	-----------	-----------	---------	------------	--------	----------	-------

State which element:

- (a) is extracted from bauxite

..... [1]

13. MAR 19/P42/Q5/d(iii)

- (a) After titanium(IV) chloride is heated with magnesium, the unreacted magnesium is removed by adding an excess of dilute hydrochloric acid to the mixture.

The dilute hydrochloric acid also dissolves the magnesium chloride.

The dilute hydrochloric acid does **not** react with the titanium or dissolve it.

- (i) Titanium does not react with the dilute hydrochloric acid or dissolve in it.
Suggest why titanium does **not** react with dilute hydrochloric acid.

..... [1]

14. O/N 18/P42/Q2/c

Magnesium, calcium and strontium are Group II elements.

(a) Calcium reacts with cold water to form two products:

- a colourless gas, **P**, which 'pops' with a lighted splint
- a weakly alkaline solution, **Q**, which turns milky when carbon dioxide is bubbled through it.

(i) Name gas **P**.

..... [1]

(ii) Identify the ion responsible for making solution **Q** alkaline.

..... [1]

(iii) Suggest the pH of solution **Q**.

..... [1]

(iv) Write a chemical equation for the reaction of calcium with cold water.

..... [2]

15. M/J 18/P42/Q2/c

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

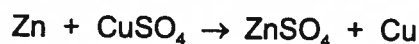
State which Period 3 element:

(a) is soft, metallic and stored in oil

..... [1]

16. M/J 18/P43/Q2/c

(a) When a piece of zinc metal is added to copper(II) sulfate solution there is an immediate reaction.



When a piece of aluminium metal is added to copper(II) sulfate solution the initial reaction is very slow.

(i) Explain why zinc metal reacts with copper(II) sulfate.

..... [1]

(ii) What type of reaction is this?

..... [1]

- (iii) Explain why the initial reaction between aluminium metal and copper(II) sulfate is very slow.

..... [1]

17. O/N 17/P41/Q3/b

- (a) The metal iron and the alloy steel are commonly used materials. A problem with them is that they rust.

- (i) How does painting iron and steel prevent rusting?

..... [1]

- (ii) Magnesium blocks can be attached to the bottom of steel boats.

Explain how the magnesium blocks prevent the whole of the bottom of the boat from rusting.

..... [2]

- (iii) Replacing the magnesium blocks with copper blocks does not prevent rusting. Explain why the copper blocks do **not** prevent rusting.

..... [1]

18. O/N 17/P41/Q5/a

- (a) Solid copper(II) carbonate undergoes thermal decomposition. One of the products of the thermal decomposition is copper(II) oxide.

- (i) State the colour change of the solid seen during the reaction.

start colour

end colour [1]

- (ii) Write a chemical equation for the thermal decomposition of copper(II) carbonate.

..... [1]

19. O/N 17/P43/Q5/a

- (a) (i) Name the products formed when sodium nitrate is heated.

..... [2]

- (ii) When copper(II) nitrate, $\text{Cu}(\text{NO}_3)_2$, undergoes thermal decomposition, three products are formed. One of the products is nitrogen dioxide, NO_2 .

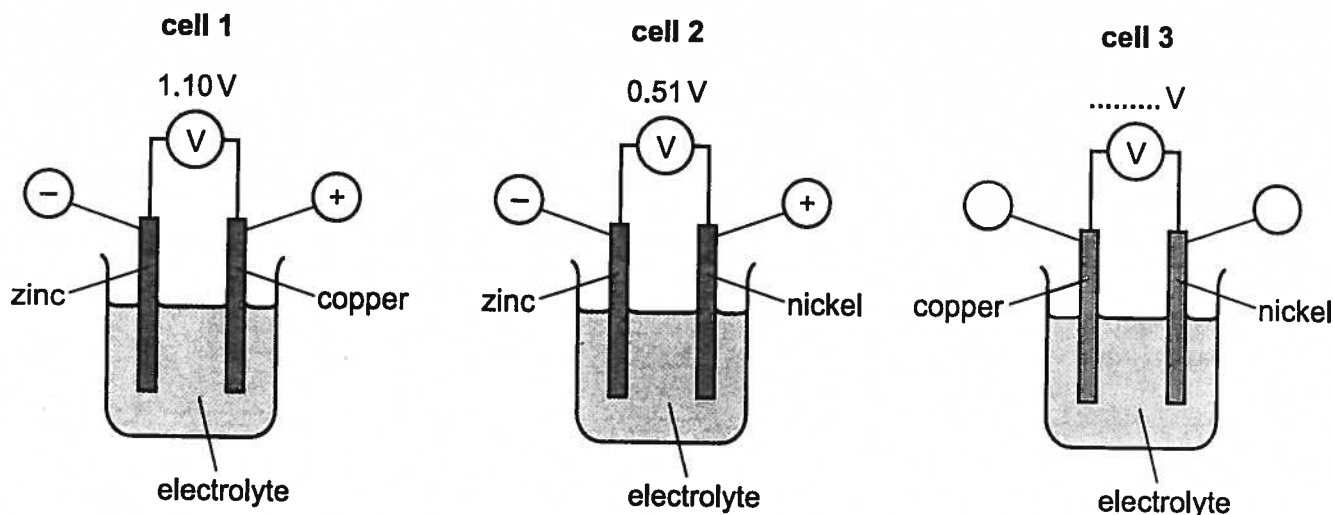
Write a chemical equation for the thermal decomposition of copper(II) nitrate.

..... [2]

20. M/J 17/P42/Q4/c

Nickel, copper and zinc are three consecutive elements in the Periodic Table.

(a) Three cells are set up each using two metals.



(i) Write the ionic half-equation for the reaction occurring at the zinc electrode in **cell 1**.

..... [2]

(ii) Put the **three** metals, copper, nickel and zinc, in order of reactivity.

most reactive

↓

.....

least reactive

[1]

(iii) Complete the labelling in **cell 3** by writing the polarity (+/–) of each electrode in the circles and calculating the reading on the voltmeter.

[2]

Answers Section

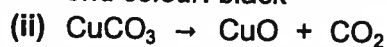
- | | |
|---|---|
| 1. M/J 22/P41/Q1 | 1 |
| (a) carbon dioxide | 1 |
| (b) aluminium oxide | |
| (c) ethanol (1) | 2 |
| carbon dioxide (1) | 1 |
| (d) ethanol | 1 |
| (e) oxygen | 1 |
| (f) carbon dioxide | 1 |
| (g) nitrogen | 1 |
| (h) diamond | 1 |
| 2. O/N 21/P41/Q1 | 1 |
| (a) (i) hydrogen | 1 |
| (ii) sodium | 1 |
| (iii) iron | 1 |
| (iv) aluminium | 1 |
| (v) zinc or copper | 1 |
| (vi) iron | 1 |
| (b) (i) zinc blende | 1 |
| (ii) bauxite | 1 |
| (c) protective oxide layer | 1 |
| (d) Any 2 from 3 | |
| • zinc | |
| • iron | |
| • copper | 2 |
| (e) (i) copper | 1 |
| (ii) zinc sulfate | 1 |
| (iii) M1 particles have more energy | |
| M2 More collisions (between particles) occur per second / per unit time | |
| M3 A greater percentage / proportion / fraction of collisions (of particles) are successful / | |
| have energy above activation energy / have energy equal to activation energy | 3 |
| (iv) Any 2 from 3: | |
| • use a catalyst | |
| • use smaller granules | |
| • increase concentration | 2 |
| 3. O/N 21/P43/Q3/g | |
| (a) M1 $\text{Mg} + \text{Pb}^{2+} \rightarrow \text{Mg}^{2+} + \text{Pb}$ (1) | 2 |
| M2 no reaction (1) | |
| 4. O/N 20/P43/Q4/a,b | 1 |
| (a) sulfur dioxide | |
| (b) (i) Any two from | |
| (coke) | |
| • releases heat or releases energy (when it reacts with oxygen or burns in air) / (acts as | |
| a) fuel / increases temperature (in the furnace) / heats (the furnace) / source of energy | |
| (coke) | |

- reduces zinc oxide / is a reducing agent / converts zinc oxide to zinc / removes oxygen from zinc oxide (coke)
- (reacts with oxygen) to produce carbon monoxide / reacts with carbon dioxide to form carbon monoxide
- carbon monoxide reduces zinc oxide / converts zinc oxide to zinc / removes oxygen from zinc oxide 2
- (ii) $\text{ZnO} + \text{CO} \rightarrow \text{Zn} + \text{CO}_2$
- OR
- $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$
- OR
- $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ 1
- (iii) temperature inside furnace is above / higher than 907°C 1
- OR temperature is above / higher than the boiling point (of zinc) **ORA**
- OR 1200°C is above / higher than the boiling point (of zinc) **ORA**
- OR 1200°C is above / higher than 907°C **ORA** 1
- (iv) condensation / condensing 1
- 5. O/N 19/P42/Q1/b
- (a) (i) potassium / K 1
- (ii) calcium / Ca 1
- 6. O/N 19/P41/Q3/c(i)
- (a) (i) (zinc is) more reactive than iron 1
- 7. O/N 19/P43/Q3/c,d
- (a) (i) $\text{Mg(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Cu(s)} + \text{Mg}^{2+}(\text{aq})$
- ionic equation correct (1)
- state symbols (1) 2
- (ii) any two from:
- ∞ solid dissolves / disappears
- ∞ blue colour of solution fades **OR** paler solution **OR** colour of solution disappears **OR** becomes colourless solution
- ∞ pink or orange or brown **AND** solid 2
- (iii) unreactive coating of aluminium oxide 1
- (b) $2\text{Al} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$
- Fe_2O_3 and Al_2O_3 both correct (anywhere) (1)
- Equation completely correct (1) 2
- 8. M/J 19/P42/Q6/b(i),c
- (a) (i) **M1** colourless (1)
- M2** to brown / orange / yellow (1) 2
- (b) **M1** two ticks for Cl_2 / KI, Br_2 / KI (1)
- M2** three crosses for Br_2 / KCl I_2 / KCl and I_2 / KBr (1) 2
- 9. M/J 19/P42/Q7/a(i),c
- (a) (i) **M1** Ni / Nickel (1)
- M2** (it) loses or donates electrons (1) 2
- (b) most reactive nickel / Ni
- lead / Pb
- least reactive silver / Ag 1

10. **M/J 19/P41/Q3/e** 2
 (a) (i) $4\text{KI} + 2\text{CuSO}_4 \rightarrow 2\text{CuI} + \text{I}_2 + 2\text{K}_2\text{SO}_4$ (2) 1
 (ii) 1+ 1
 (iii) gains electron(s) 1
 (iv) KI / potassium iodide / iodide (ions) / I^-
11. **M/J 19/P43/Q3/a**
 (a) (i) any two from:
 ∞ low melting point or low boiling point
 ∞ soft 2
 ∞ low density
 (ii) any one from:
 ∞ fizz / bubbles
 ∞ moves
 ∞ floats
 ∞ melts / forms a ball 1
 ∞ gets smaller / disappears
12. **MAR 19/P42/Q1/e** 1
 (a) aluminium
13. **MAR 19/P42/Q5/d(iii)**
 (a) (i) titanium is below hydrogen in the reactivity series **ORA**
OR titanium less reactive than hydrogen **ORA** 1
OR titanium coated with an oxide layer
14. **O/N 18/P42/Q2/c**
 (a) (i) Hydrogen [1]
 (ii) Hydroxide OR OH^- [1]
 (iii) $7 < \text{pH} < 12$ [1]
 (iv) $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{H}_2$
 M1 $\text{Ca}(\text{OH})_2$ [2]
 M2 Rest of equation
15. **M/J 18/P42/Q2/c** [1]
 (a) sodium / Na
16. **M/J 18/P43/Q2/c**
 (a) (i) zinc is more reactive than copper [1]
 (ii) displacement / redox [1]
 (iii) (aluminium) has (inert) coating of aluminium oxide [1]
17. **O/N 17/P41/Q3/b**
 (a) (i) prevents air / oxygen **AND** water from reaching the steel [1]
 (ii) magnesium is more reactive than iron / steel [1]
 the magnesium corrodes (before the iron / steel) **OR** [1]
 the magnesium corrodes preferentially [1]
 (iii) copper is less reactive than iron / steel [1]

18. O/N 17/P41/Q5/a

(a) (i) *start colour:* green
end colour: black



[1]

[1]

19. O/N 17/P43/Q5/a

(a) (i) oxygen/ O_2

sodium nitrite / sodium nitrate(III) / NaNO_2

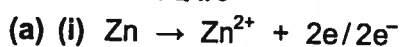
[2]



M1 CuO

M2 rest of equation fully correct

[2]

20. M/J 17/P42/Q4/c

M1 formula of Zn^{2+} on the right-hand side

M2 equation fully correct

[2]

(ii) zinc/Zn

nickel/Ni

copper/Cu

[1]

(iii) copper (+) and nickel (–)

[1]

0.59 V

[1]

10.3: Extraction of metals

1. O/N 22/P43/Q3

Aluminium is extracted from its ore by electrolysis.

(a) Name the ore of aluminium which consists mainly of aluminium oxide.

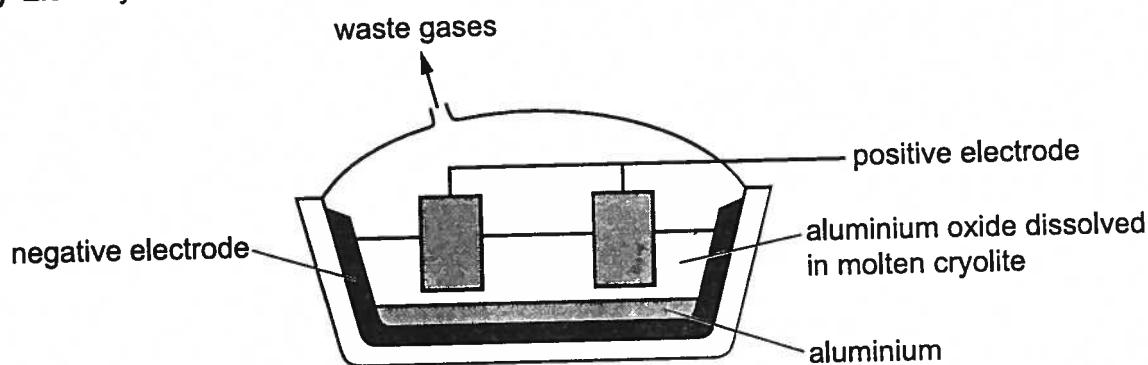
..... [1]

(b) State what is meant by the term *electrolysis*.

.....

..... [2]

(c) Electrolysis is carried out on aluminium oxide dissolved in molten cryolite.



(i) Give **two** reasons why the electrolysis is carried out on aluminium oxide dissolved in molten cryolite instead of electrolysis molten aluminium oxide only.

1

2 [2]

(ii) Write the ionic half-equation for the reaction occurring at the negative electrode.

..... [2]

(iii) The positive-electrodes are made of carbon.

Explain why the positive carbon electrodes are replaced regularly.

.....

..... [2]

(d) Aluminium is more reactive than copper.

When aluminium is added to aqueous copper(II) sulfate, no immediate reaction is seen.

Explain why.

..... [1]

(e) Aluminium reacts with oxygen to form an amphoteric oxide.

- (i) State what is meant by the term *amphoteric*.

.....
..... [1]

- (ii) The reaction between aluminium oxide and aqueous sodium hydroxide forms a salt containing the negative ion AlO_2^- . The only other product is water.

Write a chemical equation for the reaction between aluminium oxide and aqueous sodium hydroxide.

..... [2]

- (f) Gallium is in the same group as aluminium and forms similar compounds.

Predict the formulae of:

gallium(III) chloride

gallium(III) sulfate. [2]

2. M/J 22/P41/Q6

This question is about zinc and its compounds.

- (a) Zinc is extracted from its ore which is mainly zinc sulfide, ZnS .

The steps for this extraction are shown.

step 1 Zinc sulfide is converted into zinc oxide.

step 2 The zinc oxide is then reduced to zinc in a furnace. The zinc formed becomes a gas.

step 3 The zinc gas is cooled to form molten zinc.

- (i) Name the ore of zinc, which is mainly zinc sulfide.

..... [1]

- (ii) Describe how zinc sulfide is converted into zinc oxide in **step 1**.

..... [1]

- (iii) Name the reducing agent used in **step 2**.

..... [1]

- (iv) Explain why the zinc forms a gas in **step 2** inside the furnace.

..... [1]

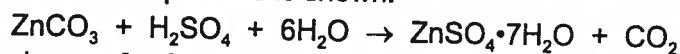
- (v) State the name of the physical change occurring when zinc gas is converted into molten zinc.

..... [1]

- (b) Zinc sulfate crystals, $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, are hydrated.

Zinc sulfate crystals are made by reacting zinc carbonate with dilute sulfuric acid.

The equation for the overall process is shown.



step 1 Large pieces of solid zinc carbonate are added to dilute sulfuric acid until the zinc carbonate is in excess. This forms aqueous zinc sulfate.

step 2 The excess zinc carbonate is separated from the aqueous zinc sulfate.

step 3 The aqueous zinc sulfate is heated until a saturated solution is formed.

step 4 The saturated solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

(i) In **step 1**, zinc carbonate is in excess when no more zinc carbonate dissolves.

State one **other** observation that indicates the zinc carbonate is in excess in **step 1**.

..... [1]

(ii) Name a different substance, other than zinc carbonate, that can be added to dilute sulfuric acid to produce aqueous zinc sulfate in **step 1**.

..... [1]

(iii) **Step 1** is repeated using powdered zinc carbonate instead of large pieces.

All other conditions are kept the same.

The rate of reaction increases.

Give a reason why the rate of reaction increases. Explain your answer in terms of particles.

.....

..... [2]

(iv) Suggest what is observed when the solution is saturated in **step 3**.

..... [1]

(v) The formula of zinc sulfate crystals is $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$.

Give the formula of the solid formed if the crystals are heated to dryness in **step 3**.

..... [1]

3. M/J 22/P43/Q1

A list of substances is shown.

aluminium oxide

carbon dioxide

carbon monoxide

chlorine

copper

glucose

iron(III) oxide

limestone

nitrogen

oxygen

Answer the questions using the substances in the list.

Each substance may be used once, more than once or not at all.

State which substance is:

(a) a product of respiration

..... [1]

(b) the main constituent of hematite

..... [1]

(c) an element which has a sulfate that is used to test for water

..... [1]

(d) a colourless toxic gas

..... [1]

(e) a reactant in fermentation

..... [1]

(f) a reducing agent in the extraction of iron

..... [1]

(g) a conductor of electricity when solid

..... [1]

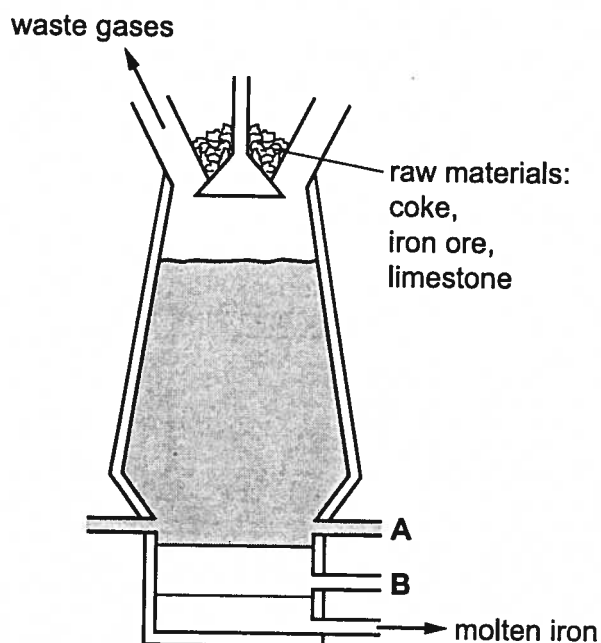
(h) a gas that is approximately 21% of clean, dry air.

..... [1]

4. **M/J 21/P41/Q6**

This question is about iron.

(a) Iron is extracted from its main ore in a blast furnace.



(i) Name the main ore of iron used in the blast furnace.

..... [1]

(ii) Name the substance that enters the blast furnace at A.

..... [1]

(iii) Name the substance that leaves the blast furnace at B.

..... [1]

(iv) Give **two** reasons for using coke in the blast furnace.

1

2 [2]

- (b) Another ore of iron is iron pyrites, FeS_2 . Iron pyrites contains the positive ion, Fe^{2+} .
Deduce the formula of the negative ion in FeS_2 .

..... [1]

- (c) Iron is a transition element.

A list of properties of iron is shown.

- Iron is a good conductor of electricity.
- Iron forms soluble salts.
- Iron forms coloured compounds.
- Iron has variable oxidation states.
- Iron acts as a catalyst.
- Iron forms a basic oxide.

- (i) Give **two** properties from the list in which iron differs from Group I elements.

1

2 [2]

- (ii) Give **two** properties from the list in which iron is similar to Group I elements.

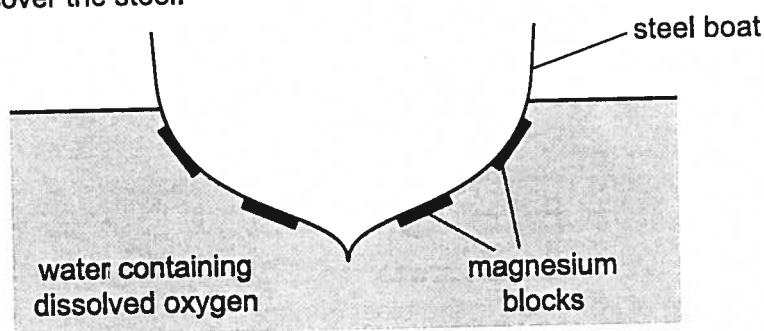
1

2 [2]

- (d) Steel consists mainly of iron.

Iron forms rust when it reacts with water and oxygen.

Magnesium blocks can be attached to the bottom of steel boats. The magnesium does not completely cover the steel.



- (i) Explain how the magnesium blocks prevent iron from rusting.

..... [2]

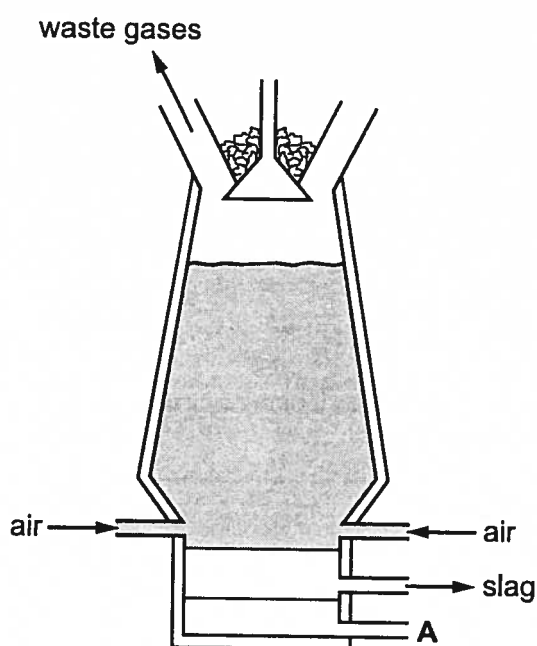
- (ii) Explain why replacing the magnesium blocks with copper blocks will **not** prevent the bottom of the boat from rusting.

..... [1]

5. M/J 21/P43/Q6

This question is about metals.

- (a) Iron is extracted from its main ore in a blast furnace.



- (i) Coke and iron ore are added at the top of the blast furnace.
Name one **other** substance that is added at the top of the blast furnace.

..... [1]

- (ii) Name the substance that leaves the blast furnace at **A**.

..... [1]

- (iii) Iron ore is mainly iron(III) oxide, Fe_2O_3 .
Name a substance that reduces iron(III) oxide to iron in the blast furnace.

..... [1]

- (iv) Temperatures inside a blast furnace can reach 2000°C .
Name **two** substances that react together, in the blast furnace, to produce this high temperature.

..... [1]

- (v) Name **two** waste gases that leave the blast furnace.

1

2 [2]

- (b) Zinc is extracted from zinc blende.

- (i) Name the main zinc compound that is present in zinc blende.

..... [1]

- (ii) When zinc is extracted, it is formed as a gas.
The gaseous zinc is then converted into molten zinc.
State the name of this physical change.

..... [1]

(c) Name the alloy that contains zinc and copper only.

..... [1]

(d) Copper has the following properties.

- It has a high melting point.
- It has a high density.
- It is a good conductor of electricity.
- It has variable oxidation states.
- It forms a basic oxide.
- It forms soluble salts.

(i) Give **two** properties from the list in which copper differs from Group I elements.

1

2 [2]

(ii) Give **two** properties from the list in which copper is similar to Group I elements.

1

2 [2]

6. O/N 20/P41/Q1/a(v)

(a) This question is about elements.

aluminium

carbon

iron

hydrogen

oxygen

silicon

sodium

sulfur

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

(i) Name the element extracted from bauxite.

..... [1]

7. O/N 20/P41/Q2/a,b

Zinc is extracted from an ore containing zinc sulfide.

(a) State the name of this zinc ore.

..... [1]

(b) This ore is converted to zinc oxide, ZnO.

Zinc oxide is then reacted with carbon.

(i) Write a chemical equation for the reaction of zinc oxide with carbon.

..... [1]

- (ii) State what type of chemical change happens to the zinc in zinc oxide in this reaction.
Explain your answer.

chemical change

explanation [2]

- (iii) Explain why aluminium is **not** extracted from aluminium oxide by heating with carbon.

..... [1]

- (iv) Suggest an alternative method for the extraction of zinc from zinc oxide.

..... [1]

8. M/J 20/P42/Q1/a(iv)

- (a) Give the name of the process that:

(i) is used to extract aluminium from aluminium oxide

..... [1]

9. M/J 20/P43/Q7/a,b,c,d

Aluminium is extracted by electrolysis. Iron is extracted from its ore by reduction with carbon.

- (a) What is meant by the term *electrolysis*?

.....

..... [2]

- (b) Name the main ore of aluminium.

..... [1]

- (c) (i) Explain why aluminium **cannot** be extracted by reduction with carbon.

..... [1]

- (ii) Describe the role of cryolite in the extraction of aluminium by electrolysis.

..... [1]

- (iii) Name the product formed at the positive electrode.

..... [1]

- (iv) Write the ionic half-equation for the reaction at the negative electrode.

..... [2]

- (d) Aluminium is used in overhead electricity cables.

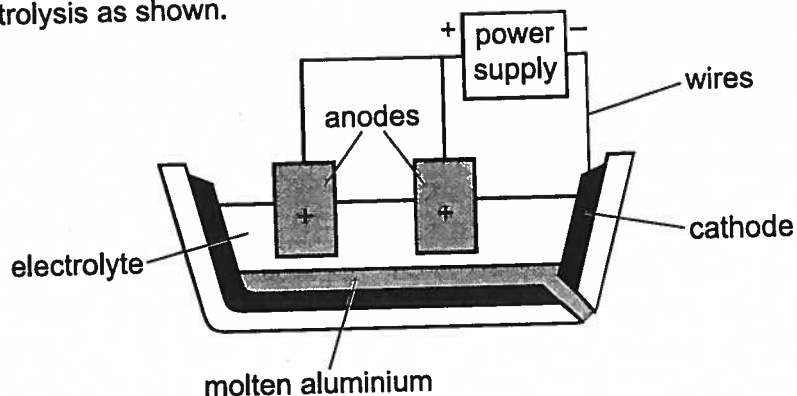
Give **two** properties of aluminium that make it suitable for use in overhead electricity cables.

1

2 [2]

10. MAR 20/P42/Q2/a,b

Aluminium is extracted from its ore. The ore is converted into pure aluminium oxide, which then undergoes electrolysis as shown.



(a) (i) Name an ore of aluminium.

..... [1]

(ii) What is meant by the term *electrolysis*?

..... [2]

(b) Aluminium oxide has a melting point of about 2000 °C, but the electrolysis process operates at about 900 °C.

(i) Name the compound added to aluminium oxide to reduce the operating temperature.

..... [1]

(ii) Suggest **one** benefit to the environment of reducing the operating temperature.

..... [1]

(iii) Write the ionic half-equation for the reaction taking place at:

the negative electrode (cathode)

the positive electrode (anode) [4]

(iv) Explain why the anodes need frequent replacement.

..... [2]

11. M/J 19/P42/Q1/c

The names of eight substances are given.

aluminium oxide

calcium oxide

ethanol

nitrogen

iron(III) oxide

methane

oxygen

silicon(IV) oxide

Answer the following questions about these substances.

Each substance may be used once, more than once or not at all.

State which substance is:

(a) the main constituent of bauxite

..... [1]

12. M/J 19/P41/Q3/a,d

Zinc and copper are elements next to each other in the Periodic Table.

(a) Zinc is obtained from zinc blende in a two-step process.

- In **step 1**, zinc blende is converted into zinc oxide.
- In **step 2**, zinc oxide is converted into zinc in a blast furnace.

Outline how each of these steps are done.

In your answer:

- give **one** chemical equation for each step
- describe how zinc is removed from the blast furnace in **step 2**.

step 1

chemical equation

step 2

chemical equation

removal of zinc in **step 2**

..... [5]

(b) A compound of copper can be used to test for water.

(i) State the full name of this compound of copper.

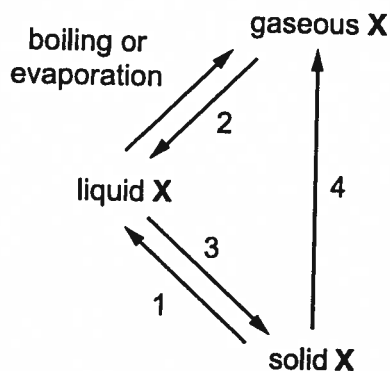
..... [1]

(ii) State the colour change that occurs when water is added to this compound of copper.

from to

13. O/N 18/P42/Q1/c

Element X can undergo the following physical changes.



[2]

- (a) Element X is a Group I metal. It burns in air to form an oxide X_2O .
Write a chemical equation for this reaction.

..... [2]

14. M/J 18/P42/Q2/b

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

- (a) is extracted from the ore bauxite

..... [1]

15. M/J 18/P41/Q3

This question is about iron.

- (a) Three of the raw materials added to a blast furnace used to extract iron from hematite are coke, hematite and limestone.

Name **one** other raw material added to the blast furnace.

..... [1]

- (b) A series of reactions occurs in a blast furnace during the extraction of iron from hematite. Describe these reactions.

Include:

- **one** chemical equation for the reduction of hematite
- **one** chemical equation for the formation of slag.

.....
.....
.....
.....
..... [5]

- (c) The iron extracted from hematite using a blast furnace is impure. Identify the main impurity in this iron and explain how it is removed in the steel-making process.

main impurity

how it is removed [3]

16. M/J 18/P43/Q2/b

- (a) Aluminium is extracted from its ore by electrolysis.

- (i) Name the main ore of aluminium.

..... [1]

(ii) Why is aluminium **not** extracted from its ore by reduction with carbon?

..... [1]

(iii) The main ore of aluminium contains aluminium oxide. Aluminium oxide is dissolved in molten cryolite before it is electrolysed.

Give **two** reasons, other than cost, why cryolite is used.

1

2 [2]

(iv) The reaction at the anode during the extraction of aluminium by electrolysis is shown.



Is this process oxidation or reduction?

Give a reason for your answer.

..... [1]

(v) During the extraction of aluminium by electrolysis, carbon dioxide is formed at the anode. Explain how carbon dioxide is formed at the anode.

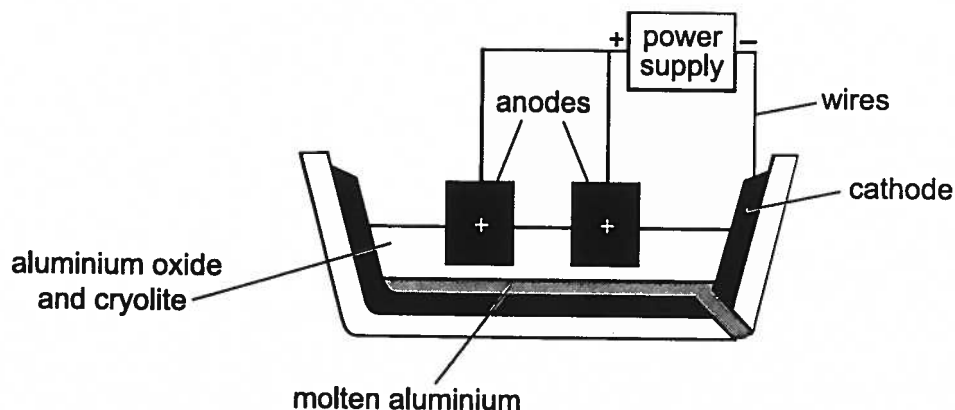
.....

..... [2]

17. O/N 17/P41/Q6/c

Aluminium is extracted from aluminium oxide by electrolysis.

(a) Aluminium can be extracted by electrolysis using the apparatus shown.



(i) Name the type of particle responsible for the transfer of charge in

the wires,

the electrolyte. [2]

(ii) Give **two** reasons why cryolite is used.

1

2 [2]

(iii) Write the ionic half-equation for the formation of aluminium during the electrolysis.

..... [1]

(iv) Explain how carbon dioxide gas is formed at the anodes.

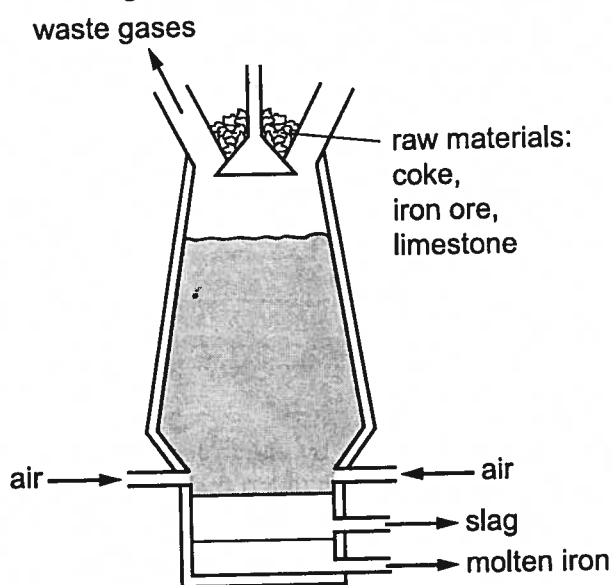
.....

.....

..... [3]

18. O/N 17/P43/Q3

Iron is extracted from its ore using coke in a blast furnace.



(a) Name the ore of iron which is mainly iron(III) oxide.

..... [1]

(b) Describe the reactions occurring in the blast furnace.
In your answer, include

- two reasons for using coke in the blast furnace,
- a chemical equation for the reduction of iron(III) oxide,
- an explanation for using limestone in the blast furnace.

.....

.....

.....

.....

.....

..... [6]

(c) (i) Describe the bonding in iron. Include a diagram in your answer.

.....
 [3]

(ii) Use your diagram in (c)(i) to explain why iron is malleable.

.....
 [2]

(iii) Iron containing a small amount of carbon is known as steel.
 Explain why steel is less malleable than iron.

.....
 [2]

(d) (i) When iron is added to dilute sulfuric acid, an aqueous solution of iron(II) sulfate is formed as one of the products.

Write a chemical equation for the reaction.

..... [1]

(ii) When iron(III) oxide is added to dilute sulfuric acid, an aqueous solution of iron(III) sulfate is formed as one of the products.

Write a chemical equation for the reaction.

..... [3]

(e) Aqueous sodium hydroxide, aqueous potassium iodide and aqueous acidified potassium manganate(VII) are added to aqueous solutions of iron(II) sulfate and iron(III) sulfate.

- Iron(II) ions, Fe^{2+} , are reducing agents in aqueous solution.
- Iron(III) ions, Fe^{3+} , are oxidising agents in aqueous solution.

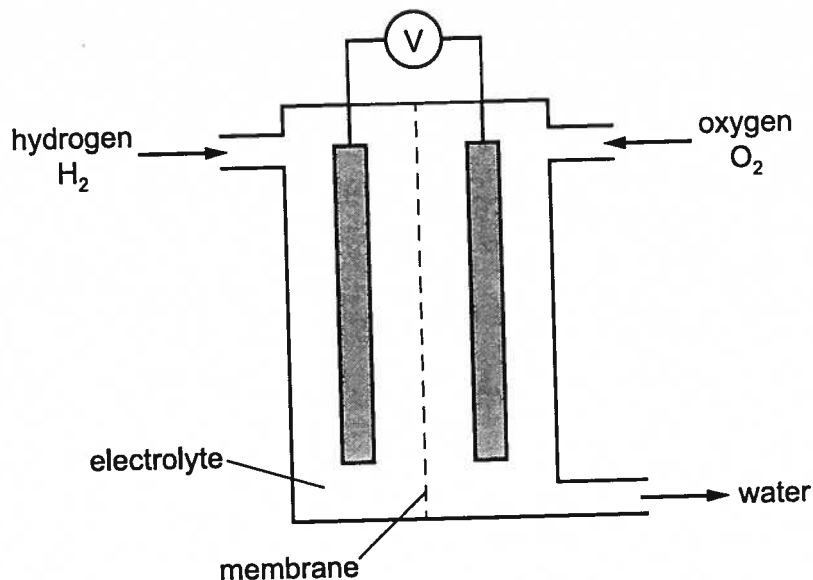
Complete the table.

reagent	observations with aqueous iron(II) sulfate	observations with aqueous iron(III) sulfate
aqueous sodium hydroxide	green precipitate	
aqueous potassium iodide		
aqueous acidified potassium manganate(VII)		no change

[4]

19. O/N 17/P43/Q4/a,b

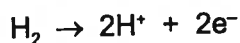
Hydrogen and oxygen react together in a hydrogen fuel cell. A hydrogen fuel cell is shown in the diagram.



(a) Name the process by which oxygen is obtained from air.

..... [1]

(b) (i) In a hydrogen fuel cell, the hydrogen molecules are converted into hydrogen ions, H^+ , according to the ionic half-equation shown.



What type of reaction does this ionic half-equation represent?

..... [1]

(ii) What **type** of substance reacts by donating hydrogen ions, H^+ ?

..... [1]

20. M/J 16/P42/Q6/a,b,c

Zinc is extracted from an ore called zinc blende, which consists mainly of zinc sulfide, ZnS .

(a) (i) The zinc sulfide in the ore is first converted into zinc oxide.

Describe how zinc oxide is made from zinc sulfide.

..... [1]

(ii) Write a chemical equation for the reaction in (a)(i).

..... [2]

(b) Zinc oxide is converted into zinc. Zinc oxide and coke are fed into a furnace. Hot air is blown into the bottom of the furnace.

Zinc has a melting point of $420^\circ C$ and a boiling point of $907^\circ C$. The temperature inside the furnace is over $1000^\circ C$.

- (i) Explain how zinc oxide is converted into zinc. Your answer should include details of how the heat is produced and equations for all the reactions you describe.

.....
.....
..... [3]

- (ii) Explain why the zinc produced inside the furnace is a gas.

..... [1]

- (iii) State the name of the physical change for conversion of gaseous zinc into molten zinc.

..... [1]

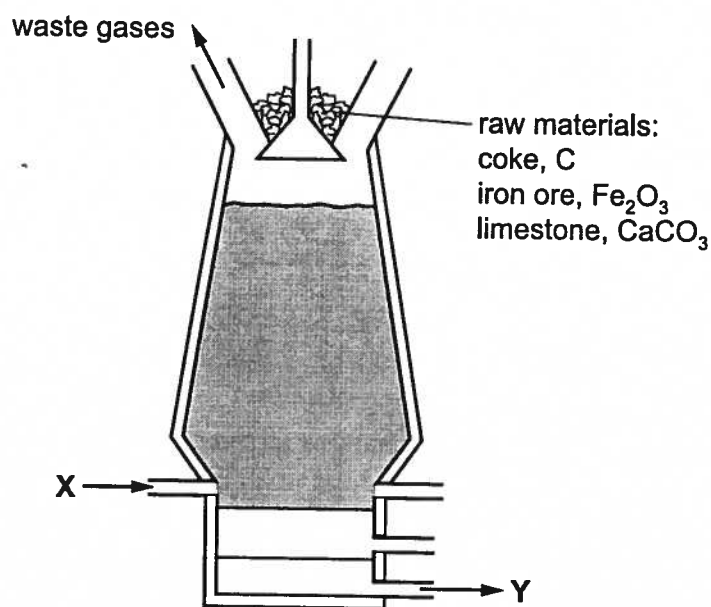
- (c) Rusting of steel can be prevented by coating the steel with a layer of zinc.

Explain, in terms of electron transfer, why steel does **not** rust even if the layer of zinc is scratched so that the steel is exposed to air and water.

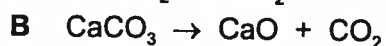
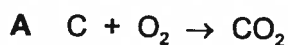
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.....
.....
..... [4]

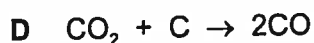
21. M/J 16/P43/Q1

The diagram shows a blast furnace.



- (a) The following equations represent reactions which take place in the blast furnace.





- (i) Which reaction is used to increase the temperature inside the blast furnace? [1]
- (ii) Which reaction is an example of thermal decomposition? [1]
- (iii) In which reaction is carbon both oxidised and reduced? [1]
- (iv) Which equation shows the removal of an impurity from the iron? [1]
- (v) Which equation shows the reaction of an acidic substance with a basic substance? [1]

(b) Use the diagram of the blast furnace to help you answer these questions.

- (i) What enters the blast furnace at X?

..... [1]

- (ii) What leaves the blast furnace at Y?

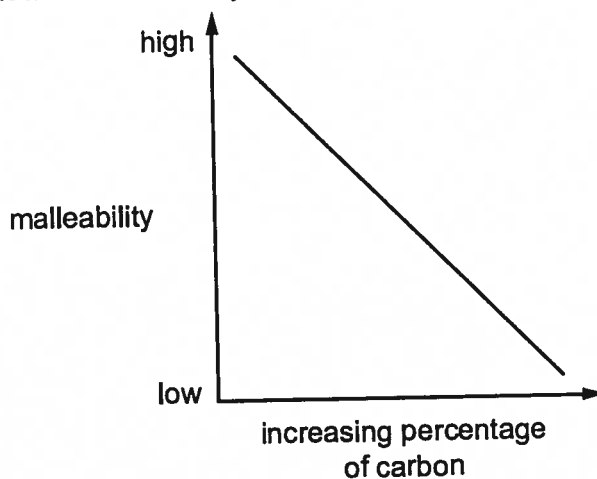
..... [1]

- (iii) Name **two** waste gases that leave the blast furnace.

1.

2. [2]

- (c) The graph shows how the malleability of iron changes as the percentage of carbon in the iron changes.



- (i) Describe how the malleability of iron changes as the percentage of carbon changes.

..... [1]

- (ii) Iron obtained from the blast furnace contains high levels of carbon.
Explain how the amount of carbon in the iron can be decreased.

..... [2]

Answers Section

1. O/N 22/P43/Q3
 - (a) bauxite 1
 - (b) **M1** breakdown by (the passage of) electricity (1) 2
M2 of an ionic compound in molten or aqueous (state) (1)
 - (c) (i) **M1** improves conductivity of the electrolyte / makes the electrolyte a better conductor (1) 2
M2 lower operating temperature (1)
 - (ii) $Al^{3+} + 3e^{-} \rightarrow Al$ 2
M1 only $Al^{3+} + (3) e^{-}$ on the left (1)
M2 equation fully correct (1)
 - (iii) **M1** anodes or carbon react(s) with oxygen (1) 2
M2 form carbon dioxide (1)
 - (d) unreactive coating of aluminium oxide (1) 1
 - (e) (i) neutralises both acids **AND** alkalis 1
(ii) $2NaOH + Al_2O_3 \rightarrow 2NaAlO_2 + H_2O$
M1 $NaAlO_2$ on the right hand side (1)
M2 equation fully correct (1) 2
 - (f) **M1** $GaCl_3$ (1)
M2 $Ga_2(SO_4)_3$ (1) 2
2. M/J 22/P41/Q6
 - (a) (i) zinc blende 1
(ii) heat(zinc sulfide) strongly in air/ roast in air 1
(iii) carbon or carbon monoxide 1
(iv) the temperature in the furnace is above or higher than the boiling point of zinc
OR
the boiling point of zinc is below or less than the temperature of the furnace 1
 - (v) condensation / condensing/ condense 1
 - (b) (i) no bubbles
or
no fizzing
or
no effervescence 1
(ii) zinc / zinc oxide / zinc hydroxide 1
(iii) (powder has) larger surface area **OR** lumps have smaller surface area (1)
(powder has) more collisions per unit time / more collision frequency
OR
lumps have fewer collisions per unit time / less collision frequency (1) 2
(iv) crystals (form on glass rod or microscope slide) 1
(v) $ZnSO_4$ 1
3. M/J 22/P43/Q1
 - (a) carbon dioxide 1
 - (b) iron(III) oxide 1
 - (c) copper 1
 - (d) carbon monoxide 1
 - (e) glucose 1
 - (f) carbon monoxide 1

- (g) copper 1
(h) oxygen 1
4. M/J 21/P41/Q6 1
- (a) (i) hematite 1
(ii) air 1
(iii) slag / calcium silicate 1
(iv) any two from:
 • (coke)
 releases heat (when it reacts with oxygen or reacts in air)
OR (acts as a) fuel
OR increases temperature (in the furnace) / heats (the furnace)
OR source of energy
 • (coke or carbon monoxide)
 reduces iron oxide
OR is a reducing agent
OR converts iron oxide to iron /
 removes oxygen from iron oxide
 • (coke)
 reacts with oxygen to **form** carbon monoxide 2
OR reacts with carbon dioxide to **form** carbon monoxide 1
- (b) S^{2-} or S
- (c) (i) any two from:
 • (iron forms) coloured compounds
 • (iron has) variable oxidation states 2
 • (iron acts as a) catalyst
- (ii) any two from:
 • (iron is) good conductor of electricity
 • (iron) forms a basic oxide 2
 • (iron salts are) soluble (in water)
- (d) (i) magnesium is **more** reactive than iron / steel **ORA** (1)
 iron is not oxidised
OR
 iron does not lose electrons
OR
 magnesium loses electrons **more easily** than or **in preference** (to iron) **ORA**
OR
 magnesium is oxidised **more easily** or reacts with oxygen **more easily**
 or corrodes **more easily** or **in preference** (to iron) 2
ORA (1)
- (ii) copper is less reactive than iron / copper is lower in the reactivity series than iron **ORA** 1
5. M/J 21/P43/Q6 1
- (a) (i) limestone 1
(ii) (molten) iron 1
(iii) coke / carbon / carbon monoxide 1
(iv) (reaction between) coke / carbon and oxygen 1
(v) nitrogen (1) 2
 carbon dioxide (1) 1
- (b) (i) zinc sulfide 1
(ii) condensation/condensing 1
- (c) brass 1

- (d) (i) any two from (copper has):
 • high density
 • high melting point
 • variable oxidation states 2
- (ii) any two from (copper):
 • good conductor of electricity
 • forms a basic oxide
 • soluble salts 2
6. O/N 20/P41/Q1/a(v)
 (a) aluminium 1
7. O/N 20/P41/Q2/a,b
 (a) zinc blende 1
- (b) (i) $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
 or
 $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$ 1
- (ii) chemical change: reduction (1)
 explanation: oxygen is lost (1) 2
- (iii) aluminium is more reactive than carbon 1
- (iv) electrolysis 1
8. M/J 20/P42/Q1/a(iv)
 (a) (i) electrolysis 1
9. M/J 20/P43/Q7/a,b,c,d
 (a) breakdown of a molten / or aqueous ionic compound by the passage of electricity 2
- (b) bauxite 1
- (c) (i) it is above carbon in the reactivity series / more reactive than carbon 1
- (ii) any **one** from:
 • aluminium oxide has high melting point / cryolite has lower melting point than aluminium oxide
 • using cryolite reduces costs / expensive to melt aluminium 1
- (iii) oxygen 1
- (iv) $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ 2
- (d) any **two** related to use as electricity cables:
 • ductile / malleable
 • conducts (electricity)
 • low density
 • protective oxide layer 2
10. MAR 20/P42/Q2/a,b
 (a) (i) bauxite 1
- (ii) **breakdown** by (the passage of) **electricity** (1)
 of an **ionic compound** in **molten / aqueous** (state) (1) 2
- (b) (i) cryolite 1
- (ii) less CO_2 emission 1
- (iii) $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
 any positive Al species gaining electron(s) (1)
 correct species and balance (1)
 $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
 any negative O species losing electron(s) (1)
 correct species and balance (1) 4
- (iv) anodes or carbon / graphite react with oxygen / O_2 (1)
 (form) carbon dioxide (1) 2

11. **M/J 19/P42/Q1/c** 1
(a) aluminium oxide
12. **M/J 19/P41/Q3/a,d**
(a) roast zinc blende (in air) (1)
 $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$ (1)
 add/react with coke (1)
 $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ OR $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$ (1)
 (zinc is) distilled (1) 5
 (b) (i) **anhydrous** copper(II) sulfate 1
 (ii) white (1)
 blue (1) 2
13. **O/N 18/P42/Q1/c**
(a) $4\text{X} + \text{O}_2 \rightarrow 2\text{X}_2\text{O}$
 M1 Species
 M2 Balance [2]
14. **M/J 18/P42/Q2/b** [1]
(a) aluminium / Al
15. **M/J 18/P41/Q3** [1]
(a) (hot) air [1]
 (b) coke is burned (to form carbon dioxide) [1]
 OR
 $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
 carbon dioxide is reduced by (more) coke to form **carbon monoxide** or **CO** [1]
 OR
 $\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$ [1]
 $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + 3\text{CO}_2$ [1]
 limestone (decomposes to) form lime / CaO / calcium oxide (and carbon dioxide) [1]
 OR
 $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
 $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$ [1]
 (c) the impurity is C [1]
 blow **into** or pass oxygen **through** (molten) iron [1]
 carbon dioxide escapes or carbon dioxide is a gas [1]
16. **M/J 18/P43/Q2/b** [1]
(a) (i) bauxite [1]
 (ii) aluminium is more reactive than carbon [1]
 (iii) to lower the operating temperature / the mixture has a lower melting point than aluminium oxide to increase the conductivity [1]
 (iv) oxidation (because)
 (the O^{2-} -ion OR 'oxide ions') electrons are lost
 OR
 (the O^{2-} -ion OR 'oxide ions') oxidation number increases [1]
 (v) electrodes/anodes are made from carbon/graphite [1]
 oxygen (made) reacts with carbon/anode [1]
17. **O/N 17/P41/Q6/c**
(a) (i) *the wires*: electrons [1]
the electrolyte: ions [1]
 (ii) any 2 from:
 ∞ increases conductivity
 ∞ as a solvent
 ∞ lowers the operating temperature [2]

- (iii) $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ [1]
 (iv) oxygen is made at the anode [1]
 the anodes are made of carbon [1]
 oxygen (made) reacts with carbon [1]

18. O/N 17/P43/Q3

- (a) hematite [1]
 (b) (coke reacts with oxygen/air) to produce heat/increase temperature/exothermically [1]
 coke is reducing agent/produces reducing agent/produces carbon monoxide
 OR
 coke reduces Fe_2O_3 /(iron) ore/hematite (producing iron) [1]
 $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ [2]
 OR
 $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
 OR
 $2\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Fe} + 3\text{CO}_2$
 M1 species correct
 M2 balanced
 limestone (decomposes to calcium oxide which) reacts with/removes acidic impurities /
 SiO_2 /sand/silica/ [1]
 silicon(IV) oxide/silicon dioxide
 limestone/calcium oxide/lime is involved in the production of slag/calcium silicate [1]
 (c) (i) positive ions/cations [1]
 sea of electrons/mobile electrons/delocalised electrons/moving electrons/
 flowing electrons attraction between positive ions and electrons [1]
 (ii) layers/rows/sheets of ions [1]
 slide/slip/shift (over each other or past each other) [1]
 (iii) particles have different sizes/radii [1]
 layers cannot slide/slip/shift [1]
 (d) (i) $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$ [1]
 (ii) $\text{Fe}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$
 M1 formula of $\text{Fe}_2(\text{SO}_4)_3$
 M2 all formulae correct (no additional species)
 M3 balanced

(e)

	observation with aqueous iron(II) sulfate	observation with aqueous iron(III) sulfate
aqueous sodium hydroxide		M3 brown precipitate
aqueous potassium iodide	M1 no change	M4 brown solution/black solid
aqueous acidified potassium manganate(VII)	M2 (pink/purple to) colourless/ decolourised	

19. O/N 17/P43/Q4/a,b

- (a) fractional distillation [1]
 (b) (i) oxidation [1]
 (ii) acid(ic) [1]

20. M/J 16/P42/Q6/a,b,c

- (a) (i) roast/heat and in air/oxygen; [1]
 (ii) $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$;
 SO_2 on right of equation; all formulae and balancing correct; [2]

- (b) (i) **M1** heat produced by carbon/coke (burning in) oxygen/air; [1]
OR
 $C + O_2 \rightarrow CO_2$ produces heat/exothermic;
OR
 $2C + O_2 \rightarrow 2CO$ produces heat/exothermic (scores **M1** and **M2**);
M2 [1]
 $C + CO_2 \rightarrow 2CO$;
OR
 $2C + O_2 \rightarrow 2CO$;
M3
 $ZnO + CO \rightarrow Zn + CO_2$;
OR
 $ZnO + C \rightarrow Zn + CO$;
OR
 $2ZnO + C \rightarrow 2Zn + CO_2$; [1]
- (ii) temperature (inside the furnace) is above 907 °C/temperature (inside the furnace) is above the boiling point (of zinc)/1000 °C is above the boiling point (of zinc); [1]
- (iii) condensation/condensing/condense; [1]
- (c) **M1** zinc is more reactive than iron/zinc is higher in the reactivity series than iron ora; [1]
M2 zinc loses electrons; [1]
M3 iron/steel/oxygen/air/water gains electrons **OR** electrons move to iron/steel/oxygen/air/water; [1]
M4 (therefore) iron does not lose electrons/get oxidised/form iron(II)/form iron(III); [1]

21. **M/J 16/P43/Q1**

- (a) (i) A; [1]
(ii) B; [1]
(iii) D; [1]
(iv) C; [1]
(v) C; [1]
- (b) (i) (hot) air; [1]
(ii) (molten) iron; [1]
(iii) any 2 from:
carbon dioxide;
carbon monoxide;
nitrogen; [2]
- (c) (i) as the percentage of carbon increases, so the malleability decreases; [1]
(ii) **M1** oxygen (gas) blown in; [1]
M2 carbon dioxide formed/ $C + O_2 \rightarrow CO_2$; [1]

10.4: Uses of metals

1. O/N 21/P41/Q4

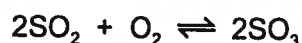
Chalcopyrite, FeCuS_2 , is used in the manufacture of sulfuric acid in the Contact process.

- (a) In the first stage of the process, chalcopyrite reacts with oxygen in the air to produce sulfur dioxide, SO_2 , iron(III) oxide and copper(II) oxide.

Complete the chemical equation for the reaction of FeCuS_2 with oxygen.



- (b) Sulfur dioxide is then converted to sulfur trioxide.



The reaction is exothermic. It is also an equilibrium.

- (i) State **two** features of an equilibrium.

1

2 [2]

- (ii) State the temperature and pressure used in this reaction.
Include units.

• temperature

• pressure [2]

- (iii) Name the catalyst used.

..... [1]

- (iv) Explain why a catalyst is used.

..... [1]

- (v) Describe and explain, in terms of equilibrium, what happens when the temperature is increased.

.....

..... [2]

- (c) Concentrated sulfuric acid is a dehydrating agent.

When glucose is dehydrated, carbon and one other product are formed.

Complete the equation to show the dehydration of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.



2. O/N 20/P41/Q2/c

- (a) Brass is an alloy of zinc.

Explain, in terms of particles, why brass is harder than pure zinc.

.....

.....
..... [3]

3. M/J 19/P41/Q3/b,c

(a) Name the alloy formed when zinc is mixed with copper.

..... [1]

(b) Copper is a transition element. It can have variable oxidation states.

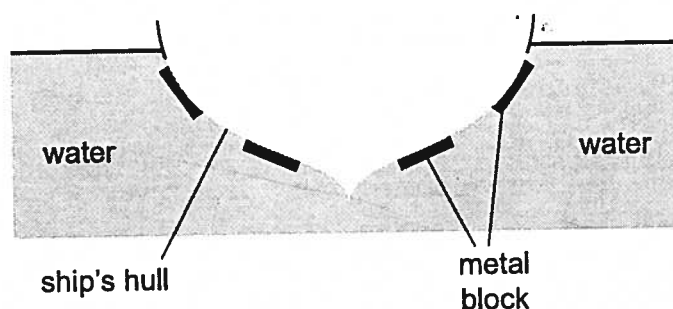
State **two** other chemical properties of transition elements which make them different from Group I elements.

1

2 [2]

4. MAR 19/P42/Q6/f,g

(a) The hull of a ship is made from steel (mainly iron). Metal blocks are placed on the ship's hull to prevent rusting.



Use your knowledge of the reactivity series to explain why:

- magnesium is suitable to use as the metal blocks
- copper is **not** suitable to use as the metal blocks.

.....
..... [2]

(b) Rust contains iron(III) oxide.

Phosphoric acid, H_3PO_4 , can be used to remove rust from an iron object and prevent further rusting.

(i) Write a chemical equation for the reaction between iron(III) oxide and phosphoric acid to form iron(III) phosphate and water.

..... [2]

(ii) Iron(III) phosphate is an insoluble salt.

Suggest how the formation of iron(III) phosphate prevents further rusting.

..... [1]

Answers Section

1. **O/N 21/P41/Q4**
 - (a) $(4\text{FeCuS } 13\text{O}_2) \rightarrow 2\text{Fe}_2\text{O}_3 + 4\text{CuO} + 8\text{SO}_2$
 Fe_2O_3 and CuO as a product (1)
 Equation fully correct (1) 2
 - (b) (i) **M1** rate of forward reaction = rate of reverse reaction (1)
M2 concentration of reactants and products are constant (1) 2
 - (ii) **M1** 450°C (1)
M2 1-2 atm (1) 2
 - (iii) vanadium(V) oxide 1
 - (iv) increase rate of reaction 1
 - (v) **M1** equilibrium shifts to left hand side (1)
M2 forward reaction is exothermic (1) 2
 - (c) $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 6\text{C} + 6\text{H}_2\text{O}$
 H_2O (1)
 balance (1) 2
2. **O/N 20/P41/Q2/c**
 - (a) exists as layers (1)
 (alloy) contains different sized (copper) atoms (1)
 makes it more difficult for layers (of atoms) to slide over each slip/shift other (1) 3
3. **M/J 19/P41/Q3/b,c**
 - (a) brass 1
 - (b) form coloured compounds / ions (1)
 act as catalysts (1) 2
4. **MAR 19/P42/Q6/f,g**
 - (a) **M1** magnesium above iron / steel in the reactivity series **ORA** / magnesium more reactive than iron / steel **ORA**(1)
M2 copper below iron / steel in the reactivity series **ORA** / copper less reactive than iron **ORA**(1) 2
 - (b) (i) $\text{Fe}_2\text{O}_3 + 2\text{H}_3\text{PO}_4 \rightarrow 2\text{FePO}_4 + 3\text{H}_2\text{O}$
M1 FePO_4 anywhere(1) **M2** The whole equation correct(1) 2
 - (ii) iron(III) phosphate acts as a barrier which prevents contact between iron and water or air / oxygen 1

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Unit 11

AIR AND WATER

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 11.1 Water and rusting
- 11.2 Air
- 11.3 Nitrogen and fertilizers

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Unit 11: Air And Water

11.1: Water and rusting

1. O/N 20/P42/Q3/d

Group I metals are very reactive. Transition elements are also metals but are less reactive than Group I metals.

- (a) When the transition element iron is added to water the iron rusts.
When an iron object is coated with a layer of zinc, rusting is prevented.

(i) Name this process of coating iron objects with a layer of zinc.

..... [1]
(ii) Explain how completely coating an iron object with a layer of zinc prevents rusting.

..... [1]
(iii) Rusting of iron ships can be prevented by attaching zinc blocks to the hull of the ship.
Explain how this prevents rusting.

.....
..... [2]

2. O/N 20/P41/Q1/b

(a) Iron rusts when it is in contact with oxygen and water.

(i) Explain how sacrificial protection prevents rusting.

.....
..... [2]

(ii) State one **other** method of rust prevention.

..... [1]

3. O/N 20/P43/Q1/f

The names of nine substances are shown.

aluminium oxide
ammonia
carbon monoxide
anhydrous cobalt(II) chloride
hydrated copper(II) sulfate
iron(III) oxide
nitrogen dioxide
silver
steel

Answer the following questions using these substances. Each substance may be used once, more than once or not at all.

Name the substance that is:

(a) used to test for the presence of water. [1]

4. O/N 19/P41/Q3/b

(a) Iron often rusts.

Name the **two** substances, other than iron, that must be present for iron to rust.

1

2 [1]

5. MAR 19/P42/Q3/e(i)

Fluorine is a Group VII element. Fluorine forms compounds with metals and non-metals.

(a) Tetrafluoromethane and lead(II) fluoride are fluorides of Group IV elements. Some properties of tetrafluoromethane and lead(II) fluoride are shown in the table.

property	tetrafluoromethane	lead(II) fluoride
formula	CF ₄	
melting point/°C	-184	855
boiling point/°C	-127	1290
conduction of electricity when solid	non-conductor	non-conductor
conduction of electricity when molten	non-conductor	good conductor

(i) What is the formula of lead(II) fluoride?

..... [1]

6. MAR 19/P42/Q6/e

(a) Iron is a transition element.

(i) Which **two** substances react with iron to form rust?

1

2 [2]

(ii) Which metal is used to galvanise iron?

..... [1]

7. M/J 18/P42/Q6/b

(a) Nitrates decompose when heated.

(i) Write a chemical equation for the decomposition of sodium nitrate when it is heated.

..... [2]

(ii) The unbalanced chemical equation for the decomposition of hydrated copper(II) nitrate crystals is shown.

Balance the chemical equation for this reaction.



[2]

(iii) When the hydrated copper(II) nitrate crystals are heated, steam is produced. When the steam condenses on a cool surface, it turns into a colourless liquid.

Anhydrous cobalt(II) chloride is used to show that the colourless liquid contains water.

How does the colour of the anhydrous cobalt(II) chloride change?

from to [2]

(iv) How would the student test to determine if the water produced in (b)(iii) is pure?

..... [1]

8. M/J 16/P41/Q2/g

Period 3 contains the elements sodium to argon. This question asks about the chemistry of each of the Period 3 elements or their compounds.

(a) Chlorine and compounds of chlorine are important in water treatment and in laboratory testing for water.

(i) Chlorine is added to water to make the water safe to drink.

Explain why adding chlorine makes water safe to drink.

..... [1]

(ii) A compound of chlorine is used in the laboratory to test for the presence of water.

Name the compound of chlorine used in this test and describe the colour change seen in a positive result of this test.

name of compound

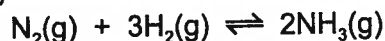
Answers Section

1. O/N 20/P42/Q3/d
- (a) (i) galvanising 1
 - (ii) prevents water or / and oxygen reaching iron 1
 - (iii) zinc more reactive (than iron) (1)
zinc corrodes / oxidises / reacts in preference to iron (1) 2
2. O/N 20/P41/Q1/b
- (a) (i) metal higher in reactivity series / metal more reactive (than iron) / allow named metal
e.g. magnesium or zinc (1)
zinc corrodes/oxidises/reacts in preference to iron (1) 2
 - (ii) any barrier method e.g. painting 1
3. O/N 20/P43/Q1/f
- (a) anhydrous cobalt(II) chloride 1
4. O/N 19/P41/Q3/b
- (a) water and oxygen / air 1
5. MAR 19/P42/Q3/e(i)
- (a) (i) PbF_2 1
6. MAR 19/P42/Q6/e
- (a) (i) M1 oxygen(1)
M2 water(1) 2
 - (ii) zinc / Zn 1
7. M/J 18/P42/Q6/b
- (a) (i) $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$
1 mark for either NaNO_2 or O_2 on the right-hand side
1 mark for fully correct equation 2
 - (ii) $2\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}(\text{s}) \rightarrow 2\text{CuO}(\text{s}) + 4\text{NO}_2(\text{s}) + \text{O}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
all 3 numbers = 2 marks
any 2 numbers = 1 mark 2
 - (iii) blue 1
pink 1
 - (iv) boiling point sharp / melting point sharp / freezing point sharp / boiling point 100 (°)C /
freezing point or melting point 0 (°)C 1
8. M/J 16/P41/Q2/g
- (a) (i) kills bacteria; 1
 - (ii) name of compound: cobalt(II) chloride;
from: blue;
to: pink; 3

11.2: Air**1. MAR 21/P42/Q3**

This question is about ammonia.

- (a) Nitrogen reacts with hydrogen to form ammonia in an industrial process.



- (i) Name this industrial process.

..... [1]

- (ii) State the meaning of the symbol $\rightleftharpoons$.

..... [1]

- (iii) State the conditions used in this industrial process. Include units.

temperature

pressure

[2]

- (iv) Name the catalyst used in this industrial process.

..... [1]

- (v) If the pressure is increased, the yield of ammonia increases.
Explain why, in terms of equilibrium.

.....
..... [2]

- (vi) If the temperature is increased, the rate of reaction increases.
Explain why, in terms of particles.

.....
.....
..... [3]

- (b) Ammonia reacts with sulfuric acid to make a compound which is used as a fertiliser.
Write the chemical equation for the reaction between ammonia and sulfuric acid.

..... [2]

2. O/N 20/P41/Q3/d(iii)

- (a) Ammonia gas is prepared at the front of a laboratory.

The pungent smell of ammonia spreads throughout the laboratory slowly.

- (i) Explain why carbon dioxide gas, CO_2 , will spread throughout the laboratory at a slower rate than ammonia gas, NH_3 .

..... [1]

3. O/N 20/P41/Q4/a,b,d,e

Air is a mixture of gases.

- (a) State the percentage of clean dry air which is oxygen. Give your answer to the nearest whole number.

..... % [1]

- (b) Oxygen and nitrogen are useful gases that can be obtained from air.

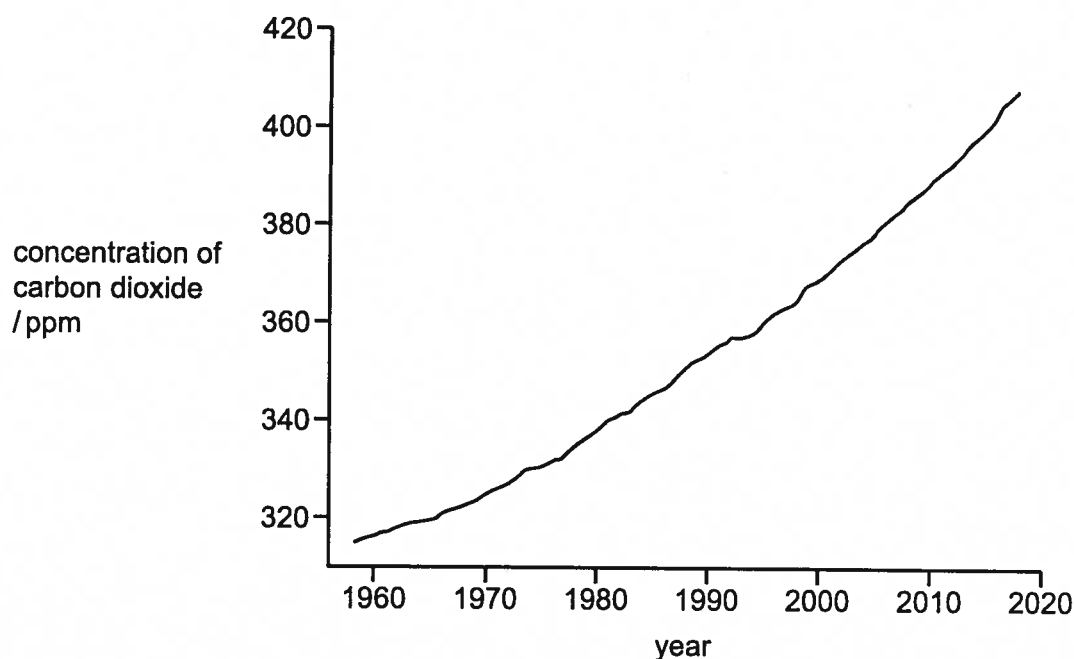
- (i) Name the process used to separate oxygen and nitrogen from liquid air.

..... [2]

- (ii) State the property of oxygen and nitrogen that allows these gases to be separated using this process.

..... [1]

- (c) The graph shows the concentration of carbon dioxide in the atmosphere over a 60-year period, measured in parts per million (ppm).



The data shown in the graph is of global concern.

Explain why.

.....
.....
..... [3]

- (d) Name the process in the carbon cycle by which plants remove carbon dioxide from the atmosphere.

..... [1]

4. O/N 20/P43/Q1/a,b,c

The names of nine substances are shown.

aluminium oxide
ammonia
carbon monoxide
anhydrous cobalt(II) chloride
hydrated copper(II) sulfate
iron(III) oxide
nitrogen dioxide
silver
steel

Answer the following questions using these substances. Each substance may be used once, more than once or not at all.

Name the substance that is:

- (a) the main constituent of hematite [1]
(b) a gas produced in car engines which causes acid rain [1]
(c) an alkaline gas [1]

5. M/J 20/P42/Q1/b(ii,iii,iv)

- (a) The symbols of the elements in Period 2 of the Periodic Table are shown.

Li Be B C N O F Ne

For each of the following, give the symbol of an element from Period 2 which matches the description.

Each element may be used once, more than once or not at all.

Which element:

- (i) makes up approximately 21% of clean, dry air
..... [1]
(ii) has atoms with only two electrons in the outer shell
..... [1]
(iii) has atoms with only seven protons
..... [1]

6. O/N 19/P42/Q2/a,b,c,d,e,f,g(i)

The gases Ar, CO₂, N₂ and O₂ are in clean, dry air.

CO, NO, NO₂ and SO₂ are gases commonly found in polluted air.

- (a) What percentage of clean, dry air is N₂?

Give your answer to the nearest whole number.

..... % [1]

- (b) Name the process used to separate O₂ from clean, dry air.

..... [2]

(c) State **one** major adverse effect of the pollutant SO_2 .

..... [1]

(d) NO and NO_2 are produced in car engines.

Describe how oxides of nitrogen form in a car engine.

..... [2]

(e) Many cars have catalytic converters in their exhaust systems. In a catalytic converter, most of the CO and NO formed in a car engine is changed into less harmful products.

Identify these products and state the metal catalyst used.

products

catalyst [3]

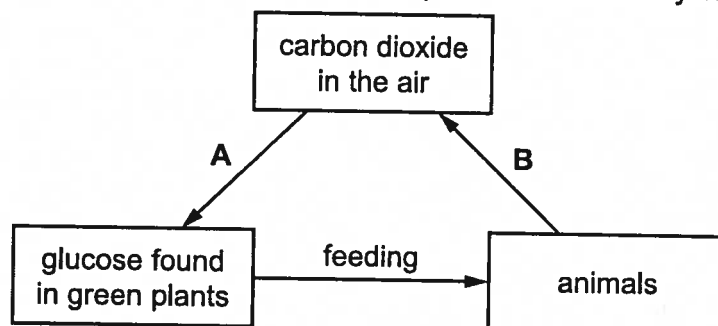
(f) CO is formed from the incomplete combustion of fossil fuels such as methane.

Write a chemical equation to show the incomplete combustion of methane.

..... [2]

(g) The CO_2 in air is part of the carbon cycle.

The scheme shows a simple representation of part of the carbon cycle.



(i) State the scientific terms for each of process **A** and process **B**.

A

B [2]

7. M/J 19/P42/Q1/b,d,e

The names of eight substances are given.

aluminium oxide

calcium oxide

ethanol

nitrogen

iron(III) oxide

methane

oxygen

silicon(IV) oxide

Answer the following questions about these substances.

Each substance may be used once, more than once or not at all.

State which substance is:

(a) a reactant in respiration

..... [1]

(b) a product of photosynthesis

..... [1]

(c) a greenhouse gas

..... [1]

8. M/J 19/P42/Q2/b(ii)

(a) Carbon is an element which exists in different forms.

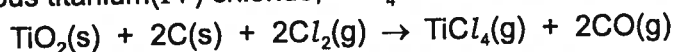
(i) Name the oxide of carbon that is a toxic gas.

..... [1]

9. MAR 19/P42/Q5/a(ii),c(iii)

Titanium is extracted from an ore called rutile. Rutile is an impure form of titanium(IV) oxide, TiO_2 .

(a) Rutile is mixed with coke and heated in a furnace through which chlorine gas is passed. The product is gaseous titanium(IV) chloride, TiCl_4 .



The gaseous titanium(IV) chloride produced is condensed into the liquid state. The titanium(IV) chloride is then separated from liquid impurities.

(i) Carbon monoxide, $\text{CO}(\text{g})$, is also produced in the reaction.

Why should carbon monoxide **not** be released into the atmosphere?

..... [1]

(b) Titanium(IV) chloride, TiCl_4 , is heated with an excess of magnesium, in an atmosphere of argon.

(i) Suggest why the reaction between titanium(IV) chloride and magnesium is done in an atmosphere of argon and **not** in air.

..... [1]

10. O/N 18/P41/Q3/b,e

(a) Nitrogen dioxide and other oxides of nitrogen are formed in car engines.

Explain how nitrogen dioxide is formed in car engines.

..... [2]

(b) Copper(II) nitrate can be made by reacting copper(II) carbonate with nitric acid. One of the products is carbon dioxide.

(i) Write a chemical equation for the reaction of copper(II) carbonate with nitric acid.

..... [2]

(ii) Carbon dioxide is added to the air by living things.

Name the chemical process by which living things add carbon dioxide to the air.

..... [1]

(iii) Carbon dioxide is removed from the air by plants.

Name the chemical process by which plants remove carbon dioxide from the air.

..... [1]

11. M/J 18/P42/Q2/d

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

- (a) is a green gas at room temperature and pressure

.....[1]

12. M/J 18/P43/Q1/a,b

The following are the symbols and formulae of some elements and compounds.

Ar Ca(OH)₂ Cl₂ CO₂ Cu Fe SO₂ V₂O₅

Answer the following questions using only the elements or compounds in the list.

Each element or compound may be used once, more than once or not at all.

State which element or compound is used:

- (a) to kill bacteria in drinking water [1]

- (b) as a food preservative [1]

13. O/N 17/P43/Q7/a

- (a) Carbon and silicon are elements in Group IV of the Periodic Table.

Carbon dioxide from the air moves into green plants and is converted into carbohydrates.

- (i) Name the process by which carbon dioxide molecules move through the air into green plants.

..... [1]

- (ii) Explain why silicon(IV) oxide **cannot** move through the air in the same way that carbon dioxide can.

..... [1]

- (iii) Name the process by which carbon dioxide is converted into glucose, C₆H₁₂O₆, in green plants. Give **two** conditions required for this process to occur. Write a chemical equation for the reaction which occurs.

name of process

condition 1

condition 2

chemical equation..... [5]

14. M/J 17/P42/Q2/d

- (a) Samples of air taken from industrial areas are found to contain small amounts of carbon monoxide.

- (i) Explain how this carbon monoxide is formed.

.....

- [2]
- (ii) State why carbon monoxide should **not** be inhaled.

..... [1]

15. O/N 16/P42/Q3

Clean, dry air contains a small amount of carbon dioxide.

- (a) The percentages of the **other** gases present in clean, dry air are shown in the table.

Complete the table by inserting the names of these gases.

name of gas	percentage present
	78
	21
	1

[2]

- (b) Oxides of nitrogen are atmospheric pollutants which can cause acid rain.
Describe the formation of oxides of nitrogen and suggest how they can cause acid rain.

.....

.....

..... [3]

- (c) Methane contributes to the greenhouse effect.
State **two** sources of methane.

1

2 [2]

- (d) Combustion and respiration add carbon dioxide to the atmosphere.
Name **one** natural process which removes carbon dioxide from the atmosphere.

..... [1]

16. M/J 16/P43/Q3

Clean dry air contains mainly nitrogen and oxygen.

- (a) Name **two** other gases that are in clean dry air.

.....

..... [2]

- (b) Air often contains pollutants.

Identify **three** common gaseous pollutants in air and state how each of these pollutants are produced.

pollutant gas 1

how it is produced

pollutant gas 2

how it is produced

pollutant gas 3

how it is produced [6]

Answers Section

1. **MAR 21/P42/Q3**
 - (a) (i) Haber 1
 - (ii) reversible / equilibrium 1
 - (iii) 450 °C (1) 2
 - 200 atm (1) 1
 - (iv) iron / Fe
 - (v) equilibrium **shifts / moves** in forwards direction (1) 2
 - fewer molecules / moles (of gas) on RHS (1)
 - (vi) particles have more energy (1)
 - more collisions (between particles) occur per second / per unit time (1)
 - a greater percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation energy / have energy equal to activation energy (1) 3
- (b) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$ 2
- $(\text{NH}_4)_2\text{SO}_4$ (1) correct equation (1)
2. **O/N 20/P41/Q3/d(iii)** 1
- (a) (i) CO_2 **molecules** are heavier (than NH_3) 1
3. **O/N 20/P41/Q4/a,b,d,e** 1
- (a) 21
- (b) (i) fractional (1) 2
- distillation (1) 1
- (ii) (different) boiling point
- (c) increase in (concentrations of) carbon dioxide 3
- (carbon dioxide is) greenhouse gas/greenhouse effect
- contributes to climate change/global warming 1
- (d) photosynthesis
4. **O/N 20/P43/Q1/a,b,c** 1
- (a) iron(III) oxide 1
- (b) nitrogen dioxide 1
- (c) ammonia
5. **M/J 20/P42/Q1/b(ii,iii,iv)** 1
- (a) (i) O 1
- (ii) Be 1
- (iii) N 1
6. **O/N 19/P42/Q2/a,b,c,d,e,f,g(i)** 1
- (a) 78
- (b) fractional (1) 2
- distillation (1) 1
- (c) acid rain
- (d) nitrogen and oxygen (from the air) react (in the engine) (1) 2
- (due to) high temperatures (1)
- (e) nitrogen (1)
- carbon dioxide (1) 3
- platinum (1)
- (f) $\text{CH}_4 + 1\frac{1}{2}\text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$
- CO and H_2O as products and methane as reactant (1) 2
- rest of the equation (1)

- (g) (i) A photosynthesis (1)
B respiration (1) 2
7. M/J 19/P42/Q1/b,d,e
(a) oxygen 1
(b) oxygen 1
(c) methane 1
8. M/J 19/P42/Q2/b(ii)
(a) carbon monoxide 1
9. MAR 19/P42/Q5/a(ii),c(iii)
(a) (i) carbon monoxide is toxic/poisonous 1
(b) (i) magnesium burns in air or oxygen OR reacts with air or oxygen / argon is unreactive or inert 1
10. O/N 18/P41/Q3/b,e
(a) M1 nitrogen and oxygen (from the air)
M2 (react) at high temperatures (in engine) or (electrical) spark (in engine) 2
(b) (i) $\text{CuCO}_3 + 2\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$
M1 carbon dioxide and water as products
M2 rest correct 2
(ii) respiration 1
(iii) photosynthesis 1
11. M/J 18/P42/Q2/d
(a) chlorine / Cl_2 / Cl 1
12. M/J 18/P43/Q1/a,b
(a) Cl_2 / chlorine 1
(b) SO_2 / sulfur dioxide 1
13. O/N 17/P43/Q7/a
(a) (i) diffusion 1
(ii) silicon(IV) oxide is a solid, whereas carbon dioxide is a gas 1
(iii) photosynthesis 1
chlorophyll / chloroplasts 1
M2 sunlight / UV (light) 1
 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ 1
M1 species correct
M2 balanced 2
14. M/J 17/P42/Q2/d
(a) (i) incomplete combustion / incomplete burning / combustion in insufficient air / oxygen of fossil fuels / named fossil fuel / named petroleum fraction / name or formula of a type of substance containing carbon 1
(ii) toxic / poisonous / combines with or binds to haemoglobin 1
15. O/N 16/P42/Q3
(a) nitrogen (78%) AND oxygen (21%)
noble gases OR argon (1%) 2
(b) nitrogen AND oxygen (from the air) react
(in the) high temperatures of a car engine
 NO_x / oxides of nitrogen react with or dissolve in water (to form an acid) 3

(c) any 2 from:

(named) ruminant animal / cattle / (anaerobic) digestion / flatulence (in animals) / animal waste / (animal) dung decomposing vegetation / animals / organisms / decaying (organic) matter / (fractional distillation / cracking of) petroleum / crude oil / hydrocarbons / natural gas / coal /

2

(d) photosynthesis

1

16. M/J 16/P43/Q3

(a) any 2 from:

carbon dioxide;
nitrogen;
any named noble gas;

2

(b) any 6 from:

carbon monoxide;
from incomplete combustion (of carbon-containing fuel);
sulfur dioxide;
from burning fossil fuels / roasting ores which contain sulphur / volcanoes;
oxides of nitrogen;
nitrogen reacting with oxygen in car engines / lightning;
methane;
from anaerobic decomposition / anaerobic decay;

6

11.3: Nitrogen and fertilisers

1. M/J 22/P43/Q3

This question is about nitrogen and compounds of nitrogen.

(a) Nitrogen molecules have the formula N_2 .

Some properties of nitrogen are shown:

- melting point of -210°C
- boiling point of -196°C
- non-conductor of electricity when solid
- insoluble in water.

(i) Name the type of bonding between the atoms in an N_2 molecule.

..... [1]

(ii) Explain, in terms of attractive forces between particles, why nitrogen has a low melting point.

..... [1]

(iii) Explain why nitrogen does **not** conduct electricity.

..... [1]

(b) Nitrogen reacts with hydrogen to form ammonia, NH_3 , in the Haber process.

State the essential conditions in the Haber process. Write an equation for the chemical reaction.

.....

.....

.....

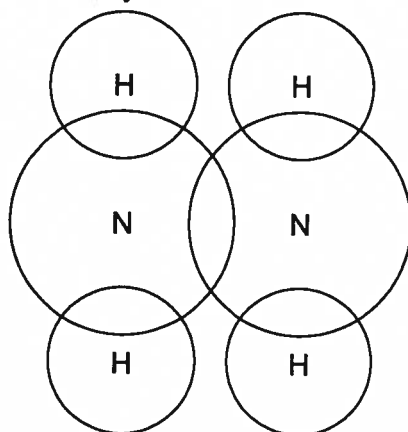
..... [4]

(c) Ammonia is made in the laboratory by heating ammonium chloride with calcium hydroxide. Balance the chemical equation for the reaction.



(d) Hydrazine, N_2H_4 , is another compound that contains nitrogen and hydrogen.

Complete the dot-and-cross diagram to show the electron arrangement in a molecule of hydrazine. Show outer electrons only.



[2]

- (e) Ammonia and hydrazine are weak bases.

The chemical equation for the reaction between one molecule of ammonia and one molecule of water is shown.



- (i) State the meaning of the term *base*.

..... [1]

- (ii) Write a chemical equation for the reaction between one molecule of hydrazine, N_2H_4 , and one molecule of water.

..... [1]

2. O/N 20/P41/Q3/c,d(i,ii)

- (a) Describe a test for ammonia gas.

test

result [2]

- (b) Ammonia gas is prepared at the front of a laboratory.

The pungent smell of ammonia spreads throughout the laboratory slowly.

- (i) Name the process that occurs when ammonia gas spreads throughout the laboratory.

..... [1]

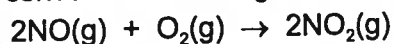
- (ii) Explain, using ideas about particles, why ammonia gas spreads throughout the laboratory.

.....

..... [2]

3. O/N 20/P43/Q3/d,e

- (a) Nitrogen monoxide, NO , is converted into nitrogen dioxide, NO_2 .



The nitrogen dioxide reacts with oxygen and water to produce nitric acid as the only product.

Write a chemical equation for this reaction.

..... [2]

- (b) Ammonium nitrate, NH_4NO_3 , is a fertiliser.

Calculate the percentage by mass of nitrogen in ammonium nitrate.

..... % [2]

4. M/J 20/P42/Q1/b(i)

- (a) The symbols of the elements in Period 2 of the Periodic Table are shown.

Li Be B C N O F Ne

For each of the following, give the symbol of an element from Period 2 which matches the description.

Each element may be used once, more than once or not at all.

Which element:

- (i) combines with hydrogen to produce ammonia

..... [1]

5. M/J 20/P41/Q4/a,b

This question is about reactions of bases and acids.

- (a) Ammonia is a gas at room temperature.

What is the test for ammonia gas? Describe the positive result of this test.

test

result

[2]

- (b) Ammonia reacts with water to form ions.



- (i) How does this equation show that ammonia, NH_3 , behaves as a base?

..... [1]

- (ii) Aqueous ammonia is described as a weak base.

Suggest the pH of aqueous ammonia.

pH = [1]

- (iii) Describe what is seen when aqueous ammonia is added to aqueous copper(II) sulfate, until no further change is seen.

.....

.....

..... [3]

6. M/J 18/P42/Q2/g

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

- (a) is non-metallic and an important component of fertilisers.

..... [1]

Answers Section

1. **M/J 22/P43/Q3** 1
 - (a) (i) covalent 1
 - (ii) weak force(s) of attraction between **molecules**
 - (iii) no ions 1
 - OR no mobile electrons
 - (b) 450 °C (1)
 - 200 atmospheres (1)
 - iron (catalyst) (1)
 - $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (1) 4
 - (c) $2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + 2H_2O$ 1
 - (d) all 4 NH dot and cross bonds (1)
 - single bonding pair between N's **and** two non-bonding electrons on each N **and** no non-bonding e on H and nitrogen octet complete (1) 2
 - (e) (i) proton acceptor 1
 - (ii) $N_2H_4 + H_2O \rightleftharpoons N_2H_5^+ + OH^-$ 1
2. **O/N 20/P41/Q3/c,d(i,ii)**
 - (a) test: (damp red) litmus paper (1) 2
 - result: (litmus goes) blue (1) 1
 - (b) (i) diffusion
 - (ii) particles move from an area of high to low concentration 2
 - particles move randomly
3. **O/N 20/P43/Q3/d,e**
 - (a) $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$
 - all formulae (1) 2
 - equation fully correct(1)
 - (b) (M_r of NH_4NO_3 =) 80 (1) 2
 - 35% (1)
4. **M/J 20/P42/Q1/b(i)** 1
 - (a) (i) N
5. **M/J 20/P41/Q4/a,b** 1
 - (a) (damp) litmus 1
 - (turns) blue 1
 - (b) (i) proton acceptor 1
 - (ii) Above pH 7 up to 11 1
 - (iii) blue precipitate 1
 - precipitate dissolves 1
 - deep blue solution remains 1
6. **M/J 18/P42/Q2/g** [1]
 - (a) phosphorus / P



Unit 12

SULFUR

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

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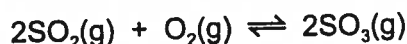
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Unit 12: Sulfur

1. O/N 22/P42/Q3

Sulfuric acid is manufactured by an industrial process. Sulfur is obtained from sulfur-containing metal ores.

The sulfur in the metal ore is converted to sulfur dioxide which is then oxidised to sulfur trioxide as shown.



(a) Name a metal ore which contains sulfur.

..... [1]

(b) Describe the process which converts metal ores to sulfur dioxide.

..... [1]

(c) Name the industrial process used to manufacture sulfuric acid.

..... [1]

(d) The reaction that produces sulfur trioxide is an equilibrium. The forward reaction is exothermic.

(i) State the temperature and pressure used to make sulfur trioxide.

temperature °C

pressure atm [2]

(ii) Name the catalyst used.

..... [1]

(iii) Describe **two** features of an equilibrium.

1

2 [2]

(iv) State the effect, if any, on the position of equilibrium when the following changes are made.

Explain your answers.

temperature is increased

.....

pressure is increased

..... [4]

(v) Explain, in terms of particles, what happens to the rate of reaction when the temperature is increased.

.....

.....

 [3]

(e) Name the compound formed when sulfuric acid reacts with ammonia. .

..... [1]

2. O/N 22/P43/Q5

This question is about sulfuric acid, H_2SO_4 , and salts that can be made from sulfuric acid.

(a) Sulfuric acid is manufactured by the Contact process.

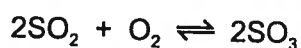
stage 1 Molten sulfur burns in air to produce sulfur dioxide.

stage 2 Sulfur dioxide reacts with oxygen to form sulfur trioxide, SO_3 .

stage 3 Sulfur trioxide reacts with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum is converted into sulfuric acid.

(i) The equation for the reaction in **stage 2** is shown.



State the temperature and pressure used in **stage 2**.

Name the catalyst used in **stage 2**.

temperature °C

pressure atm

catalyst [3]

(ii) Write the chemical equation for the reaction in **stage 3**.

..... [1]

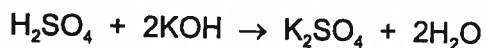
(iii) Name the substance that reacts with oleum in **stage 4**.

..... [1]

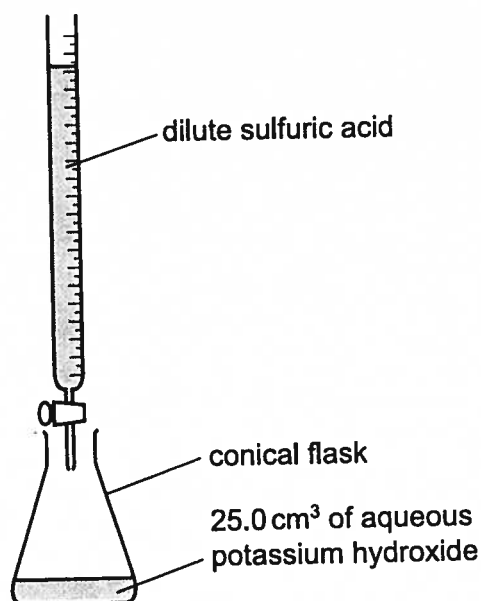
(b) Name the black solid that is produced when concentrated sulfuric acid is added to sugar, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$.

..... [1]

(c) Dilute sulfuric acid and aqueous potassium hydroxide are used to make aqueous potassium sulfate.



The method includes use of the following apparatus.



- (i) Calculate the volume of 0.0625 mol/dm^3 dilute sulfuric acid, H_2SO_4 , that completely reacts with 25.0 cm^3 of 0.100 mol/dm^3 potassium hydroxide, KOH , to produce aqueous potassium sulfate.

Use the following steps.

- Calculate the number of moles of KOH in 25.0 cm^3 of 0.100 mol/dm^3 KOH .

= mol

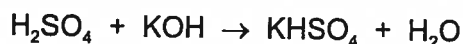
- Deduce the number of moles of H_2SO_4 that react with KOH .

= mol

- Calculate the volume of H_2SO_4 required.

volume = cm^3 [3]

- (ii) The experiment is repeated using the same volume and concentration of potassium hydroxide and the same concentration of dilute sulfuric acid. In this second experiment, the product is aqueous potassium hydrogensulfate, KHSO_4 .



Use your answer to (c)(i) and the equation to deduce the volume of H_2SO_4 required.

volume = cm^3 [1]

- (d) Aqueous potassium hydrogensulfate, $\text{KHSO}_4(\text{aq})$, contains the ions $\text{K}^+(\text{aq})$, $\text{H}^+(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$.

Describe the observations in the following tests.

- (i) A flame test is carried out on aqueous potassium hydrogensulfate.

..... [1]

- (ii) Solid copper(II) carbonate is added to aqueous potassium hydrogensulfate.

.....

..... [2]

- (iii) An acidic solution containing aqueous barium ions, $\text{Ba}^{2+}(\text{aq})$, is added to aqueous potassium hydrogensulfate.

..... [1]

- (e) Write the ionic equation for the reaction in (d)(iii).

Include state symbols.

..... [3]

3. M/J 22/P41/Q4

This question is about compounds of sulfur.

- (a) Sulfuric acid, H_2SO_4 , is manufactured using the Contact process. This manufacture involves four stages.

stage 1 Molten sulfur burns in air to produce sulfur dioxide.

stage 2 Sulfur dioxide reacts with oxygen to form sulfur trioxide.

stage 3 Sulfur trioxide combines with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum reacts to form concentrated sulfuric acid.

- (i) Write a chemical equation for the reaction occurring in **stage 1**.

..... [1]

- (ii) State the essential conditions that are necessary for **stage 2**. Write an equation for the chemical reaction that occurs.

.....

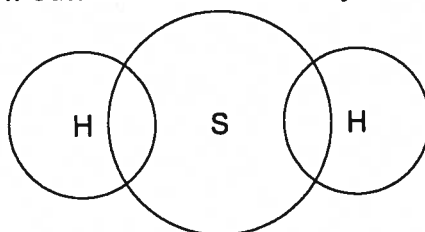
.....

.....[4]
 (iii) Write a chemical equation for the reaction occurring in **stage 3**.

.....[1]
 (iv) Name the substance that reacts with oleum in **stage 4**.

..... [1]
 (b) Hydrogen sulfide has the formula H_2S .

(i) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of hydrogen sulfide. Show outer shell electrons only.



..... [2]
 (ii) Balance the chemical equation for the reaction of hydrogen sulfide with sulfur dioxide shown.
 $.....H_2S + SO_2 \rightarrowS +H_2O$ [1]

4. O/N 20/P41/Q5/b(i),c

(a) Sulfuric acid is manufactured using the Contact process. This manufacture involves four stages.

(i) **Stage 1** involves the combustion of sulfur to form sulfur dioxide.
 Write the chemical equation for **stage 1**.

.....[1]
 (b) In **stage 3** sulfur trioxide, SO_3 , is converted to oleum, $H_2S_2O_7$.
 In **stage 4** oleum reacts to form sulfuric acid, H_2SO_4 .
 State what oleum reacts with in **stage 4**.

5. M/J 20/P42/Q3/b,c,e,f

(a) Sulfur trioxide is converted into oleum, $H_2S_2O_7$, in **stage 3**.
 What is sulfur trioxide reacted with to convert it into oleum?

.....[1]
 (b) Oleum is converted into sulfuric acid in **stage 4**.
 Write a chemical equation for the conversion of oleum, $H_2S_2O_7$, into sulfuric acid.

.....[2]
 (c) Sulfur dioxide is a reducing agent.
 Give the colour change that occurs when excess sulfur dioxide is bubbled into acidified aqueous potassium manganate(VII).

starting colour of the solution

final colour of the solution[1]

- (d) When sulfuric acid reacts with ammonia the salt produced is ammonium sulfate.

Write the chemical equation for this reaction.

.....[2]

6. M/J 20/P41/Q3/a

Sulfur dioxide, SO_2 , is used in the manufacture of sulfuric acid.

- (a) In the first stage of the process, sulfur dioxide is obtained from sulfur-containing ores.

Name **one** of these ores.

.....[1]

7. O/N 19/P42/Q3/b

Ammonia is an important chemical.

- (a) Ammonia is a base and reacts with sulfuric acid to form the salt, ammonium sulfate.

(i) What is meant by the term *base*?

.....[1]

(ii) Name the industrial process used to manufacture sulfuric acid.

.....[1]

(iii) Write a chemical equation for the reaction between ammonia and sulfuric acid.

.....[2]

8. O/N 19/P41/Q2/c,d

- (a) The manufacture of sulfuric acid by the Contact process occurs in four stages.

stage 1 Molten sulfur is burned in air to produce sulfur dioxide gas.

stage 2 Sulfur dioxide is reacted with oxygen to form sulfur trioxide.

stage 3 Sulfur trioxide is combined with concentrated sulfuric acid to form oleum, $\text{H}_2\text{S}_2\text{O}_7$.

stage 4 Oleum is added to water to form sulfuric acid.

- (i) Complete the chemical equation for **stage 1** by adding the appropriate state symbols.



- (ii) Name the catalyst used in **stage 2** and state the temperature used.

catalyst

temperature °C [2]

- (iii) Write chemical equations for the reactions in **stage 3** and **stage 4**.

stage 3

stage 4[2]

- (b) Sulfur dioxide is a toxic gas.

- (i) State one **environmental** reason why sulfur dioxide should **not** be released into the atmosphere.

..... [1]

(ii) Describe the test for sulfur dioxide.

test

observations[2]

9. O/N 18/P42/Q3/a,b

Sulfur is an important element.

(a) Explain how burning fossil fuels containing sulfur leads to the formation of acid rain.

.....
..... [2]

(b) Sulfuric acid is manufactured by the Contact process. One step in the Contact process involves a reversible reaction in which sulfur trioxide, SO_3 , is formed.

(i) Write a chemical equation for this reversible reaction. Include the correct symbol to show that the reaction is reversible.

..... [2]

(ii) State the conditions and name the catalyst used in this reversible reaction.

temperature

pressure

catalyst[2]

(iii) Describe how the sulfur trioxide formed is converted into sulfuric acid in the next steps of the Contact process.

.....
..... [2]

10. O/N 18/P41/Q4/a,b

(a) Sulfuric acid is made industrially by a four-step process.

step 1 Sulfur is burned in air to produce sulfur dioxide.

step 2 Sulfur dioxide is converted into sulfur trioxide.

step 3 Sulfur trioxide is reacted with concentrated sulfuric acid to produce oleum.

step 4 Oleum is reacted with water to produce concentrated sulfuric acid.

(i) Some sulfur is obtained by mining.

Name **one** other major source of sulfur.

..... [1]

(ii) What is the name of the process by which sulfuric acid is made industrially?

..... [1]

(iii) Describe the conversion of sulfur dioxide into sulfur trioxide in **step 2**.

In your answer, include:

- a chemical equation for the reaction
- the essential reaction conditions.

.....

 [5]

(b) When concentrated sulfuric acid is added to glucose, $C_6H_{12}O_6$, a black solid is produced. The concentrated sulfuric acid acts as a dehydrating agent.

(i) What is removed from the glucose in this reaction?

..... [1]

(ii) Name the black solid produced in this reaction.

..... [1]

11. M/J 18/P42/Q2/f

This question is about the elements in Period 3 of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar
----	----	----	----	---	---	----	----

For each of the following, identify a Period 3 element which matches the description. Each element may be used once, more than once or not at all.

State which Period 3 element:

(a) forms **two** different oxides during the Contact process

..... [1]

12. O/N 16/P43/Q5/a,b,c,d,e

Sulfuric acid can be manufactured from the raw materials sulfur, air and water. The process can be divided into four stages.

stage 1 converting sulfur into sulfur dioxide

stage 2 converting sulfur dioxide into sulfur trioxide

stage 3 converting sulfur trioxide into oleum, $H_2S_2O_7$

stage 4 converting oleum into sulfuric acid

stage 1

(a) (i) Describe how sulfur is converted into sulfur dioxide.

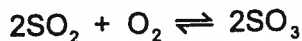
..... [1]

(ii) Write a chemical equation for the conversion of sulfur into sulfur dioxide.

..... [1]

stage 2

- (b) Sulfur dioxide is converted into sulfur trioxide according to the following equation.



The reaction is carried out at a temperature of 450 °C and a pressure of 1–2 atmospheres using a catalyst. The energy change, ΔH , for the reaction is –196 kJ/mol.

- (i) What is the meaning of the symbol $\rightleftharpoons$?

..... [1]

- (ii) Name the catalyst used in this reaction.

..... [1]

- (iii) Why is a catalyst used?

..... [1]

- (iv) If a temperature higher than 450 °C were used, what would happen to the amount of sulfur trioxide produced? Give a reason for your answer.

.....

..... [2]

- (v) Suggest a reason why a temperature lower than 450 °C is **not** used.

..... [1]

- (vi) If a pressure higher than 1–2 atmospheres were used, what would happen to the amount of sulfur trioxide produced? Give a reason for your answer.

.....

..... [2]

stage 3

- (c) (i) What is added to sulfur trioxide to convert it into oleum?

..... [1]

- (ii) Write a chemical equation for the conversion of sulfur trioxide into oleum.

..... [1]

stage 4

- (d) (i) What is added to oleum to convert it into sulfuric acid?

..... [1]

- (ii) Write a chemical equation for the conversion of oleum into sulfuric acid.

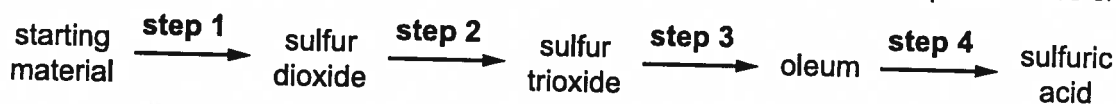
..... [1]

- (e) Give one use of sulfuric acid.

..... [1]

13. M/J 16/P41/Q5

Sulfuric acid is produced by the Contact process. The steps of the Contact process are shown.



- (a) Sulfur is a common starting material for the Contact process.
Name a source of sulfur.

..... [1]

- (b) Describe **step 2**, giving reaction conditions and a chemical equation. Reference to reaction rate and yield is not required.

.....

- (c) **Step 3** involves adding sulfur trioxide to concentrated sulfuric acid to form oleum.
Complete the chemical equation for this reaction.



- (d) Dilute sulfuric acid is a typical acid.

A student adds excess dilute sulfuric acid to a sample of solid copper(II) carbonate in a test-tube.

- (i) Give **three** observations the student would make.

.....
 [2]

- (ii) Give the **names** of all products formed.

..... [1]

- (e) Concentrated sulfuric acid has different properties to dilute sulfuric acid.

When concentrated sulfuric acid is added to glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, steam is given off and a black solid is formed.

- (i) Name the black solid.

..... [1]

- (ii) What type of reaction has occurred?

..... [1]

Answers Section

1. O/N 22/P42/Q3 1
- (a) zinc blende 1
- (b) strong heating in air / roasting in air 1
- (c) contact
- (d) (i) M1 450 (°C) (1) 2
 M2 1–2 (atm) (1) 1
- (ii) vanadium(V) oxide
- (iii) M1 the rate of forward reaction equals (the rate of the) reverse reaction (1) 2
 M2 concentrations of reactants and products are constant (1)
- (iv) *increased temperature:*
 M1 (position of) equilibrium moves to left-hand side (1)
 M2 reaction is exothermic (1)
 increased pressure:
 M3 (position of) equilibrium moves to right-hand side (1) 4
 M4 more (gaseous) moles / molecules on left-hand side (1)
- (v) M1 rate increases **and** particles have more energy (1)
 M2 more collisions (between particles) occur per second / per unit time (1)
 M3 more of the particles / collisions have energy equal to or above the activation energy (1)
 or
 more of the particles / collisions have sufficient energy to react
 or
 a higher percentage / proportion / fraction of collisions (of particles) are successful 3
 / have energy equal to or above activation energy 1
- (e) ammonium sulfate
2. O/N 22/P43/Q5
- (a) (i) M1 450 (1)
 M2 1-2 (1) 3
 M3 vanadium (V) oxide (1) 1
- (ii) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ 1
- (iii) water 1
- (b) carbon
- (c) (i) M1 $2.5 \times 10^{-3} / 0.0025$ (1)
 M2 $1.25 \times 10^{-3} / 0.00125$ (1) 3
 M3 20 (1) 1
- (ii) 40 1
- (d) (i) lilac flame
- (ii) **Any Two from**
 • solid disappears
 • blue solution 2
 • bubbles / effervescence / fizzing 1
- (iii) white precipitate
- (e) $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
 M1 BaSO_4 on the right (1)

- M2 ONLY** Ba^{2+} and SO_4^{2-} on the left (1)
M3 state symbols (1) 3
3. **M/J 22/P41/Q4**
 (a) (i) $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$ 1
 (ii) (temperature) 450°C (1)
 (pressure) 1–2 atmosphere(s) (1)
 vanadium(V) oxide catalyst (1)
 $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ (1) 4
 (iii) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ 1
 (iv) water 1
- (b) (i) 2 bonding pairs as **one dot and cross** each (1)
 2 lone pairs on S (and no additional electrons on Hs) to complete the outer shell on S and both Hs (1) 2
 (ii) $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$ 1
4. **O/N 20/P41/Q5/b(i),c**
 (a) $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$ 1
 (b) water / H_2O 1
5. **M/J 20/P42/Q3/b,c,e,f**
 (a) concentrated sulfuric acid 1
 (b) $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$ 2
 (c) purple to colourless 1
 (d) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$ 2
6. **M/J 20/P41/Q3/a**
 (a) zinc blende 1
7. **O/N 19/P42/Q3/b**
 (a) (i) proton acceptor 1
 (ii) Contact (process) 1
 (iii) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$
 $(\text{NH}_4)_2\text{SO}_4$ (1)
 rest of the equation (1) 2
8. **O/N 19/P41/Q2/c,d**
 (a) (i) l.....g.....g. 1
 (ii) vanadium (V) oxide **or** vanadium pentoxide (1)
 450°C (1) 2
 (iii) $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ (1)
 $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$ (1) 2
 (b) (i) (it causes) acid rain 1
 (ii) test – (aqueous) potassium manganate (VII) (1)
 (purple to) colourless (1) 2
9. **O/N 18/P42/Q3/a,b**
 (a) M1 Sulfur dioxide / SO_2 is formed
 M2 SO_2 reacts with (atmospheric) water (vapour) / rain 2
 (b) (i) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
 M1 Balanced equation
 M2 reversible arrow 2

- (ii) M1 450°C (units required)
M2 1–5 atmospheres (units required)
M3 Vanadium (V) oxide or vanadium pentoxide or V_2O_5 3
- (iii) M1 SO_3 added to (concentrated) H_2SO_4
M2 (Oleum) diluted with / added to water 2
10. O/N 18/P41/Q4/a,b 1
- (a) (i) from petroleum or (crude) oil or fossil fuels 1
(ii) Contact (process) 1
(iii) M1 vanadium pentoxide or vanadium(V) oxide or V_2O_5 (catalyst); 1
M2 1–5 atmospheres; (Units required) 1
M3 450°C; units required 1
M4 $2SO_2 + O_2 \rightarrow 2SO_3$; 1
M5 equilibrium / reversible reaction in equation or text 1
- (b) (i) water / H_2O 1
(ii) carbon / C 1
11. M/J 18/P42/Q2/f 1
- (a) sulfur / S 1
12. O/N 16/P43/Q5/a,b,c,d,e 1
- (a) (i) burned / heated in air 1
(ii) $S + O_2 \rightarrow SO_2$ 1
- (b) (i) equilibrium / reversible 1
(ii) vanadium(V) oxide / vanadium pentoxide 1
(iii) increase rate (of reaction) / allow lower temperature to be used /
allow lower pressure to be used 1
(iv) less SO_3 1
forward reaction is exothermic / it is exothermic / reverse reaction is endothermic 1
(v) rate too low / reaction too slow / slower 1
(vi) more SO_3 1
fewer moles or molecules (of gas) on right-hand side /
more moles or molecules (of gas) on left-hand side 1
- (c) (i) concentrated sulfuric acid / concentrated H_2SO_4 1
(ii) $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$ 1
- (d) (i) water 1
(ii) $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$ 1
- (e) detergents / car batteries / dyes / paints / synthetic resins /
printing inks / metal extraction / cleaning metals / 1
13. M/J 16/P41/Q5 1
- (a) (sulfur-containing) fossil fuels; 1
- (b) M1 vanadium pentoxide / vanadium(V) oxide / V_2O_5 (catalyst);
M2 1–5 atmospheres (units required);
M3 450°C (units required);
M4 $2SO_2 + O_2 \rightarrow 2SO_3$; 5
M5 equilibrium / reversible reaction; 1
- (c) $H_2S_2O_7$; 1

- (d) (i) 3 correct (2 marks)
2 correct (1 mark)
bubbles/effervescence/fizzing;
dissolves/disappears/*forms* a solution;
blue (solution); 2
- (ii) carbon dioxide and water and copper(II) sulfate; 1
- (e) (i) carbon; 1
- (ii) dehydration; 1

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Unit 13

ORGANIC CHEMISTRY

IGCSE Chemistry P-4
Topical Workbook
Article Number: 714

Editorial Board

Syllabus

- 13.1 Names of compounds
- 13.2 Fuels
- 13.3 Homologous series
- 13.4 Alkanes
- 13.5 Alkenes
- 13.6 Alcohols
- 13.7 Carboxylic acids
- 13.8 Polymers

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Unit 13: Organic Chemistry

13.1: Names of compounds

1. O/N 19/P42/Q1/c

(a) One element in the first 36 elements is used as the fuel in a fuel cell.

(i) Name this element.

.....[1]

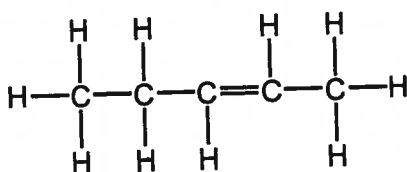
(ii) Write the overall chemical equation for the reaction which occurs when the element in (c)(i) reacts in a fuel cell.

.....[2]

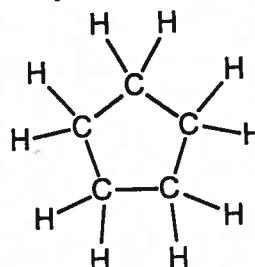
2. M/J 19/P43/Q6/a(i)

Two hydrocarbons have the structures shown.

hydrocarbon A



hydrocarbon B



(a) (i) Why are these two compounds *hydrocarbons*?

.....
.....[2]

3. M/J 18/P42/Q7/e

Many organic compounds, such as alcohols, carboxylic acids and esters, contain the elements carbon, hydrogen and oxygen only.

(e) Alkanes and alkenes are hydrocarbons.

(i) What is meant by the term *hydrocarbon*?

.....
..... [2]

(ii) State the general formula of:

alkanes

alkenes [2]

4. M/J 18/P41/Q6

The table shows the structures of four hydrocarbons.

P	Q	R	S
CH_3-CH_3	$\text{CH}_2=\text{CH}_2$	$\text{CH}_2=\text{CH}-\text{CH}_3$	$\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_3$

- (a) Why are compounds **P**, **Q**, **R** and **S** known as hydrocarbons?

.....
..... [2]

- (b) Compound **P** is saturated.

What is meant by the term *saturated*?

..... [1]

- (c) Compound **P** undergoes a substitution reaction with chlorine.

- (i) What is meant by the term *substitution reaction*?

..... [1]

- (ii) State a condition required for this reaction to occur.

..... [1]

- (iii) Write a chemical equation for this reaction.

..... [2]

- (d) Compound **R** undergoes an addition reaction with bromine.

- (i) Why is this reaction an addition reaction?

..... [1]

- (ii) A compound containing bromine is formed in this reaction.

Draw the structure of this compound. Show all of the atoms and all of the bonds.

[1]

- (e) Draw the structure of an unbranched isomer of compound **S**. Show all of the atoms and all of the bonds. Name this unbranched isomer of compound **S**.
structure

name..... [2]

- (f) Compound **Q** undergoes polymerisation.

- (i) Name the polymer formed.

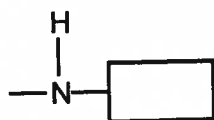
..... [1]

- (ii) Complete the chemical equation to show the polymerisation of compound **Q**.



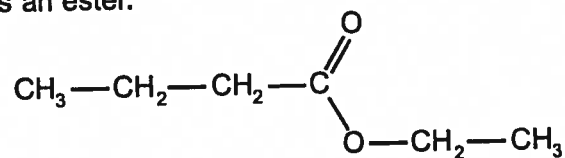
- (g) Amino acids undergo polymerisation to form proteins. Part of a protein molecule with the linkages missing is shown.

Draw the linkages on the diagram. Show all of the atoms and all of the bonds.



[2]

(h) The structure shows an ester.



Write the word equation for a reaction which could be used to make this ester.

.....[3]

Answers Section

1. O/N 19/P42/Q1/c
 - (a) (i) hydrogen / H 1
 - (ii) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
 water as product from reaction of hydrogen and oxygen (1)
 balanced (1) 2
2. M/J 19/P43/Q6/a(i)
 - (a) M1 contain hydrogen and carbon
 M2 only 2
3. M/J 18/P42/Q7/e
 - (a) (i) (they contain) carbon and hydrogen (atoms)
 only 1
 - (ii) alkane: $\text{C}_n\text{H}_{2n+2}$ 1
 alkene: C_nH_{2n} 1
4. M/J 18/P41/Q6
 - (a) (they contain) carbon and hydrogen (atoms)
 only 1
 - (b) (all) the (C—C) bonds are single 1
 - (c) (i) (one) atom or group is replaced by another (atom or group) 1
 (ii) ultra-violet light OR sunlight 1
 (iii) $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$
 1 mark for $\text{C}_2\text{H}_5\text{Cl}$
 1 mark for the rest of the equation 2
 - (d) (i) only one product (compound) forms 1
 (ii) fully displayed formula of 1,2-dibromopropane 1
 - (e) fully displayed formula of but-2-ene 1
 but-2-ene 1
 - (f) (i) poly(ethene) 1
 (ii) single bond between two C atoms 1
 fully correct answer 1
 - (g) any one correct amide link showing all bonds 1
 both amide links shown in the correct orientation for three amino acids 1
 - (h) ethanol + butanoic acid $\rightarrow$ ethyl butanoate + water 1
 1 mark for the names of the reactants
 1 mark for the name of the ester
 1 mark for water as a product 3

13.2: Fuels

1. O/N 20/P41/Q1/a(i)

(a) This question is about elements.

aluminium
carbon
iron
hydrogen
oxygen
silicon
sodium
sulfur

Answer the following questions about these elements.

Each element may be used once, more than once or not at all.

(i) Name the element that can be used as a fuel.

..... [1]

2. O/N 20/P43/Q1/e

The names of nine substances are shown.

aluminium oxide
ammonia
carbon monoxide
anhydrous cobalt(II) chloride
hydrated copper(II) sulfate
iron(III) oxide
nitrogen dioxide
silver
steel

Answer the following questions using these substances. Each substance may be used once, more than once or not at all.

Name the substance that is:

(a) a gas formed by the incomplete combustion of fossil fuels [1]

3. O/N 20/P43/Q6/a(i)

(a) A carboxylic acid and an ester are structural isomers.

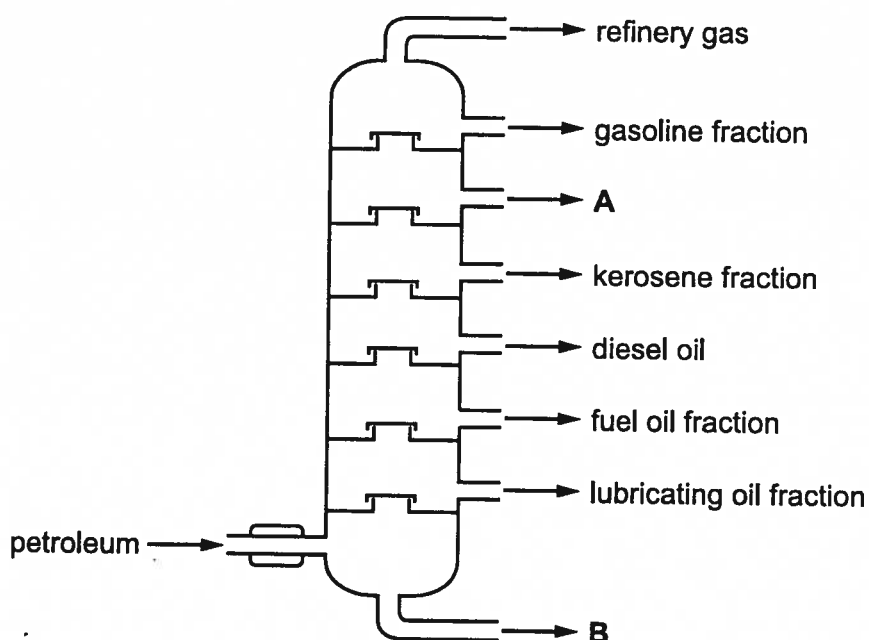
(i) State the meaning of the term *structural isomers*.

.....
..... [2]

4. MAR 20/P42/Q1/a,b,c,d

Petroleum is a useful natural resource.

The diagram shows how petroleum can be separated into useful substances.



(a) What is the name of the separation process shown in the diagram?

.....[2]

(b) Name the fraction leaving at:

A

B[2]

(c) Refinery gas is a mixture of hydrocarbons.

One refinery gas is butane, C_4H_{10} .

(i) Suggest the names of **two** other refinery gases.

..... and [2]

(ii) Write the chemical equation for the complete combustion of butane.

..... [2]

(iii) Name the toxic gas produced by the incomplete combustion of butane.

..... [1]

(d) Gasoline and kerosene are both fuels. They have different properties.

(i) Describe the differences in the properties given.

viscosity of the fuel

flammability of the fuel [2]

(ii) What difference in the molecules of gasoline and kerosene causes these differences in properties?

..... [1]

5. M/J 18/P41/Q6/d,e

(a) Compound **R** undergoes an addition reaction with bromine.

(i) Why is this reaction an addition reaction?

..... [1]

(ii) A compound containing bromine is formed in this reaction.

Draw the structure of this compound. Show all of the atoms and all of the bonds.

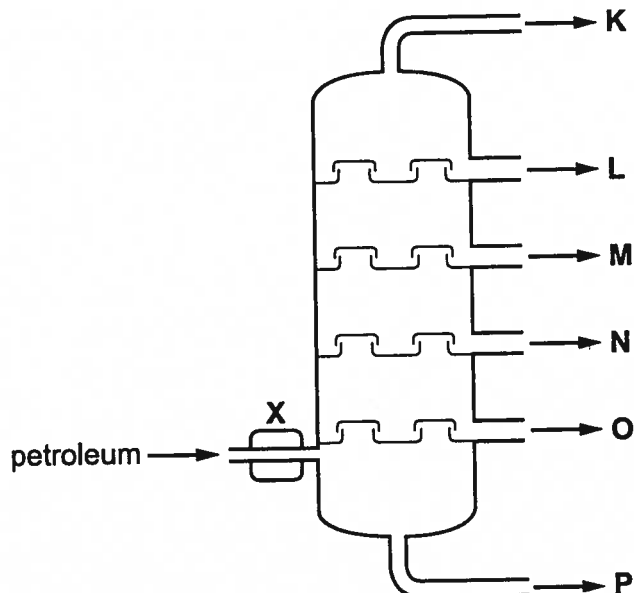
[1]

(b) Draw the structure of an unbranched isomer of compound **S**. Show all of the atoms and all of the bonds. Name this unbranched isomer of compound **S**.
structure

name.....[2]

6. O/N 16/P41/Q3/a

(a) Petroleum is a mixture of hydrocarbons. It is separated into useful fractions by fractional distillation. This can be done using the fractionating column shown.



(i) What happens to the petroleum at point **X**, before it enters the fractionating column?

..... [1]

(ii) State **two** ways in which fraction **O** differs from fraction **L**.

.....
..... [2]

7. M/J 16/P41/Q6/a,b,c

Petroleum is a source of many important chemicals.

- (a) Name **two** industrial processes which must take place to produce alkenes from petroleum.

.....

..... [2]

- (b) Ethene, $\text{CH}_2=\text{CH}_2$, and propene, $\text{CH}_2=\text{CHCH}_3$, can both be converted into polymers.

- (i) What type of polymerisation takes place when ethene forms a polymer?

..... [1]

- (ii) What is the empirical formula of the polymer formed from ethene?

..... [1]

- (iii) Propene has the structural formula $\text{CH}_2=\text{CHCH}_3$.

Draw **two** repeat units of the polymer made from propene.

[2]

- (c) Ethene will react with steam to form ethanol.

Propene will react with steam to form two isomers, both of which are alcohols.
Suggest the structures of these alcohols.

[2]

Answers Section

- | | |
|--|---------------------------------|
| 1. O/N 20/P41/Q1/a(i)
(a) hydrogen / carbon | 1 |
| 2. O/N 20/P43/Q1/e
(a) carbon monoxide | 1 |
| 3. O/N 20/P43/Q6/a(i)
(a) (i) compounds with the same molecular formula (1)
different structural formula (1) | 2 |
| 4. MAR 20/P42/Q1/a,b,c,d
(a) distillation (1)
fractional (1)
(b) naphtha (1)
bitumen (1)
(c) (i) any two from:
methane ethane propane
(ii) $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$
CO ₂ and H ₂ O as products (1)
correct equation (1)
(iii) carbon monoxide
(d) (i) kerosene is more viscous
gasoline is more flammable
(ii) kerosene molecules with a longer carbon chain | 2
2
2
2
1
2
1 |
| 5. M/J 18/P41/Q6/d,e
(a) (i) only one product (compound) forms
(ii) fully displayed formula of 1,2-dibromopropane
(b) fully displayed formula of but-2-ene
but-2-ene | 1
1
1
1 |
| 6. O/N 16/P41/Q3/a
(a) (i) heated / evaporated / boiled
(ii) any 2 from:
(O is) more viscous / thicker
(O is) darker
(O has) longer / bigger molecules / more carbon atoms
(O has a) higher boiling point OR melting point
(O is) less flammable | 1
2 |
| 7. M/J 16/P41/Q6/a,b,c
(a) fractional distillation;
cracking;
(b) (i) addition;
(ii) CH ₂ ;
(iii) | 2
1
1 |
- ```

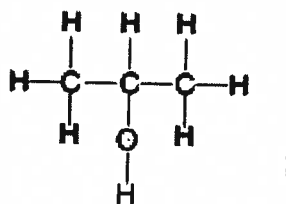
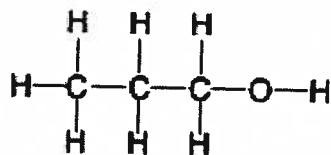
 H H H H
 | | | |
 — C — C — C — C —
 | | | |
 CH3 H CH3 H

```

**M1** chain of 4 carbon atoms with single bonds and continuation bonds;  
**M2** correctly positioned CH<sub>3</sub> side chains;

2

(c)



2

### 13.3: Homologous series

1. O/N 20/P42/Q4/a

Alkenes and alkanes are homologous series of compounds containing carbon and hydrogen atoms.

(a) State the name of the type of compound made from carbon and hydrogen atoms only.

..... [1]

2. O/N 20/P41/Q6/a,e(i)

(a) Ethane, propane and butane are members of the same homologous series.

(i) Name this homologous series.

..... [1]

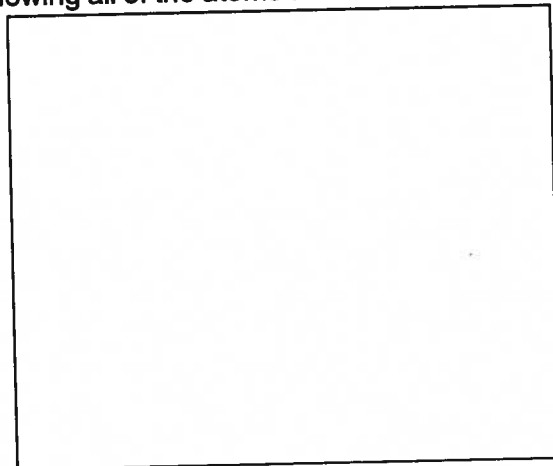
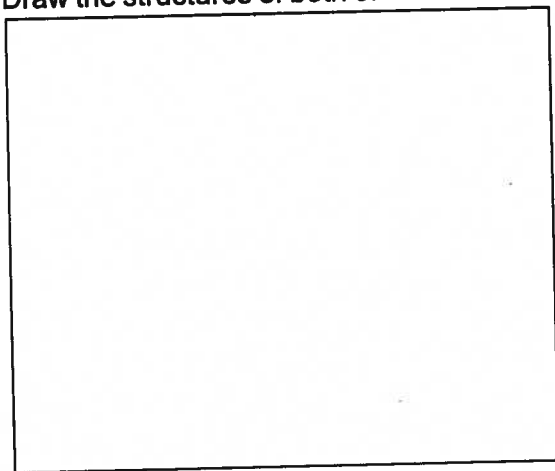
(ii) State **two** ways members of the same homologous series are similar.

1 .....

2 ..... [2]

(b) There are two structural isomers with the formula  $C_4H_{10}$ .

(i) Draw the structures of both of these isomers, showing all of the atoms and all of the bonds.

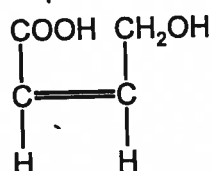


[2]

3. M/J 20/P42/Q6/c

(a) There are three functional groups in compound A.

compound A



(i) Name the homologous series of compounds that contains the following structures.

$C=C$  .....

$-OH$  .....

$-COOH$  ..... [3]

(ii) What would you observe when compound **A** is added to:

aqueous bromine .....

aqueous sodium carbonate? .....

[2]

4. **M/J 19/P42/Q1/a**

The names of eight substances are given.

aluminium oxide

calcium oxide

ethanol

nitrogen

iron(III) oxide

methane

oxygen

silicon(IV) oxide

Answer the following questions about these substances.

Each substance may be used once, more than once or not at all.

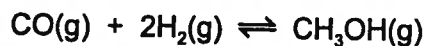
State which substance is:

(a) the main constituent of natural gas

..... [1]

5. **M/J 19/P42/Q4/c(i,ii)**

Methanol is made industrially by reacting carbon monoxide with hydrogen. The gases react at a temperature of 250 °C and a pressure of 75 atmospheres.



The forward reaction is exothermic.

(a) Methanol is a member of the homologous series of alcohols.

(i) State **two** general characteristics of a homologous series.

1 .....

2 .....

[2]

(ii) Draw the structures of **two** different alcohols, each containing **three** carbon atoms. Show all of the atoms and all of the bonds.

Name these **two** alcohols.

name .....

name .....

[4]

**6. M/J 17/P42/Q6/a**

(a) An homologous series is a 'family' of organic compounds whose names have the same ending.

(i) Name the homologous series for which the names of the organic compounds end in *-ene* and *-oic acid*.

*-ene* ..... [1]

*-oic acid* ..... [1]

(ii) State **two** characteristics of an homologous series.

.....  
..... [2]

**7. M/J 17/P41/Q6/a,b**

The alkenes and alkanes are both examples of homologous series which are hydrocarbons.

(a) What is meant by the term *hydrocarbon*?

.....  
..... [2]

(b) Give **three** characteristics of an homologous series.

1 .....  
2 .....  
3 ..... [3]

**8. O/N 16/P41/Q3/b**

(a) Most of the hydrocarbons obtained from petroleum are alkanes. The alkanes are an homologous series of saturated hydrocarbons with the general formula  $C_nH_{2n+2}$ .

Give **two** characteristics, other than having the same general formula, of members of an homologous series.

.....  
..... [2]

## Answers Section

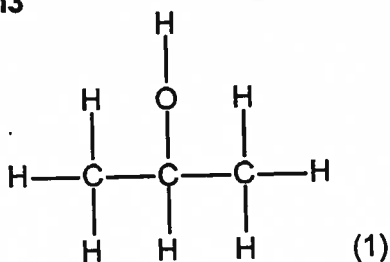
1. O/N 20/P42/Q4/a  
(a) hydrocarbon 1
2. O/N 20/P41/Q6/a,e(i)  
(a) (i) alkanes 1  
(ii) one mark each for any two of:  
  - same chemical properties
  - same functional group
  - same general formula
  - (consecutive members) differ by  $\text{CH}_2$
  - common (allow similar) methods of preparation
  - physical properties vary in predictable manner / show trends / gradually change OR example of a physical property variation i.e. melting point / boiling point / volatility 2
- (b) (i)
- $$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & | & & | & & | & & | \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & | & & | & & | & & | \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \end{array}$$

(1)

$$\begin{array}{ccccc} & \text{H} & & \text{H} & & \text{H} \\ & | & & | & & | \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & | & & | & & | \\ & \text{H} & & \text{C} & & \text{H} \\ & & & | & & \\ & & & \text{H} & & \end{array}$$

(1)
- 2
3. M/J 20/P42/Q6/c  
(a) (i) alkene (1)  
 alcohols / alkanols (1)  
 carboxylic acids/alkanoic acids (1)  
 (ii) turns colourless / decolourised (1)  
 bubbles / fizzing / effervescence (1) 3  
 2
4. M/J 19/P42/Q1/a  
(a) methane 1
5. M/J 19/P42/Q4/c(i,ii)  
(a) (i) any 2 from:  
 ∞ same or **similar** chemical properties or reactions (1)  
 ∞ (same) general formula (1)  
 ∞ (consecutive members) differ by  $\text{CH}_2$  (1)  
 ∞ same functional group (1)  
 ∞ common (allow **similar**) methods of preparation (1)  
 ∞ physical properties vary in predictable manner / show trends / gradually change OR  
 example of a physical property variation i.e. melting point / boiling point / volatility 2  
 (ii) M1
- $$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} \\ & | & & | & & | \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{O} - \text{H} \\ & | & & | & & | \\ & \text{H} & & \text{H} & & \text{H} \end{array}$$

(1)
- M2 propan-1-ol (1)

**M3****M4** propan-2-ol (1)

4

**6. M/J 17/P42/Q6/a**

1

- (a) (i) alkene  
carboxylic acid

1

- (ii) any 2 from:

- ∞ same / similar chemical properties
- ∞ (same) general formula
- ∞ (consecutive members) differ by  $\text{CH}_2$
- ∞ same functional group
- ∞ common (allow similar) methods of preparation
- ∞ physical properties vary in predictable manner / show trends / gradually change / example of a physical property variation

2

**7. M/J 17/P41/Q6/a,b**

1

- (a) (compound that) contains carbon and hydrogen  
and no other elements / only

1

- (b) any 3 from:

- ∞ same / similar chemical properties
- ∞ (same) general formula
- ∞ (consecutive members) differ by  $\text{CH}_2$
- ∞ same functional group
- ∞ common (allow similar) methods of preparation
- ∞ physical properties vary in predictable manner / show trends / gradually change / example of a physical property variation

3

**8. O/N 16/P41/Q3/b**

- (a) any 2 from:

- similar / same chemical properties
- same functional group
- trend / pattern in physical properties
- (neighbouring members) differ by  $\text{CH}_2$
- common methods of preparation

2



## 13.4: Alkanes

### 1. M/J 21/P42/Q2/f

(a) Members of one homologous series only react with chlorine in the presence of sunlight.

(i) Name a member of this homologous series.

..... [1]

(ii) Name **two** products that form when the compound in (i) reacts with chlorine.

1 .....

2 ..... [2]

### 2. O/N 20/P42/Q4/c

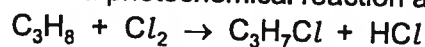
(a) Propane undergoes a substitution reaction with chlorine.

Write the chemical equation for the reaction between one molecule of propane and one molecule of chlorine.

..... [2]

### 3. M/J 20/P42/Q6/a,b(i)

(a) Propane reacts with chlorine in a photochemical reaction as shown.



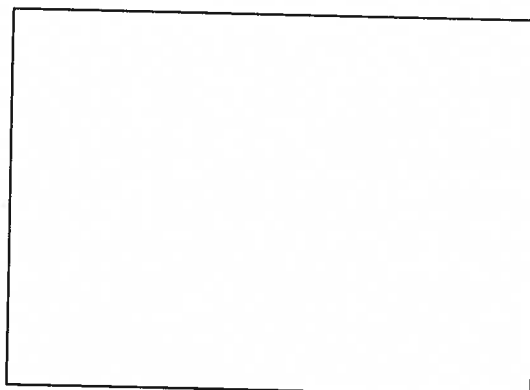
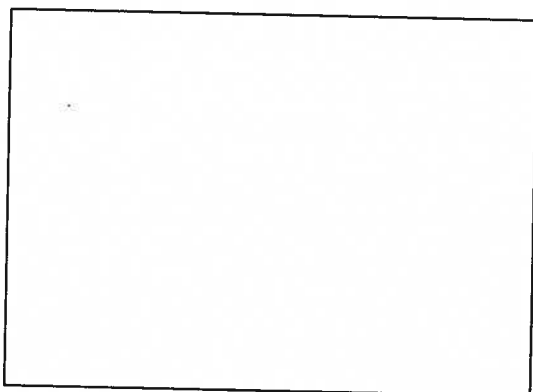
(i) What type of reaction is this?

..... [1]

(ii) What condition is needed for this photochemical reaction to occur?

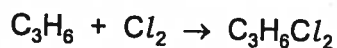
..... [1]

(iii) Draw **two** structural isomers of compounds with the formula  $\text{C}_3\text{H}_7\text{Cl}$   
Show all of the atoms and all of the bonds.



[2]

(b) Propene reacts with chlorine in an addition reaction as shown.



(i) State why this is an addition reaction.

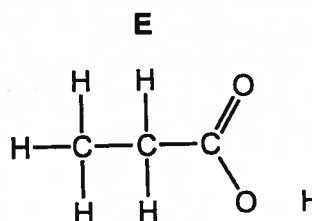
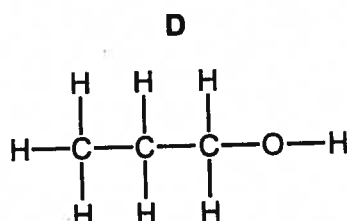
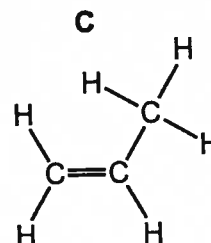
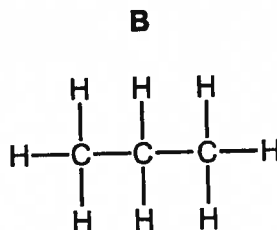
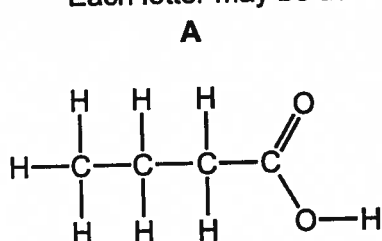
..... [1]

## 4. M/J 20/P43/Q1/a(ii,iii,iv,viii)

(a) The structures of five organic compounds, A, B, C, D and E, are shown.

Answer the questions that follow.

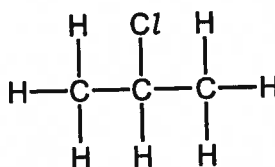
Each letter may be used once, more than once or not at all.

(i) Give the letter of the compound that has the empirical formula  $\text{CH}_2$ —

..... [1]

(ii) Give the letter of **one** compound that reacts with bromine in an addition reaction.

..... [1]

(iii) Give the letter of **one** compound that reacts with chlorine to form the compound shown.

..... [1]

(iv) Draw a structural isomer of compound D.

Show all of the atoms and all of the bonds.

[1]

## 5. M/J 17/P42/Q2/a

Carbon and silicon are elements in Group IV of the Periodic Table. Both carbon and silicon exist as more than one isotope.

(a) Define the term *isotopes*.

.....

..... [2]

**6. M/J 17/P42/Q4/a,b**

Nickel, copper and zinc are three consecutive elements in the Periodic Table.

(a) Nickel and copper are transition elements.

State **three** chemical properties of transition elements.

.....

.....

..... [3]

(b) Copper(II) oxide is a basic oxide but zinc oxide is an amphoteric oxide. Both oxides are insoluble in water.

You are provided with a mixture of solid copper(II) oxide and solid zinc oxide. Describe how you would obtain a sample of copper(II) oxide from this mixture.

.....

.....

..... [3]

**7. O/N 16/P42/Q5/c,d**

Chlorine, bromine and iodine are halogens.

(a) When one mole of bromine,  $\text{Br}_2$ , reacts with one mole of propene, one organic product is formed.

(i) Which part of the propene molecule reacts with bromine?

..... [1]

(ii) What is the name of the type of reaction which takes place between bromine and propene?

..... [1]

(b) When one mole of chlorine,  $\text{Cl}_2$ , reacts with one mole of propane, a mixture of two structural isomers is formed.

(i) What is the name of the type of reaction which takes place between chlorine and propane?

..... [1]

(ii) Explain what is meant by the term *structural isomers*.

.....

..... [2]

(iii) Draw the structure of **two** structural isomers formed when **one** mole of chlorine reacts with **one** mole of propane.

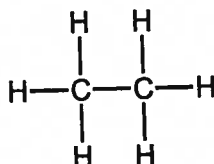
[2]

8. O/N 16/P41/Q3/c,d

- (a) The alkane with the molecular formula  $C_5H_{12}$  can exist as a number of structural isomers. Draw the structures of **two** isomers with the formula  $C_5H_{12}$ .

[2]

- (b) The alkane ethane has the structure shown.



When a mixture of ethane and chlorine is exposed to ultraviolet light a substitution reaction takes place.

Draw the structure of **one** organic product from this substitution reaction.

[1]

## Answers Section

1. **M/J 21/P42/Q2/f**
  - (a) (i) any **named** alkane 1
  - (ii) name of organic product derived from **2(f)(i)** (1)  
hydrogen chloride /  $\text{HCl}$  (1) 2
2. **O/N 20/P42/Q4/c**
  - (a)  $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$   
 $\text{HCl}$  as product  
rest of equation 2
3. **M/J 20/P42/Q6/a,b(i)**
  - (a) (i) substitution 1
  - (ii) ultraviolet light / ultraviolet radiation 1
  - (iii)

$$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{Cl} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$$

$$\begin{array}{c} \text{H} & \text{H} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{Cl} & \text{H} \end{array}$$
  - (b) only one product / double bond becomes single bond / two molecules join (to make one molecule) 2
4. **M/J 20/P43/Q1/a(ii,iii,iv,viii)**
  - (a) (i) C 1
  - (ii) C 1
  - (iii) B 1
  - (iv)

$$\begin{array}{c} & \text{H} & \\ & | & \\ \text{H} & \text{O} & \text{H} \\ | & | & | \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ | & | & | \\ \text{H} & \text{H} & \text{H} \end{array}$$

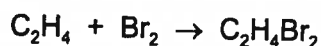
1
5. **M/J 17/P42/Q2/a**
  - (a) atoms of the same element / atoms with the same proton number /  
atoms with the same atomic number 1  
different neutron **number** / different nucleon **number** / different mass number 1
6. **M/J 17/P42/Q4/a,b**
  - (a) any 3 from: 3
    - catalyst
    - more than one / variable oxidation state / oxidation number / valency
    - form coloured **compounds** / coloured **ions**
    - forms complex ions / complexes
  - (b) add sodium hydroxide (solution) /  $\text{NaOH}$  / potassium hydroxide (solution) /  $\text{KOH}$  1  
zinc oxide dissolves / reacts **OR** copper(II) oxide does not dissolve / react 1  
filter / decant / centrifuge (copper(II) oxide) 1
7. **O/N 16/P42/Q5/c,d**
  - (a) (i) the  $(\text{C}=\text{C})$  double bond 1
  - (ii) addition **OR** bromination 1
  - (b) (i) substitution 1
  - (ii) (compounds with the) same molecular formula

- |                                             |   |
|---------------------------------------------|---|
| different structural formulae or structures | 2 |
| (iii) structure of 1-chloropropane          |   |
| structure of 2-chloropropane                | 2 |
8. O/N 16/P41/Q3/c,d
- (a) any 2 structures from:
- |                 |   |
|-----------------|---|
| pentane         |   |
| methylbutane    |   |
| dimethylpropane | 2 |
- (b) correct structure with any number from 1 to 6 of the hydrogen atoms replaced by chlorine atoms
- |  |   |
|--|---|
|  | 1 |
|--|---|

## 13.5: Alkenes

### 1. O/N 22/P41/Q5

Ethene is an alkene which reacts with bromine as shown in the equation.



(a) Write the general formula of alkenes.

..... [1]

(b) Describe the colour change seen when ethene is bubbled through aqueous bromine.

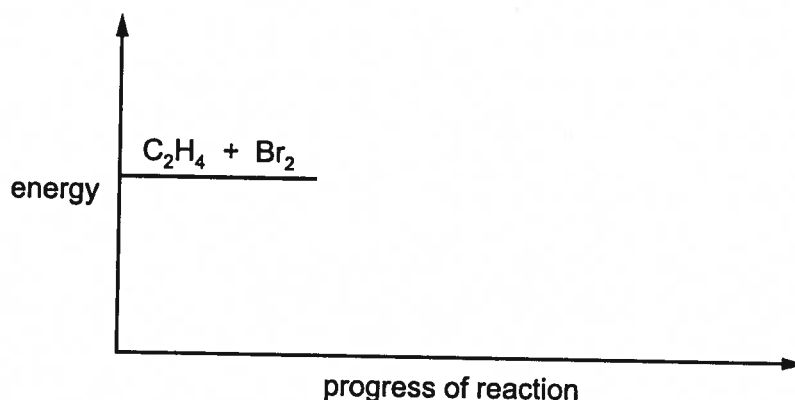
from ..... to ..... [1]

(c) In this reaction only one product is formed from two reactants.

Name this type of organic reaction.

..... [1]

(d) Part of the energy profile diagram of this reaction is shown.

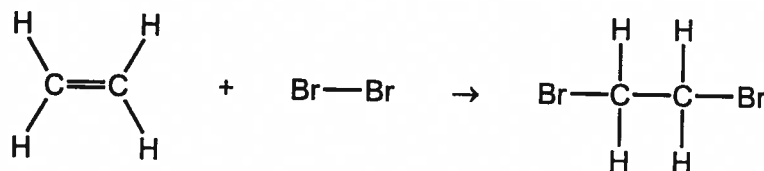


(i) The reaction is exothermic.

Complete the energy profile diagram for this reaction.  
Include:

- the position of the products
- an arrow to show the activation energy, labelled as A
- an arrow to show the energy change for the reaction.

(ii) The chemical equation for the reaction can be represented as shown. [3]



Some bond energies are given.

| bond  | bond energy / kJ mol |
|-------|----------------------|
| C-H   | 410                  |
| C=C   | 610                  |
| Br-Br | 190                  |
| C-C   | 350                  |
| C-Br  | 290                  |

Use the bond energies in the table to calculate the energy change in this reaction.  
Use the following steps.

- Calculate the energy needed to break bonds.

energy = ..... kJ

- Calculate the energy released in making bonds.

energy = ..... kJ

- Determine the energy change in this reaction.

energy change in this reaction = ..... kJ/mol [3]

## 2. O/N 21/P41/Q5

Alkenes and carboxylic acids are both families of similar compounds with similar chemical properties. Alkenes and carboxylic acids have different reactions.

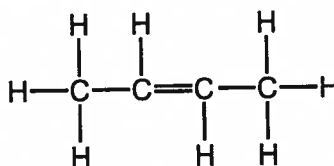
(a) State the term used for a 'family' of similar compounds.

..... [1]

(b) State the general formula of alkenes.

..... [1]

(c) The structure of but-2-ene is shown.



(i) But-2-ene reacts with aqueous bromine in an addition reaction.

Describe the colour change seen when but-2-ene is added to aqueous bromine.



from ..... to ..... [1]

(ii) State what is meant by the term *addition reaction*.

..... [1]

(iii) Write the chemical equation for the reaction between but-2-ene and bromine.

..... [2]

(iv) But-2-ene forms a polymer.

Suggest the name of the polymer formed from but-2-ene.

..... [1]

(v) Name and draw a structural isomer of but-2-ene.

Show all of the atoms and all of the bonds.

name .....

structure

[2]

(d) Butanoic acid,  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ , is a carboxylic acid.

(i) Deduce the empirical formula of butanoic acid.

..... [1]

(ii) Complete the chemical equation for the reaction of butanoic acid and sodium carbonate,  $\text{Na}_2\text{CO}_3$ .

$2\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow \dots + \dots + \dots$  [2]

(iii) Butanoic acid reacts with methanol to form an organic compound and water.

- Name the organic compound formed.

..... [1]

- Draw the structure of the organic compound formed.  
Show all of the atoms and all of the bonds.

[2]

3. O/N 20/P42/Q4/b

(a) Alkenes take part in addition reactions.

(i) Describe what is meant by the term *addition reaction*.

..... [1]

- (ii) Draw the structure of the product made in the addition reaction between propene and bromine. Show all of the atoms and all of the bonds.

[2]

- (iii) Describe the colour change seen when propene is added to aqueous bromine.

from ..... to ..... [2]

- (iv) Draw the structures of molecules of **two** different alkenes which both undergo an addition reaction with steam to form butan-2-ol. Show all of the atoms and all of the bonds.

[2]

4. O/N 20/P41/Q6/d,e(ii)

- (a) (i) Aqueous bromine was added to a sample of ethene.  
Give the colour change seen.

from ..... to ..... [2]

- (ii) Explain, in terms of bonding, why there is no colour change when aqueous bromine is added to ethane.

..... [1]

- (b) There are two structural isomers with the formula  $C_4H_{10}$ .

- (i) Butane is formed when longer chain hydrocarbons are cracked.

Complete the chemical equation to show the other product when butane is formed by cracking.



5. M/J 20/P41/Q3/d

- (a) When propan-1-ol is heated with concentrated sulfuric acid as a catalyst an unsaturated hydrocarbon of relative molecular mass 42 is formed and one other product.

- (i) What is meant by the term *unsaturated*?

..... [1]

- (ii) Write the chemical equation for this reaction.

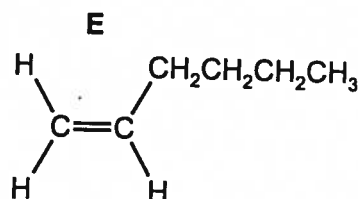
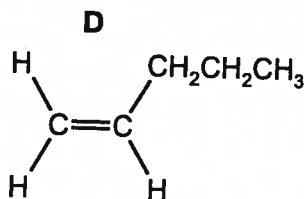
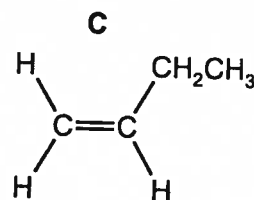
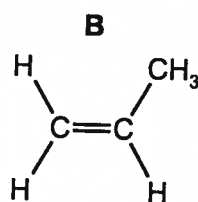
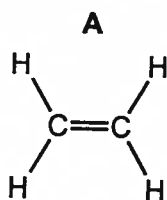
..... [2]

- (iii) Name the unsaturated hydrocarbon formed.

..... [1]

## 6. M/J 19/P41/Q5/a,b,c,d,e,f(ii)

The structures of five alkenes, **A**, **B**, **C**, **D** and **E**, are shown.



(a) What is the general formula of alkenes?

..... [1]

(b) What is the molecular formula of alkene **D**?

..... [1]

(c) Predict which alkene, **A**, **B**, **C**, **D** or **E**, has the highest boiling point.  
Explain your answer.

alkene .....

explanation .....

..... [2]

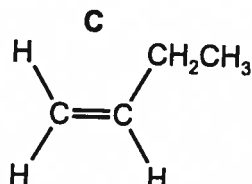
(d) Which alkene, **A**, **B**, **C**, **D** or **E**, diffuses most quickly?  
Explain your answer.

alkene .....

explanation .....

..... [2]

(e) A student added aqueous bromine to alkene **C**.

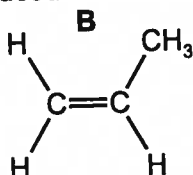


Describe the colour change seen and draw the structure of the product. Show all of the atoms and all of the bonds.

colour change from ..... to .....  
structure .....

[2]

- (f) Two different alcohols can be produced from alkene **B** by an addition reaction.



- (i) State the reagent and conditions needed to produce an alcohol from alkene **B**.

reagent .....

conditions .....

[3]

7. MAR 19/P42/Q3/f(i,ii)

- (a) Tetrafluoroethene is an unsaturated compound with the formula  $C_2F_4$ .

Tetrafluoroethene is the monomer used to make the polymer poly(tetrafluoroethene).

- (i) What is meant by the term *unsaturated*?

..... [1]

- (ii) Describe a test to show that tetrafluoroethene is unsaturated.

test .....

observations..... [2]

8. O/N 18/P42/Q5/a,b,c,d

Alkynes are a homologous series of unsaturated hydrocarbons.

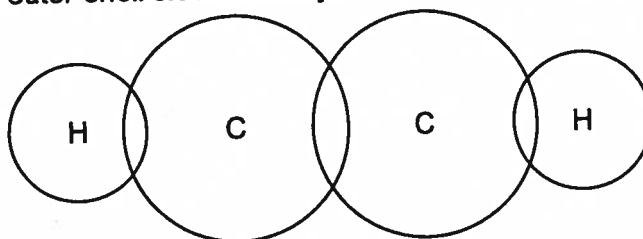
All members contain a  $C \equiv C$  triple bond.

- (a) Complete the table showing information about the first three alkynes.

|           |                  |                     |                          |
|-----------|------------------|---------------------|--------------------------|
| formula   | $C_2H_2$         | $C_3H_4$            |                          |
| structure | $H-C \equiv C-H$ | $H-C \equiv C-CH_3$ | $H-C \equiv C-CH_2-CH_3$ |
| name      | ethyne           |                     | butyne                   |

[2]

- (b) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethyne,  $H-C \equiv C-H$ . Show outer shell electrons only.



[2]

(c) Compounds in the same homologous series have the same general formula.

(i) Give **two** other characteristics of members of a homologous series.

1 .....

2 ..... [2]

(ii) Use the information in the table in (a) to deduce the general formula of alkynes.

..... [1]

(d) Alkynes are unsaturated.

Describe a test for unsaturation.

test .....

result ..... [2]

9. M/J 18/P42/Q7/e

(a) Alkanes and alkenes are hydrocarbons.

(i) What is meant by the term *hydrocarbon*?

.....

..... [2]

(ii) State the general formula of:

alkanes .....

alkenes ..... [2]

10. M/J 18/P41/Q6/a,b,c,d,e,f

The table shows the structures of four hydrocarbons.

| P                          | Q                         | R                             | S                                          |
|----------------------------|---------------------------|-------------------------------|--------------------------------------------|
| $\text{CH}_3\text{--CH}_3$ | $\text{CH}_2\text{=CH}_2$ | $\text{CH}_2\text{=CH--CH}_3$ | $\text{CH}_2\text{=CH--CH}_2\text{--CH}_3$ |

(a) Why are compounds **P**, **Q**, **R** and **S** known as hydrocarbons?

.....

..... [2]

(b) Compound **P** is saturated.

What is meant by the term *saturated*?

..... [1]

(c) Compound **P** undergoes a substitution reaction with chlorine.

(i) What is meant by the term *substitution reaction*?

..... [1]

(ii) State a condition required for this reaction to occur.

..... [1]

(iii) Write a chemical equation for this reaction.

..... [2]

(d) Compound **R** undergoes an addition reaction with bromine.

(i) Why is this reaction an addition reaction?

..... [1]

(ii) A compound containing bromine is formed in this reaction.

Draw the structure of this compound. Show all of the atoms and all of the bonds.

[1]

(e) Draw the structure of an unbranched isomer of compound **S**. Show all of the atoms and all of the bonds. Name this unbranched isomer of compound **S**.  
structure

name..... [2]

(f) Compound **Q** undergoes polymerisation.

(i) Name the polymer formed.

..... [1]

(ii) Complete the chemical equation to show the polymerisation of compound **Q**.

$n \text{CH}_2=\text{CH}_2 \rightarrow$  [2]

11. O/N 17/P43/Q6

(a) Alkanes and alkenes are two homologous series of hydrocarbons.

(i) What is meant by the term *hydrocarbon*?

..... [1]

(ii) What is the general formula of the homologous series of

alkanes, .....

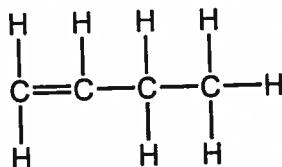
alkenes? ..... [2]

(iii) Other than having a general formula, state **two** characteristics of a homologous series.

1 .....

2 ..... [2]

(iv) The structure of an alkene molecule with the molecular formula  $\text{C}_4\text{H}_8$  is shown.



Draw the structure of a different alkene molecule with the molecular formula  $C_4H_8$ . Show all of the atoms and all of the bonds.

[1]

- (v) What term describes molecules with the same molecular formula but different structural formulae?

[1]

- (b)  $25\text{ cm}^3$  of a gaseous hydrocarbon,  $C_xH_y$ , were burnt in  $150\text{ cm}^3$  of oxygen. This was an excess of oxygen.

After cooling, the volume of the gases remaining was  $100\text{ cm}^3$ . This consisted of  $75\text{ cm}^3$  of carbon dioxide and  $25\text{ cm}^3$  of unreacted oxygen. The water that was produced in the reaction was liquid.

All volumes were measured at the same temperature and pressure.

- (i) What is meant by an excess of oxygen?

[1]

- (ii) What was the volume of oxygen that reacted with the hydrocarbon?

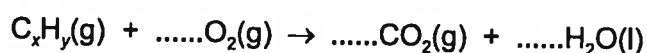
.....  $\text{cm}^3$  [1]

- (iii) Complete the table to show the smallest whole number ratio of volumes.

|                                                       | volume of<br>hydrocarbon reacted | : | volume of<br>oxygen reacted | : | volume of<br>carbon dioxide<br>produced |
|-------------------------------------------------------|----------------------------------|---|-----------------------------|---|-----------------------------------------|
| <b>smallest whole<br/>number ratio of<br/>volumes</b> |                                  | : |                             | : |                                         |

[1]

- (iv) Use your answer to (b)(iii) to balance the chemical equation. Deduce the formula of the hydrocarbon.



formula of the hydrocarbon ..... [2]

## 12. M/J 17/P41/Q6/c,d

The alkenes and alkanes are both examples of homologous series which are hydrocarbons.

- (a) Name and draw the structure of the second member of the alkene homologous series.

Show all of the atoms and all of the bonds.

name .....

structure

[2]

(b) Alcohols can be made from alkenes.

Name the reagent and conditions needed to convert an alkene into an alcohol.

.....

..... [2]

**13. O/N 16/P41/Q3/e**

(a) Isoprene is a naturally occurring hydrocarbon.

(i) Explain how the name of isoprene suggests that it contains a C=C double bond.

..... [1]

(ii) A sample of isoprene had the following composition by mass: C, 88.24%; H, 11.76%.

Calculate the empirical formula of isoprene. Show all your working.

empirical formula = ..... [3]

(iii) What additional information would be required to calculate the molecular formula of isoprene?

..... [1]

**14. M/J 16/P41/Q6/a**

Petroleum is a source of many important chemicals.

(a) Name **two** industrial processes which must take place to produce alkenes from petroleum.

.....

..... [2]



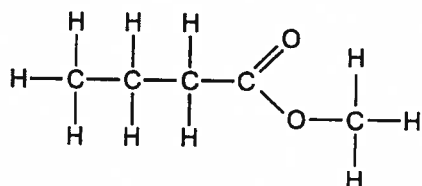
## Answers Section

### 1. O/N 22/P41/Q5

- (a)  $C_nH_{2n}$  1
- (b) orange to colourless 1
- (c) addition
- (d) (i) **M1** horizontal line below energy level to right hand side of reactants line and labelled  $C_2H_4Br_2$  (1) 1
- M2** activation energy 'hump' with upward arrow labelled **A** from the reactants level (1)
- M3** one downward arrow starting from the energy level of the reactants and finishing at the energy level of the products (1) 3
- (ii) **M1** energy needed to break bonds
- $$4 \times C-H + C=C + Br-Br = 4 \times 410 + 610 + 190 = 2440 \text{ (kJ)} \text{ (1)}$$
- M2** energy released in making bonds
- $$4 \times C-H + C-C + 2 \times C-Br = 4 \times 410 + 350 + 2 \times 290 = 2570 \text{ (kJ)} \text{ (1)}$$
- M3** energy change  $M1 - M2 = -130 \text{ (kJ/mol)}$  (1) 3

### 2. O/N 21/P41/Q5

- (a) homologous series 1
- (b)  $C_nH_{2n}$  1
- (c) (i) orange to colourless 1
- (ii) (only) one product is formed 1
- (iii)  $C_4H_8 + Br_2 \rightarrow C_4H_8Br_2$
- $C_4H_8Br_2$  (1)
- equation fully correct (1) 2
- (iv) (poly) but-2-ene 1
- (v) but-1-ene (1)
- structure of but-1-ene
- $$\begin{array}{ccccccc}
 H & H & H & H & & & \\
 | & | & | & | & & & \\
 C & = & C & - & C & - & C & - & H \\
 | & & & & | & & | & & \\
 H & & & & H & & H & & 
 \end{array}
 \quad (1)$$
- 2
- (d) (i)  $C_2H_4O$  1
- (ii)  $2C_3H_7COOH + Na_2CO_3 \rightarrow 2C_3H_7COONa + H_2O + CO_2$
- $C_3H_7COONa$  (1)
- equation fully correct (1) 2
- (iii) methyl butanoate (1)



ester link (1)

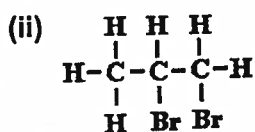
rest of structure (1)

3

## 3. O/N 20/P42/Q4/b

(a) (i) (only) one product is formed

1



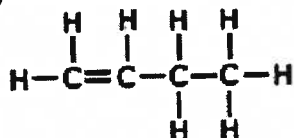
3 single bonded C atoms  
entire molecule correct

2

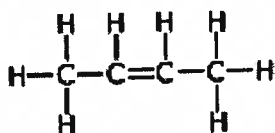
(iii) orange (1) to colourless (1)

2

(iv) structure of but-1-ene



structure of but-2-ene



2

## 4. O/N 20/P41/Q6/d,e(ii)

(a) (i) from: orange (1)

2

to: colourless (1)

1

(ii) contains no double bonds/ethane **only** contains single bonds

1

(b) (i)  $C_2H_4$ 

## 5. M/J 20/P41/Q3/d

(a) (i) not all (C-C) bonds are single

1

(ii)  $C_3H_7OH \rightarrow C_3H_6 + H_2O$ M1  $C_3H_6$ 

2

M2 rest of the equation

1

(iii) propene

## 6. M/J 19/P41/Q5/a,b,c,d,e,f(ii)

(a)  $C_nH_{2n}$  (1)

1

(b)  $C_5H_{10}$  (1)

1

(c) E (1)

2

it has the longest carbon chain (1)

(d) A (1)

2

it has the lowest  $M_r$  (1)

(e) orange to colourless (1)

2

structure of 1,2-dibromobutane (1)

(f) (i) steam (1)

catalyst (1)

one other condition: either 60 atm pressure **OR** 300 °C (1)

3

## 7. MAR 19/P42/Q3/f(i,ii)

(a) (i) not all the bonds are single bonds

1

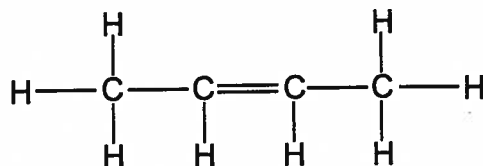
(ii) M1 bromine / bromine water (1)

M2 turns colourless / decolourises (1)

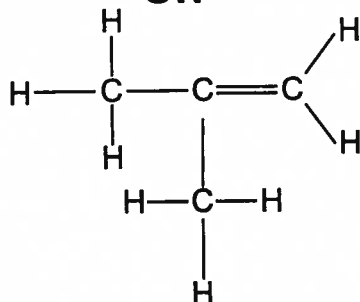
2

8. **O/N 18/P42/Q5/a,b,c,d**
- (a)  $C_4H_6$   
Propyne 2
- (b) M1 one shared pair between each H and C  
M2 three shared pairs of electrons between the C atoms and no other unpaired electrons 2
- (c) (i) Any two from:  
same or similar chemical properties  
(contain) the same functional group  
(show) a trend or gradual change in physical properties  
(consecutive) members differ by  $CH_2$   
common methods of preparation 2
- (ii)  $C_nH_{2n-2}$  1
- (d) M1 Bromine water or aqueous bromine  
M2 Changes to colourless or decolourises 2
9. **M/J 18/P42/Q7/e**
- (a) (i) (they contain) carbon and hydrogen (atoms) only 1  
1
- (ii) alkane:  $C_nH_{2n+2}$  1  
alkene:  $C_nH_{2n}$  1
10. **M/J 18/P41/Q6/a,b,c,d,e,f**
- (a) (they contain) carbon and hydrogen (atoms) only 1  
1
- (b) (all) the (C–C) bonds are single 1
- (c) (i) (one) atom or group is replaced by another (atom or group) 1  
(ii) ultra-violet light OR sunlight 1
- (iii)  $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$   
1 mark for  $C_2H_5Cl$   
1 mark for the rest of the equation 2
- (d) (i) only one product (compound) forms 1  
(ii) fully displayed formula of 1,2-dibromopropane 1
- (e) fully displayed formula of but-2-ene 1  
but-2-ene 1
- (f) (i) poly(ethene) 1  
(ii) single bond between two C atoms 1  
fully correct answer 1
11. **O/N 17/P43/Q6**
- (a) (i) compounds containing carbon and hydrogen only 1  
(ii) alkanes:  $C_nH_{2n+2}$  1  
alkenes:  $C_nH_{2n}$  1
- (iii) any 2 from:  
∞ same or similar chemical properties  
∞ (consecutive members) differ by  $CH_2$   
∞ same functional group  
∞ common (allow similar) methods of preparation  
∞ physical properties vary in predictable manner/ show trends/  
gradually change OR example of a physical property variation 2

(iv)



OR



(v) structural isomers

(b) (i) more than enough oxygen to react with **all** of the hydrocarbon(ii) 125 (cm<sup>3</sup>)

(iii) 1:5:3

(iv) C<sub>3</sub>H<sub>8</sub>If full credit is not awarded, allow 1 mark for C<sub>x</sub>H<sub>y</sub>(g) + 5O<sub>2</sub>(g) → 3CO<sub>2</sub>(g) + 4H<sub>2</sub>O(l)

12. M/J 17/P41/Q6/c,d

(a) propene

structure correctly shown

(b) steam

catalyst

13. O/N 16/P41/Q3/e

(a) (i) (ends in) ene

(ii) M1 88.24 / 12 AND 11.76 / 1

M2 7.353 / 7.353 (= 1) AND 11.76 / 7.353 = (1.6)

M3 C<sub>5</sub>H<sub>8</sub>

(iii) relative molecular mass

14. M/J 16/P41/Q6/a

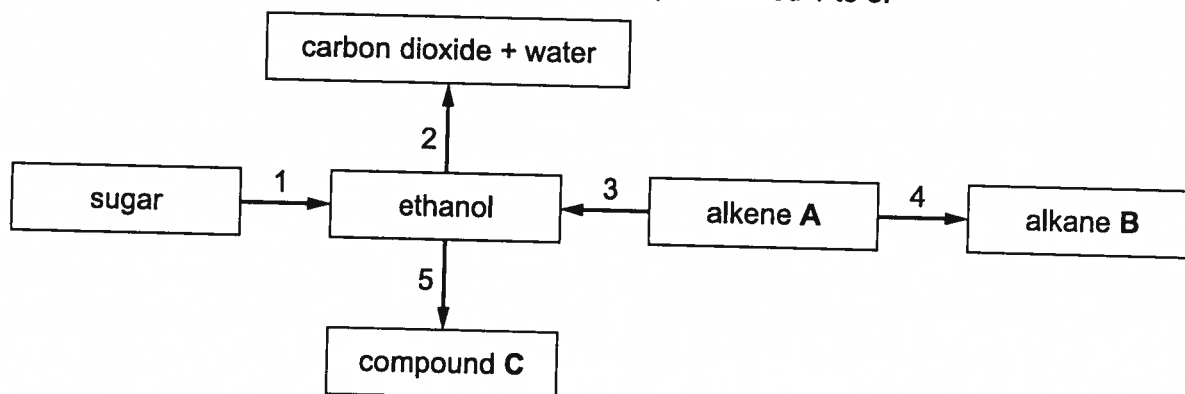
(a) fractional distillation;

cracking;

## 13.6: Alcohols

### 1. M/J 22/P42/Q5

The reaction scheme shows five organic reactions, numbered 1 to 5.



(a) Name reaction 1.

..... [1]

(b) Name reaction 2 and write the chemical equation for this reaction.

name .....

equation ..... [3]

(c) Reaction 3 forms ethanol from alkene A.

(i) Identify alkene A.

..... [1]

(ii) State the type of reaction that occurs during reaction 3.

..... [1]

(iii) State the reagents and conditions needed for reaction 3.

.....

..... [2]

(d) Alkene A is converted into alkane B in reaction 4.

(i) State the reagent and conditions for reaction 4.

.....

..... [3]

(ii) State the general formula of alkanes.

..... [1]

- (e) Ethanol is oxidised in reaction 5 by heating it with dilute sulfuric acid and one other reagent.  
(i) Identify the other reagent in reaction 5.

..... [1]

- (ii) Name the homologous series compound **C** belongs to.

..... [1]

- (iii) Draw the structure of compound **C**.  
Show all of the atoms and all of the bonds.

[1]

2. O/N 21/P42/Q6

Ethanol,  $C_2H_5OH$ , belongs to the homologous series called alcohols.

- (a) Write the general formula of alcohols.

..... [1]

- (b) Explain why ethanol **cannot** be described as a hydrocarbon.

..... [1]

- (c) Ethanol can be manufactured from different substances by reaction with steam or by fermentation.

- (i) Give the formula of the substance which reacts with steam to form ethanol.

..... [1]

- (ii) Name a substance which will undergo fermentation to form ethanol.

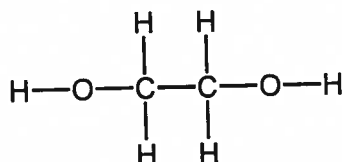
..... [1]

- (d) Ethanol is a fuel.

Write the chemical equation for the complete combustion of ethanol.

..... [2]

- (e) Ethane-1,2-diol has two alcohol functional groups.



One molecule of ethane-1,2-diol will react with two molecules of ethanoic acid to form molecule **X**.

**X** has two ester functional groups and a molecular formula of  $C_6H_{10}O_4$ .

- (i) State the empirical formula of **X**.

..... [1]

- (ii) Draw the structure of **X**.  
Show all of the atoms and all of the bonds.

[2]

- (iii) Name the **other** substance formed in this reaction.

..... [1]

- (f) Each alcohol functional group in ethane-1,2-diol reacts with acidified potassium manganate(VII) to form a different organic compound, **Y**.

- (i) Name the functional groups formed in **Y**.

..... [1]

- (ii) Draw the structure of **Y**.

Show all of the atoms and all of the bonds.

[1]

3. O/N 21/P43/Q7

- (a) Ethanol is a member of the homologous series of alcohols.  
Give **two** characteristics of members of a homologous series.

1 .....

2 ..... [2]

- (b) Ethanol can be manufactured from ethene.

Ethene can be made from long chain hydrocarbons such as decane,  $C_{10}H_{22}$ .  
Ethene is then converted into ethanol.

- (i) Name the process used to obtain ethene from long chain hydrocarbons such as decane,  $C_{10}H_{22}$ .

..... [1]

- (ii) Complete the chemical equation to show the formation of ethene from decane,  $C_{10}H_{22}$ .



- (iii) Write the chemical equation for the conversion of ethene into ethanol.

..... [1]

- (iv) Name the type of reaction occurring when ethene is converted into ethanol.

..... [1]

- (v) Give **one** condition for the reaction in which ethene is converted into ethanol.

..... [1]

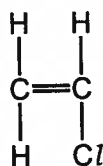
(c) Ethanol can also be produced by fermentation of carbohydrates such as glucose.

Give **two** advantages of manufacturing ethanol by fermentation compared to manufacturing ethanol from ethene.

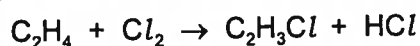
1 .....

2 ..... [2]

(d) (i) Under certain conditions ethene can react with chlorine to produce chloroethene. The structure of chloroethene is shown.



The equation for the chemical reaction is shown.



State the type of chemical reaction between ethene and chlorine that this equation shows.

..... [1]

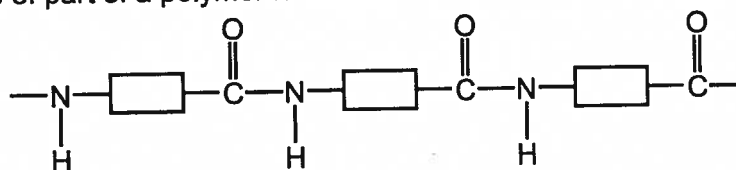
(ii) Chloroethene monomers can be converted into a polymer called poly(chloroethene). State the type of polymerisation that produces poly(chloroethene) from chloroethene.

..... [1]

(iii) Draw a section of the poly(chloroethene) molecule made from **two** monomer molecules.

[2]

(e) The structure of part of a polymer is shown.



This polymer is made from one type of monomer only.

Complete the diagram to show the structure of the monomer used to produce this polymer. Show all of the atoms and all of the bonds in the functional groups.



[2]

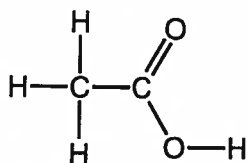
4. O/N 20/P42/Q5/a(iii),c(i)

This question is about alcohols, carboxylic acids and esters.

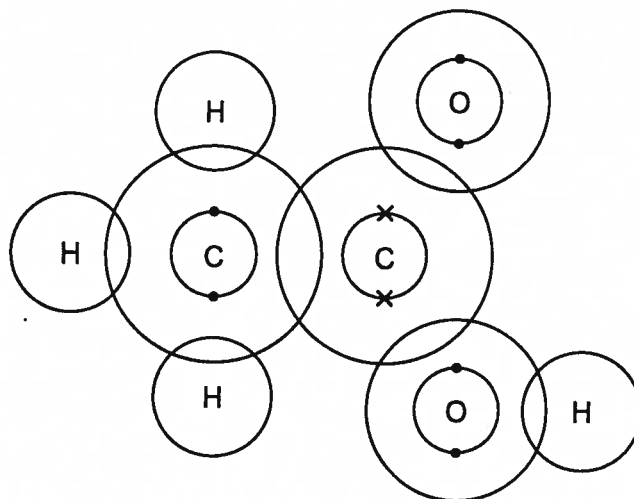
(a) Ethanol will react with hot aqueous potassium manganate(VII) to form ethanoic acid.



- (i) The structure of ethanoic acid is shown.

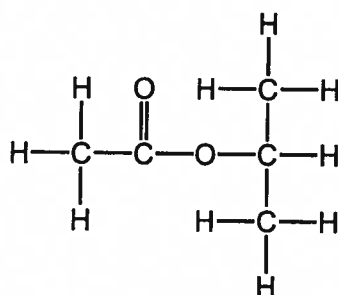


Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanoic acid.



[3]

- (b) Ethanoic acid will react with an alcohol to form the ester shown.



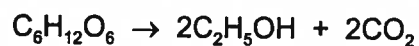
- (i) Name the **other** product formed when ethanoic acid reacts with an alcohol to make this ester.

..... [1]

5. O/N 20/P41/Q7/a,b,d

- (a) Ethanol can be manufactured by two different methods.

Method 1: fermentation of a sugar,  $\text{C}_6\text{H}_{12}\text{O}_6$



Method 2: reaction of ethene with steam



- (i) Give **one** advantage of using fermentation compared with Method 2.

..... [1]

- (ii) Give one disadvantage of using fermentation compared with Method 2.

..... [1]

- (b) Ethanol reacts with acidified potassium manganate(VII) to form water and a product that turns litmus red.

- (i) State the name of the product that turns the litmus red.

..... [1]

- (ii) State the type of reaction that ethanol undergoes when it reacts with acidified potassium manganate(VII).

..... [1]

- (c) The table shows the melting points of ethanol and sodium chloride.

| substance       | melting point/°C |
|-----------------|------------------|
| ethanol         | -114             |
| sodium chloride | 801              |

The difference in melting points is due to differences in attractive forces between particles in these substances.

Name the type of attractive force in each substance, which is responsible for the difference in melting points.

ethanol .....

sodium chloride ..... [2]

#### 6. M/J 20/P41/Q5/a,b,d,f(i)

Ethanol is manufactured by two different processes.

- (a) For each process, name the organic reactant and state the type of reaction.

organic reactant ..... type of reaction .....

organic reactant ..... type of reaction ..... [4]

- (b) Alcohols can be oxidised to form carboxylic acids.

Name a suitable oxidising agent for this reaction.

..... [1]

- (c) The table shows some information about aldehydes.

- (i) Complete the table.

| name              | .....             | ethanal                         | propanal                        | butanal |
|-------------------|-------------------|---------------------------------|---------------------------------|---------|
| molecular formula | CH <sub>2</sub> O | C <sub>2</sub> H <sub>4</sub> O | C <sub>3</sub> H <sub>6</sub> O | .....   |

[2]

- (ii) Deduce the general formula of aldehydes.

..... [1]

(d) Propanone belongs to a homologous series called ketones. Ketones have the same C=O group as aldehydes but the C=O group is not at the end of the carbon chain. Propanone has the same molecular formula as propanal,  $C_3H_6O$ .

- (i) What term is used to describe molecules with different structures but with the same molecular formula?

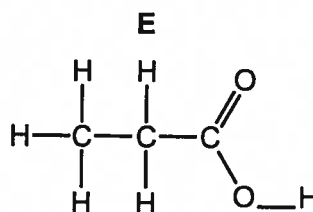
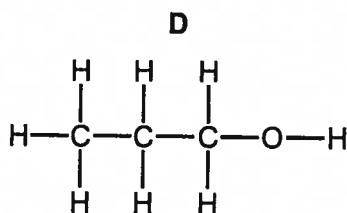
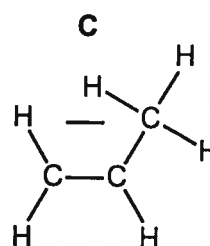
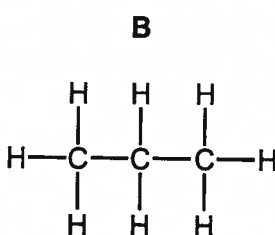
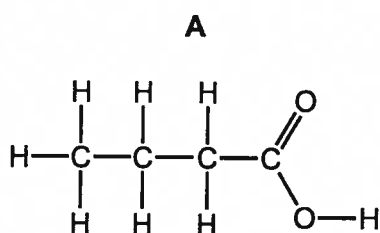
..... [1]

**7. M/J 20/P43/Q1/a(i)**

- (a) The structures of five organic compounds, **A**, **B**, **C**, **D** and **E**, are shown.

Answer the questions that follow.

Each letter may be used once, more than once or not at all.



- (i) Give the letter of the compound that is propan-1-ol.

.....[1]

**8. O/N 19/P42/Q5/a**

Methanol,  $\text{CH}_3\text{OH}$ , is a member of the homologous series of alcohols.

- (a)** Methanol can be made from methane in a two-step process.

**step 1** Methane is reacted with chlorine gas to produce chloromethane,  $\text{CH}_3\text{Cl}$ .

**step 2**  $\text{CH}_3\text{Cl}$  is reacted with sodium hydroxide to produce  $\text{CH}_3\text{OH}$  and one other product.

- (i) What conditions are needed in **step 1**?

..... [1]

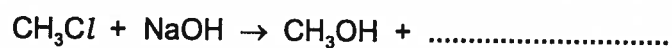
- (ii) Write the chemical equation for the reaction which occurs in step 1.**

..... [1]

- (iii) State the type of organic reaction occurring in step 1.**

..... [1]

- (iv) Complete the chemical equation for step 2.**



[1]

## 9. O/N 19/P41/Q7/a,c(iii)

This question is about ethanol.

- (a) Ethanol that is suitable for use as a fuel can be manufactured from sugars such as glucose,  $C_6H_{12}O_6$ , by a two-step process.

Describe how this can be done. In your answer, include:

- an equation for the reaction in which ethanol is formed
- the essential conditions for the reaction in which ethanol is formed
- the name of the process used to obtain ethanol that is pure enough to use as a fuel from the reaction mixture.

.....  
 .....  
 .....  
 .....  
 ..... [5]

- (b) Ethanol can be oxidised by hydrogen peroxide to form ethanal,  $CH_3CHO$ . A catalyst for this reaction is  $Fe^{3+}$

- (i) The table gives the boiling points of ethanal and ethanol.

| substance | boiling point/ $^{\circ}C$ |
|-----------|----------------------------|
| ethanal   | 20                         |
| ethanol   | 78                         |

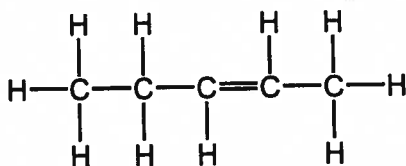
In terms of attractive forces between particles, suggest why ethanal has a lower boiling point than ethanol.

..... [1]

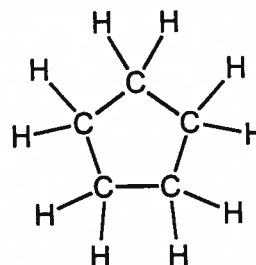
## 10. M/J 19/P43/Q6/a(ii),b

- (a) Two hydrocarbons have the structures shown.

hydrocarbon A



hydrocarbon B



- (i) Hydrocarbon B reacts in the same way as a typical alkane.  
 Describe a chemical test to tell the difference between hydrocarbon A and hydrocarbon B.  
 State the name of the reagent you would use and the result you would obtain with hydrocarbon A and hydrocarbon B.

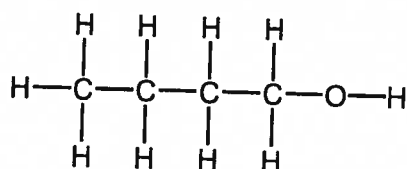
reagent .....

result with hydrocarbon A .....

result with hydrocarbon **B** ..... [3]

- (b) Alkenes react with steam to form alcohols.  
Compound **C** is an alcohol.

compound **C**



Draw the structure of the alkene which could be reacted with steam to make compound **C**.  
Show all of the atoms and all of the bonds.

11. M/J 19/P42/Q4/c(ii)

[1]

- (a) Methanol is a member of the homologous series of alcohols.

- (i) Draw the structures of **two** different alcohols, each containing **three** carbon atoms. Show all of the atoms and all of the bonds.

Name these **two** alcohols.

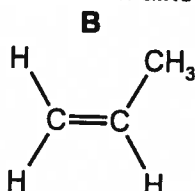
name .....

name .....

12. M/J 19/P41/Q5/f(i)

[4]

- (a) Two different alcohols can be produced from alkene **B** by an addition reaction.



- (i) Draw the structures of the **two** alcohols. Show all of the atoms and all of the bonds.

[2]

**13. MAR 18/P42/Q5/a,b,c,d**

Alcohols are a 'family' of organic molecules which have the same general formula.

- (a) What is the name given to any 'family' of organic molecules which have the same general formula and similar chemical properties?

..... [1]

- (b) Give the general formula of alcohols.

..... [1]

- (c) Propan-1-ol can be made from propene.

- (i) Name the reagent and give the conditions needed to convert propene into propan-1-ol.

reagent .....

conditions ..... [2]

- (ii) Write a chemical equation for the complete combustion of propan-1-ol.

..... [2]

- (d) A simple sugar can be represented as shown.



Simple sugars can be polymerised to make more complex carbohydrates.

- (i) Complete the diagram to show part of a carbohydrate **polymer** made from the simple sugar shown.



[2]

- (ii) Name the chemical process which occurs when a carbohydrate polymer is broken down into simple sugars.

..... [1]

- (iii) What conditions are needed for this process to occur?

..... [1]

**14. O/N 18/P41/Q4/b**

- (a) When concentrated sulfuric acid is added to glucose,  $\text{C}_6\text{H}_{12}\text{O}_6$ , a black solid is produced. The concentrated sulfuric acid acts as a dehydrating agent.

- (i) What is removed from the glucose in this reaction?

..... [1]

- (ii) Name the black solid produced in this reaction.

..... [1]

**15. O/N 18/P41/Q6/a**

- (a) Ethane,  $\text{C}_2\text{H}_6$ , is a member of the homologous series called alkanes.

Ethanol,  $\text{C}_2\text{H}_5\text{OH}$ , is a member of the homologous series called alcohols.

- (i) Alkanes are hydrocarbons.

What is meant by the term *hydrocarbon*?

.....  
..... [2]

- (ii) All members of a homologous series can be represented by a general formula.  
State the general formula of:

- alkanes .....
- alcohols ..... [2]

- (iii) State **two** characteristics, other than having the same general formula, of members of a homologous series.

1 .....

2 ..... [2]

16. M/J 18/P42/Q7/e,f

- (a) Alkanes and alkenes are hydrocarbons.

- (i) What is meant by the term *hydrocarbon*?

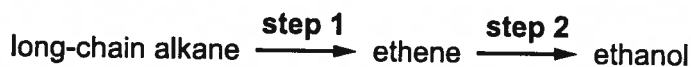
.....  
..... [2]

- (ii) State the general formula of:

alkanes .....

alkenes ..... [2]

- (b) Ethanol can be produced from long-chain alkanes as shown.



Describe the **two-stage** manufacture of ethanol from the long-chain alkane octane,  $\text{C}_8\text{H}_{18}$ .  
Include:

- the names of the types of chemical reactions that occur
- reaction equations
- reaction conditions.

**step 1** .....

.....

**step 2** .....

.....

..... [5]

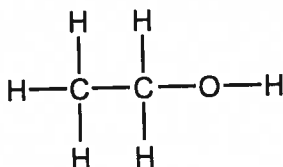
## 17. M/J 18/P43/Q4/a,b,c

Ethanol is a member of the homologous series of alcohols.

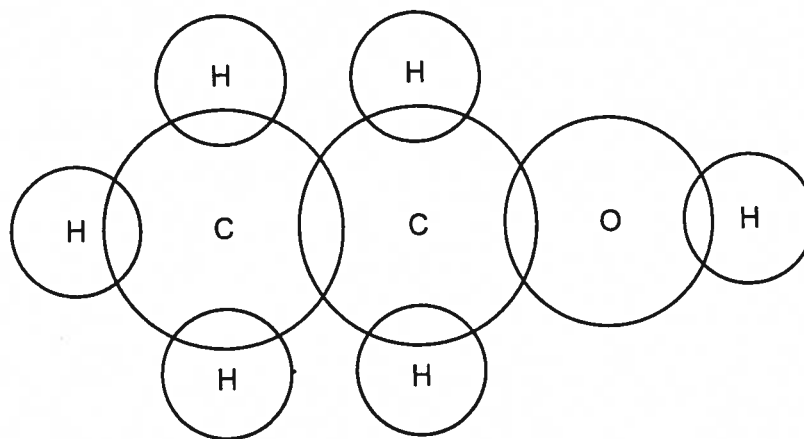
(a) Give **two** characteristics of members of a homologous series.

- 1 .....  
2 ..... [2]

(b) The structure of ethanol is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanol. Show outer shell electrons only.



[2]

(c) Ethanol can be produced by the catalytic addition of steam to ethene or by the fermentation of glucose.

(i) Write a chemical equation for the production of ethanol by the catalytic addition of steam to ethene.

..... [1]

(ii) Write a chemical equation for the production of ethanol by the fermentation of glucose,  $\text{C}_6\text{H}_{12}\text{O}_6$ .

..... [1]

(iii) State **one** advantage of producing ethanol by the catalytic addition of steam to ethene. Your answer must **not** refer to cost.

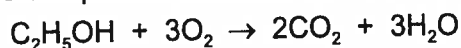
..... [1]

(iv) State **one** advantage of producing ethanol by the fermentation of glucose. Your answer must **not** refer to cost.

..... [1]

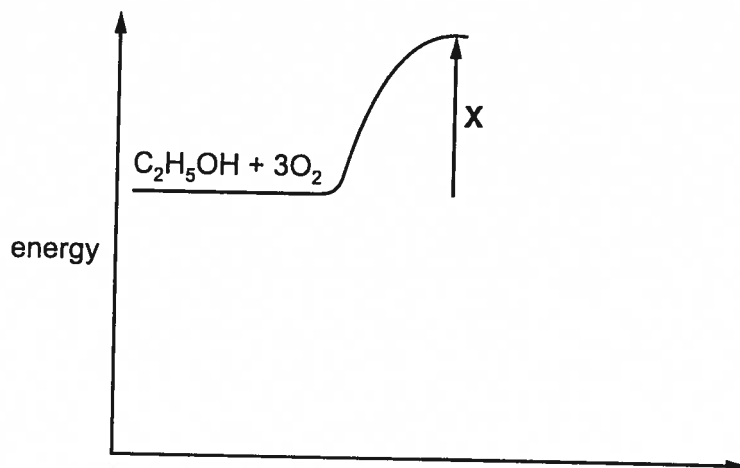
## 18. O/N 17/P42/Q3/d,e

The chemical equation for the complete combustion of ethanol,  $\text{C}_2\text{H}_5\text{OH}$ , is shown.





The energy released when one mole of ethanol undergoes complete combustion is 1280 kJ. Part of the energy level diagram for this reaction is shown.



- (a) Dodecane is an alkane containing 12 carbon atoms. Ethanol can be manufactured from dodecane in a two-stage process.
- In **stage 1**, each molecule of dodecane is converted into three molecules of ethene and one molecule of another hydrocarbon.
- (i) Name the process which occurs in **stage 1**.
- ..... [1]
- (ii) Write a chemical equation for the reaction which occurs in **stage 1**.
- ..... [2]
- In **stage 2**, ethene reacts with steam to produce ethanol.
- (iii) State **two** conditions needed for **stage 2**.
- 1 ..... [2]
- 2 ..... [2]
- (iv) Name the type of reaction which occurs in **stage 2**.
- ..... [1]
- (v) Suggest how to test the purity of the ethanol produced.
- ..... [2]
- (b) Ethanol can also be manufactured by the fermentation of glucose,  $C_6H_{12}O_6$ .
- (i) State **two** conditions needed for the fermentation of glucose.
- 1 ..... [2]
- 2 ..... [2]

- (ii) Complete the chemical equation for the fermentation of glucose.



- (iii) One disadvantage of fermentation is that the maximum concentration of ethanol produced is about 15%.

Suggest why the concentration of ethanol produced by fermentation does **not** exceed 15%.

..... [1]

- (iv) Give **one** other disadvantage of manufacturing ethanol by fermentation.

..... [1]

- (v) Give **one** advantage, other than cost, of manufacturing ethanol by fermentation.

..... [1]

- (vi) Suggest the name of a process to obtain ethanol from a mixture of ethanol and water.

..... [1]

**19. O/N 17/P41/Q4/a,b**

- (a) Ethanol,  $\text{C}_2\text{H}_5\text{OH}$ , can be made by fermentation.

- (i) Complete the chemical equation for the formation of ethanol by fermentation.

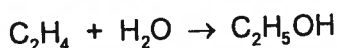


- (ii) State **two** conditions required for fermentation.

1 .....

2 ..... [2]

- (b) Ethanol can also be made by the catalytic hydration of ethene. The equation for the reaction is shown.



- (i) Name a suitable catalyst for this reaction.

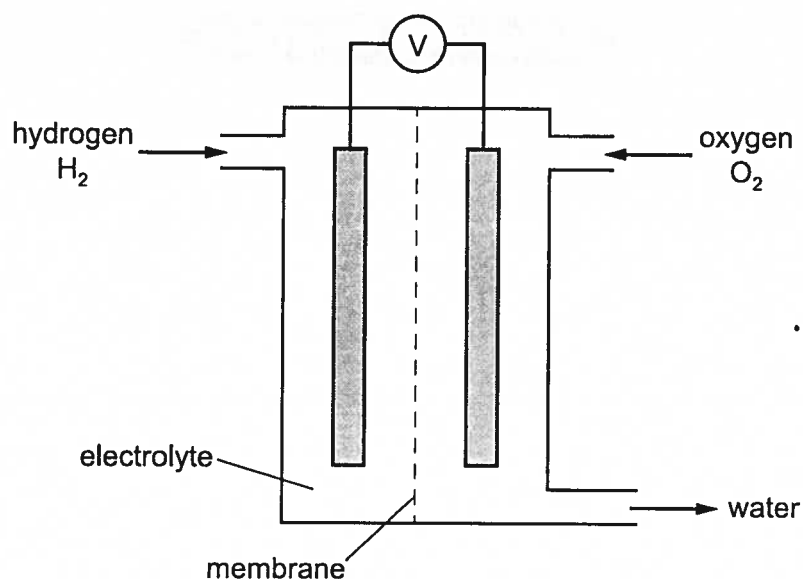
..... [1]

- (ii) Calculate the maximum mass of ethanol that can be made from 56 g of ethene.

maximum mass of ethanol ..... g [2]

**20. O/N 17/P43/Q4/e**

Hydrogen and oxygen react together in a hydrogen fuel cell. A hydrogen fuel cell is shown in the diagram.



(a) Some fuel cells use ethanol,  $\text{C}_2\text{H}_5\text{OH}$ , instead of hydrogen. Carbon dioxide and water are products of the reaction in an ethanol fuel cell.

(i) Write a chemical equation for the overall reaction occurring in an ethanol fuel cell.

..... [2]

(ii) State an environmental problem caused by the release of carbon dioxide into the atmosphere.

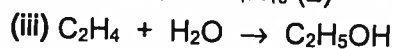
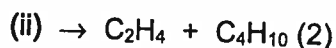
..... [1]

(iii) Name the process by which ethanol can be manufactured from a renewable resource.

..... [1]

## Answers Section

- 1. M/J 22/P42/Q5**
- (a) fermentation 1
- (b) (complete) combustion  
 $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 3\text{H}_2\text{O} + 2\text{CO}_2$   
 reactant species 3  
 fully correct equation 1
- (c) (i) ethene 1  
 (ii) addition 1  
 (iii) steam 2  
 catalyst + 60 atm or catalyst + 300 °C
- (d) (i) hydrogen 3  
 catalyst 1  
 high temperature 1  
 (ii)  $\text{C}_n\text{H}_{2n+2}$  1
- (e) (i) potassium manganate(VII) 1  
 (ii) carboxylic acids 1  
 (iii) fully displayed formula of ethanoic acid 1
- 2. O/N 21/P42/Q6**
- (a)  $\text{C}_n\text{H}_{2n+1}\text{OH}$  (1) 1
- (b) not all atoms are C or H 1
- (c) (i)  $\text{C}_2\text{H}_4$  1  
 (ii) glucose 1
- (d)  $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 3\text{H}_2\text{O} + 2\text{CO}_2$  2  
 species (1) correct equation (1) 1
- (e) (i)  $\text{C}_3\text{H}_5\text{O}_2$  1  
 (ii) any one ester link (1) 2  
 rest of molecule (1) 1  
 (iii) water 1
- (f) (i) carboxylic acid 1  
 (ii) structure of ethanedioic acid 1
- 3. O/N 21/P43/Q7**
- (a) 1 for each for any 2 of:
- (same) general formula
  - similar chemical properties
  - (consecutive members) differ by  $\text{CH}_2$
  - same functional group
  - common (allow similar or same) methods of preparation
  - physical properties and vary in predictable manner or show trends or gradually change
- OR example of a physical property variation, e.g.
- melting points increase
  - boiling points increase
  - volatility decreases
- (b) (i) cracking 2  
 1



(iv) hydration / addition

(v) 1 mark for **ANY 1** of:

- 300 °C
- 60 atmospheres
- $\text{H}_3\text{PO}_4$

2

1

1

1

(c) 1 mark each for any 2 of:

- **carbohydrates** are renewable
- **fossil fuels** are non-renewable **OR** fossil fuels conserved
- lower temperature **OR** lower energy
- hydration is equilibrium meaning lower yield
- lower pressure (used)

2

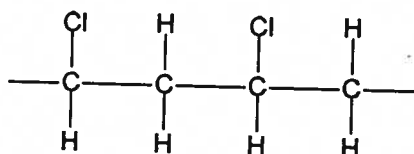
(d) (i) substitution

1

(ii) addition

1

(iii)

**M1** any one repeat unit with extension bonds (1)**M2** both units fully correct (1)

(e) 1 mark for each functional group fully correct (2)

2

2

## 4. O/N 20/P42/Q5/a(iii),c(i)

- (a) (i) all single bonding dot and cross pairs correct (1)  
double C=O bond dot and cross pairs are correct (1)  
Complete diagram is correct (1)

(b) (i) water

3

1

## 5. O/N 20/P41/Q7/a,b,d

- (a) (i) sugar or  $\text{C}_6\text{H}_{12}\text{O}_6$ , is renewable / sustainable  
(ii) slow(er) process

1

(b) (i) ethanoic acid

1

(ii) oxidation

1

(c) ethanol: (forces of attraction) between molecules (1)

1

sodium chloride: (force of attraction) between positive and negative ions/ionic bonding (1) 2

## 6. M/J 20/P41/Q5/a,b,d,f(i)

(a) M1 sugar(s)

M2 fermentation

M3 ethene

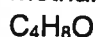
M4 hydration

4

(b) (acidified) potassium manganate(VII)

1

(c) (i) methanal



1

(ii)  $\text{C}_n\text{H}_{2n}\text{O}$ 

1

(d) (i) (structural) isomers

1

1

7. **M/J 20/P43/Q1/a(i)** 1  
(a) (i) D
8. **O/N 19/P42/Q5/a** 1  
(a) (i) ultraviolet light 1  
(ii)  $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$  1  
(iii) substitution 1  
(iv) NaCl
9. **O/N 19/P41/Q7/a,c(iii)**  
(a)  $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{CO}_2 + 2\text{C}_2\text{H}_5\text{OH}$  (1)  
any three from  
∞ anaerobic  
∞ 30 °C  
∞ yeast  
∞ glucose aqueous 5  
fractional distillation (of aqueous ethanol) (1) 1  
(b) (i) (attractive) forces between molecules weaker in ethanal
10. **M/J 19/P43/Q6/a(ii),b**  
(a)(i) **M1** (Reagent): Bromine (water / solution)  
**M2** (Result with hydrocarbon A): becomes colourless / decolourised 3  
**M3** (Result with hydrocarbon B): no change / stays orange  
(b)  

$$\begin{array}{ccccccc}
 & \text{H} & & \text{H} & & & \text{H} \\
 & | & & | & & & | \\
 \text{H} & - \text{C} & - & \text{C} & - & \text{C} = & \text{C} \\
 & | & & | & & | & | \\
 & \text{H} & & \text{H} & & \text{H} & \text{H}
 \end{array}$$
 1
11. **M/J 19/P42/Q4/c(ii)**  
(a) (i) **M1**  

$$\begin{array}{ccccccc}
 & \text{H} & & \text{H} & & \text{H} & \\
 & | & & | & & | & \\
 \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{O} - \text{H} \\
 & | & & | & & | & \\
 & \text{H} & & \text{H} & & \text{H} & 
 \end{array}$$
 (1)  
**M2** propan-1-ol (1)  
**M3**  

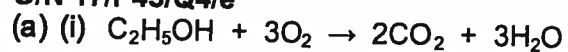
$$\begin{array}{ccccccc}
 & & & \text{H} & & & \\
 & & & | & & & \\
 & & & \text{O} & & & \\
 & & & | & & & \\
 & \text{H} & & \text{O} & & \text{H} & \\
 & | & & | & & | & \\
 \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\
 & | & & | & & | & \\
 & \text{H} & & \text{H} & & \text{H} & 
 \end{array}$$
 (1)  
**M4** propan-2-ol (1) 4
12. **M/J 19/P41/Q5/f(i)**  
(a) (i) structure of propan-1-ol (1) structure of propan-2-ol (1)  

$$\begin{array}{ccccccc}
 & \text{H} & \text{H} & \text{H} & & \text{H} & \\
 & | & | & | & & | & \\
 \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{O} & - \text{H} & \\
 & | & | & | & & & \\
 & \text{H} & \text{H} & \text{H} & & & 
 \end{array}
 \quad
 \begin{array}{ccccccc}
 & & & \text{H} & & & \\
 & & & | & & & \\
 & & & \text{O} & & & \\
 & & & | & & & \\
 & \text{H} & & \text{O} & & \text{H} & \\
 & | & & | & & | & \\
 \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{H} & & \\
 & | & & | & & | & \\
 & \text{H} & & \text{H} & & \text{H} & 
 \end{array}$$
 2

- 13. MAR 18/P42/Q5/a,b,c,d**
- (a) homologous series 1
- (b)  $C_nH_{2n+2}O$  OR  $C_nH_{2n+1}OH$  1
- (c) (i) M1 steam  
M2 catalyst 2
- (ii)  $2C_3H_7OH + 9O_2 \rightarrow 6CO_2 + 8H_2O$   
M1 species  
M2 fully correct equation 2
- (d) (i) M1 at least one  $-O-$  link between two blocks  
M2 correct structure including continuation bonds 2
- (ii) hydrolysis 1
- (iii) enzyme OR heat + acid 1
- 14. O/N 18/P41/Q4/b**
- (a) (i) water /  $H_2O$  1
- (ii) carbon / C 1
- 15. O/N 18/P41/Q6/a**
- (a) (i) M1 (compound that) contains carbon and hydrogen  
M2 and no other elements / only 1
- (ii) Alkanes:  $C_nH_{2n+2}$  1
- Alcohols  $C_nH_{2n+1}OH$  OR  $C_nH_{2n+2}O$  1
- (iii) any **two** from:  
Similar / same chemical properties  
Same functional group  
Trend or gradual change in physical properties  
(Neighbouring) members differ by  $CH_2$  2
- 16. M/J 18/P42/Q7/e,f**
- (a) (i) (they contain) carbon and hydrogen (atoms)  
only 1
- (ii) alkane:  $C_nH_{2n+2}$  1
- alkene:  $C_nH_{2n}$  1
- (b) (step 1) crack / cracking (of octane) 1
- (step 1) equation with only  $C_8H_{18}$  on left hand side and  $C_2H_4$   
+ other correct product(s) on right hand side 1
- e.g.  $C_8H_{18} \rightarrow C_2H_4 + C_6H_{14}$
- (step 2) hydration / addition 1
- (step 2) one correct condition for **either** process required
- (cracking):  $450 (^\circ)C$  to  $800 (^\circ)C$  / zeolites / aluminosilicates / silica /  $SiO_2$  /  
aluminium oxide /  $Al_2O_3$  / alumina / china / broken
- pot / chromium oxide /  $Cr_2O_3$  / up to 70 atmospheres
- (hydration): phosphoric acid /  $H_3PO_4$  /  $300 (^\circ)C$  / 60 atmospheres 1
- $C_2H_4 + H_2O \rightarrow C_2H_5OH$  /  $CH_3CH_2OH$  1
- 17. M/J 18/P43/Q4/a,b,c**
- (a) any **two** from:  
trend in physical properties  
same/similar chemical properties  
(same) general formula  
successive members differ by  $CH_2$   
same functional group 2

- (b) all bonding pairs correct and no extra incorrect non-bonding electrons 1  
4 non-bonding electrons on O completing oxygen octet 1
- (c) (i)  $C_2H_4 + H_2O \rightarrow C_2H_5OH$  1  
(ii)  $C_6H_{12}O_6 \rightarrow 2CO_2 + 2C_2H_5OH$  1  
(iii) any **one** from:  
pure(r) product  
fast(er) reaction  
continuous process 1  
(iv) any **one** from:  
renewable feedstock  
lower temperature  
lower pressure 1
18. O/N 17/P42/Q3/d,e 1
- (a) (i) cracking 1  
(ii)  $C_{12}H_{26} \rightarrow 3C_2H_4 + C_6H_{14}$   
M1  $C_{12}H_{26}$  2  
M2 rest of equation 1  
(iii) phosphoric acid 1  
heat 1  
(iv) addition / hydration 1  
(v) measure its boiling temperature 1  
compare to (known) data 1
- (b) (i) any 2 from:  
 $\infty$  37 °C  
 $\infty$  anaerobic  
 $\infty$  glucose is aqueous  
 $\infty$  yeast 2  
(ii)  $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$   
M1  $CO_2$  as a product 2  
M2 Rest of equation 1  
(iii) yeast is killed by the ethanol 1  
(iv) slow rate of reaction 1  
(v) uses renewable resources / does not use a finite resource 1  
(vi) fractional distillation 1
19. O/N 17/P41/Q4/a,b
- (a) (i)  $\rightarrow 2(C_2H_5OH) + 2CO_2$   
M1 carbon dioxide made as product 2  
M2 balanced  
(ii) any 2 from:  
 $\infty$  37 °C  
 $\infty$  anaerobic  
 $\infty$  glucose is aqueous 2  
 $\infty$  yeast 1
- (b) (i) (concentrated) phosphoric acid 1  
(ii) 92  
If full credit is not awarded, allow 1 mark for  $M_r$  of ethene = 28 2



**20. O/N 17/P43/Q4/e**

**M1** species correct

**M2** balanced

(ii) climate change/greenhouse effect/consequence of climate change

(iii) fermentation

2

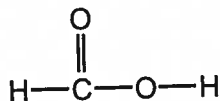
1

1

## 13.7: Carboxylic acids

### 1. O/N 22/P42/Q6

A carboxylic acid **Y** has the structure shown.



(a) State the general formula of carboxylic acids.

..... [1]

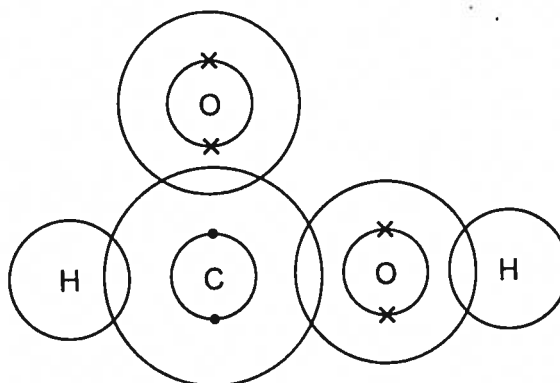
(b) Name carboxylic acid **Y**.

..... [1]

(c) Write the molecular formula of carboxylic acid **Y**.

..... [1]

(d) Complete the dot-and-cross diagram to show the arrangement of electrons in a molecule of carboxylic acid **Y**.



[3]

(e) Carboxylic acid **Y** will react with propan-1-ol,  $\text{C}_3\text{H}_7\text{OH}$ , to form ester **Z** and one other product.

(i) Name and draw the structure of ester **Z**.

Show all of the atoms and all of the bonds.

name .....

structure

[3]

(ii) Name the other product formed when carboxylic acid **Y** reacts with propan-1-ol.

..... [1]

(iii) Name:

- an ester which is a structural isomer of ester Z

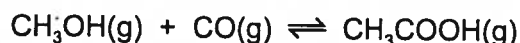
- a carboxylic acid which is a structural isomer of ester Z.

[2]

**2. M/J 22/P41/Q5**

Ethanoic acid is manufactured by the reaction of methanol with carbon monoxide.

An equilibrium mixture is produced.



(a) State **two** characteristics of an equilibrium.

1 .....

2 ..... [2]

(b) The purpose of the industrial process is to produce a high yield of ethanoic acid at a high rate of reaction.

The manufacture is carried out at a temperature of 300 °C.

The forward reaction is exothermic.

Use this information to state why the manufacture is **not** carried out at temperatures:

- **below** 300 °C

.....

- **above** 300 °C.

[2]

(c) Complete the table using only the words *increases*, *decreases* or *no change*.

|                         | effect on the rate of the forward reaction | effect on the equilibrium yield of $\text{CH}_3\text{COOH}(\text{g})$ |
|-------------------------|--------------------------------------------|-----------------------------------------------------------------------|
| adding a catalyst       |                                            | no change                                                             |
| decreasing the pressure |                                            |                                                                       |

[3]

(d) Suggest which of the following metals is a suitable catalyst for the reaction. Give a reason for your answer.

aluminium

calcium

cobalt

magnesium

potassium

suitable catalyst .....

reason ..... [2]

(e) Ethanoic acid is a member of the homologous series of carboxylic acids.

State the general formula of this homologous series.

..... [1]

- (f) Draw the structure of the carboxylic acid containing three carbon atoms. Show all of the atoms and all of the bonds.

[2]

- (g) When carboxylic acids react with alcohols, esters are produced.

The formula of ester X is  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_3$ .

- (i) Name ester X.

..... [1]

- (ii) Give the name of the carboxylic acid and the alcohol that react together to produce ester X.

carboxylic acid .....

alcohol ..... [2]

- (h) Ester Y has the following composition by mass:

C, 48.65%; H, 8.11%; O, 43.24%.

Calculate the empirical formula of ester Y.

empirical formula = ..... [3]

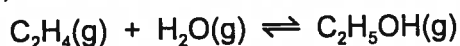
- (i) Ester Z has the empirical formula  $\text{C}_2\text{H}_4\text{O}$  and a relative molecular mass of 88.

Determine the molecular formula of ester Z.

molecular formula = ..... [1]

### 3. M/J 22/P43/Q4

Ethanol is made industrially by the reaction of ethene with steam. The reaction occurs at a temperature of  $300^\circ\text{C}$  and a pressure of 60 atmospheres.



A catalyst is used in this reaction.

The forward reaction is exothermic.

- (a) State what is meant by the term *catalyst*.

.....

..... [2]

- (b) Complete the table using only the words *increases*, *decreases* or *no change*.

|                            | effect on the rate of the forward reaction | effect on the equilibrium yield of $C_2H_5OH(g)$ |
|----------------------------|--------------------------------------------|--------------------------------------------------|
| increasing the temperature |                                            |                                                  |
| decreasing the pressure    |                                            |                                                  |

[4]

- (c) Ethanol is a member of the alcohol homologous series.  
Members of the same homologous series have the same general formula.

(i) State the general formula of alcohols.

..... [1]

(ii) State two **general** characteristics, other than the same general formula, of all homologous series.

1 .....

2 ..... [2]

(iii) One alcohol containing three carbon atoms is propan-1-ol.

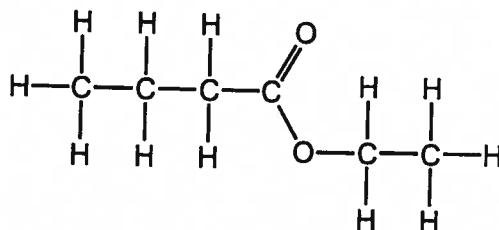
Draw the structure of one **other** alcohol containing three carbon atoms. Show all of the atoms and all of the bonds.

Name the alcohol you have drawn.

name ..... [2]

(d) When alcohols react with carboxylic acids, esters are produced.

(i) The structure of ester X is shown.



Name ester X.

..... [1]

(ii) Give the name of the alcohol and the carboxylic acid that react together to produce ester X.

alcohol .....

carboxylic acid ..... [2]

(e) Ester Y has the following composition by mass:

C, 58.82%; H, 9.80%; O, 31.37%.

Calculate the empirical formula of ester Y.

empirical formula = ..... [3]

- (f) Ester **Z** has the empirical formula  $C_3H_6O$  and a relative molecular mass of 116.  
Calculate the molecular formula of ester **Z**.

molecular formula = ..... [1]

**4. MAR 21/P42/Q5**The table shows the names or structures of organic compounds **P** to **U**.

| <b>P</b>                                                                                                 | <b>Q</b>         | <b>R</b>                                                                                                    |
|----------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------|
| <pre>       H   H   H                 H — C — C — C — H                       H   H   H           </pre> | propanoic acid   | but-1-ene                                                                                                   |
| <b>S</b>                                                                                                 | <b>T</b>         | <b>U</b>                                                                                                    |
| propan-1-ol                                                                                              | methyl butanoate | <pre>       H   H   H        \     /         C = C — C — H        /              H       H           </pre> |

- (a) Give the letters of the organic compounds, **P** to **U**, that are unsaturated hydrocarbons.

..... [2]

- (b) Describe the test for an unsaturated hydrocarbon.

test .....

observations ..... [2]

- (c) But-1-ene is an unbranched molecule.

- (i) Name the unbranched isomer of but-1-ene.

..... [1]

- (ii) Draw the structure of a branched isomer of but-1-ene. Show all of the atoms and all of the bonds.

[1]

(d) Dodecane is an alkane with 12 carbon atoms. Dodecane can be cracked.

(i) Write the formula of dodecane.

.....[1]

(ii) Give the letters of all the organic compounds, **P** to **U**, that can be formed when dodecane is cracked.

.....[2]

(e) Name the reagent and suggest the conditions needed to convert organic compound **U** into organic compound **S**.

reagent .....

conditions ..... [3]

(f) Organic compound **S** can be converted to organic compound **Q** by reaction with an acidified reagent.

(i) Name the type of chemical change that happens to organic compound **S**.

..... [1]

(ii) Name the acidified reagent added to organic compound **S**.

..... [1]

(g) Organic compound **T** is made by reacting two compounds together.

(i) Name the homologous series that organic compound **T** belongs to.

..... [1]

(ii) Name the **two** compounds which react together to make organic compound **T**.

Draw the structures of each compound you have named. Show all of the atoms and all of the bonds.

name .....  
structure .....

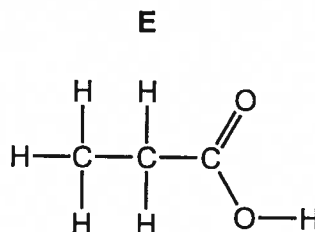
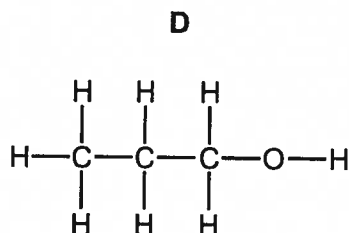
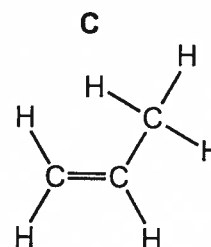
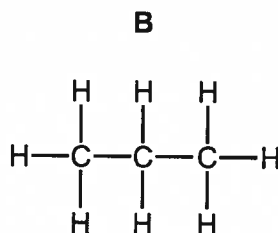
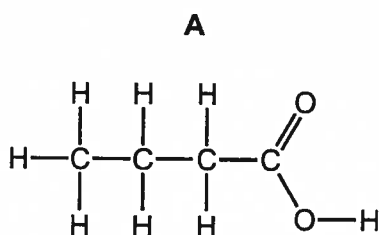
name .....  
structure .....

(iii) Deduce the molecular formula of organic compound **T**. [4]

..... [1]





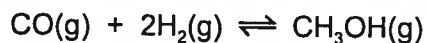


- (i) Give the letter of **one** compound that is an acid.

..... [1]

**9. M/J 19/P42/Q4/d**

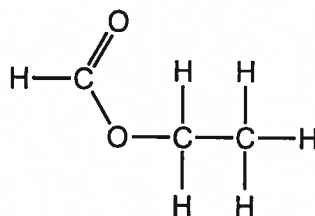
Methanol is made industrially by reacting carbon monoxide with hydrogen. The gases react at a temperature of 250 °C and a pressure of 75 atmospheres.



The forward reaction is exothermic.

- (a) Alcohols react with carboxylic acids to produce esters.

- (i) The structure of ester **X** is shown.



Name ester **X**.

..... [1]

- (ii) Give the name of the carboxylic acid and the alcohol that react together to produce ester **X**.

carboxylic acid .....

alcohol ..... [2]

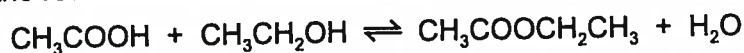
- (iii) Ester **Y** is different from ester **X** but also has the formula  $\text{C}_3\text{H}_6\text{O}_2$ .

Draw the structure of ester **Y**. Show all of the atoms and all of the bonds.

..... [2]

**10. M/J 19/P43/Q5/a(i)**

Carboxylic acids react with alcohols to form esters. The reaction is reversible. The equation for the reaction between ethanoic acid and ethanol is shown.



(a) (i) What is the name of the ester formed in this reaction?

..... [1]

**11. MAR 19/P42/Q4/d**

(a) Ethanoic acid reacts with methanol to form an ester.

(i) State **two** conditions required for this reaction.

1 .....

2 ..... [2]

(ii) Draw the structure of the ester formed when ethanoic acid reacts with methanol. Show all of the atoms and all of the bonds. Name the ester.  
structure

name ..... [3]

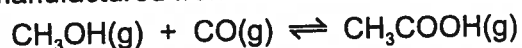
(iii) Name an ester which is a structural isomer of the ester in (d)(ii).

..... [1]

**12. Mar 19/P42/Q4**

This question is about ethanoic acid,  $\text{CH}_3\text{COOH}$ .

(a) Ethanoic acid is manufactured from methanol and carbon monoxide.



The process is done at  $200^\circ\text{C}$  and 30 atmospheres pressure.

The forward reaction is exothermic.

Complete the table using only the words *increases*, *decreases* or *no change*.

|                            | effect on the rate of the forward reaction | effect on the equilibrium yield of $\text{CH}_3\text{COOH(g)}$ |
|----------------------------|--------------------------------------------|----------------------------------------------------------------|
| adding a catalyst          |                                            | no change                                                      |
| increasing the temperature |                                            |                                                                |
| decreasing the pressure    | decreases                                  |                                                                |

[4]

(b) How would you show that an aqueous solution of ethanoic acid is an acid **without** using an indicator or measuring the pH?

State the reagent you would use and give the expected observations. Write a chemical equation for the reaction that you describe.

- reagent

.....

- expected observations

.....

- chemical equation

..... [3]

(c) Ethanoic acid is a weak acid.

(i) What is meant by the term *acid*?

..... [1]

(ii) Why is ethanoic acid described as *weak*?

..... [1]

(d) Ethanoic acid reacts with methanol to form an ester.

(i) State **two** conditions required for this reaction.

1 .....

2 ..... [2]

(ii) Draw the structure of the ester formed when ethanoic acid reacts with methanol. Show all of the atoms and all of the bonds. Name the ester.  
structure

name ..... [3]

(iii) Name an ester which is a structural isomer of the ester in (d)(ii).

..... [1]

**13. O/N 18/P42/Q5/e,f**

Alkynes are a homologous series of unsaturated hydrocarbons.

All members contain a  $\text{C}\equiv\text{C}$  triple bond.

(a) (i) Name an oxidising agent which can be used to oxidise ethanol to ethanoic acid.

..... [2]

(ii) Draw the structure of ethanoic acid. Show all of the atoms and all of the bonds.

[1]

(b) Carboxylic acids can be converted into esters.

(i) The ester formed by reacting propanoic acid and methanol has the molecular formula  $\text{C}_4\text{H}_8\text{O}_2$ .

Name this ester and draw its structure. Show all of the atoms and all of the bonds.

name of the ester .....

structure of the ester

[2]

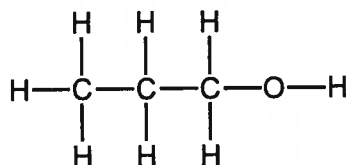
(ii) Name another ester with the molecular formula  $C_4H_8O_2$ .

..... [1]

**14. O/N 18/P41/Q6/c,d**

(a) Propan-1-ol is an alcohol.

The structure of propan-1-ol is shown.

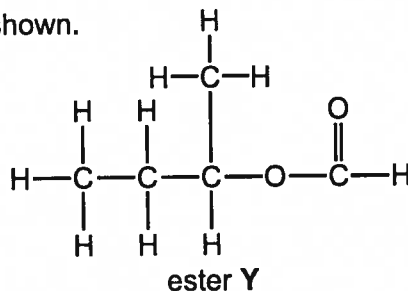


Propan-1-ol reacts with ethanoic acid to form an ester.

Give the name of the ester formed in this reaction.

..... [1]

(b) Ester Y has the structure shown.



(i) Give the molecular formula of ester Y.

..... [1]

(ii) Draw the structures of the carboxylic acid and the alcohol used to make ester Y. Show all of the atoms and all of the bonds. Give the name of the carboxylic acid and the alcohol.

structure of the carboxylic acid

name of the carboxylic acid .....

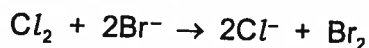
structure of the alcohol

name of the alcohol ..... [4]

**15. M/J 18/P42/Q4/f**

Potassium reacts with bromine at room temperature to form potassium bromide.

- (a) When chlorine gas is passed through aqueous potassium bromide, a redox reaction occurs. The ionic equation is shown.



- (i) Write an ionic half-equation showing what happens to the chlorine molecules,  $\text{Cl}_2$ , in this reaction.

..... [1]

- (ii) Explain why the bromide ions,  $\text{Br}^-$ , act as reducing agents in this reaction.

..... [1]

**16. M/J 18/P42/Q7/c,d**

Many organic compounds, such as alcohols, carboxylic acids and esters, contain the elements carbon, hydrogen and oxygen only.

- (a) Compounds **T** and **V** both have the molecular formula  $\text{C}_3\text{H}_6\text{O}_2$ .

- Compound **T** produces bubbles of carbon dioxide gas when it is added to aqueous sodium carbonate.
- Compound **V** is an ester.

- (i) What is the name given to compounds with the same molecular formula but different structures?

..... [1]

- (ii) Draw the structures of compounds **T** and **V**. Show all of the atoms and all of the bonds.  
compound **T**

compound **V**

[2]

- (iii) All compounds with the molecular formula  $\text{C}_3\text{H}_6\text{O}_2$  can undergo complete combustion in an excess of oxygen.

Complete the chemical equation for this reaction.



[2]

- (b) Compound **W** has the molecular formula  $\text{C}_2\text{H}_6\text{O}$ . Compound **W** reacts when heated with ethanoic acid and a catalyst to produce a sweet-smelling liquid.

- (i) Give the name of the homologous series to which compound **W** belongs.

..... [1]

(ii) Draw the structure of compound **W**. Show all of the atoms and all of the bonds.

[1]

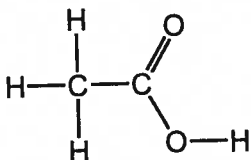
**17. O/N 17/P41/Q4/c,d,e**

(a) Ethanol can be oxidised to form ethanoic acid.

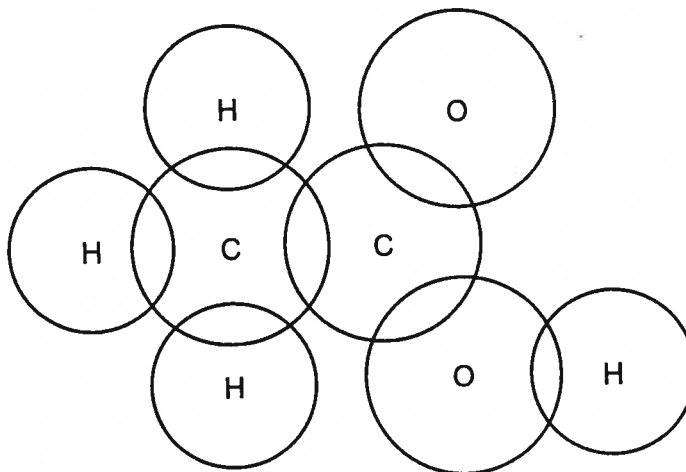
(i) Name a suitable oxidising agent for this reaction.

..... [1]

(ii) A molecule of ethanoic acid has the structure shown.



Complete the dot-and-cross diagram to show the electron arrangement in ethanoic acid.  
Show outer shell electrons only.



[3]

(b) Ethanoic acid is a weak acid.

(i) When referring to an acid, what is meant by the term *weak*?

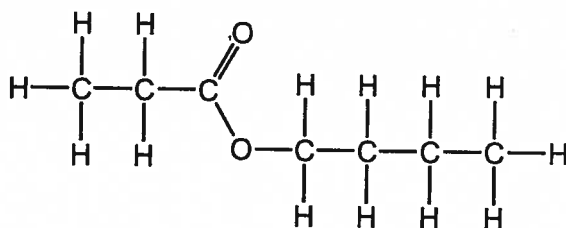
..... [1]

(ii) Describe how you could show that ethanoic acid is a weaker acid than hydrochloric acid.

.....  
.....  
..... [3]

- (c) Carboxylic acids react with alcohols to make esters.

The structure of an ester is shown.



Draw the structures of the carboxylic acid and alcohol from which this ester can be made.  
Give the names of the carboxylic acid and alcohol.

structure of the carboxylic acid .....

name of the carboxylic acid .....

structure of the alcohol .....

name of the alcohol .....[4]

**18. M/J 17/P42/Q6/b,c(i)**

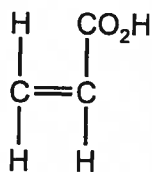
- (a) Propan-1-ol is a member of the homologous series of alcohols. It reacts in the same way as ethanol with acidified potassium manganate(VII) and with carboxylic acids.

Name the **type** of compound that is formed when propan-1-ol is heated with

acidified potassium manganate(VII), .....

ethanoic acid and a suitable catalyst. ....[2]

- (b) The structure of prop-2-enoic (acrylic) acid is shown.



- (i) What would you see if prop-2-enoic acid were added to

aqueous bromine, .....

a solution of sodium carbonate. ....[2]

**19. M/J 17/P41/Q6/e,f**

The alkenes and alkanes are both examples of homologous series which are hydrocarbons.

- (a) The alcohol butanol,  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ , can be converted into a carboxylic acid with four carbon atoms.

- (i) Name the carboxylic acid formed from butanol and draw its structure. Show all of the atoms and all of the bonds.

name .....

structure

[2]

- (ii) Ethanoic acid can be formed from ethanol by fermentation. It can also be formed by the addition of a suitable chemical reagent.

Name the reagent needed to convert ethanol into ethanoic acid.

[2]

- (iii) State the type of chemical change which occurs when ethanol is converted into ethanoic acid.

[1]

- (b) Describe how a student could prepare the ester methyl ethanoate in a school laboratory. In your description give

- the names of the **two** starting organic chemicals,
- the essential reaction conditions needed,
- a chemical equation for the reaction.

[5]

**20. M/J 16/P41/Q6/d(i,ii,iii)**

Petroleum is a source of many important chemicals.

- (a) Esters are organic chemicals noted for their characteristic smells. Ethanoic acid and methanol will react to form an ester.

- (i) Name the catalyst needed to form an ester from ethanoic acid and methanol.

[1]

- (ii) Name the ester formed when ethanoic acid reacts with methanol.

[1]

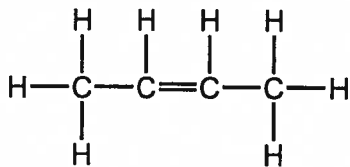
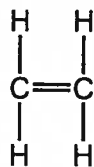
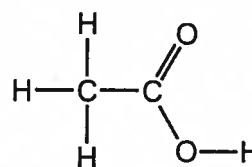
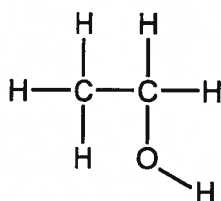
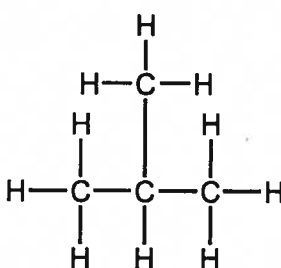
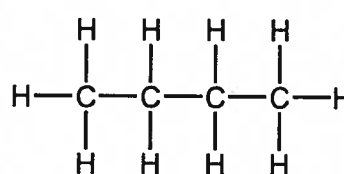
- (iii) Draw the structure of the ester formed when ethanoic acid reacts with methanol. Show all bonds.

[2]



**21. M/J 16/P43/Q2/a,b,c**

The structures of six organic compounds are shown.

**A****B****C****D****E****F**

(a) Give the name of **F**.

..... [1]

(b) Identify **two** of the compounds that are members of the same homologous series.

Give the general formula of this homologous series.

compounds .....

general formula ..... [2]

(c) Which **two** compounds are isomers of each other?

Explain why they are isomers.

compounds .....

explanation .....

..... [3]

## Answers Section

**1. O/N 22/P42/Q6**

- (a)  $C_nH_{2n+1}COOH$  1
- (b) methanoic acid 1
- (c)  $CH_2O_2$  1
- (d) **M1** all single bonding dot and cross pairs correct (1)  
**M2** double  $C=O$  bond dot and cross pairs are correct (1)  
**M3** four non-bonding cross electrons on each O atom (1) 3
- (e) (i) **M1** propyl methanoate (1)  
**M2** ester linkage with all atoms and all bonds (1)  
**M3** correct structure of propyl methanoate (1) 3
- (ii) water 1
- (iii) **M1** methyl propanoate (1)  
**M2** butanoic acid (1) 2

**2. M/J 22/P41/Q5**

- (a) the **rate** of forward reaction equals the rate of the reverse reaction (1)  
**concentrations** of reactants and products are constant (1) 2
- (b) reaction too slow (1)  
yield of ethanoic acid too low (1) 2
- (c) 

|               |               |
|---------------|---------------|
| increases (1) |               |
| decreases (1) | decreases (1) |

 3
- (d) cobalt (1)  
transition element (1) 2
- (e)  $C_nH_{2n+1}COOH$  1
- (f)  $COOH$  fully displayed (1)
- $\begin{array}{c} O \\ // \\ C \\ \backslash \\ O-H \end{array}$
- whole molecule completely correct (1) 2
- (g) (i) methyl butanoate 1
- (ii) butanoic acid (1)  
methanol (1) 2
- (h) C 48.65 / 12  
H 8.11 / 1  
O 43.24 / 16  
**OR**  
4.05:8.11:2.70 (1)  
fractions shown dividing all by smallest  
**OR**  
1.5:3:1  
**OR**  
3:6:2 (1)  
 $C_3H_6O_2$  (1) 3

(i)  $C_4H_8O_2$ 

1

## 3. M/J 22/P43/Q4

(a) (a substance which) increases the rate of a reaction  
remains unchanged or unaffected or without changing mass

2

(b)

|               |               |
|---------------|---------------|
| increases (1) | decreases (1) |
| decreases (1) | decreases (1) |

4

(c) (i)  $C_nH_{2n+1}OH$ 

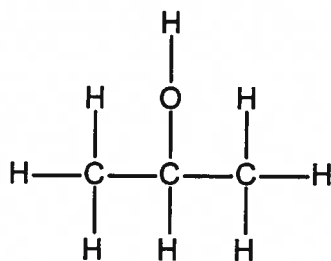
1

(ii) any 2 from:

- same or **similar** chemical properties or reactions
- (contain) the same functional group
- (consecutive members) differ by  $CH_2$
- physical properties vary in predictable manner / show trends / gradually change

2

(iii)



(1)

propan-2-ol (1)

2

(d) (i) ethyl butanoate

1

(ii) ethanol (1)

butanoic acid (1)

2

(e) C 58.82 / 12 H 9.80 / 1 O 31.37 / 16 OR

4.90:9.80:1.96 (1) OR

2.5:5:1

5:10:2 (1)

 $C_5H_{10}O_2$ (1)

3

(f)  $C_6H_{12}O_2$ 

1

## 4. MAR 21/P42/Q5

(a) R and U only

2

(b) aqueous bromine (1)

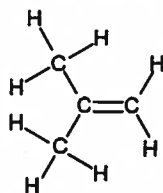
turns colourless / decolourises (1)

2

(c) (i) but-2-ene

1

(ii) structure of methyl propene



1

(d) (i)  $C_{12}H_{26}$ 

1

(ii) P, R and U

2

- (e) steam (1)  
catalyst (1)  
One other condition (1):  
either 60 atm or 300 °C

3

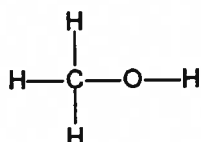
- (f) (i) oxidation  
(ii) potassium manganate(VII)

1

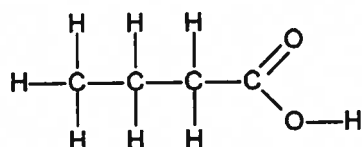
1

- (g) (i) ester  
(ii) name = methanol structure =

1



name = butanoic acid structure =



4

(iii)  $\text{C}_5\text{H}_{10}\text{O}_2$

1

5. O/N 20/P42/Q5/a(i,ii)

- (a) (i) presence of an acid  
(ii) oxidation

1

1

6. O/N 20/P41/Q7/c(i)

- (a) ethyl methanoate

1

7. O/N 20/P43/Q6/a(ii)

(a)

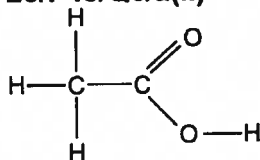


diagram (1)  
ethanoic acid (1)

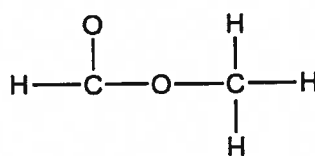


diagram (1)  
methyl methanoate (1)

4

8. M/J 20/P43/Q1/a,(vii)

- (a) A or E

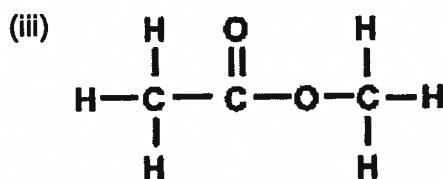
1

9. M/J 19/P42/Q4/d

- (a) (i) ethyl methanoate  
(ii) M1 methanoic acid (1)  
M2 ethanol (1)

1

2



M1 correct displayed ester linkage (1)  
M2 whole molecule fully correct (1)

2

10. M/J 19/P43/Q5/a(i)

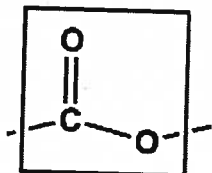
- (a) ethyl ethanoate

1

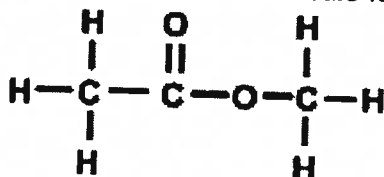
## 11. MAR 19/P42/Q4/d

- (a) (i) M1 heat(1)  
M2 catalyst / concentrated sulfuric acid(1)
- (ii) M1 Correct ester linkage(1)

2



M1 and M2 whole molecule fully correct(2)



M3 methyl ethanoate (1)

- (iii) ethyl methanoate

3

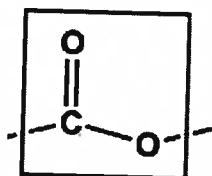
1

## 12. MAR 19/P42/Q4

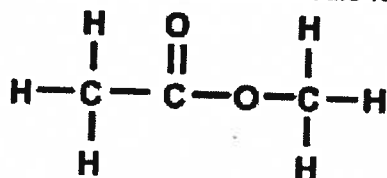
(a)

|                 |                 |
|-----------------|-----------------|
| M1 increases(1) | No change       |
| M2 increases(1) | M3 decreases(1) |
| decreases       | M4 decreases(1) |

- (b) M1 Suitable metal e.g. magnesium / any carbonate / any base(1) [4]  
M2 suitable observation e.g. insoluble base / insoluble carbonate/metal dissolve or disappear or metal / carbonate bubbles(1)  
M3 balanced equation fully correct(1)
- (c) (i) proton donor [3]  
(ii) ionises / dissociates partially or incompletely [1]
- (d) (i) M1 heat(1) [1]  
M2 catalyst / concentrated sulfuric acid(1)  
(ii) M1 Correct ester linkage(1) [2]



M1 and M2 whole molecule fully correct(2)



M3 methyl ethanoate (1)

- (iii) ethyl methanoate

[3]

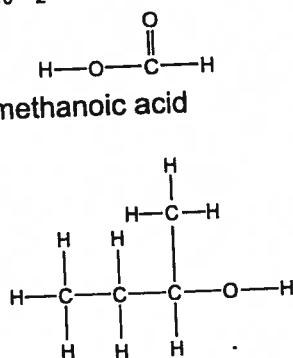
[1]

## 13. O/N 18/P42/Q5/e,f

- (a) (i) M1 Acidified; [2]  
M2 (Potassium) manganate (VII) [1]  
(ii) Diagram of ethanoic acid
- (b) (i) M1 Methyl propanoate [2]  
M2 Diagram of methyl propanoate [1]  
(ii) Any four carbon ester not named in 5(f)(i)

## 14. O/N 18/P41/Q6/c,d

- (a) propyl ethanoate [1]  
(b) (i)  $C_5H_{10}O_2$  [1]  
(ii) M1 [1]  
M2 methanoic acid [1]  
M3



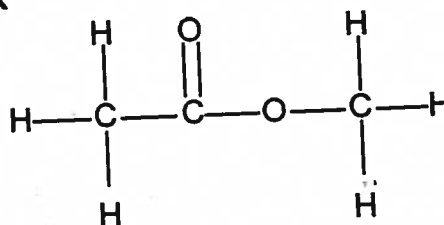
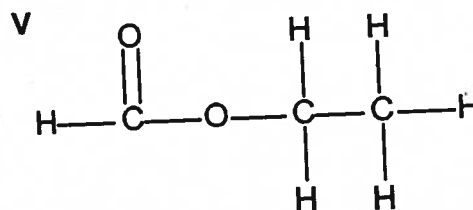
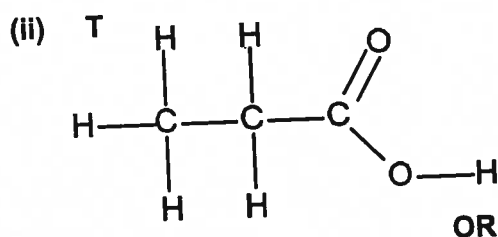
M4 butan-2-ol

## 15. M/J 18/P42/Q4/f

- (a) (i)  $Cl_2 + 2e(-) \rightarrow 2Cl^-$  [1]  
(ii) (bromide ions) lose electrons / donate electrons / are oxidised [1]

## 16. M/J 18/P42/Q7/c,d

- (a) (i) structural isomers [1]

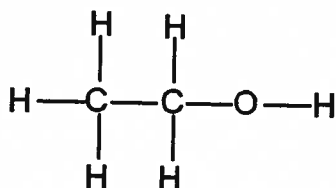


- (iii)  $C_3H_6O_2 + 3\frac{1}{2} O_2 \rightarrow 3CO_2 + 3H_2O$

1 mark for all formulae correct [2]  
1 mark for correct balancing [1]

- (b) (i) alcohol / alkanol

(ii)



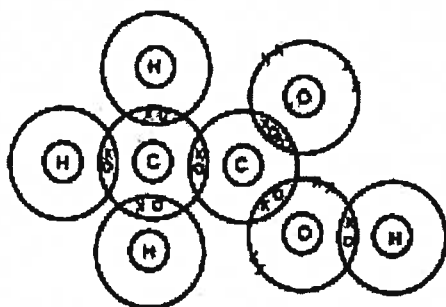
[1]

17. O/N 17/P41/Q4/c,d,e

(a) (i) (acidified) potassium manganate(VII) OR potassium (di)chromate(VI)

[1]

(ii)



M1 all shared pairs of electrons correct for single bonds

M2 2 shared pairs of electrons for the C=O bond

M3 total of 8 electrons on each O including 4 non-bonding electrons and no additional non-bonding electrons

[3]

(b) (i) partially ionised / dissociated

[1]

(ii) M1 (acids) have same concentration

[1]

M2: measure pH OR describe how to measure pH (such as use Universal Indicator)

M3: lower pH corresponds to the stronger acid / hydrochloric acid

OR

M2: add calcium / magnesium / zinc / iron

M3: faster rate of forming bubbles corresponds to the stronger acid / hydrochloric acid

OR

M2: rate of reaction with (metal) carbonate

M3: faster rate of forming bubbles corresponds to the stronger acid / hydrochloric acid

OR

M2: rate of reaction with (named) metal oxide

M3: dissolves faster means that reaction is with the stronger acid / hydrochloric acid

OR

M2: electrical conductivity

M3: greater conductivity corresponds to the stronger acid / hydrochloric acid

OR

M2: add sodium hydroxide (or other named alkali)

M3: greater temperature change corresponds to the stronger acid / hydrochloric acid [2]

(c) structure of propanoic acid

[1]

propanoic acid

[1]

structure of butan-1-ol

[1]

butan-1-ol

[1]

18. M/J 17/P42/Q6/b,c(i)

(a) carboxylic acid / aldehyde  
ester

[1]

[1]

(b) (i) colourless / decolourised  
bubbles / fizzing / effervescence

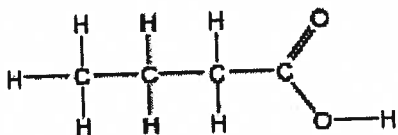
[1]

[1]

## 19. M/J 17/P41/Q6/e,f

(a) (i) butanoic acid

[1]



[1]

(ii) acidified  
(potassium) manganate(VII)

[1]

[1]

(iii) oxidation

[1]

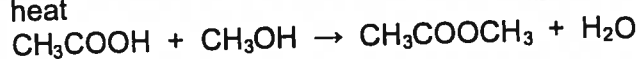
(b) methanol  
ethanoic acid  
catalyst  
heat

[1]

[1]

[1]

[1]



[1]

## 20. M/J 16/P41/Q6/d(i,ii,iii)

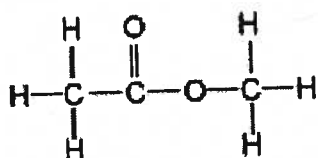
(a) (i) (concentrated) sulfuric acid;

[1]

(ii) methyl ethanoate;

[1]

(iii)



M1 ester link;

[2]

M2 rest of molecule;

## 21. M/J 16/P43/Q2/a,b,c

(a) butane;

[1]

(b) compounds: E and F;

[1]

general formula:  $\text{C}_n\text{H}_{2n+2}$ ;

[1]

OR

compounds: A and B;

[1]

general formula:  $\text{C}_n\text{H}_{2n}$ ;

[1]

(c) compounds: E and F;

[1]

explanation: same molecular formula / contain the same number of atoms each element;

different structures different structural formulae / different arrangement of atoms;

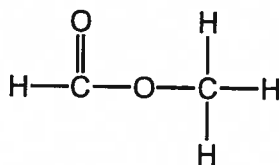
[2]



## 13.8: Polymers

### 1. O/N 22/P41/Q6

Ester Y has the structure shown.



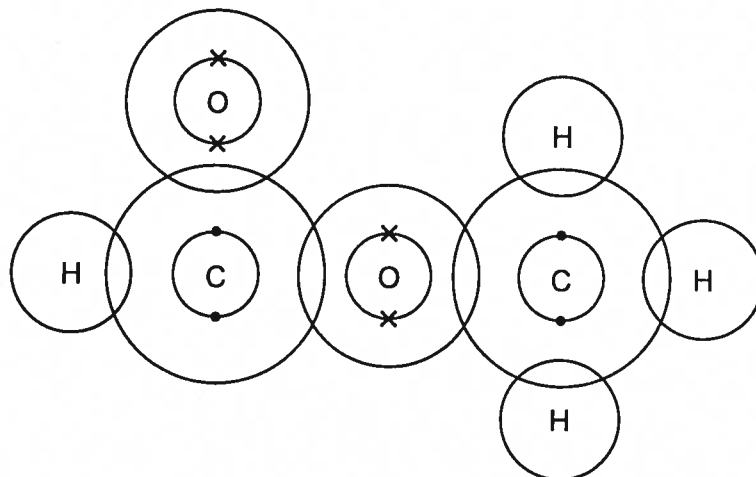
(a) (i) Name ester Y.

..... [1]

(ii) Deduce the empirical formula of ester Y.

..... [1]

(b) Complete the dot-and-cross diagram to show the arrangement of electrons in a molecule of ester Y.



[3]

(c) Ester Y can be made by reacting two organic compounds together.

Name the compounds and draw their structures.

Show all of the atoms and all of the bonds.

name .....

structure

name .....

structure

[4]

(d) (i) Describe what is meant by the term *structural isomer*.

.....  
 ..... [2]

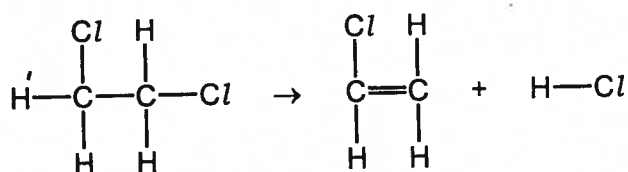
(ii) Name a carboxylic acid which is a structural isomer of ester Y.

..... [1]

2. O/N 22/P43/Q6

(a) Chloroethene ( $\text{CH}_2=\text{CHCl}$ ) can be manufactured from 1,2-dichloroethane ( $\text{CH}_2\text{ClCH}_2\text{Cl}$ ).

The equation can be represented as shown.



(i) Some bond energies are given.

| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| C—C  | 350                      |
| C=C  | 610                      |
| C—Cl | 340                      |
| C—H  | 410                      |
| H—Cl | 430                      |

Use the bond energies in the table to calculate the energy change, in kJ/mol, of the reaction.

Use the following steps.

- Calculate the energy needed to break bonds.

energy = ..... kJ

- Calculate the energy released when bonds form.

energy = ..... kJ

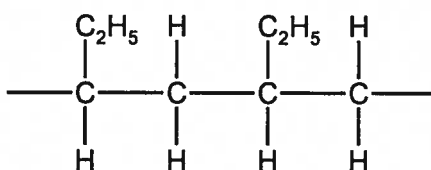
- Calculate the energy change of the reaction.

energy change of the reaction = ..... kJ/mol [3]

- (ii) Deduce whether the energy change for this reaction is exothermic or endothermic.  
Give a reason for your answer.

..... [1]

- (b) Part of a synthetic polymer is shown.



- (i) State the number of monomer units that are needed to make the part of the polymer shown.

..... [1]

- (ii) Name and draw the structure of the monomer used to make this polymer. Show all of the atoms and all of the bonds.

name .....

structure

[3]

- (iii) State the empirical formula of the polymer.

..... [1]

- (c) Proteins are natural polymers.

Proteins are broken down into amino acids. The process is similar to how complex carbohydrates are broken down to give simple sugars.

- (i) Name the type of reaction in which proteins are broken down into amino acids.

..... [1]

- (ii) Name **two** types of substance that are used to break down proteins into amino acids.

1 .....

2 ..... [2]

(iii) Amino acids are colourless.

A sample containing a mixture of amino acids is separated. Each amino acid is detected and identified.

- Name the process used to separate the amino acids.

.....

- Name the type of substance used to detect the amino acids.

.....

- Give the symbol of the value used to determine the identity of each amino acid after separation and detection.

..... [3]

(d) Proteins are natural polymers. Proteins contain amide linkages.

Synthetic polyamides also contain amide linkages.

(i) Name a synthetic polyamide.

..... [1]

(ii) Identify the **two** functional groups present in the monomers used to produce synthetic polyamides.

1 .....

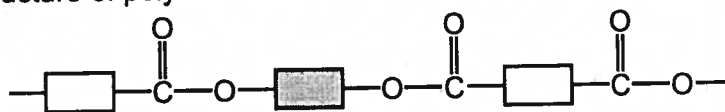
2 ..... [2]

### 3. M/J 22/P42/Q6

This question is about polymers.

(a) Polymer X is a condensation polymer.

Part of the structure of polymer X is shown.



(i) How many molecules of water are produced when this part of polymer X is formed from its monomers?

.....[1]

(ii) Complete the structures of the **two** monomers used to make polymer X.  
Show all of the atoms and all of the bonds in the functional groups.



and



[2]

(iii) What type of condensation polymer is **X**?

.....[1]

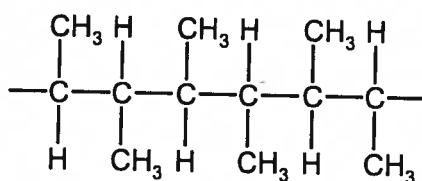
(b) Part of polymer **Y** has the structure shown.



State the number of different types of monomer needed to make polymer **Y**.

.....[1]

(c) Part of polymer **Z** has the structure shown.



(i) Draw and name the structure of the monomer which forms polymer **Z**.  
Show all of the atoms and all of the bonds.

name .....

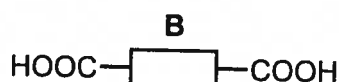
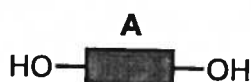
[3]

(ii) Name the chemical process used to make the monomer that forms polymer **Z**.

.....[1]

#### 4. M/J 21/P42/Q6

Molecules **A** and **B** can form condensation polymers.



(a) Each molecule has two identical functional groups.

(i) Name the functional group in **B**.

.....[1]

(ii) Draw the part of the structure of the synthetic polymer that would form when two molecules of **A** and two molecules of **B** combine. Show all of the bonds in the linkages.

[3]

(iii) Name the other product formed when molecules of **A** and **B** undergo polymerisation.

.....[1]

(b) Molecule **A** is a simple sugar unit which can be made by hydrolysis of complex carbohydrates.

- (i) Draw part of the complex carbohydrate that could be hydrolysed to make molecules of **A**.  
Include **one** linkage and show all of the bonds in the linkage.

[1]

- (ii) State **two** sets of conditions which could be used to hydrolyse the complex carbohydrate to form **A**.

1 .....

2 ..... [2]

- (iii) Name the technique used to identify the individual sugar units made by the hydrolysis of a complex carbohydrate.

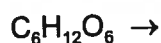
..... [1]

(c) Ethanol can be made from the simple sugar glucose,  $C_6H_{12}O_6$ .

- (i) State the name of this process.

..... [1]

- (ii) Complete the chemical equation for this reaction.



[2]

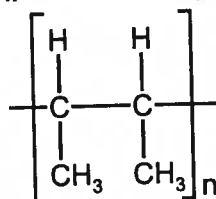
### 5. MAR 21/P42/Q6

Polymers are large molecules built up from small molecules.

- (a) State the name given to the small molecules from which polymers are made.

..... [1]

- (b) The formula of a polymer is shown.



- (i) Draw the structure of the small molecule from which this polymer is made. Show all of the atoms and all of the bonds.

[2]

- (ii) State the type of polymerisation used to make this polymer.

..... [1]

(c) Three amino acids are shown. They combine to form part of a natural polymer.



(i) Name the type of natural polymer formed when amino acids combine.

..... [1]

(ii) Complete the diagram to show part of the structure of the natural polymer that forms when these three amino acids combine. Show all of the bonds in the linkages.



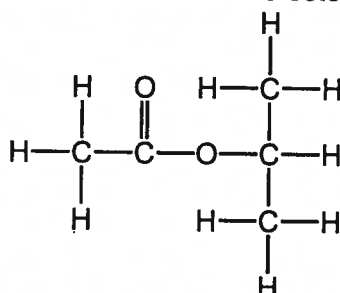
[3]

(iii) Name the type of chemical reaction that takes place when this natural polymer is converted back to amino acids.

..... [1]

6. O/N 20/P42/Q5/c(ii,iii),d

(a) Ethanoic acid will react with an alcohol to form the ester shown.



(i) Give **one** condition needed when ethanoic acid reacts with the alcohol to make this ester.

..... [1]

(iii) Draw the structure of the alcohol which was added to ethanoic acid to make this ester. Show all of the atoms and all of the bonds.

[2]

(b) Polyesters can be manufactured from carboxylic acids and alcohols.

Hexanedioic acid has the structure:  $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$ .

This structure can be simplified as shown.



Ethanediol has the structure:  $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$ .

This structure can be simplified as shown.



The functional groups are found at the end of each molecule.

(i) State what is meant by the term *functional group*.

..... [1]

(ii) Determine the empirical formula of hexanedioic acid.

..... [1]

(iii) Calculate the percentage by mass of oxygen present in ethanediol.

Give your answer to the nearest whole number.

..... % [2]

(iv) Complete the diagram to show a section of polyester manufactured from hexanedioic acid and ethanediol. Include all of the atoms and all of the bonds in the linkages.



(v) State the name of a polyester.

..... [1]

7. O/N 20/P41/Q7/c(ii)

(a) Ethanol reacts with methanoic acid to form an ester.

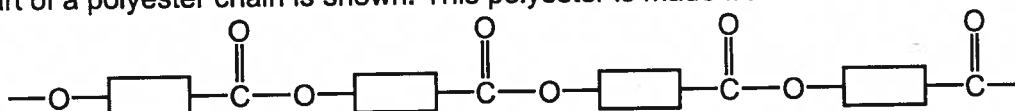
(i) Draw the structure of the ester formed.

Show all of the atoms and all of the bonds.

[1]

8. O/N 20/P43/Q6/b,c,d

(a) Part of a polyester chain is shown. This polyester is made from one monomer.



(i) On the diagram draw a ring around one unit of the polymer that is repeated.

[1]

(ii) Name the type of polymerisation that produces polyesters.

..... [1]

(iii) Complete the diagram to show the structure of the monomer used to produce this polyester. Show all of the atoms and all of the bonds in the functional groups.



(b) A polyamide is made from the two monomers shown.



Complete the diagram to show a section of the polyamide made from the two monomers. Show all of the atoms and all of the bonds in the linkages.



(c) Naturally occurring polyamides are constituents of food.

(i) State the name given to naturally occurring polyamides.

..... [1]



(ii) Name the monomers which form naturally occurring polyamides.

.....[1]

9. M/J 20/P42/Q6/d

(a) Compound A can be used as a single monomer to produce two different polymers.

(i) Draw **one** repeat unit of the addition polymer formed from compound A.

[2]

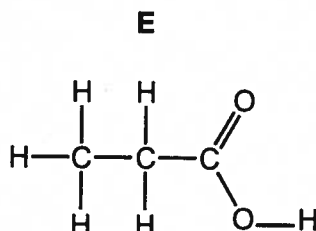
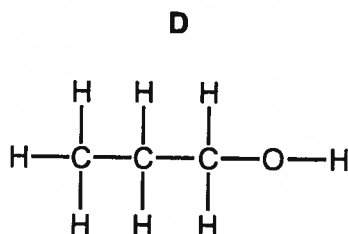
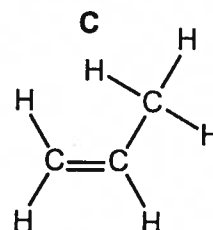
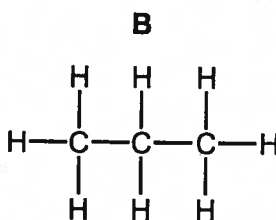
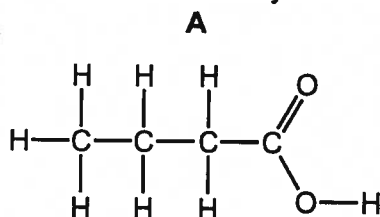
(ii) What type of condensation polymer is formed from compound A?

.....[1]

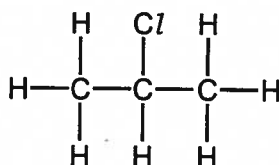
10. M/J 20/P43/Q1/a(iv,v)

(a) The structures of five organic compounds, A, B, C, D and E, are shown.  
Answer the questions that follow.

Each letter may be used once, more than once or not at all.



(i) Give the letter of **one** compound that reacts with chlorine to form the compound shown.



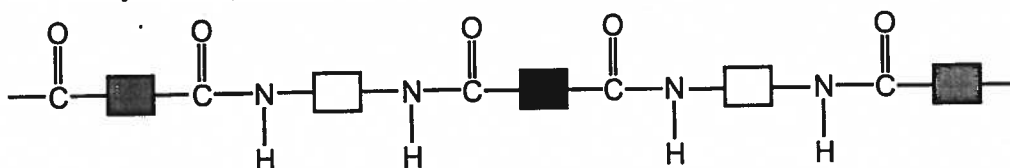
.....[1]

(ii) Give the letters of **two** compounds that can react with each other to form an ester.

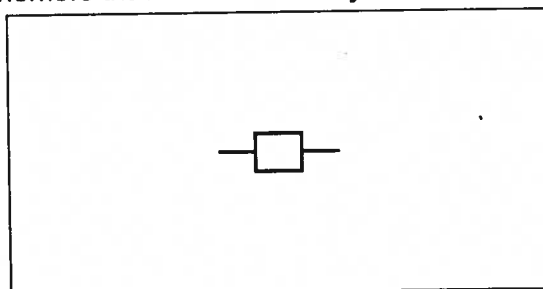
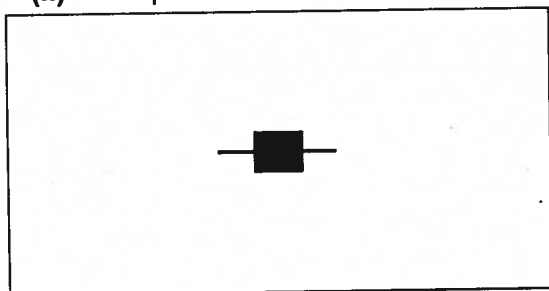
..... and .....[1]

## 11. M/J 20/P43/Q8

(a) Part of the synthetic polymer, nylon, is shown in the diagram.

(i) Circle **one** amide linkage on the diagram.

[1]

(ii) Complete the structures of the **two** monomers that react to form nylon.

[2]

(iii) Name the other product formed when nylon is produced.

.....[1]

(b) Items made from nylon are often disposed of by burying them in the ground. This is called landfill.

Why is the disposal of nylon using landfill a problem?

.....[1]

(c) Give the name of a natural polymer.

.....[1]

## 12. MAR 20/P42/Q5

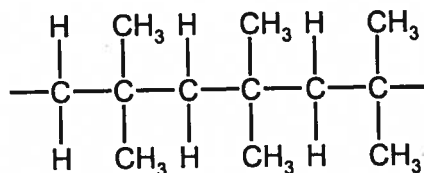
There are two types of polymers.

(a) Addition polymers are made from many identical small units.

(i) What is the term used to describe these small units?

.....[1]

(ii) A section of an addition polymer is shown.

Draw the structure of the small unit used to make this addition polymer.  
Show all of the atoms and all of the bonds.

[2]

(b) Polyamides are condensation polymers.

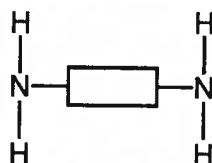
What does the term *condensation* mean when used to describe this type of polymer?

- .....[1]
- (c) A polyamide can be made from two different molecules.

A simplified structure of octanedioic acid is shown.



A simplified structure of 1,6-diaminohexane is shown.



- (i) Complete the diagram to show a section of polyamide manufactured from octanedioic acid and 1,6-diaminohexane. Include all of the atoms and all of the bonds in the linkages.



[3]

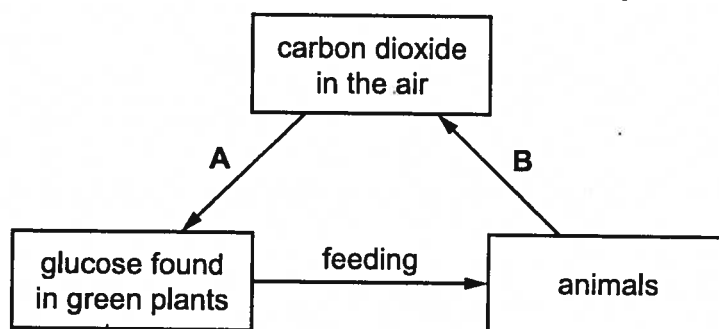
- (ii) State the name of a synthetic polyamide.

.....[1]

### 13. O/N 19/P42/Q2/g(ii,iii,iv)

- (a) The  $\text{CO}_2$  in air is part of the carbon cycle.

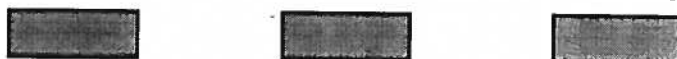
The scheme shows a simple representation of part of the carbon cycle.



- (i) Plants convert glucose into complex carbohydrates.

A unit of glucose can be represented as  $\text{HO}-\text{[ ]}-\text{OH}$ .

Complete the diagram to show the complex carbohydrate formed from **three** units of glucose. Show all of the atoms and all of the bonds in the linkages.



[2]

- (ii) Complex carbohydrates break down to form simple sugars.

State **two** ways that complex carbohydrates can be broken down into simple sugars.

1.....

2.....[2]

- (iii) Name a suitable technique for separating and identifying the individual sugars formed when complex carbohydrates are broken down.

.....[1]

14. O/N 19/P42/Q5/c(i,ii,iv)

- (a) Methanol reacts with propanoic acid to form an ester with a molecular formula  $C_4H_8O_2$ .

- (i) Name the ester formed when methanol reacts with propanoic acid.

.....[1]

- (ii) Name **one** other substance formed when methanol reacts with propanoic acid.

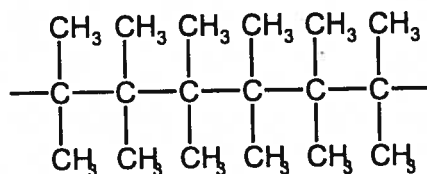
.....[1]

- (iii) State the conditions needed to form an ester from a carboxylic acid and an alcohol.

.....[1]

15. O/N 19/P41/Q5

- (a) Part of the structure of synthetic polymer **A** is shown.



- (i) What type of synthetic polymer is **A**?

.....[1]

- (ii) Deduce the empirical formula of polymer **A**.

.....[1]

- (iii) Draw the structure of the monomer from which polymer **A** is made.

[2]

- (b) The formula  $C_4H_{10}$  represents two different structural isomers.

- (i) What is meant by the term *structural isomers*?

.....

.....[2]

- (ii) Draw the structures of **two** structural isomers with the formula  $C_4H_{10}$ .  
Show all of the atoms and all of the bonds.

[2]

- (iii) All structural isomers of  $C_4H_{10}$  are flammable.

Write a chemical equation for the **incomplete** combustion of  $C_4H_{10}$ .

.....[2]

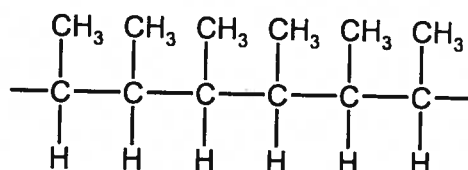
#### 16. O/N 19/P43/Q7

Addition polymerisation and condensation polymerisation are two types of polymerisation.

- (a) Which functional group is present in all the monomers which are used to make addition polymers?

.....[1]

- (b) Part of an addition polymer is shown.



- (i) How many monomer units are needed to make the part of the addition polymer shown?

..... [1]

- (ii) Draw the structure of the monomer that is used to make this addition polymer. Show all of the atoms and all of the bonds.

Name the monomer.

name ..... [2]

- (iii) State the empirical formula of:

the monomer .....

the polymer. .... [2]

- (c) Complex carbohydrates are natural condensation polymers. They can be broken down into colourless monomers which can then be separated and identified.

**X** is a complex carbohydrate.

Starting with a sample of **X**, describe how to produce, separate, detect and identify the monomers which make it up.

Your answer should include:

- the name of the process used to break down **X** into its monomers
- **two** types of substance that can be used to break down **X**
- the name of the process used to **separate** the monomers
- the method used to **detect** the monomers after they have been separated
- the method used to **identify** the monomers after they have been separated and detected.

.....

.....

.....

.....

- .....[6]
- (d) Synthetic polyamides are condensation polymers.

(i) Name a synthetic polyamide.

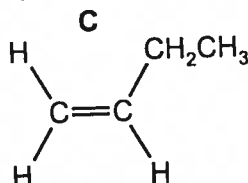
..... [1]

- (ii) Synthetic polyamides can be made by reacting carboxylic acids with amines.  
Name the other substance that is produced in this reaction.

..... [1]

17. M/J 19/P41/Q5/g

- (a) Alkene C can be converted into a polymer.



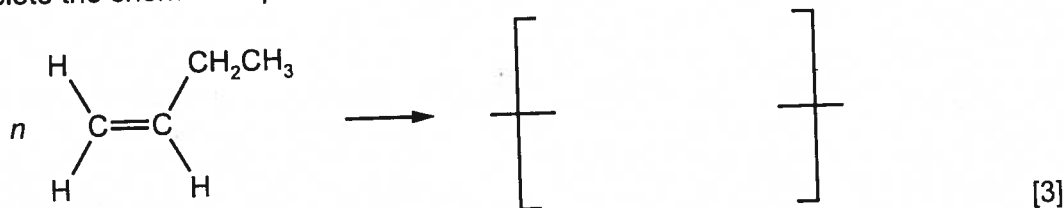
- (i) What type of polymerisation occurs?

..... [1]

- (ii) Suggest the name of the polymer formed.

..... [1]

- (iii) Complete the chemical equation to show this polymerisation.



[3]

- (iv) State the empirical formula of the polymer formed.

..... [1]

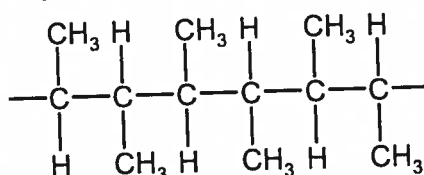
18. M/J 19/P43/Q6/c,d

- (a) Alkenes can form polymers.

- (i) What type of polymerisation occurs when alkenes form polymers?

..... [1]

- (ii) Part of the structure of a polymer is shown.

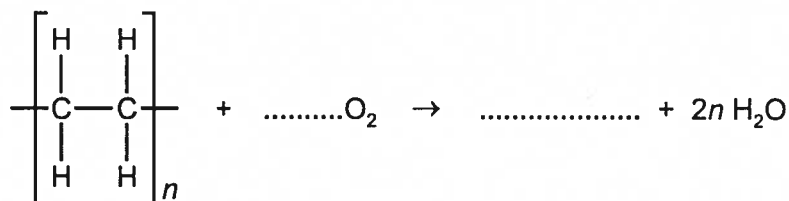


Draw the structure of the alkene from which this polymer can be made. Show all of the atoms and all of the bonds.

[1]

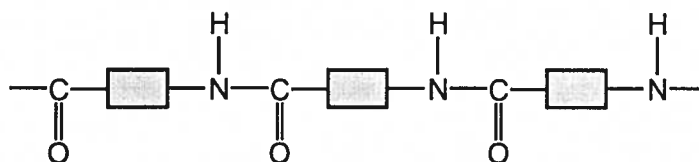
(iii) Polymers can undergo incomplete combustion to form carbon monoxide.

Complete the chemical equation for the incomplete combustion of poly(ethene). The only carbon-containing product is carbon monoxide.



[2]

(b) Part of the structure of a polyamide is shown.



This polyamide is formed from identical monomers. Complete the diagram to show the structure of **one** monomer. Show all of the atoms and all of the bonds.



[2]

### 19. MAR 19/P42/Q3/f(iv)

(a) Tetrafluoroethene is an unsaturated compound with the formula  $\text{C}_2\text{F}_4$ .

Tetrafluoroethene is the monomer used to make the polymer poly(tetrafluoroethene).

(i) Tetrafluoroethene can be polymerised to form poly(tetrafluoroethene).

Draw **one** repeat unit of poly(tetrafluoroethene). Show all of the atoms and all of the bonds.

[2]

### 20. M/J 16/P41/Q6/d

Petroleum is a source of many important chemicals.

(a) Esters are organic chemicals noted for their characteristic smells. Ethanoic acid and methanol will react to form an ester.

(i) Name the catalyst needed to form an ester from ethanoic acid and methanol.

..... [1]

(ii) Name the ester formed when ethanoic acid reacts with methanol.

..... [1]

(iii) Draw the structure of the ester formed when ethanoic acid reacts with methanol. Show all bonds.

[2]

(iv) Give the name of a polyester.

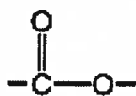
..... [1]



## Answers Section

1. O/N 22/P41/Q6
  - (a) (i) methyl methanoate 1
  - (a) (ii) CH<sub>2</sub>O 1
  - (b) M1 all single bonding dot and cross pairs correct (1)
  - M2 double C=O bond dot and cross pairs are correct (1)
  - M3 complete diagram is correct (1) 3
  - (c) M1 methanoic acid (1)
  - M2 structure of methanoic acid (1)
  - M3 methanol (1)
  - M4 structure of methanol (1) 4
  - (d) (i) M1 same (molecular) formula (1)
  - M2 different structural formula (1) 2
  - (ii) ethanoic acid 1
2. O/N 22/P43/Q6
  - (a) (i) M1 2670 (1)
  - M2 2610 (1)
  - M3 (+) 60 (1) 3
  - (ii) endothermic **AND** energy released when bonds form is less than energy absorbed to break bonds **OR** endothermic **AND** overall energy change has a positive sign 1
  - (b) (i) 2 1
  - (ii) M1 but-1-ene (1)
  - M2 C=C (both C atoms should have 4 bonds) (1)
  - M3 Displayed formula of but-1-ene (1) 3
  - (iii) CH<sub>2</sub> 1
  - (c) (i) hydrolysis 1
  - (ii) M1 acids (1)
  - M2 enzymes (1) 2
  - (iii) M1 chromatography (1)
  - M2 locating agent (1)
  - M3 R<sub>f</sub> (1) 3
  - (d) (i) nylon 1
  - (ii) M1 carboxylic acid (1)
  - M2 amine (1) 2
3. M/J 22/P42/Q6
  - (a) (i) 3 1
  - (ii) diagram of dioic acid showing two fully displayed carboxylic acid groups on unshaded block
  - diagram of diol showing two fully displayed hydroxyl groups on shaded block 2
  - (iii) polyester 1

- (b) one 1
- (c) (i) any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1) 3  
structure of but-2-ene (1) 1  
but-2-ene 1  
(ii) cracking 1
4. **M/J 21/P42/Q6** 1
- (a) (i) carboxylic acid 1  
(ii) any correct displayed ester link between any two blocks showing all atoms and all bonds (1) correct orientation of three displayed inter-block ester links with correct orientation (1) continuation bonds on polyester (1) 3  
(iii) water 1
- (b) (i) two blocks linked by -O- 1  
(ii) acid (and heat) (1) 2  
enzymes (ignore names) (1) 1  
(iii) chromatography 1
- (c) (i) fermentation 1  
(ii)  $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$   
 $C_2H_5OH$  (1) 2  
rest of equation (1) 1
5. **MAR 21/P42/Q6** 1
- (a) monomers 1
- (b) (i) any hydrocarbon with one C=C bond (with both C atoms having 4 bonds) (1)  
structure of but-2-ene (1)
- $$\begin{array}{ccccccc}
 & H & & H & & H & & H \\
 & | & & | & & | & & | \\
 H & -C & - & C & = & C & - & C & -H \\
 & | & & & & & & | \\
 & H & & & & & & H
 \end{array}$$
- (ii) addition 2  
1
- (c) (i) protein 1
- (ii)
- $$\begin{array}{ccccccc}
 & H & & O & & O & & O \\
 & | & & || & & || & & || \\
 -N & - & \blacksquare & -C & -N & - & \blacksquare & -C & -N & - & \square & -C & - \\
 & & & & | & & & & | & & & & \\
 & & & & H & & & & H & & & & 
 \end{array}$$
- any correct amide link between any two blocks showing all atoms and all bonds (1)  
identical orientation of both inter-block amide links including terminal groups with  
correct orientation (if shown) (1) continuation bonds on polymer (1) 3  
1
- (iii) hydrolysis (1)
6. **O/N 20/P42/Q5/c(ii,iii),d** 1
- (a) (i) acid (catalyst) 1  
(ii) any one alcohol group on a molecule (1) 2  
structure of propan-2-ol (1) 1
- (b) (i) the atoms/group of atoms which give (any molecule its) chemical properties 1  
(ii)  $C_3H_5O_2$  1  
(iii)  $M_r$  of  $HO-CH_2-CH_2-OH = 62$  (1) 2  
 $\%ge = 100 \times 32/62 = 52\%$  (1)
- (iv) any correct ester link between any two blocks showing all atoms and all bonds (1)

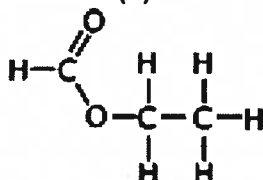


- correct orientation of 3 inter-block ester links and continuation bonds (1)  
(v) Terylene

2  
1

## 7. O/N 20/P41/Q7/c(ii)

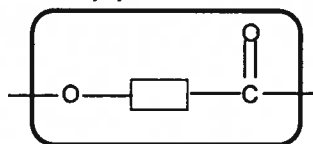
(a) (i)



1

## 8. O/N 20/P43/Q6/b,c,d

(a) (i)

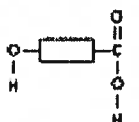


1

(ii) condensation

1

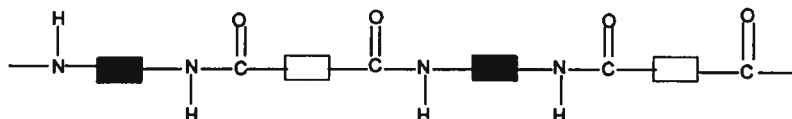
(iii)



1 for each correct functional group

2

(b)



one correct linkage fully displayed (1)  
the whole structure fully correct (1)

2

(c) (i) proteins

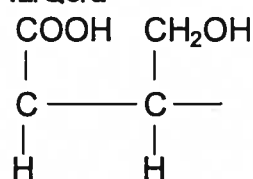
1

(ii) amino acids

1

## 9. M/J 20/P42/Q6/d

(a) (i)



C-C (1)

2

each C bonded to  $-\text{CH}_2\text{OH}$  /  $-\text{COOH}$  and H with no other atoms + extension bonds (1)

(ii) polyester

1

## 10. M/J 20/P43/Q1/a(iv,v)

(a) (i) B

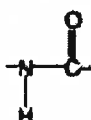
1

(ii) D AND E or A

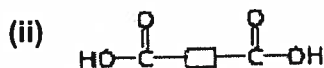
1

## 11. M/J 20/P43/Q8

(a) (i)



1



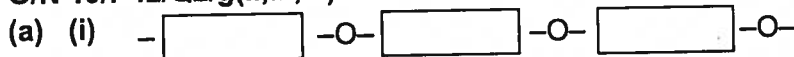
(iii) water

- (b) non-biodegradable / running out of space / toxic / leaching  
(c) (complex) carbohydrate / protein

12. MAR 20/P42/Q5

- (a) (i) monomer  
(ii) any **hydrocarbon** with one C=C bond (with both C atoms having 4 bonds) (1)  
structure of methylpropene (1)  
(b) water is a product (when polymer is made)  
(c) (i) any correct amide link between any two blocks showing all atoms and all bonds (1)  
correct orientation of 3 inter-block amide links (1)  
continuation bonds (1)  
(ii) nylon

13. O/N 19/P42/Q2/g(ii,iii,iv)



two -O- link between three 'blocks' (1)  
three complete units with continuation bonds (1)

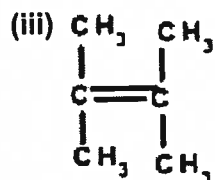
- (ii) acid (and heat) (1)  
enzymes (ignore names) (1)  
(iii) chromatography

14. O/N 19/P42/Q5/c(i,ii,iv)

- (a) (i) methyl propanoate  
(ii) water  
(iii) (acid) catalyst

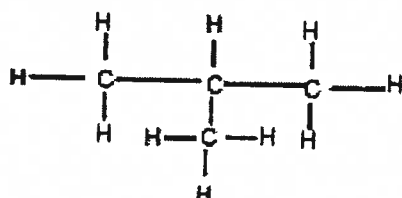
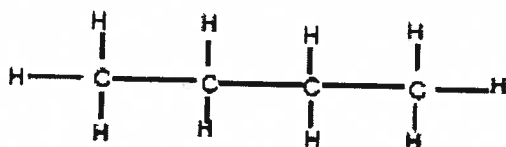
15. O/N 19/P41/Q5

- (a) (i) addition  
(ii) CH<sub>2</sub>

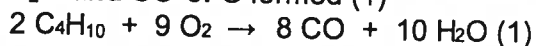


- one C=C (1)  
fully correct structure (1)  
(b) (i) (compounds / molecules with) the same molecular formula (1)  
different structural formulae (1)

(ii)



2

(iii)  $\text{H}_2\text{O}$  and  $\text{CO}$  or  $\text{C}$  formed (1)

2

## 16. O/N 19/P43/Q7

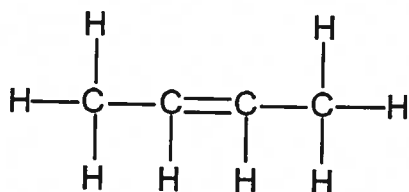
(a) carbon-carbon double bond /  $\text{C} = \text{C}$ 

1

(b) (i) 3

1

(ii)



but-2-ene (1)

2

(iii)  $\text{CH}_2$  $\text{CH}_2$ 

2

(c) (broken down by) hydrolysis (1)

acid (used to break down) (1)

enzymes (used to break down) (1)

chromatography (used to separate) (1)

locating agent / (view under) UV light (used to detect) (1)

measure  $R_f$  (values) or retention factor / compare with standards (used to identify) (1)

6

(d) (i) Nylon / Kevlar

1

(ii) water

1

## 17. M/J 19/P41/Q5/g

(a) (i) addition

1

(ii) poly(but-1-ene)

1

(iii) M1 2 C atoms (only) with a single bond between them linked to the continuation bonds shown

M2 correct repeat unit showing one  $\text{C}_2\text{H}_5/\text{CH}_2\text{CH}_3$  side chain attached to one of the C atoms in M1

M3 correct use of 'n'

3

(iv)  $\text{CH}_2$ 

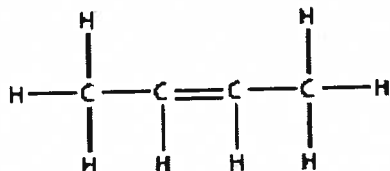
1

18. M/J 19/P43/Q6/c,d

1

(a) (i) addition

(ii)

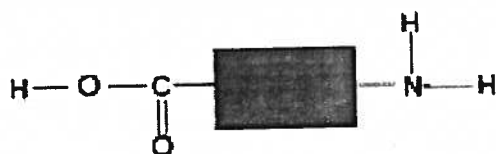


1

(iii) M1 CO on right  
M2  $2n \text{ O}_2 \rightarrow 2n(\text{CO})$ 

2

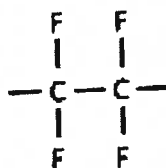
(b)

M1  $\text{-NH}_2$  group drawn as displayed on one end  
M2 carboxylic acid group drawn as displayed on the other end

2

19. MAR 19/P42/Q3/f(iv)

(a) (i)



M1 C-C(1)

M2 each C bonded to 2 F and no other atoms + extension bonds(1)

2

20. M/J 16/P41/Q6/d

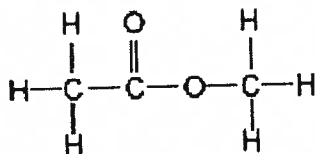
(a) (i) (concentrated) sulfuric acid;

[1]

(ii) methyl ethanoate;

[1]

(iii)



M1 ester link;

M2 rest of molecule;

[2]

(iv) terylene;

[1]